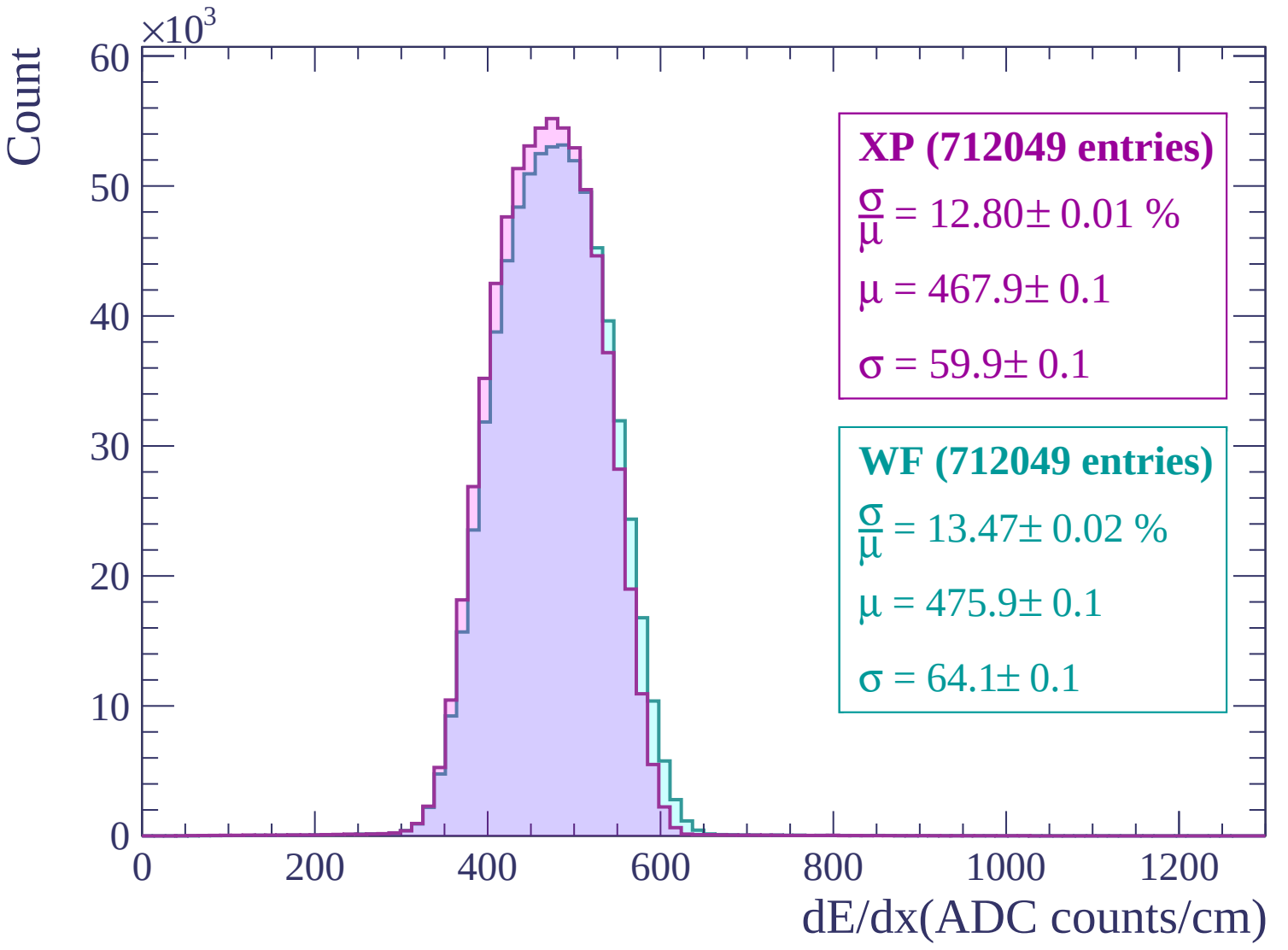
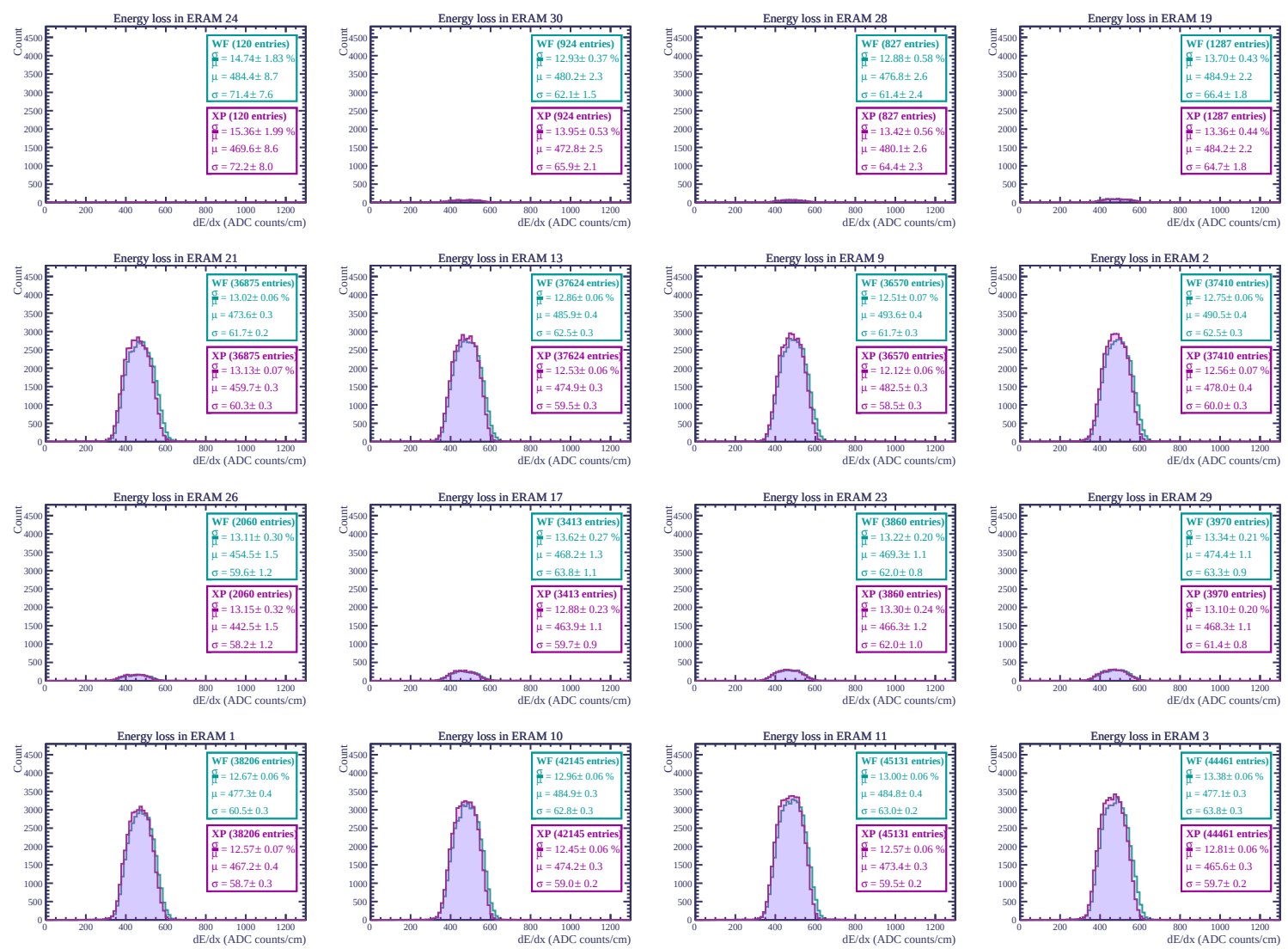
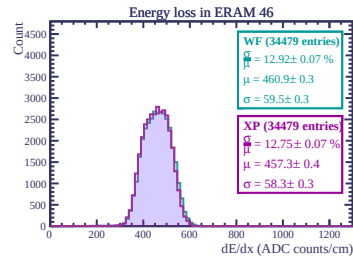
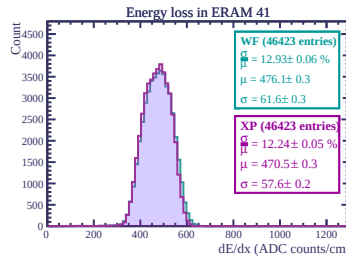
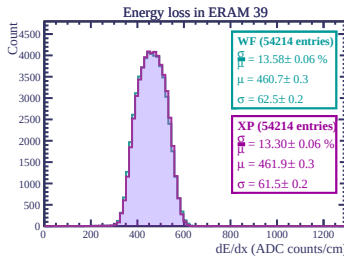
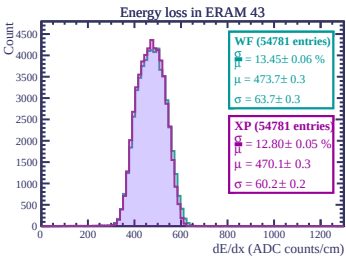
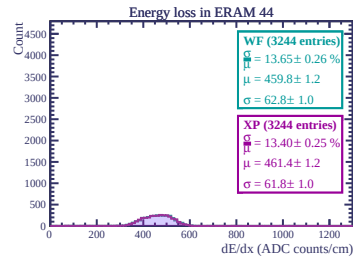
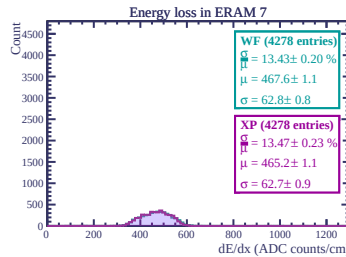
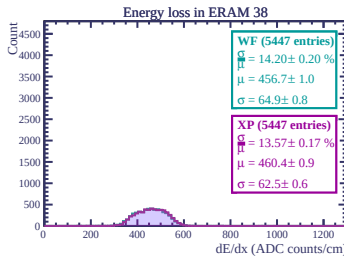
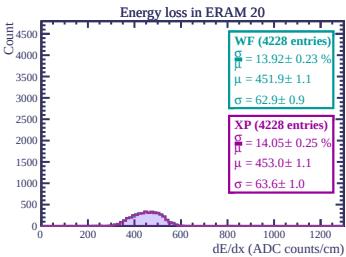
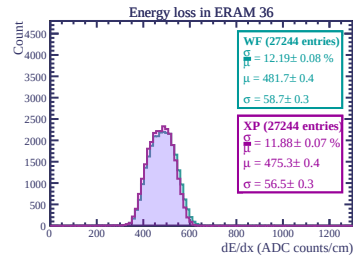
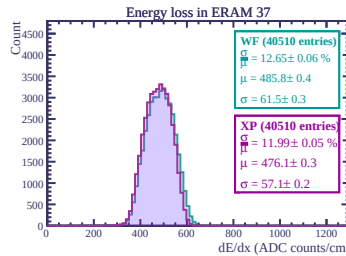
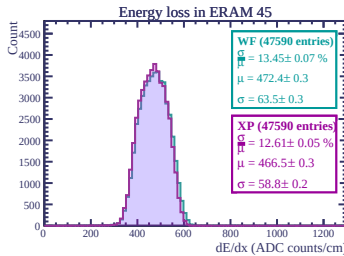
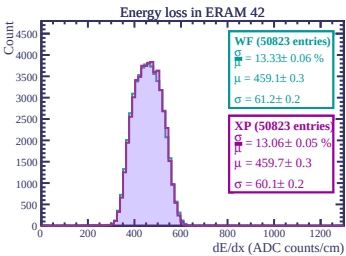
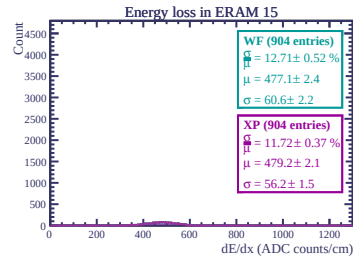
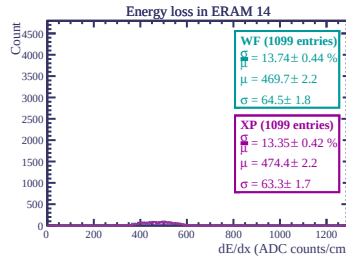
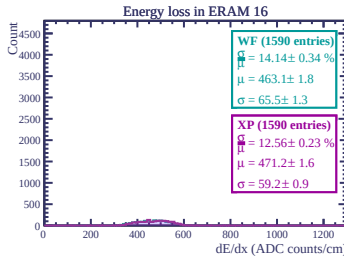
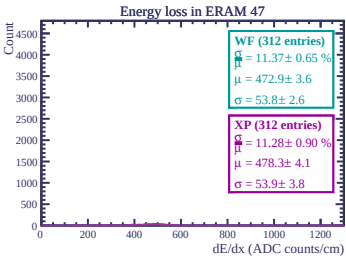
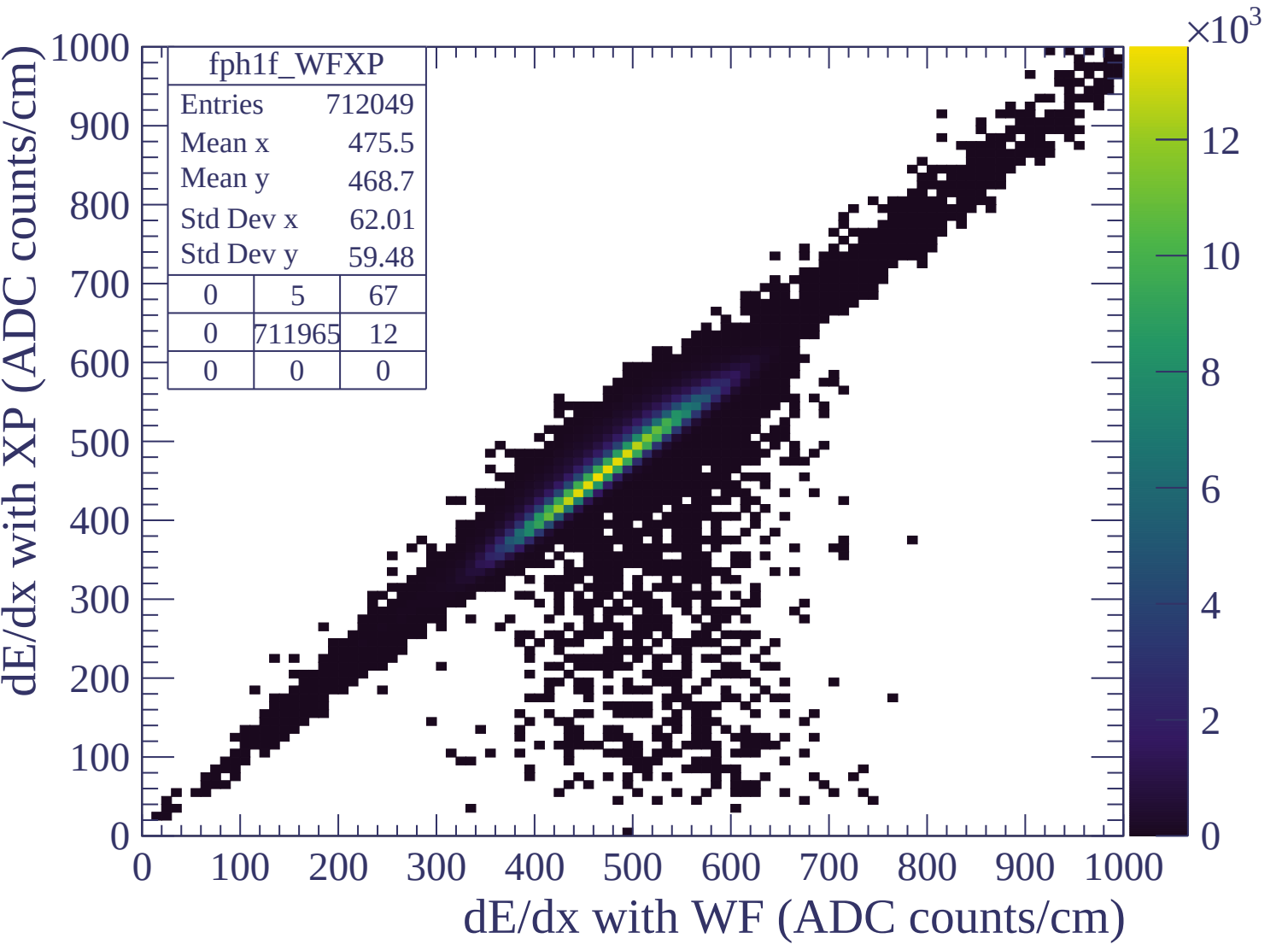
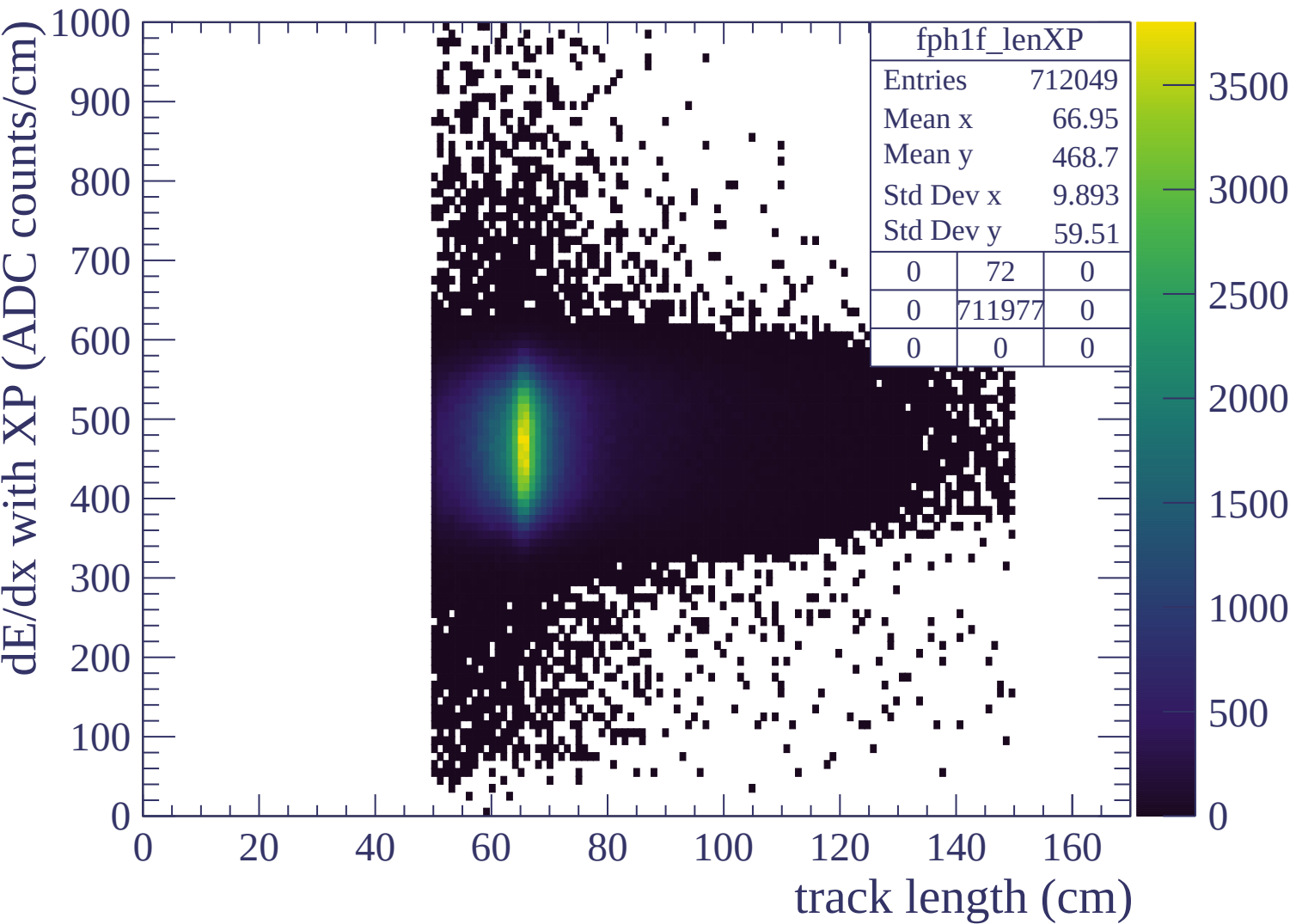


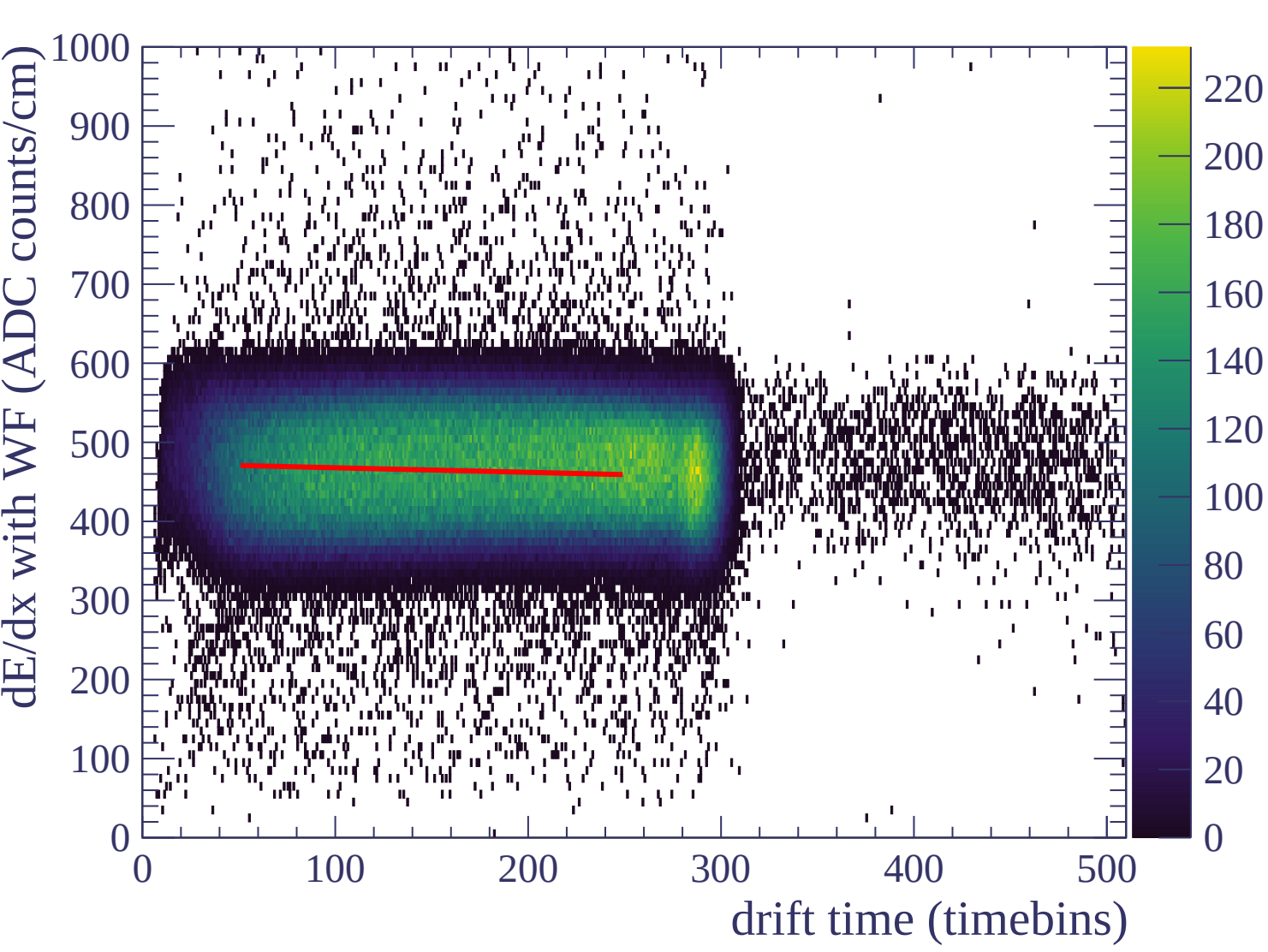


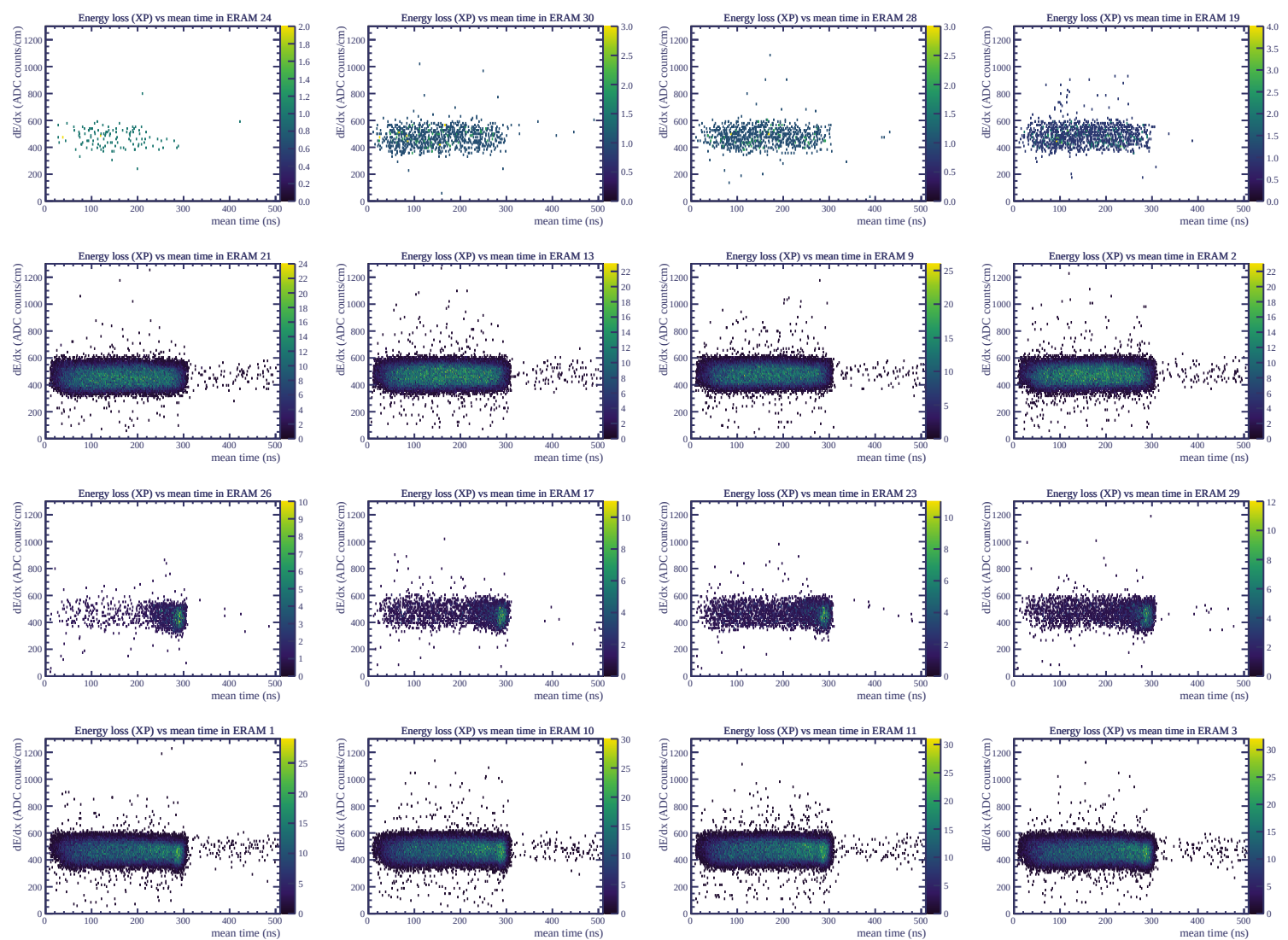


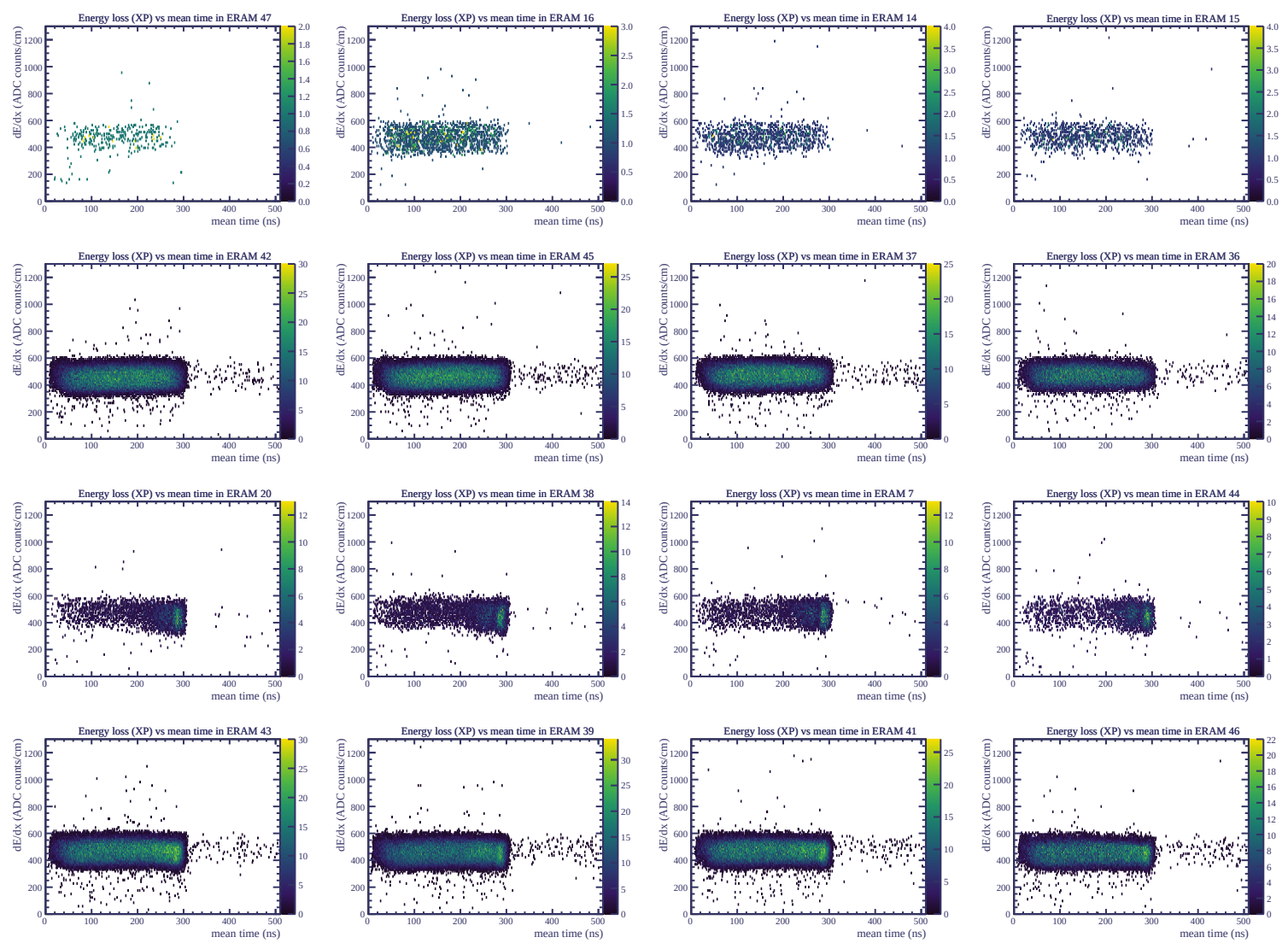




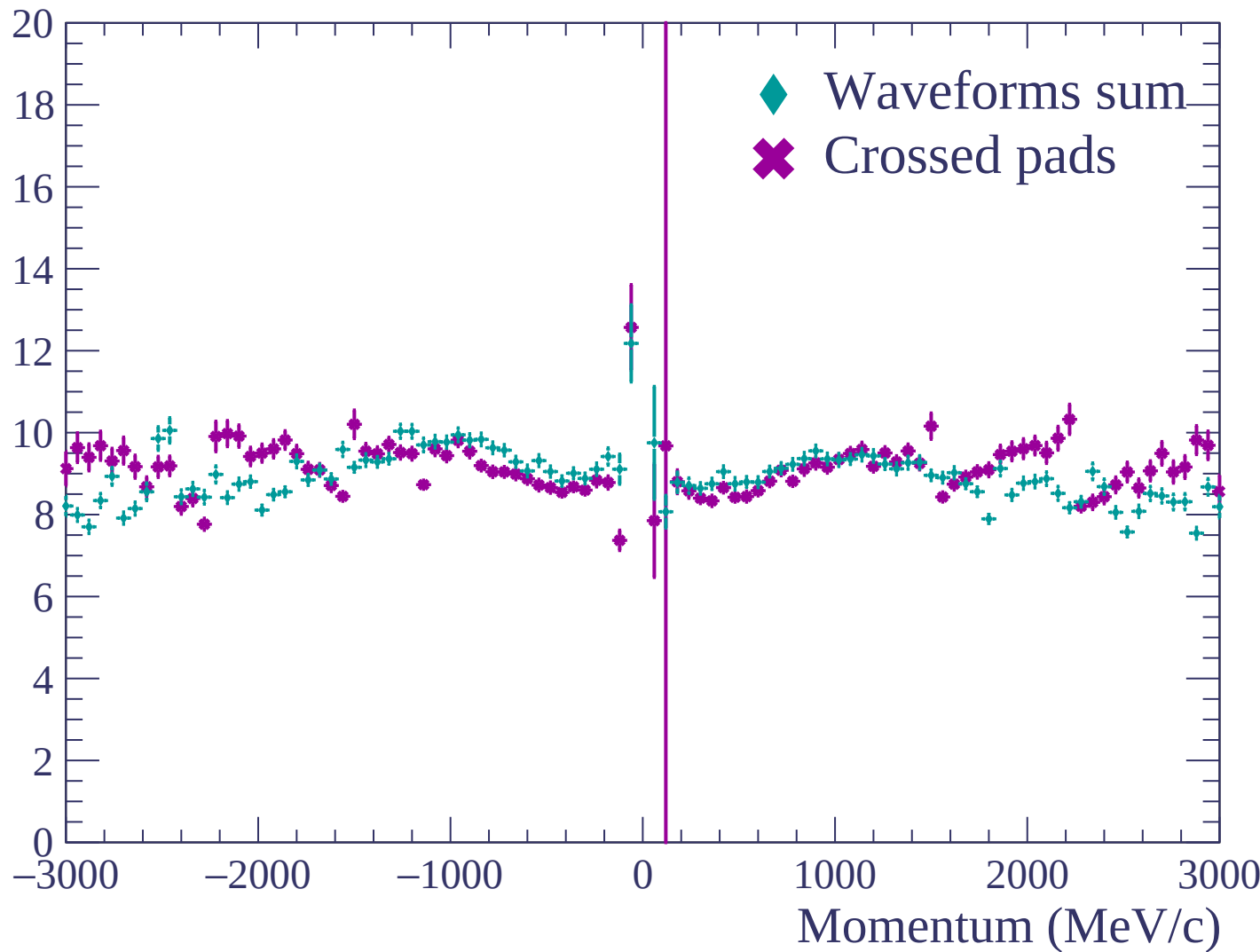


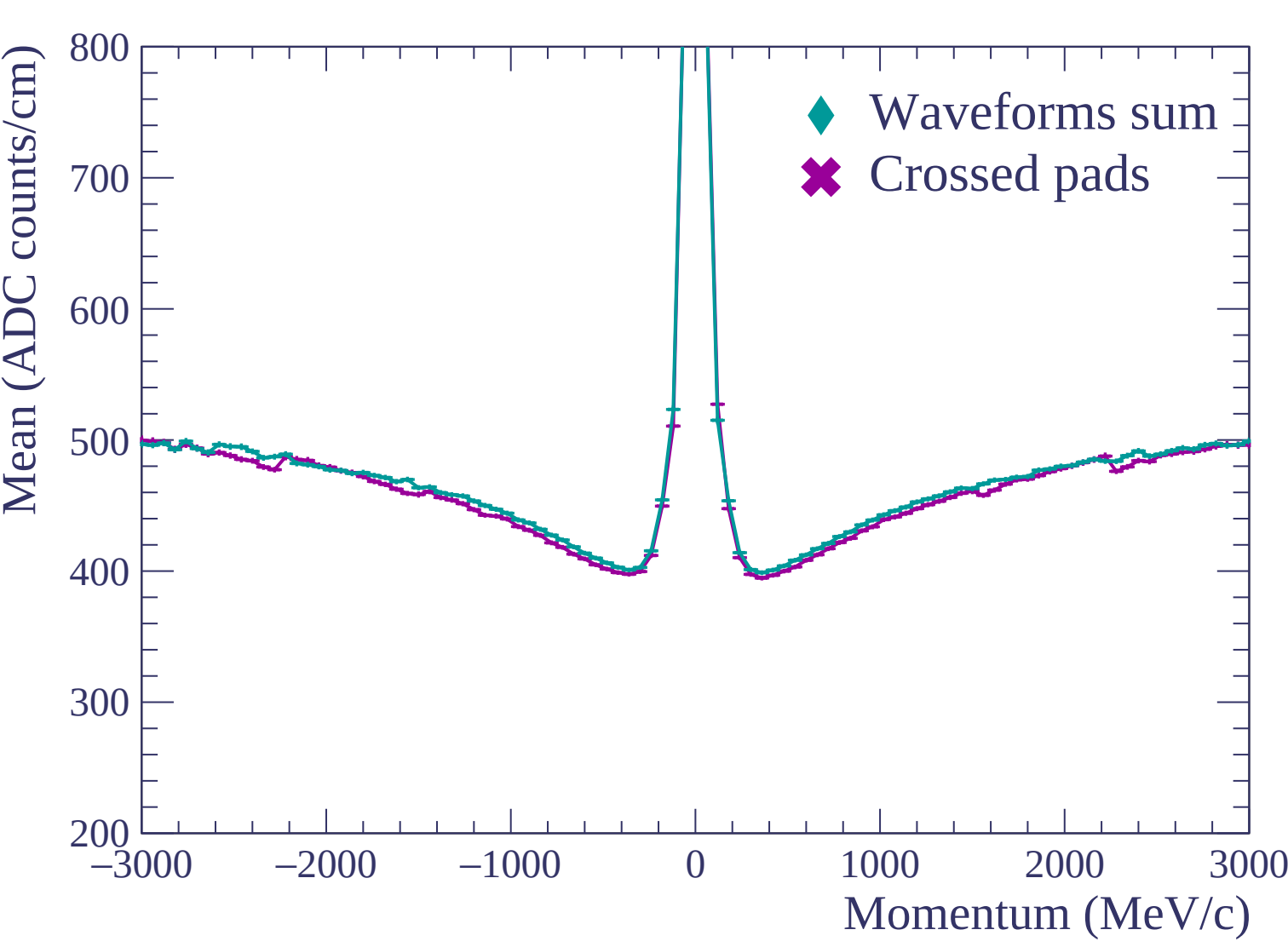


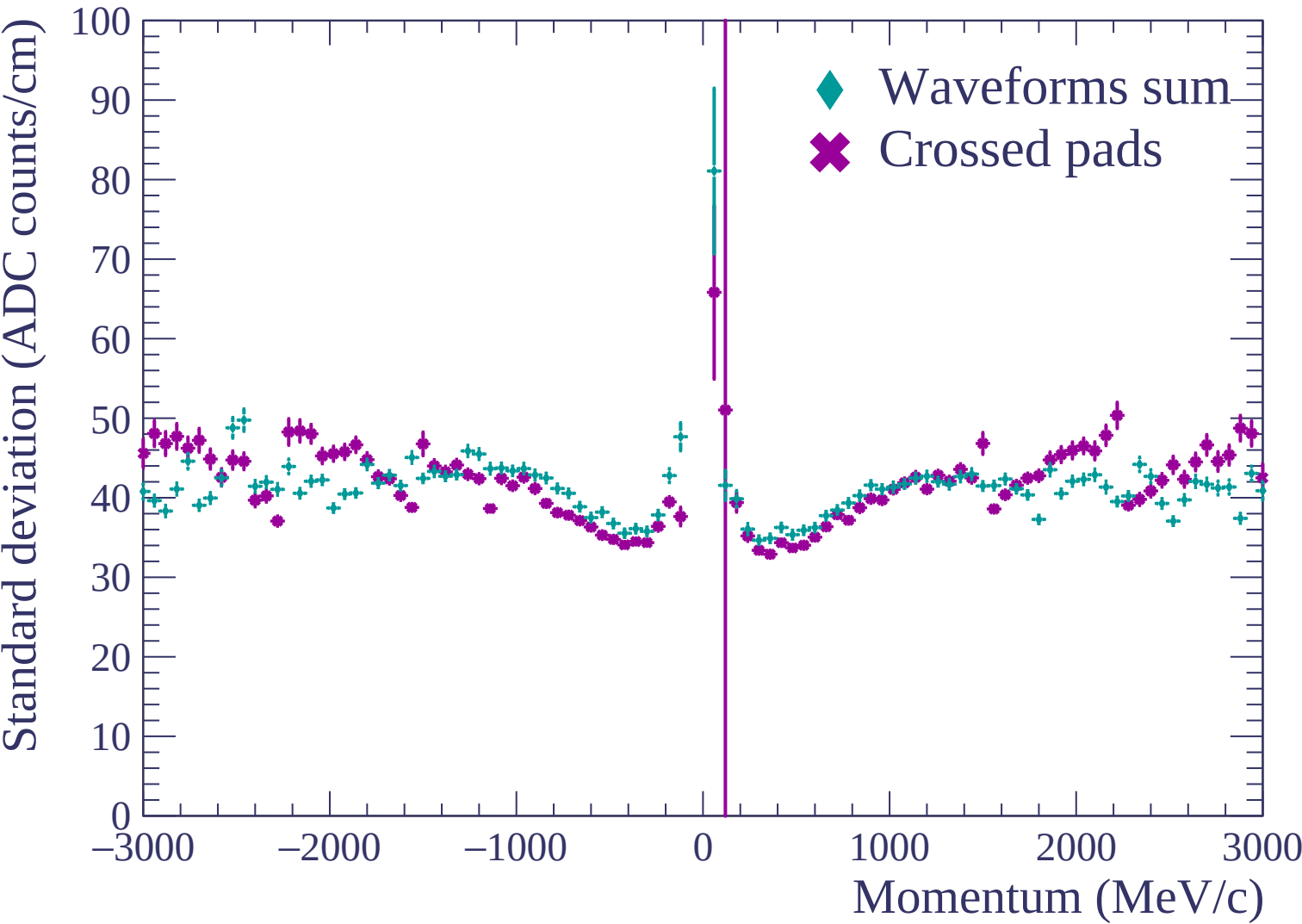


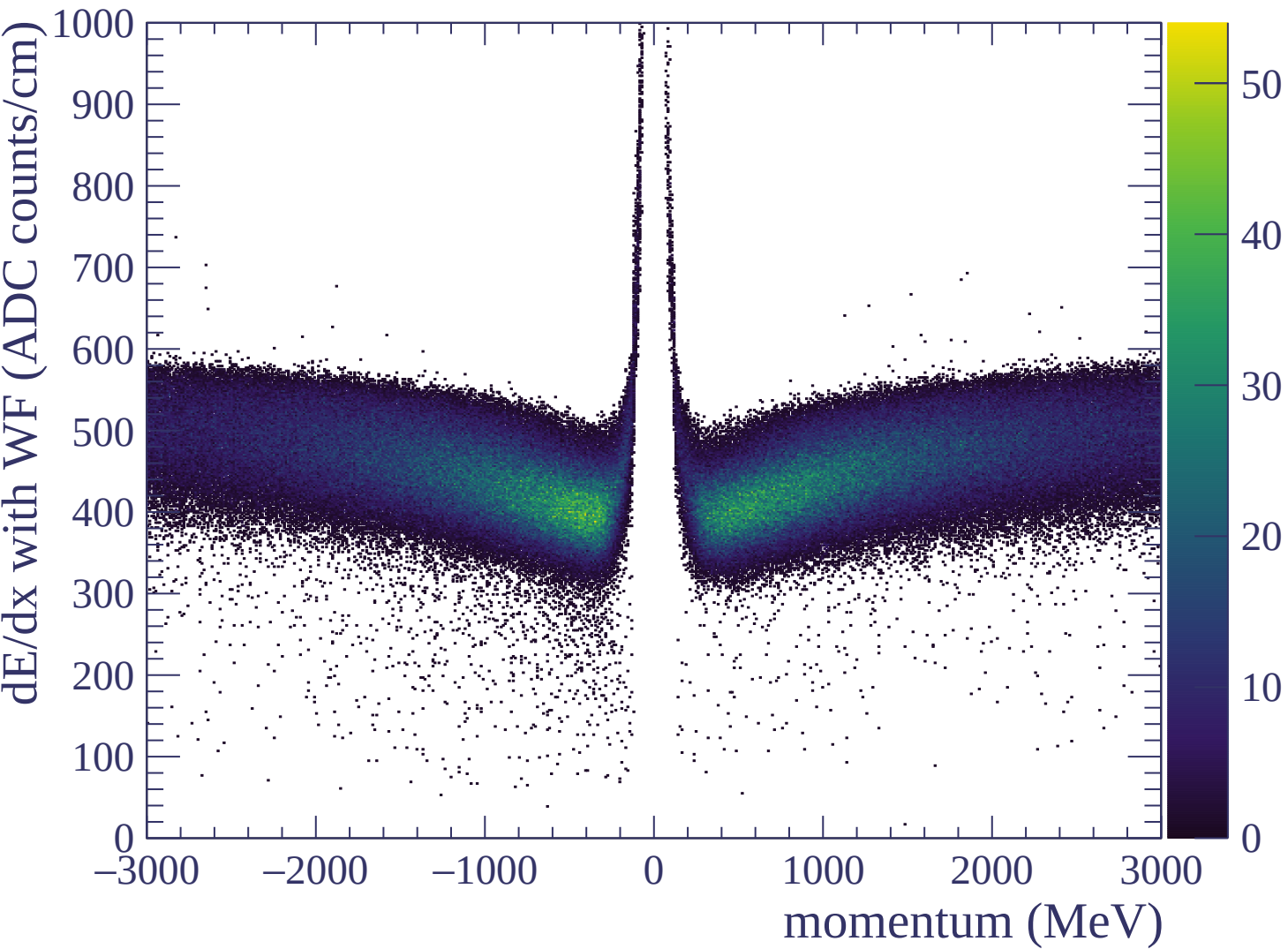


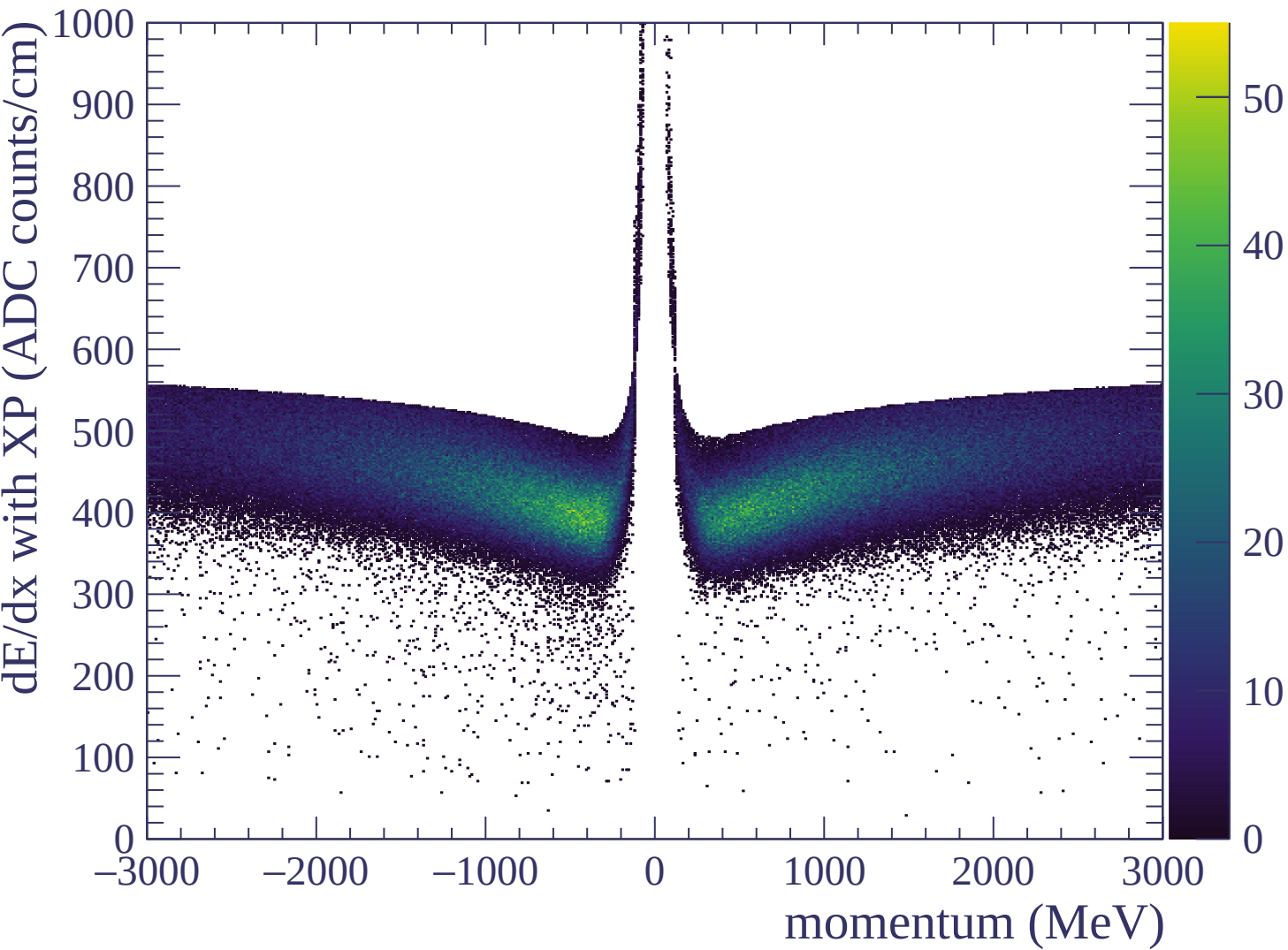
Resolution (%)

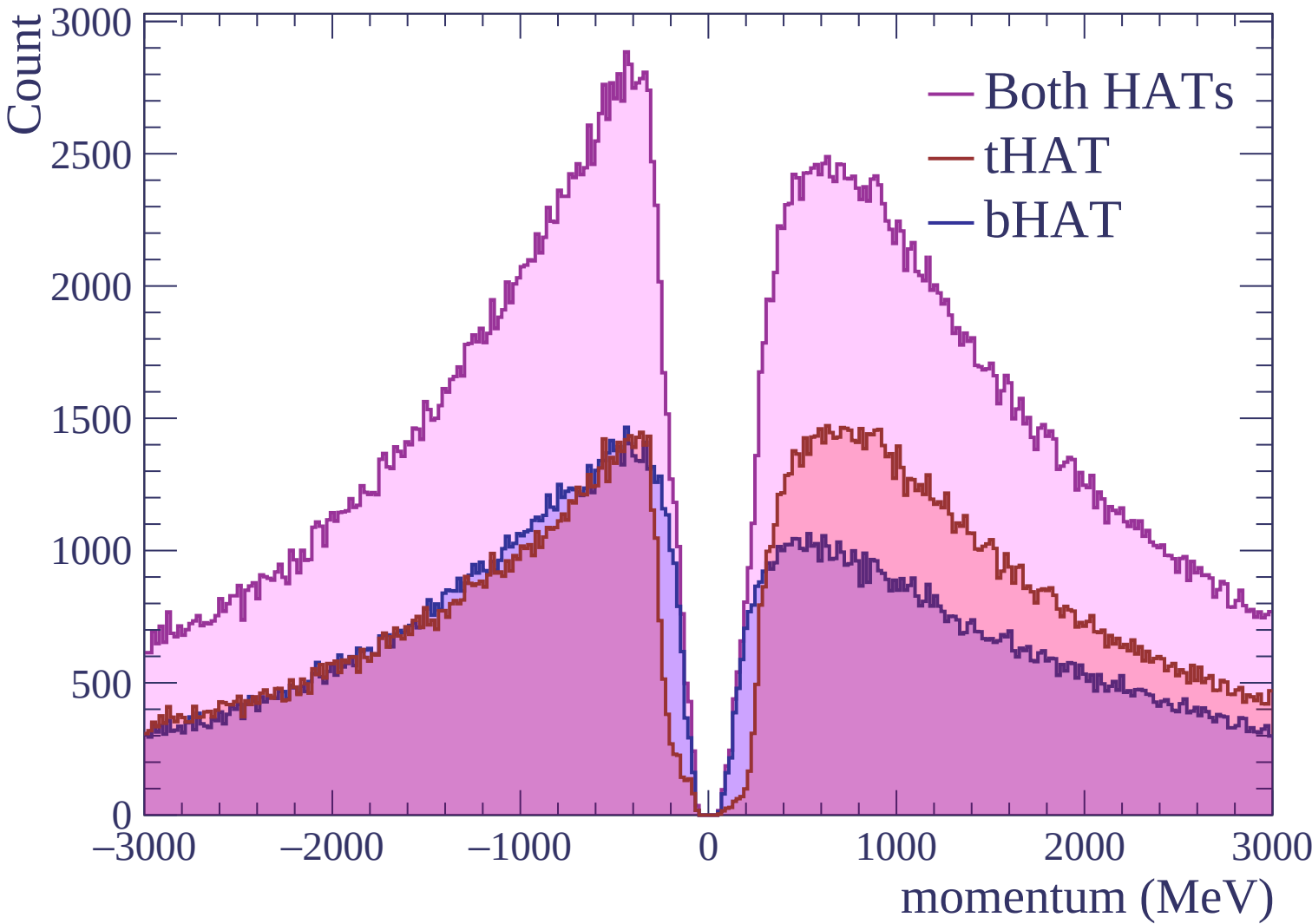


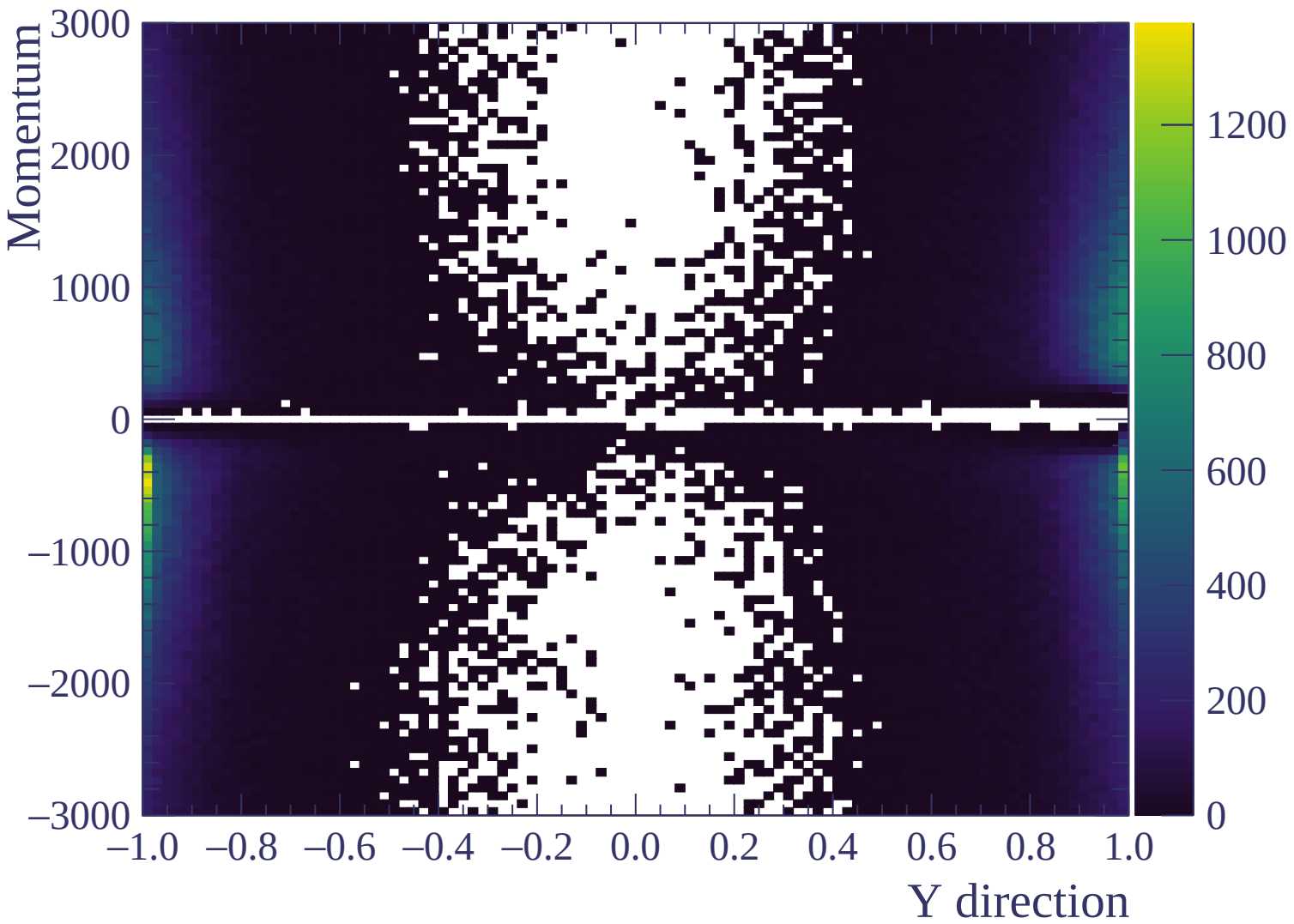


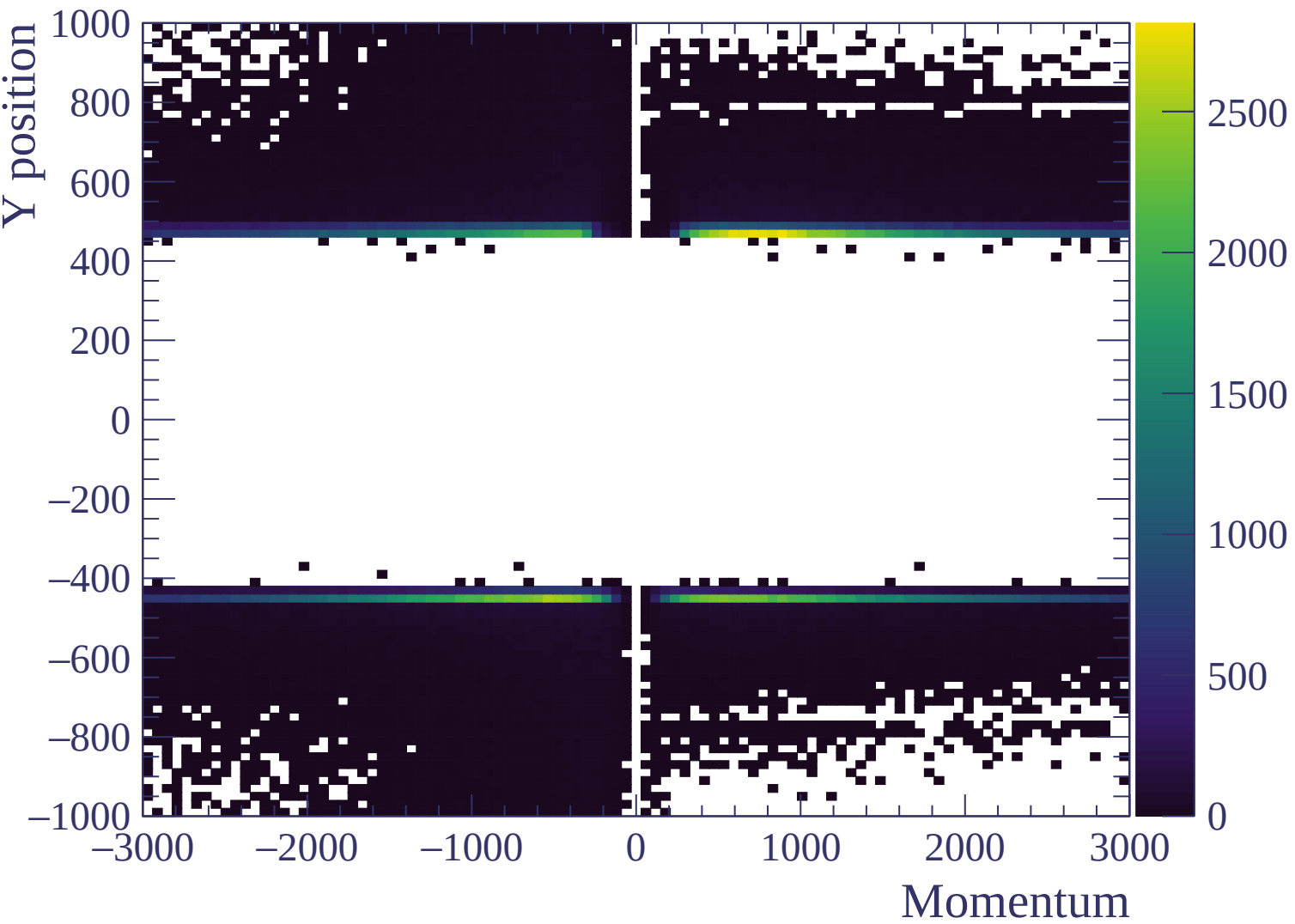


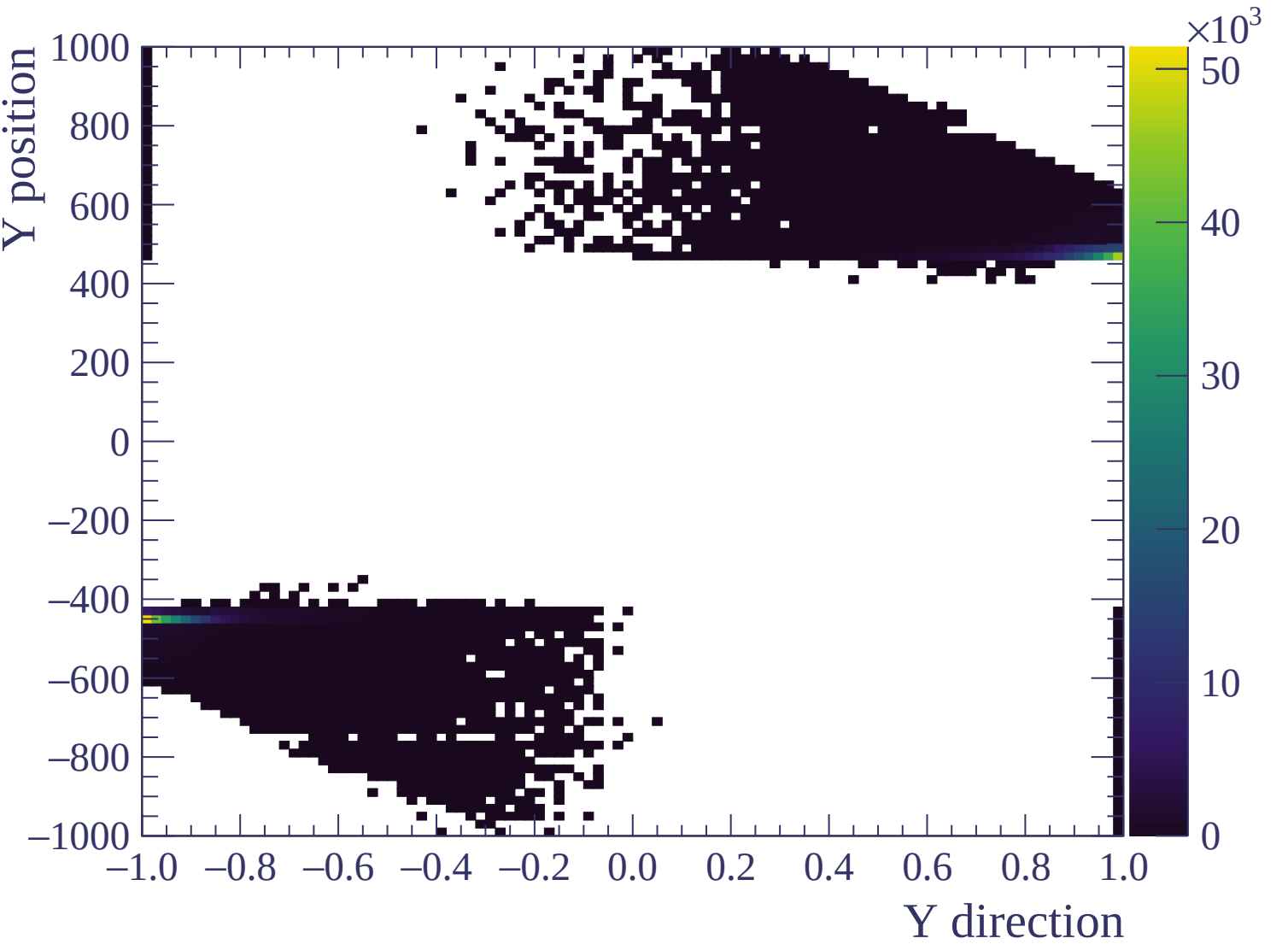




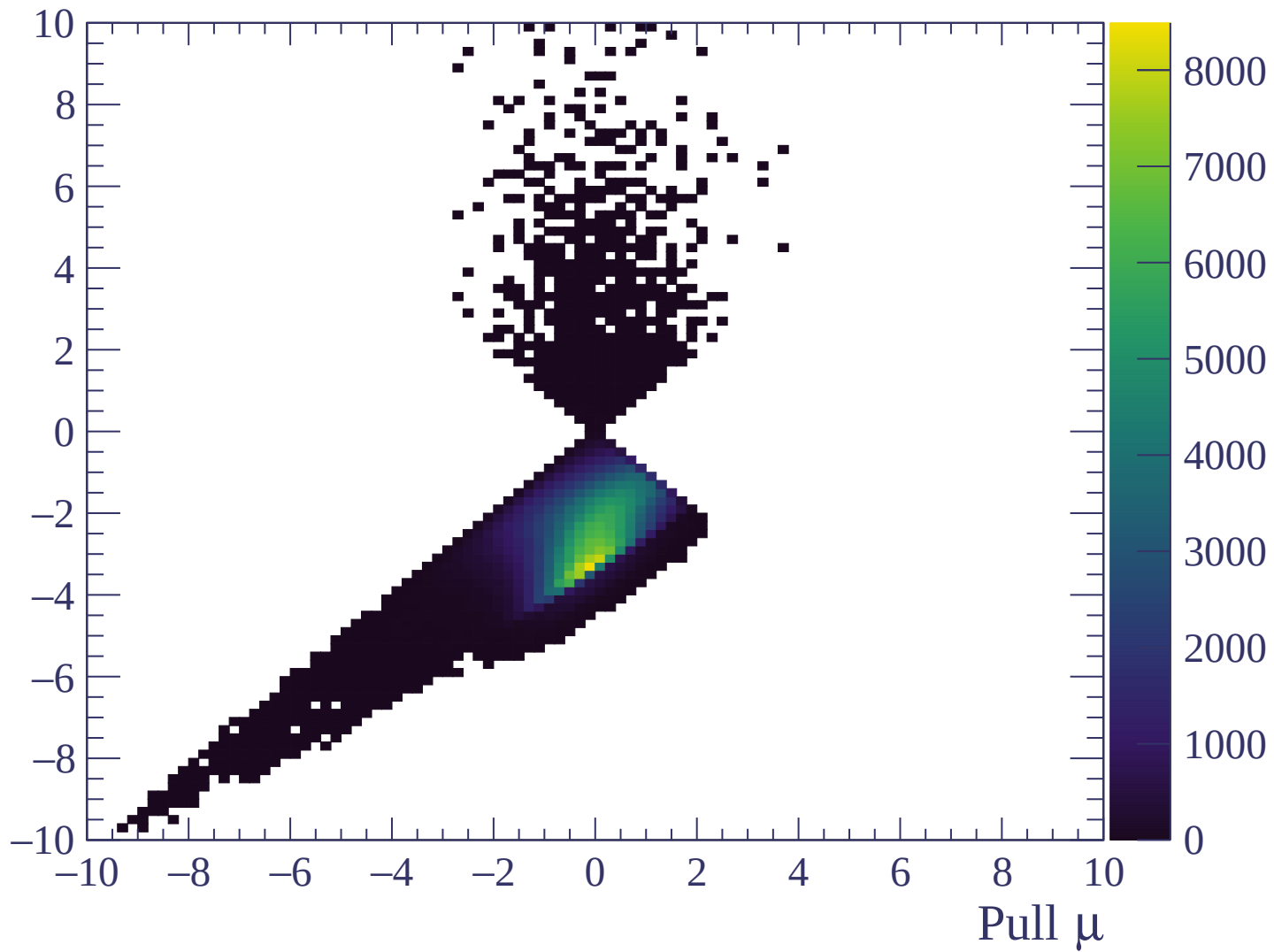






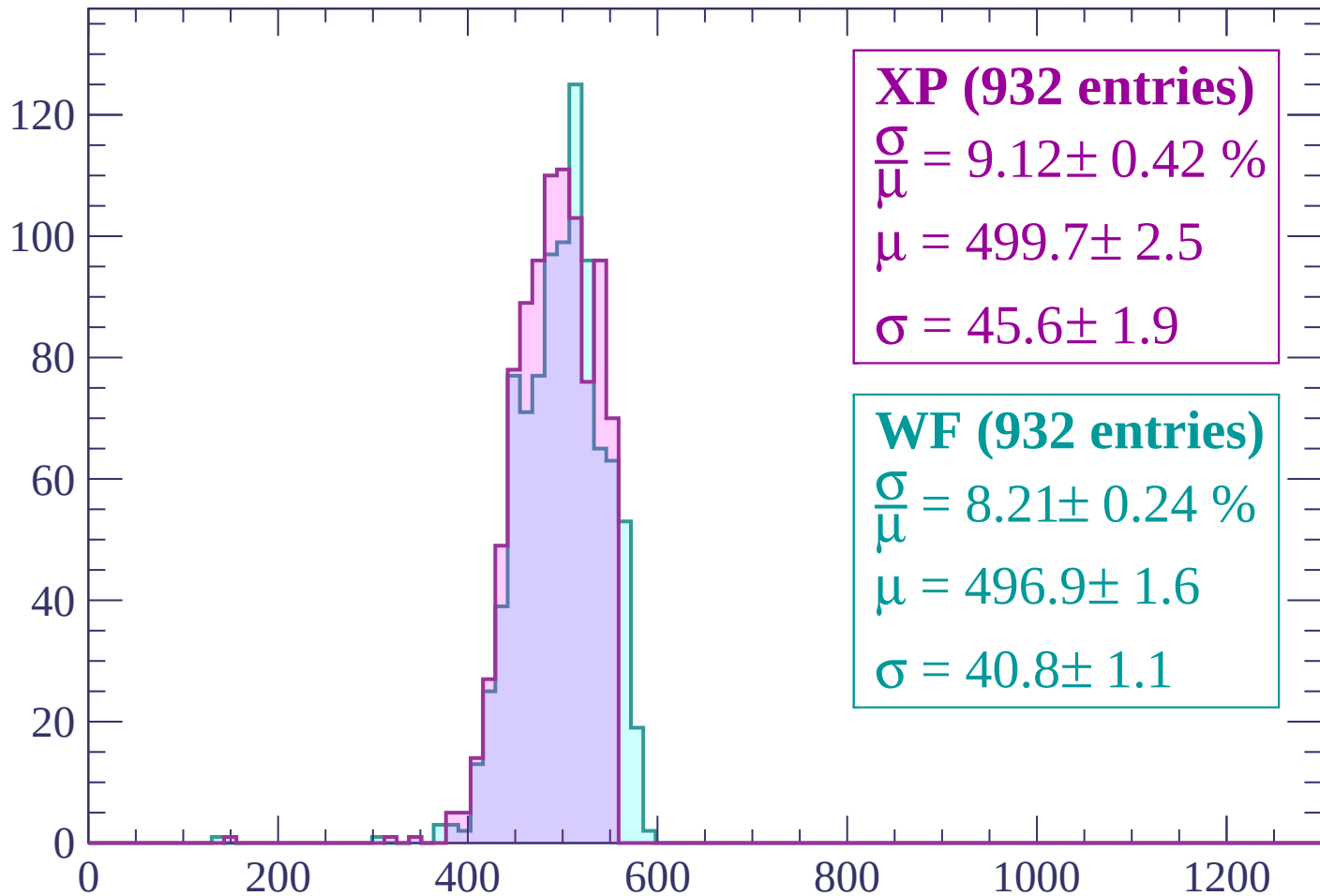


Pull e



Energy loss | $-3000 < p < -2940$

Count



XP (932 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 0.42 \%$$

$$\mu = 499.7 \pm 2.5$$

$$\sigma = 45.6 \pm 1.9$$

WF (932 entries)

$$\frac{\sigma}{\mu} = 8.21 \pm 0.24 \%$$

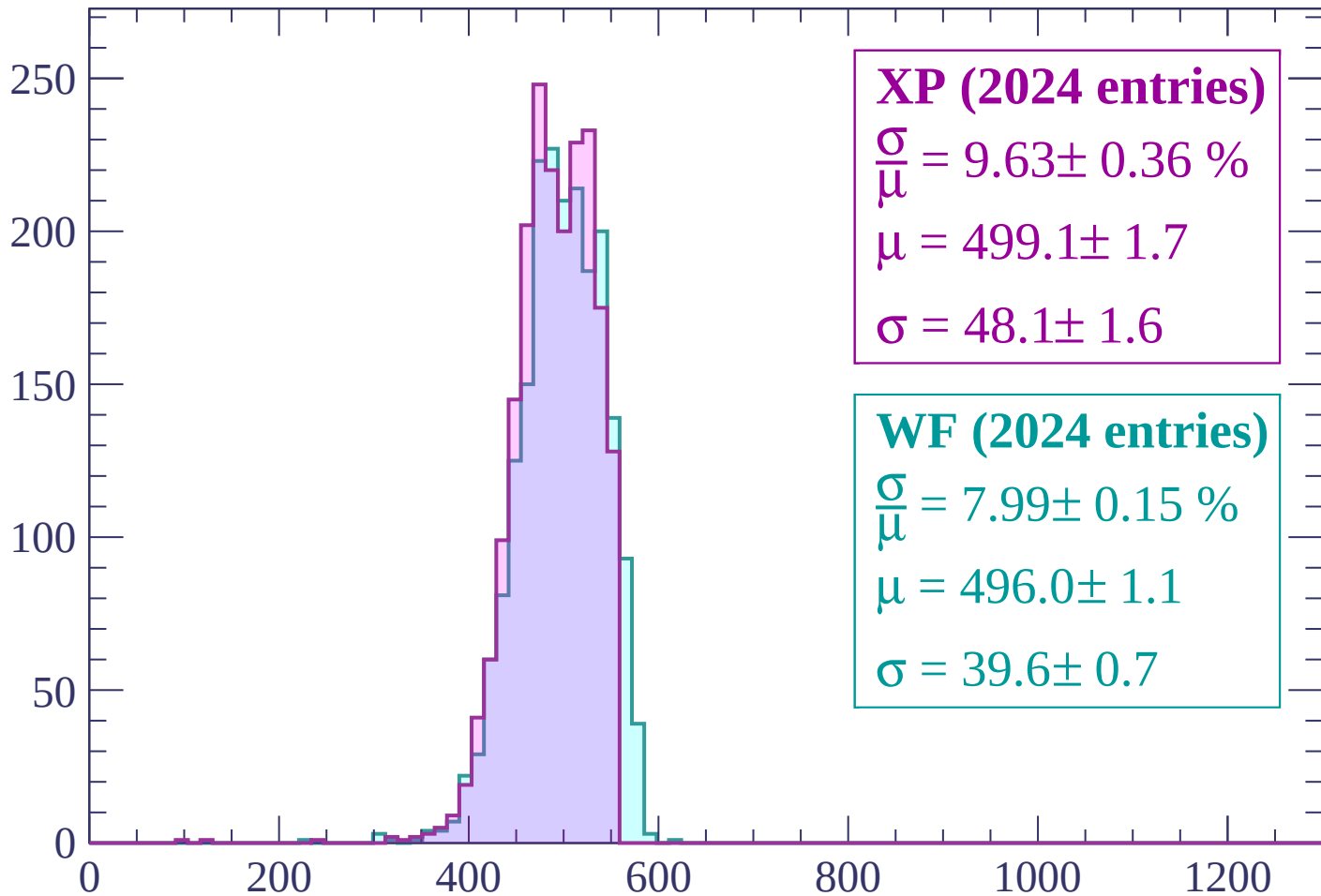
$$\mu = 496.9 \pm 1.6$$

$$\sigma = 40.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2940 < p < -2880

Count



XP (2024 entries)

$$\frac{\sigma}{\mu} = 9.63 \pm 0.36 \%$$

$$\mu = 499.1 \pm 1.7$$

$$\sigma = 48.1 \pm 1.6$$

WF (2024 entries)

$$\frac{\sigma}{\mu} = 7.99 \pm 0.15 \%$$

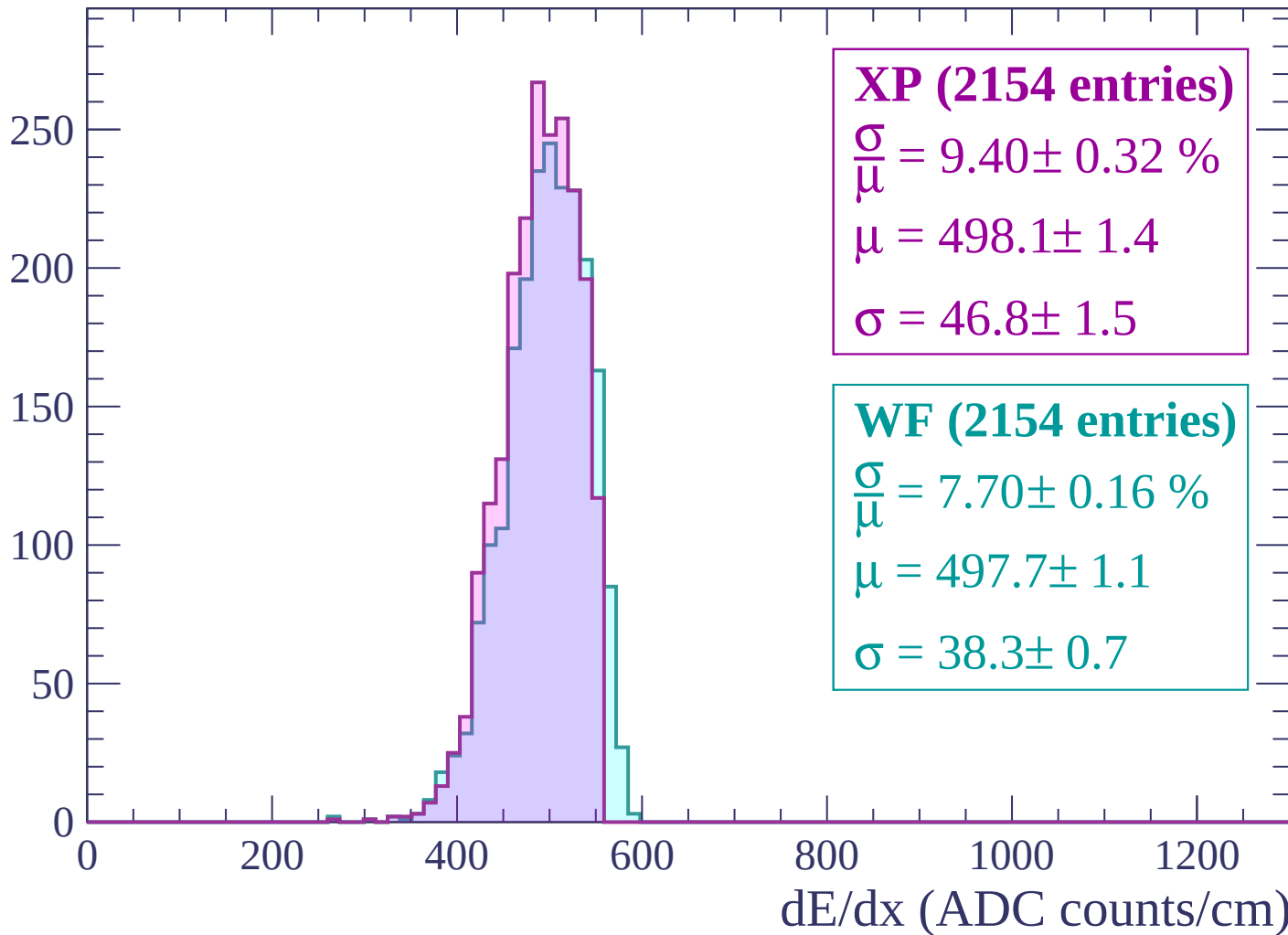
$$\mu = 496.0 \pm 1.1$$

$$\sigma = 39.6 \pm 0.7$$

dE/dx (ADC counts/cm)

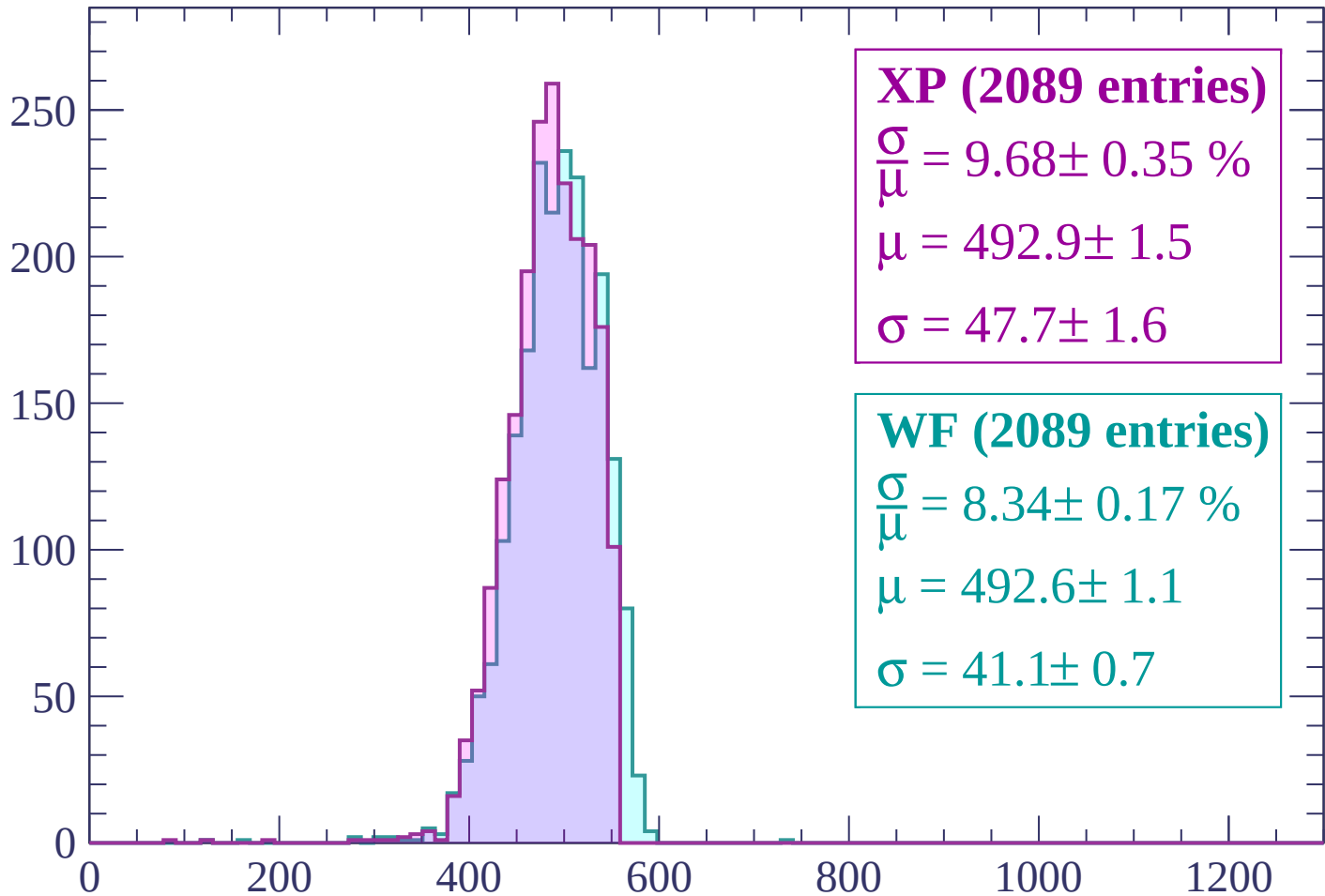
Energy loss | $-2880 < p < -2820$

Count



Energy loss | -2820 < p < -2760

Count



XP (2089 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.35 \%$$

$$\mu = 492.9 \pm 1.5$$

$$\sigma = 47.7 \pm 1.6$$

WF (2089 entries)

$$\frac{\sigma}{\mu} = 8.34 \pm 0.17 \%$$

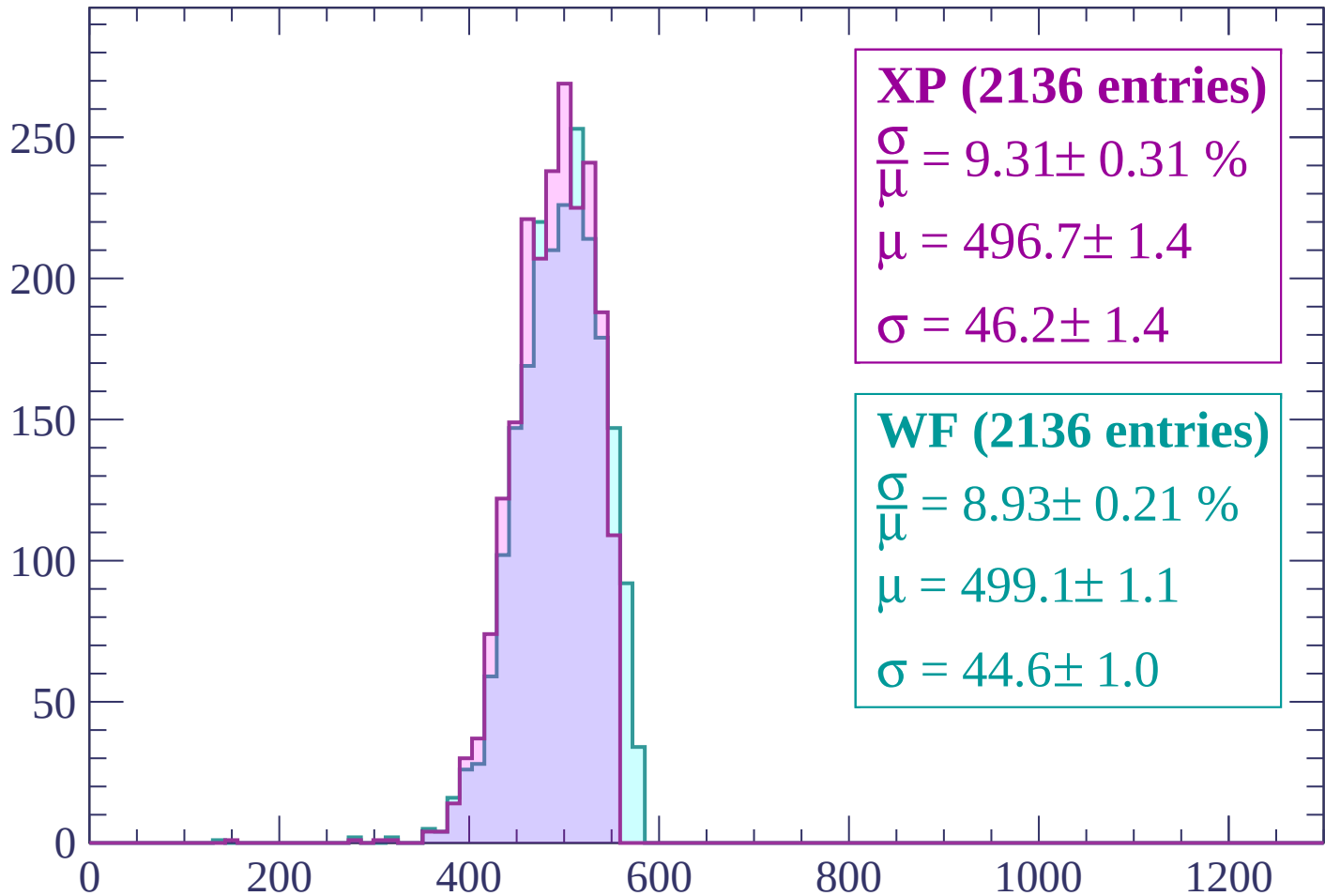
$$\mu = 492.6 \pm 1.1$$

$$\sigma = 41.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700

Count



XP (2136 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.31 \%$$

$$\mu = 496.7 \pm 1.4$$

$$\sigma = 46.2 \pm 1.4$$

WF (2136 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.21 \%$$

$$\mu = 499.1 \pm 1.1$$

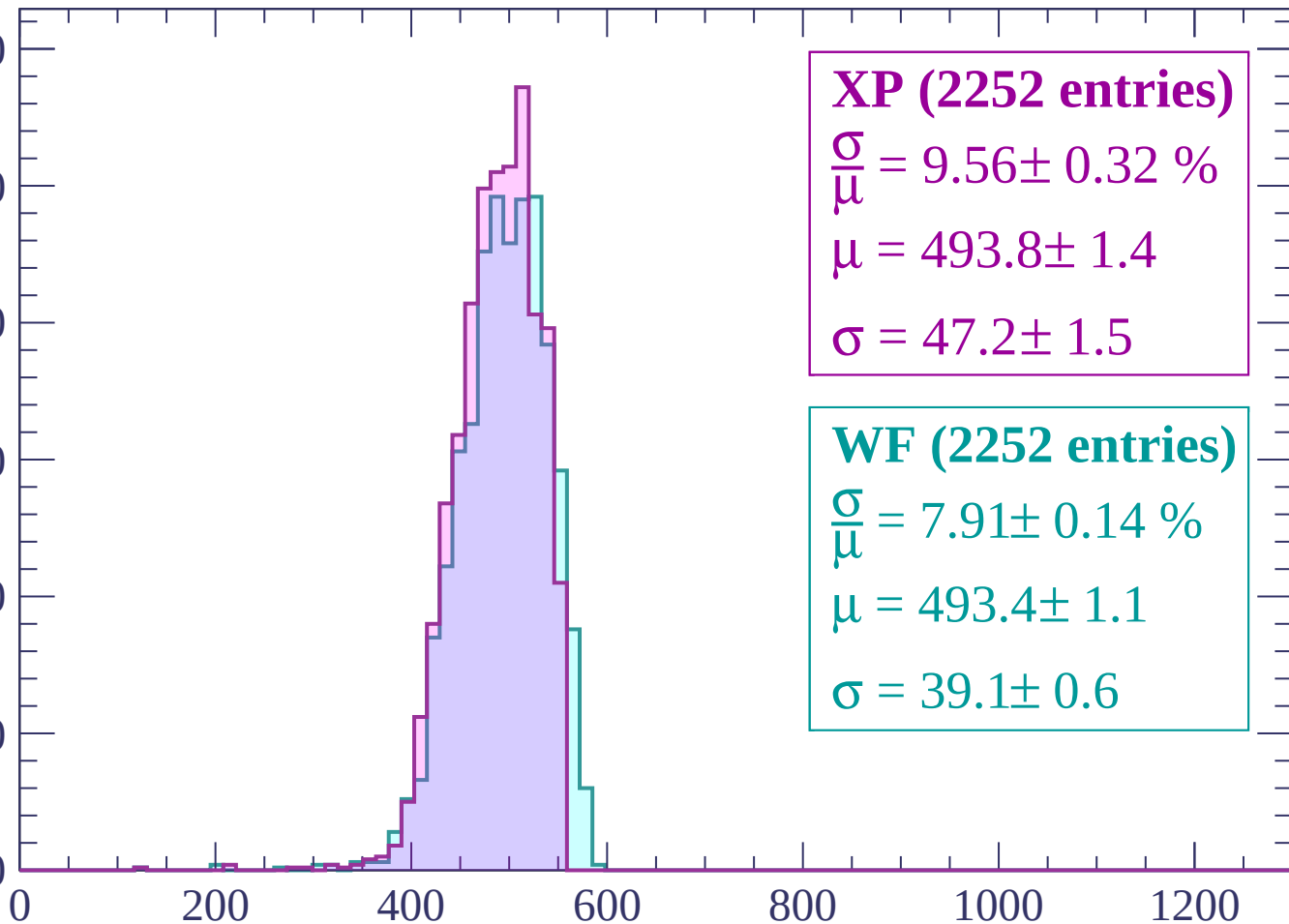
$$\sigma = 44.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2700 < p < -2640

Count

300
250
200
150
100
50
0



XP (2252 entries)

$\frac{\sigma}{\mu} = 9.56 \pm 0.32 \%$

$\mu = 493.8 \pm 1.4$

$\sigma = 47.2 \pm 1.5$

WF (2252 entries)

$\frac{\sigma}{\mu} = 7.91 \pm 0.14 \%$

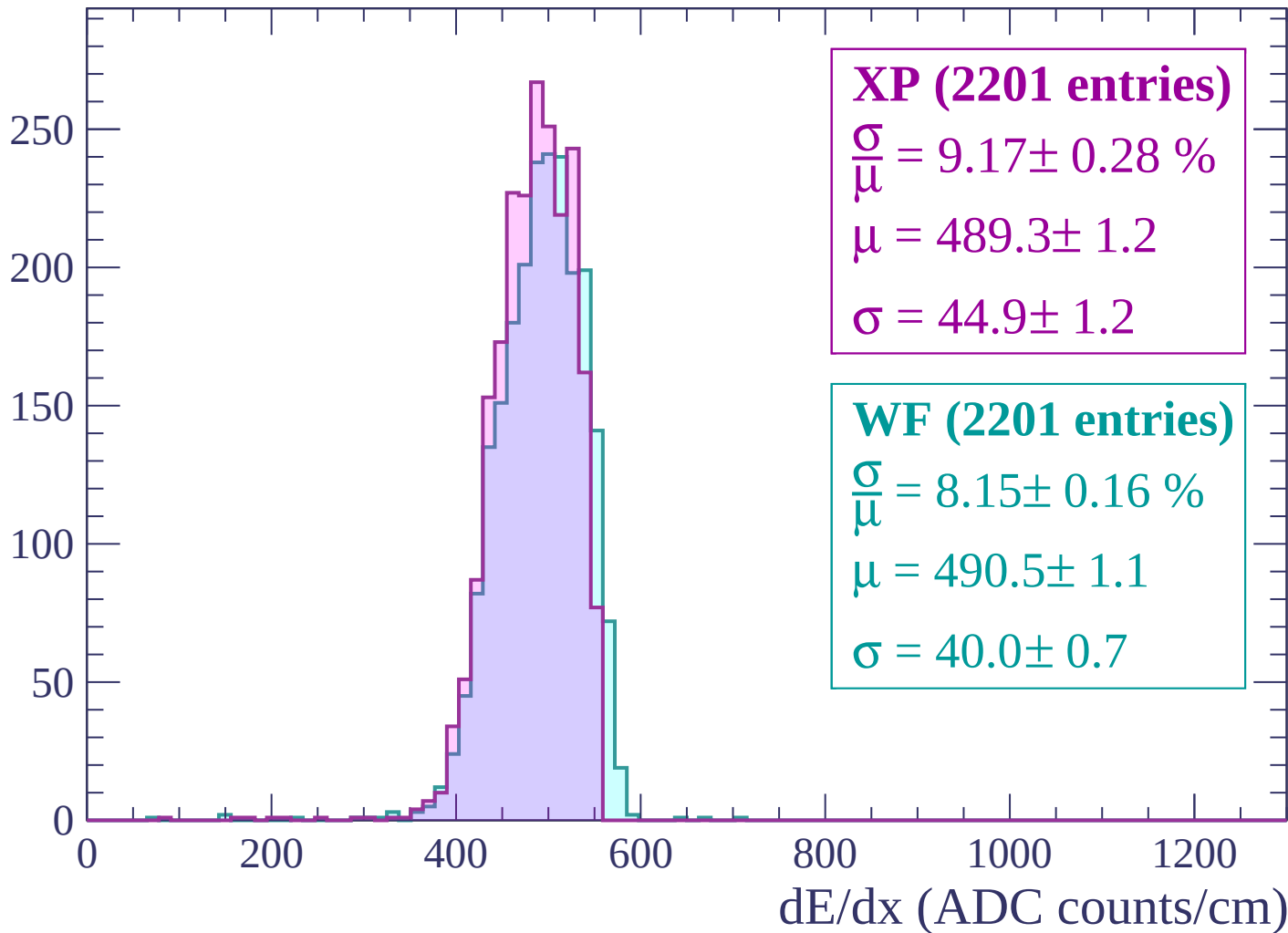
$\mu = 493.4 \pm 1.1$

$\sigma = 39.1 \pm 0.6$

dE/dx (ADC counts/cm)

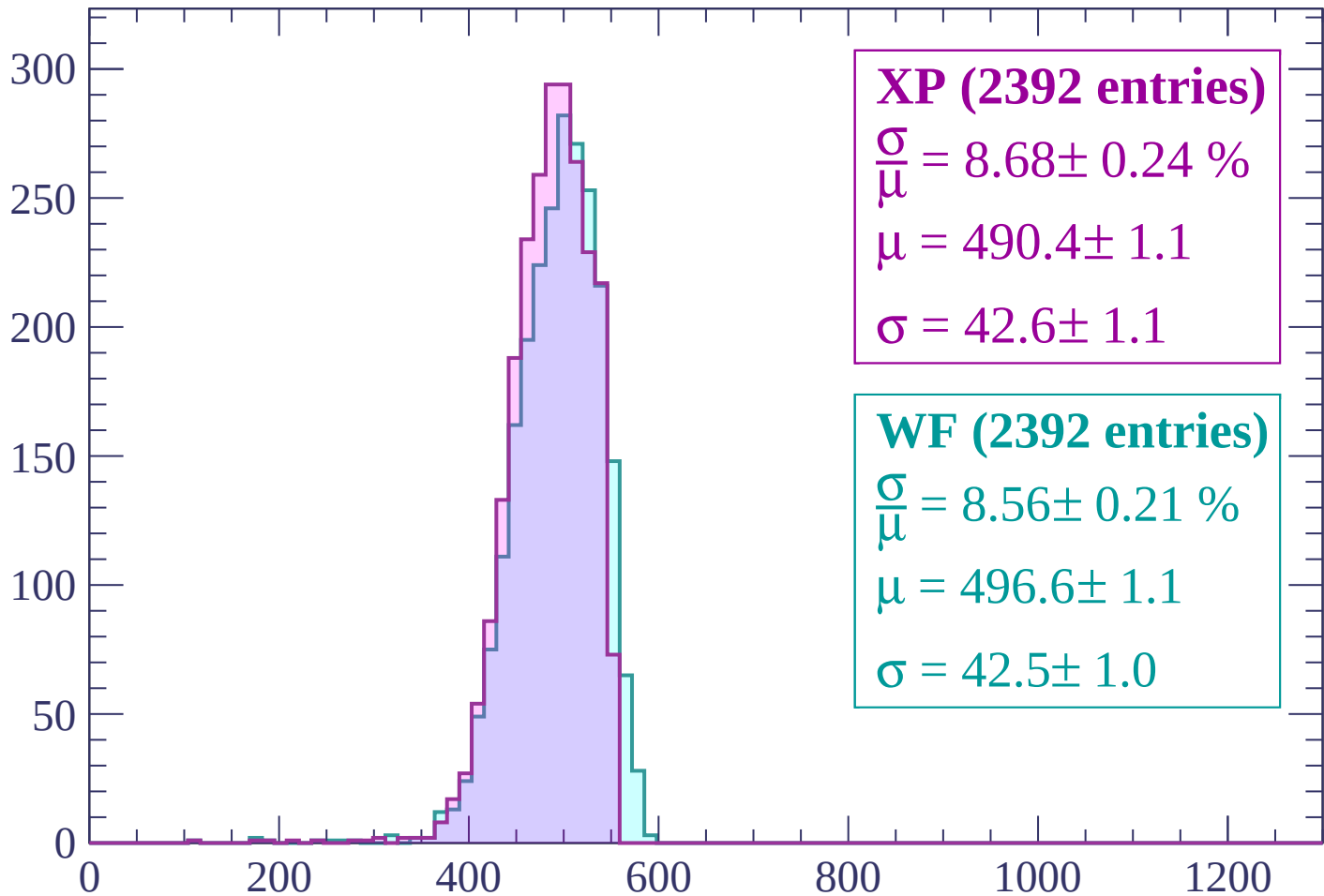
Energy loss | -2640 < p < -2580

Count



Energy loss | -2580 < p < -2520

Count



XP (2392 entries)

$$\frac{\sigma}{\mu} = 8.68 \pm 0.24 \%$$

$$\mu = 490.4 \pm 1.1$$

$$\sigma = 42.6 \pm 1.1$$

WF (2392 entries)

$$\frac{\sigma}{\mu} = 8.56 \pm 0.21 \%$$

$$\mu = 496.6 \pm 1.1$$

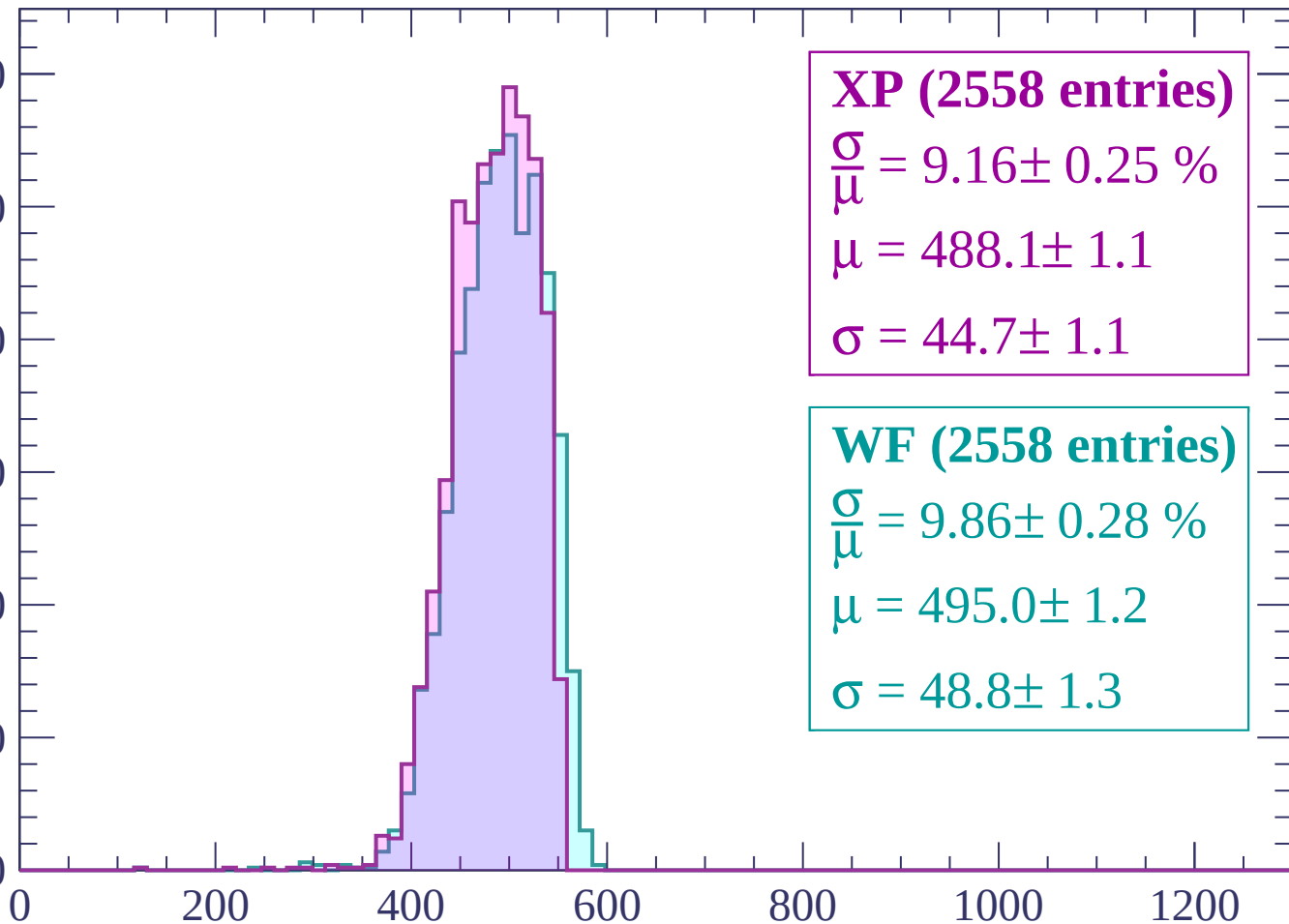
$$\sigma = 42.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count

300
250
200
150
100
50
0



XP (2558 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.25 \%$$

$$\mu = 488.1 \pm 1.1$$

$$\sigma = 44.7 \pm 1.1$$

WF (2558 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.28 \%$$

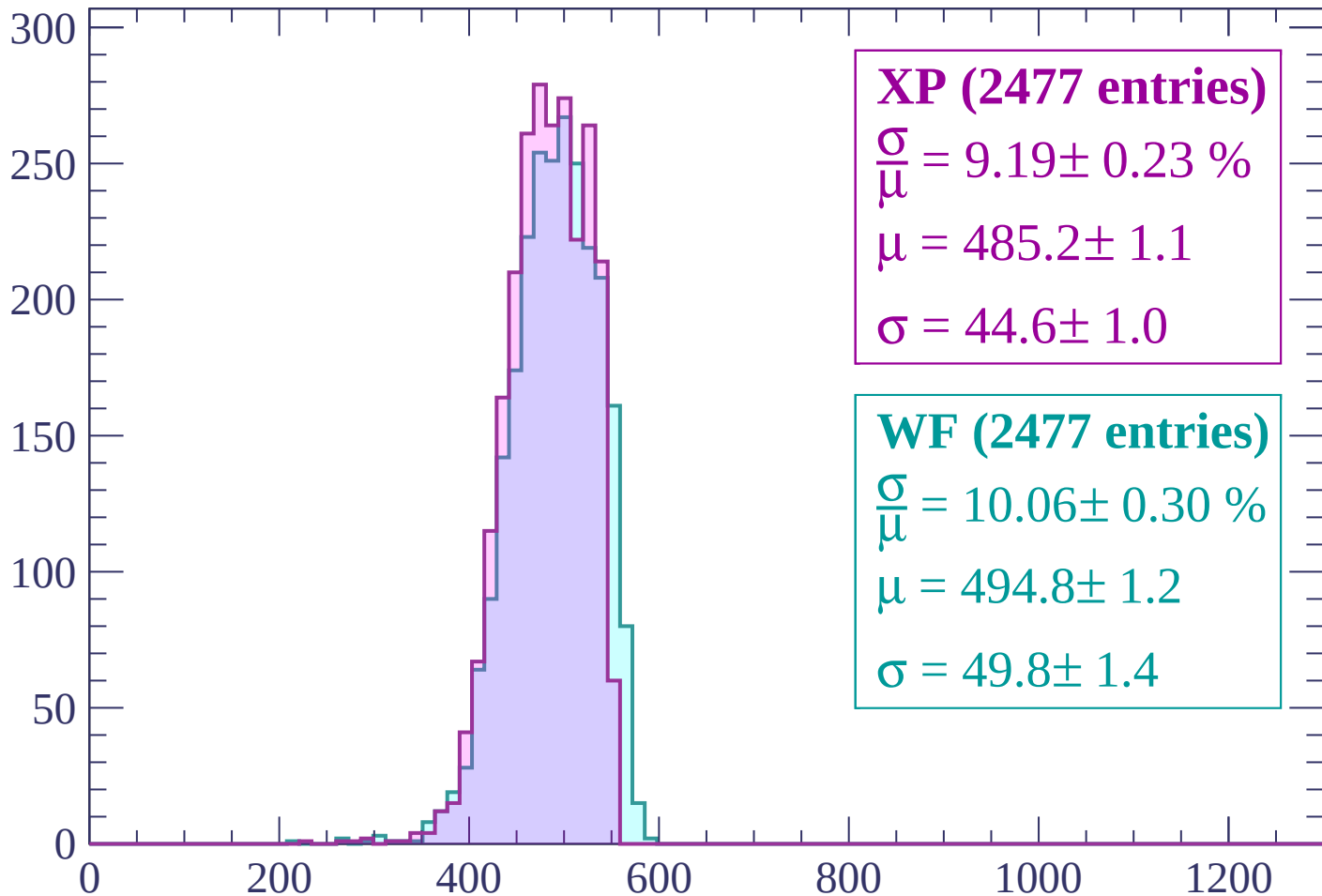
$$\mu = 495.0 \pm 1.2$$

$$\sigma = 48.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -2460 < p < -2400

Count



XP (2477 entries)

$$\frac{\sigma}{\mu} = 9.19 \pm 0.23 \%$$

$$\mu = 485.2 \pm 1.1$$

$$\sigma = 44.6 \pm 1.0$$

WF (2477 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.30 \%$$

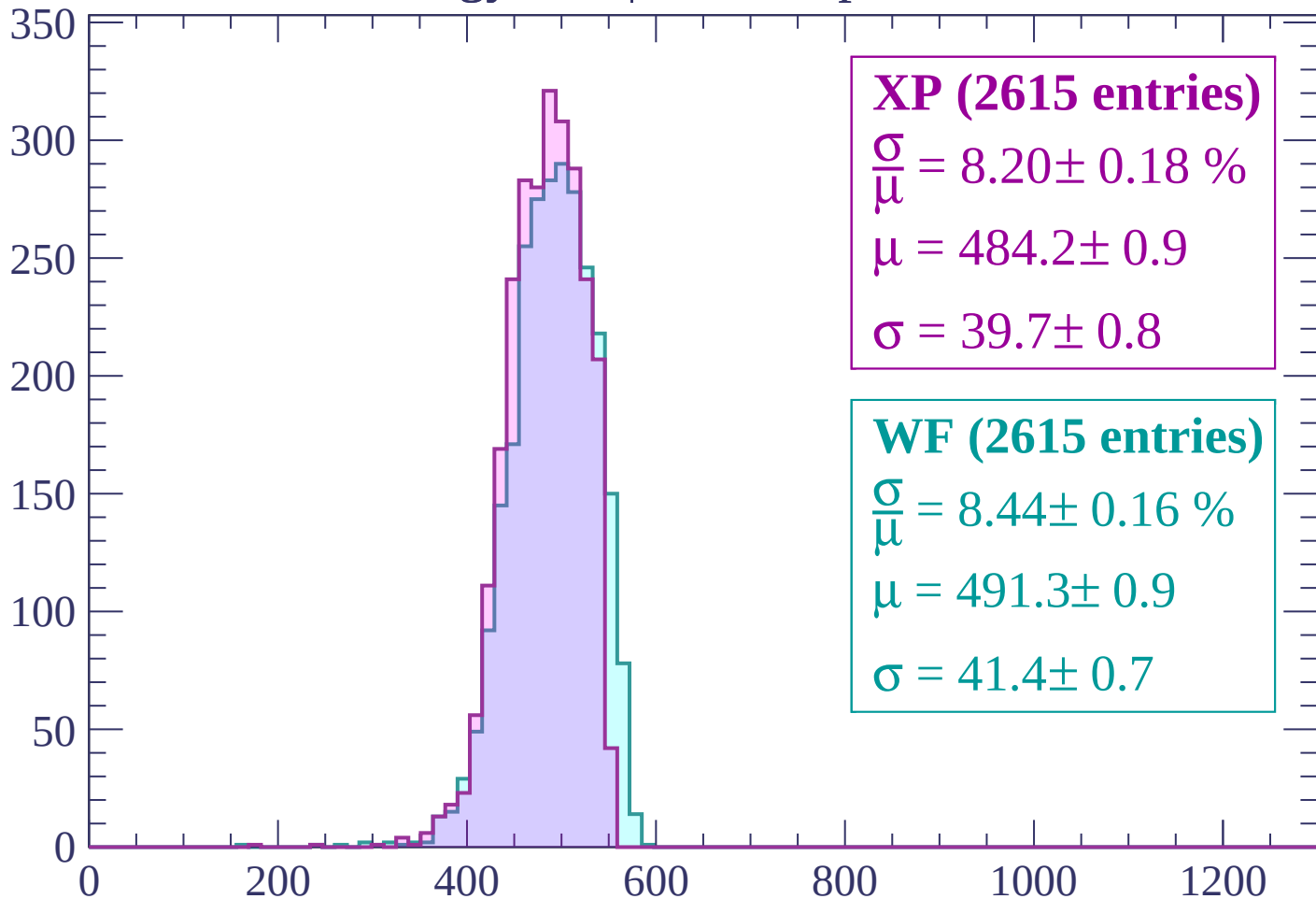
$$\mu = 494.8 \pm 1.2$$

$$\sigma = 49.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-2400 < p < -2340$

Count



XP (2615 entries)

$$\frac{\sigma}{\mu} = 8.20 \pm 0.18 \%$$

$$\mu = 484.2 \pm 0.9$$

$$\sigma = 39.7 \pm 0.8$$

WF (2615 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.16 \%$$

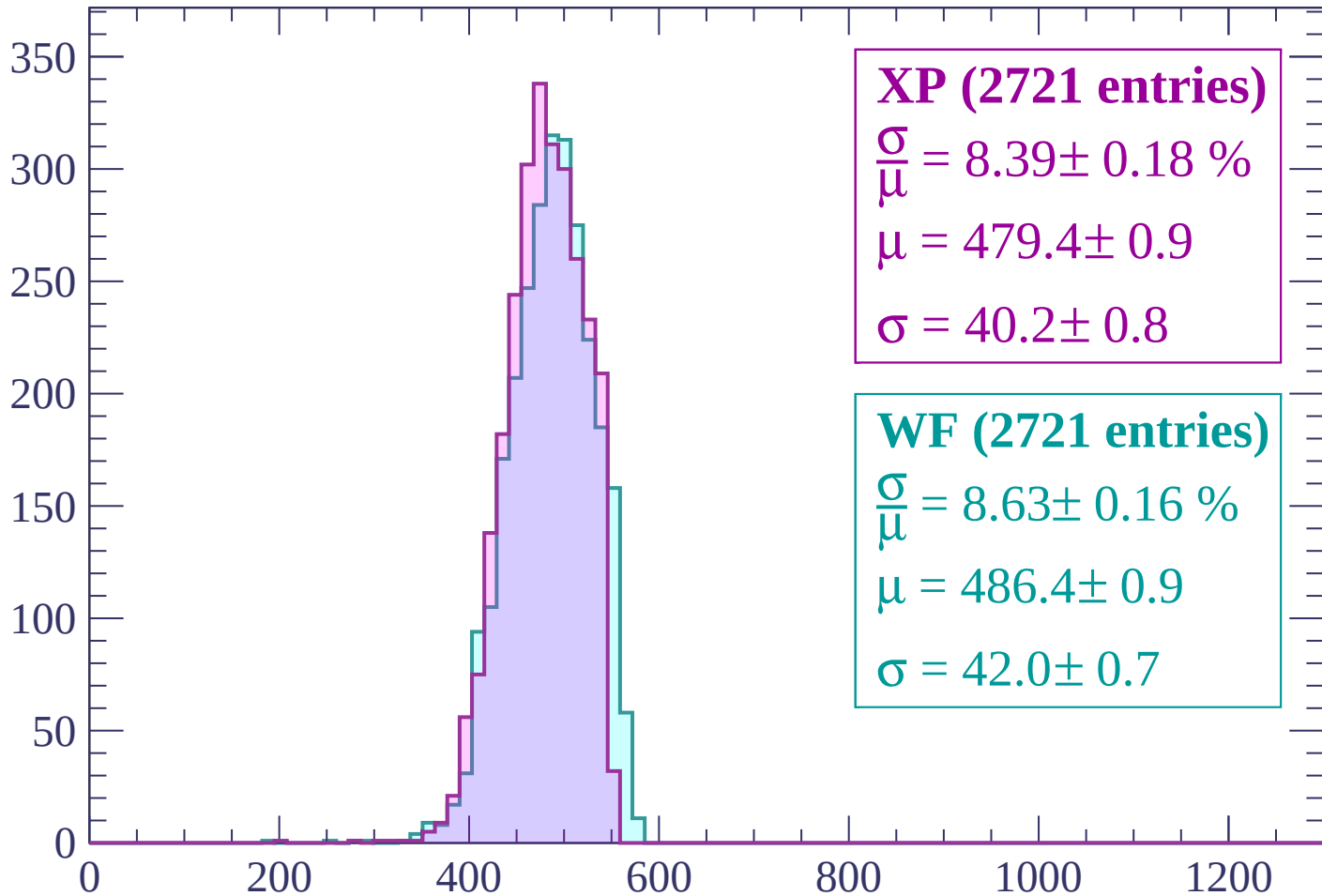
$$\mu = 491.3 \pm 0.9$$

$$\sigma = 41.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-2340 < p < -2280$

Count



XP (2721 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.18 \%$$

$$\mu = 479.4 \pm 0.9$$

$$\sigma = 40.2 \pm 0.8$$

WF (2721 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.16 \%$$

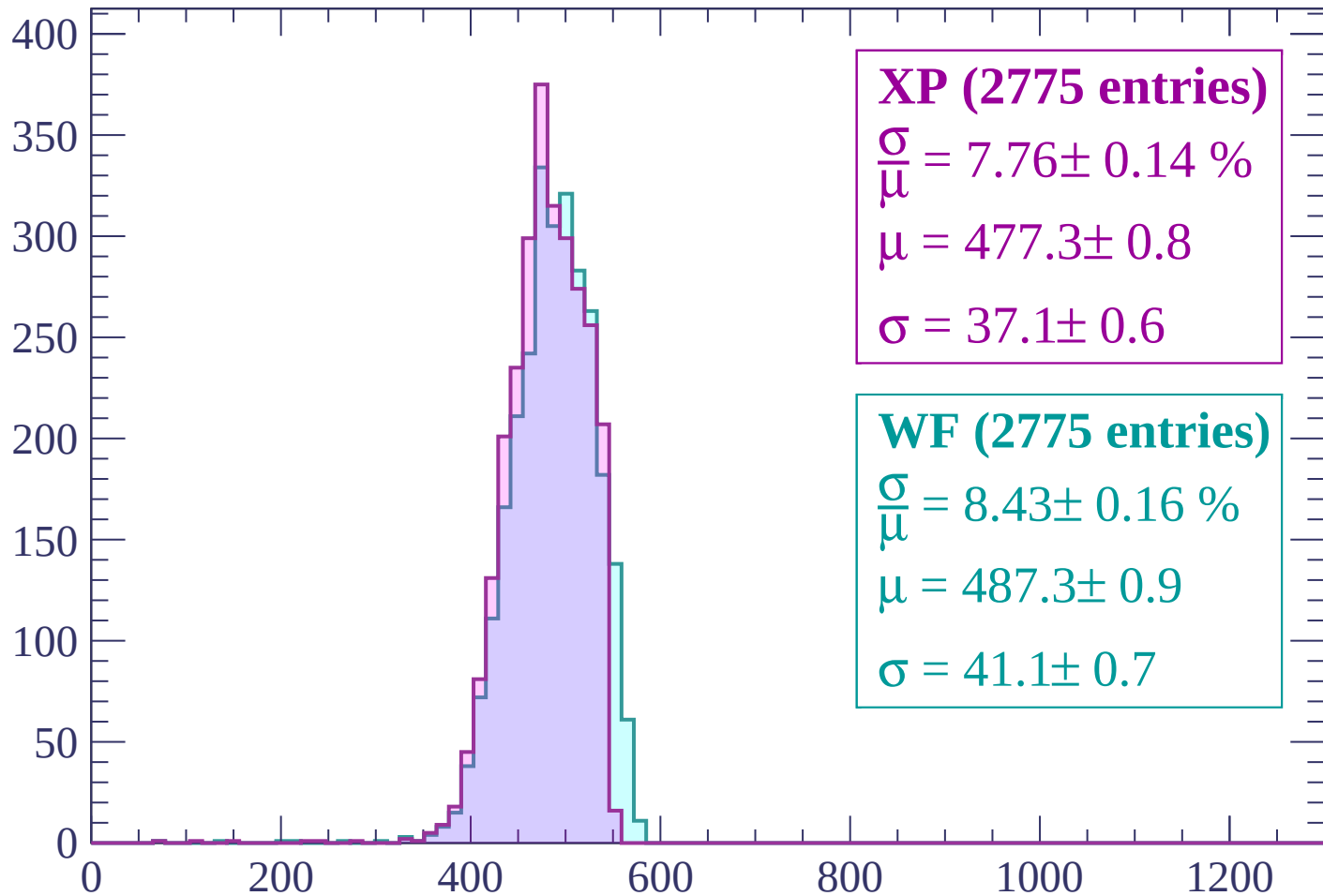
$$\mu = 486.4 \pm 0.9$$

$$\sigma = 42.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-2280 < p < -2220$

Count



XP (2775 entries)

$$\frac{\sigma}{\mu} = 7.76 \pm 0.14 \%$$

$$\mu = 477.3 \pm 0.8$$

$$\sigma = 37.1 \pm 0.6$$

WF (2775 entries)

$$\frac{\sigma}{\mu} = 8.43 \pm 0.16 \%$$

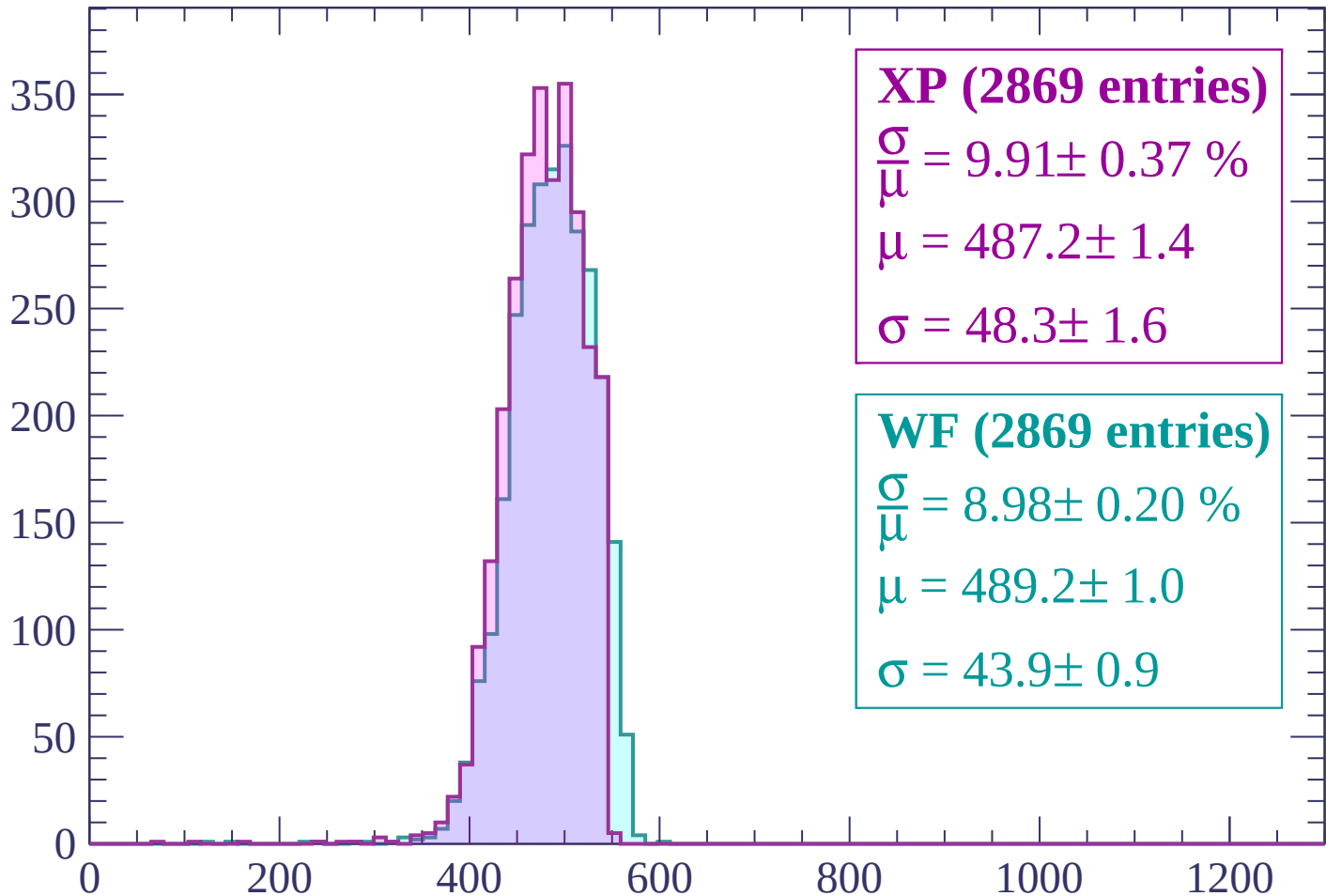
$$\mu = 487.3 \pm 0.9$$

$$\sigma = 41.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count



XP (2869 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.37 \%$$

$$\mu = 487.2 \pm 1.4$$

$$\sigma = 48.3 \pm 1.6$$

WF (2869 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.20 \%$$

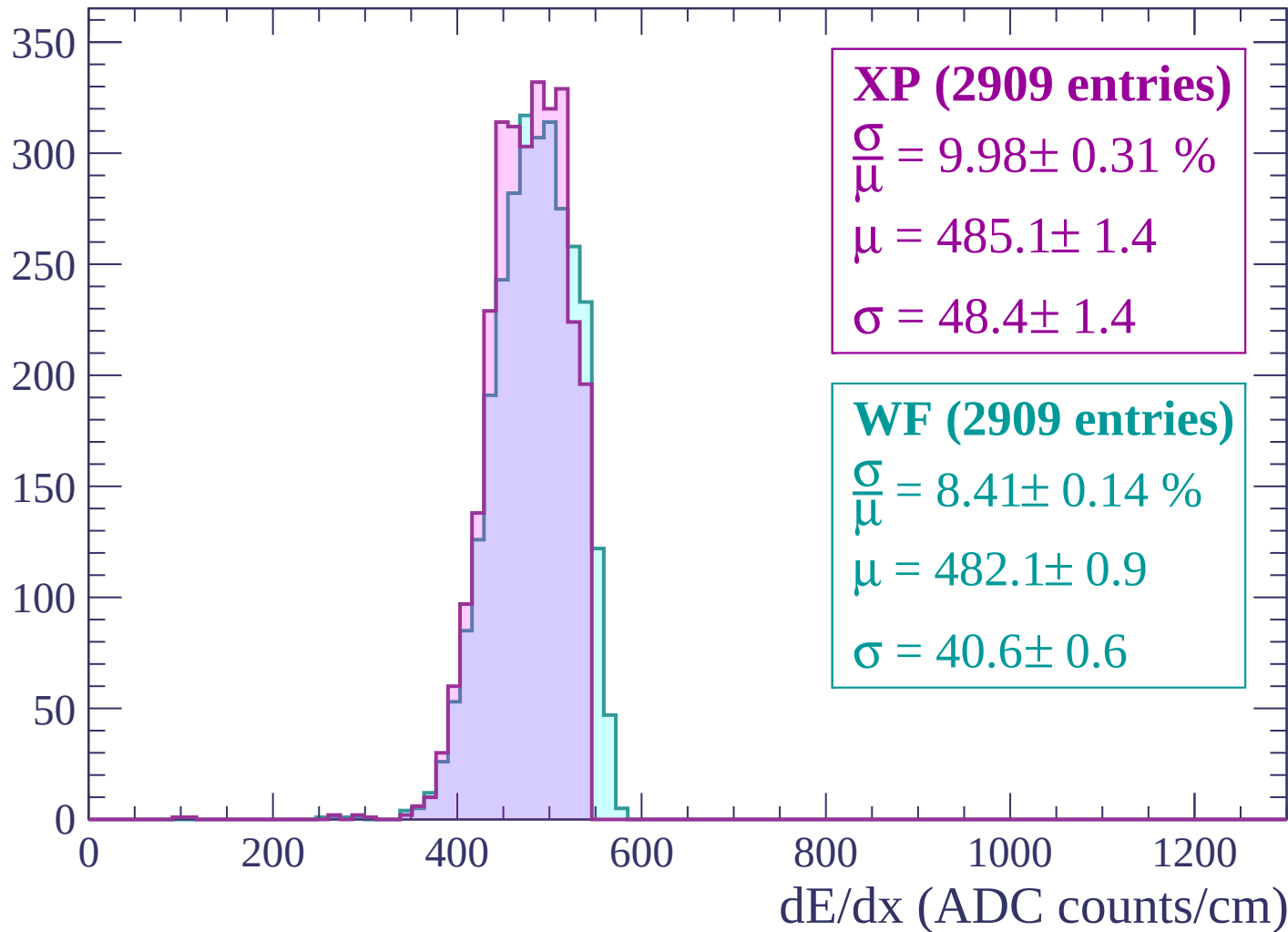
$$\mu = 489.2 \pm 1.0$$

$$\sigma = 43.9 \pm 0.9$$

dE/dx (ADC counts/cm)

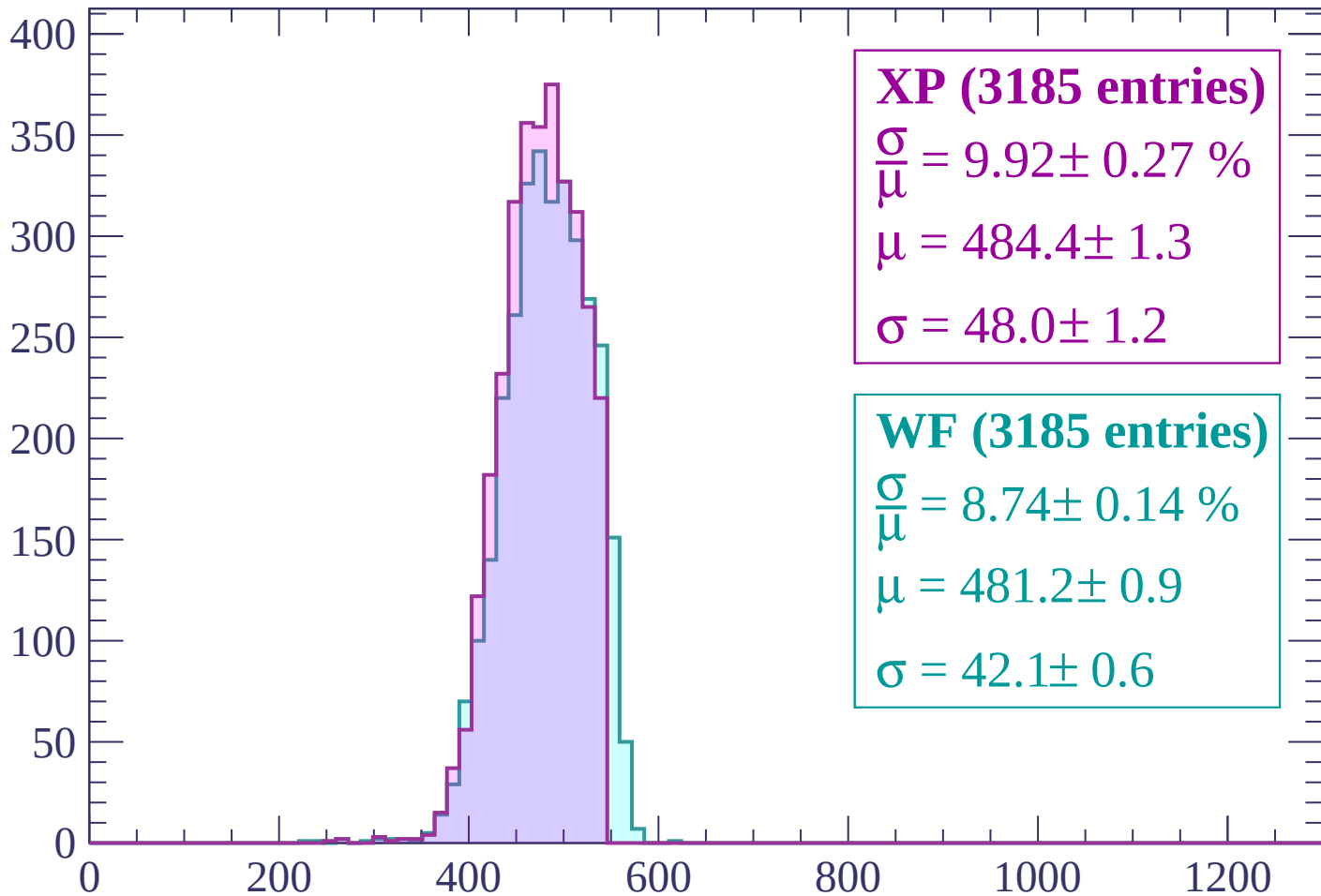
Energy loss | -2160 < p < -2100

Count



Energy loss | -2100 < p < -2040

Count



XP (3185 entries)

$$\frac{\sigma}{\mu} = 9.92 \pm 0.27 \%$$

$$\mu = 484.4 \pm 1.3$$

$$\sigma = 48.0 \pm 1.2$$

WF (3185 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.14 \%$$

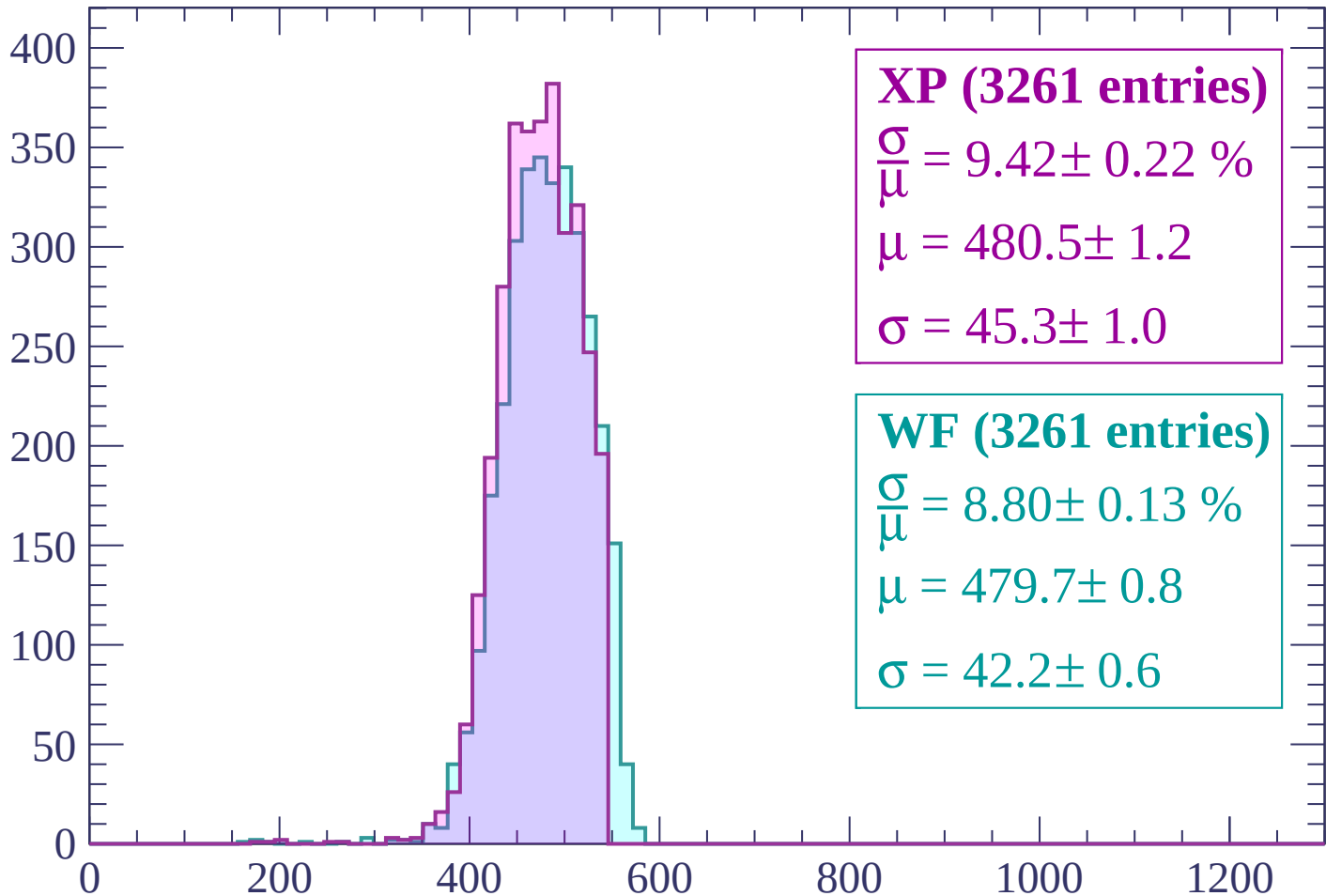
$$\mu = 481.2 \pm 0.9$$

$$\sigma = 42.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980

Count



XP (3261 entries)

$$\frac{\sigma}{\mu} = 9.42 \pm 0.22 \%$$

$$\mu = 480.5 \pm 1.2$$

$$\sigma = 45.3 \pm 1.0$$

WF (3261 entries)

$$\frac{\sigma}{\mu} = 8.80 \pm 0.13 \%$$

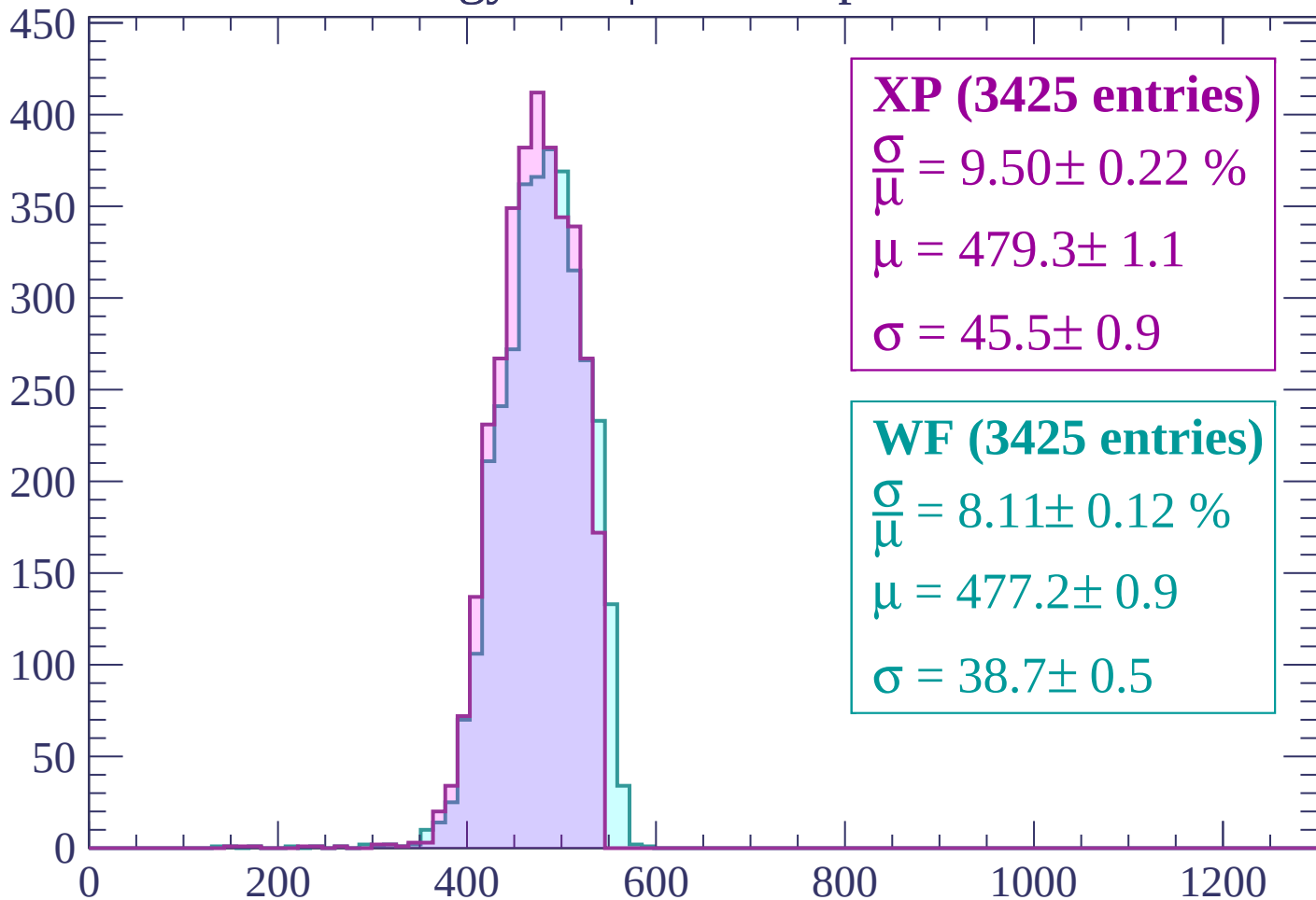
$$\mu = 479.7 \pm 0.8$$

$$\sigma = 42.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count



XP (3425 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.22 \%$$

$$\mu = 479.3 \pm 1.1$$

$$\sigma = 45.5 \pm 0.9$$

WF (3425 entries)

$$\frac{\sigma}{\mu} = 8.11 \pm 0.12 \%$$

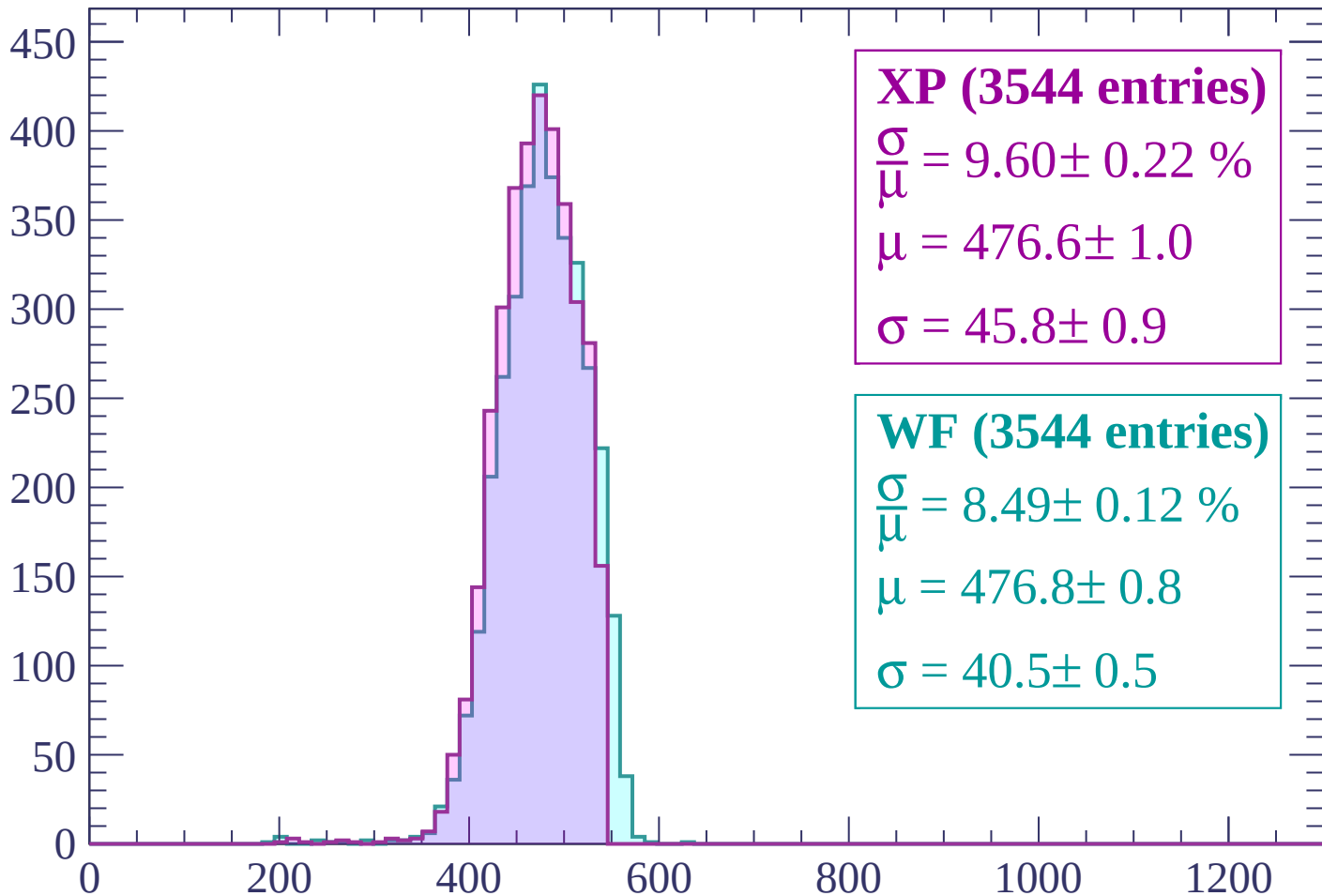
$$\mu = 477.2 \pm 0.9$$

$$\sigma = 38.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count



XP (3544 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.22 \%$$

$$\mu = 476.6 \pm 1.0$$

$$\sigma = 45.8 \pm 0.9$$

WF (3544 entries)

$$\frac{\sigma}{\mu} = 8.49 \pm 0.12 \%$$

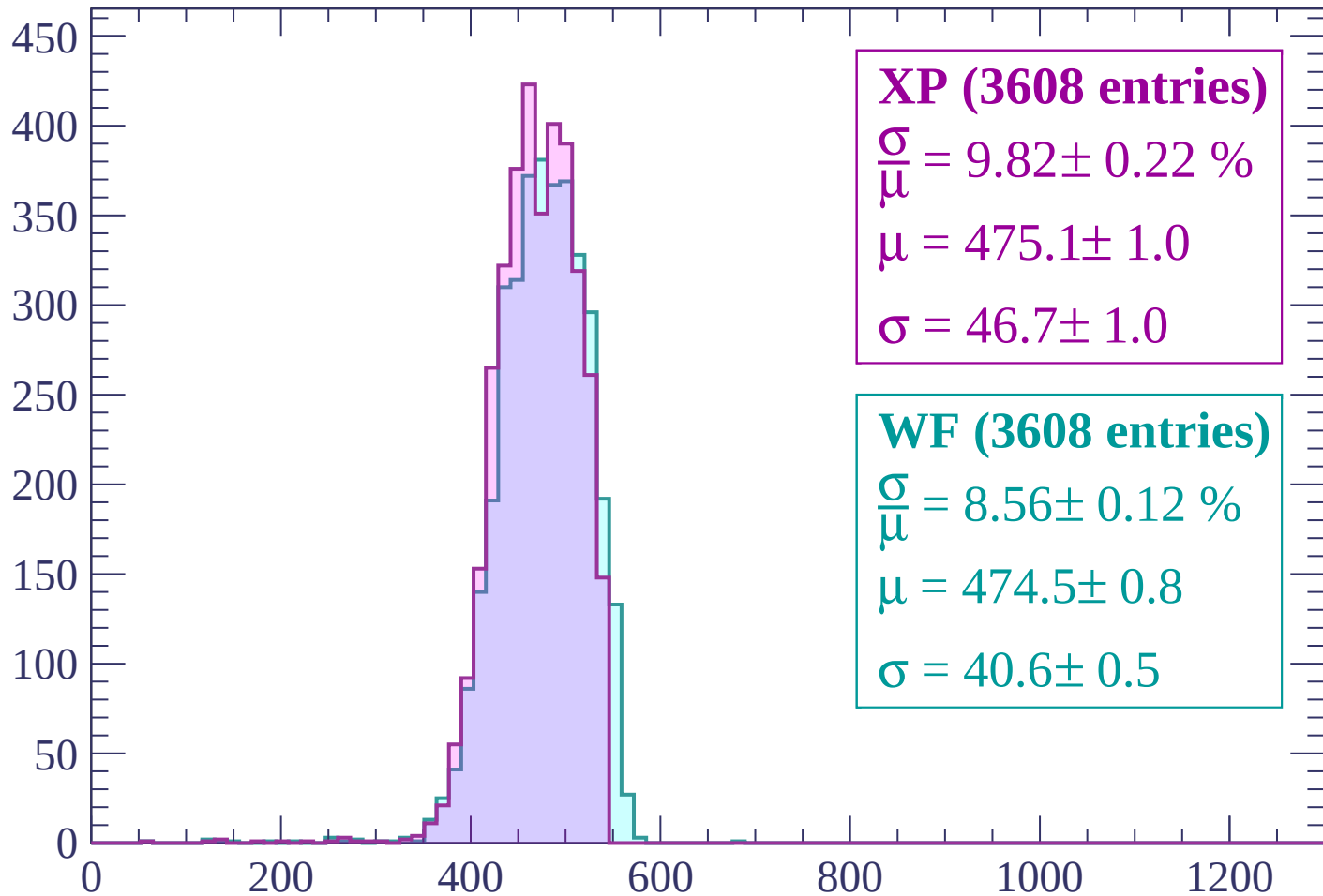
$$\mu = 476.8 \pm 0.8$$

$$\sigma = 40.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1860 < p < -1800$

Count



XP (3608 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.22 \%$$

$$\mu = 475.1 \pm 1.0$$

$$\sigma = 46.7 \pm 1.0$$

WF (3608 entries)

$$\frac{\sigma}{\mu} = 8.56 \pm 0.12 \%$$

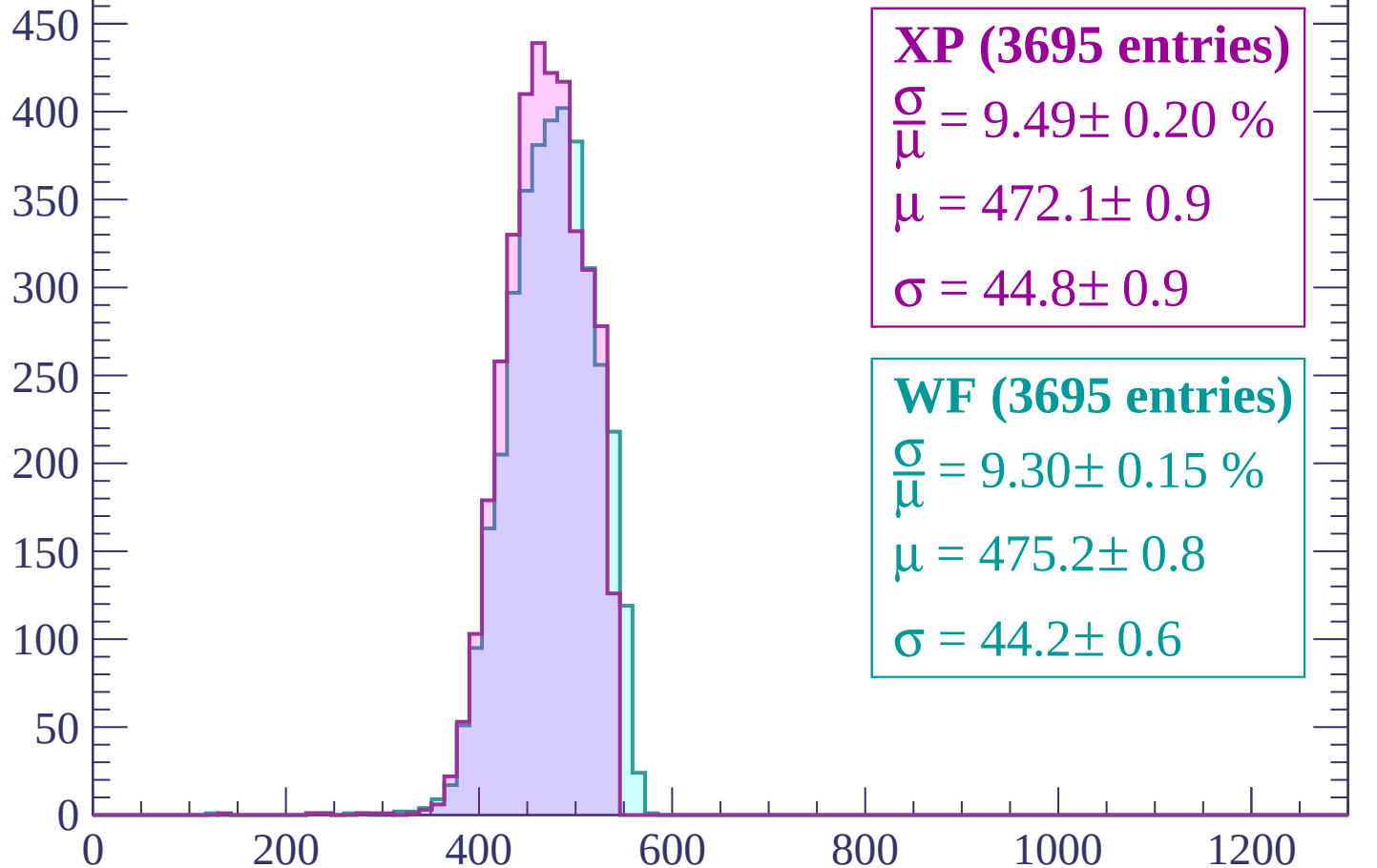
$$\mu = 474.5 \pm 0.8$$

$$\sigma = 40.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1740$

Count



XP (3695 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.20 \%$$

$$\mu = 472.1 \pm 0.9$$

$$\sigma = 44.8 \pm 0.9$$

WF (3695 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.15 \%$$

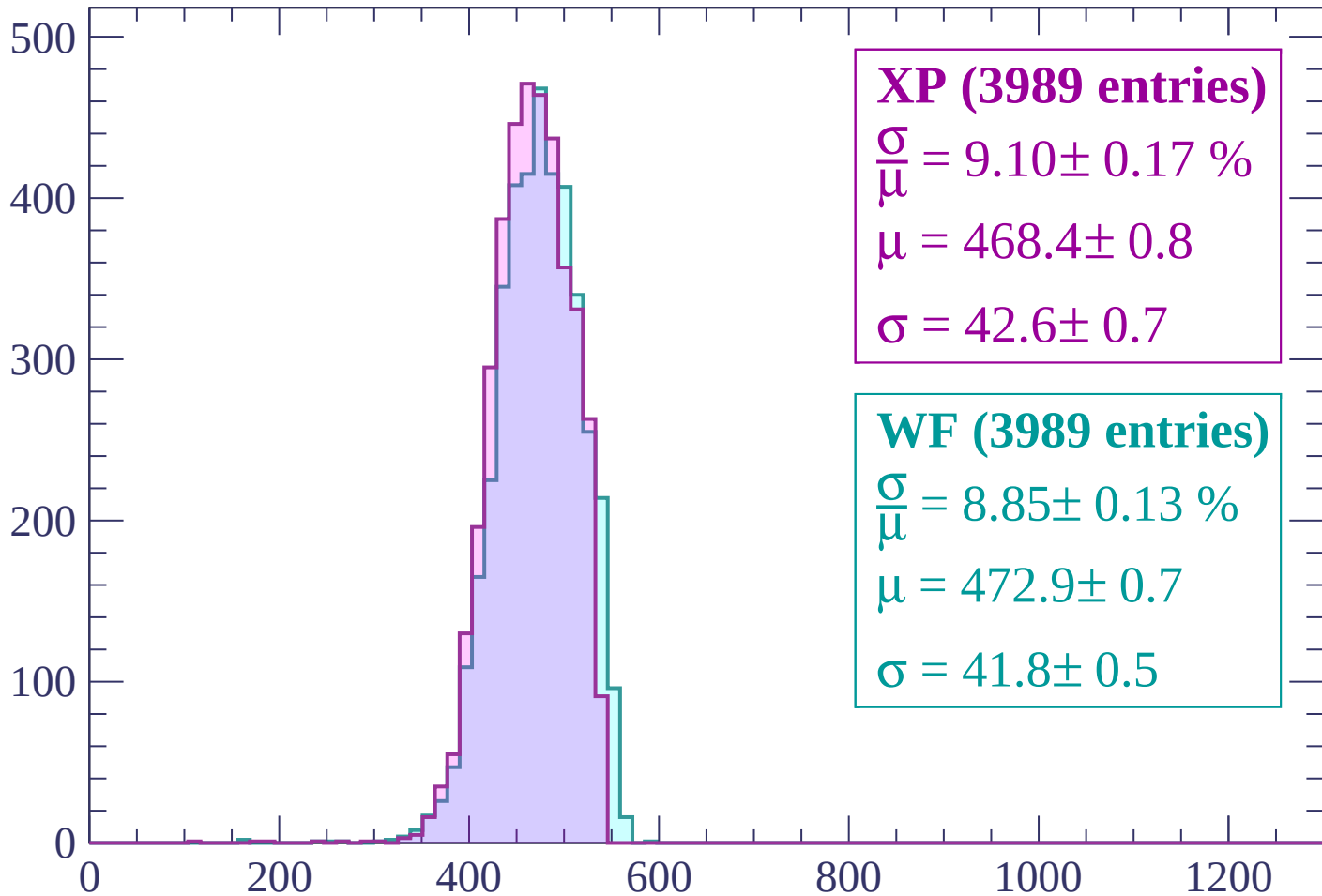
$$\mu = 475.2 \pm 0.8$$

$$\sigma = 44.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (3989 entries)

$\frac{\sigma}{\mu} = 9.10 \pm 0.17 \%$

$\mu = 468.4 \pm 0.8$

$\sigma = 42.6 \pm 0.7$

WF (3989 entries)

$\frac{\sigma}{\mu} = 8.85 \pm 0.13 \%$

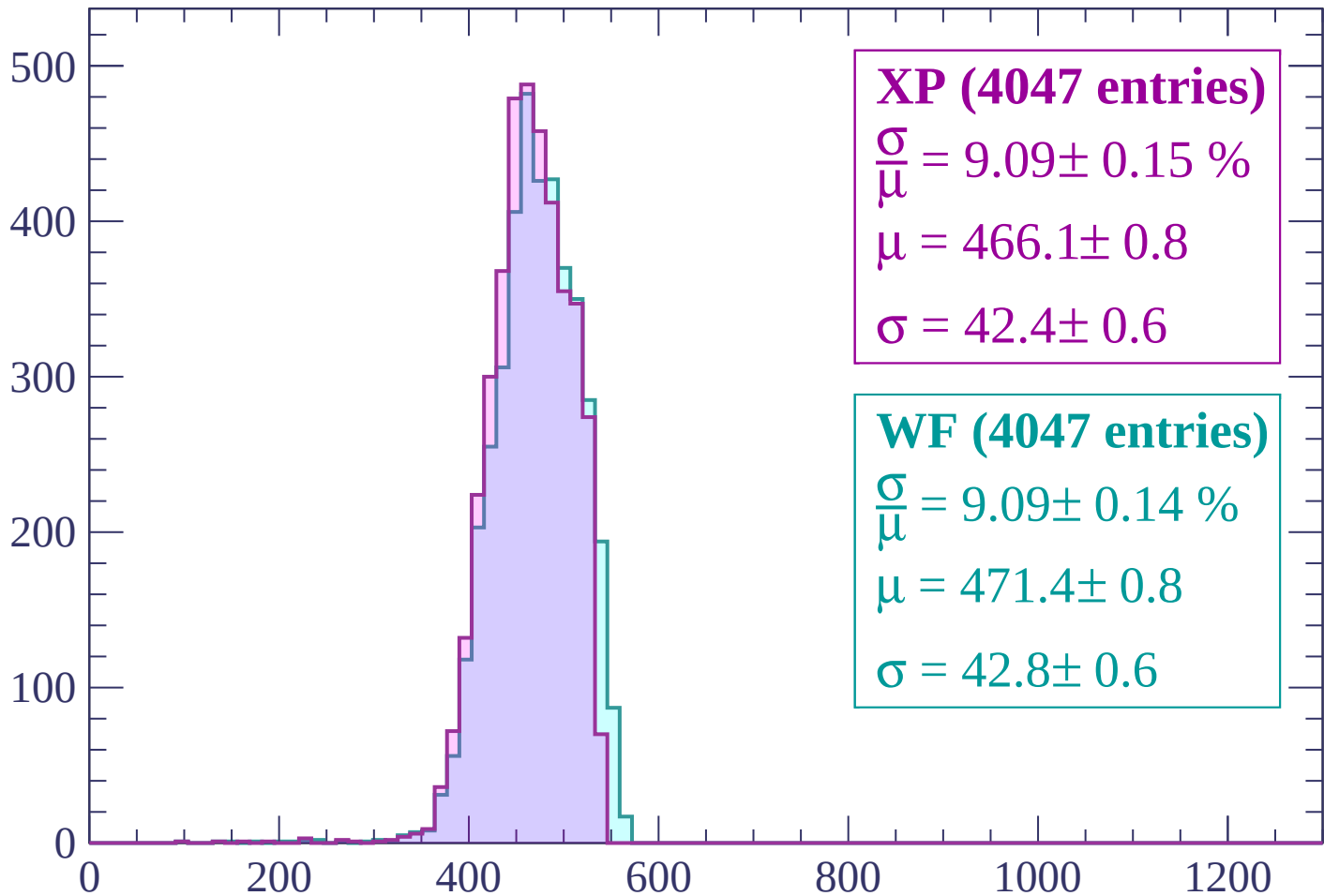
$\mu = 472.9 \pm 0.7$

$\sigma = 41.8 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$

Count



XP (4047 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.15 \%$$

$$\mu = 466.1 \pm 0.8$$

$$\sigma = 42.4 \pm 0.6$$

WF (4047 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.14 \%$$

$$\mu = 471.4 \pm 0.8$$

$$\sigma = 42.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (4185 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.14 \%$$

$$\mu = 462.5 \pm 0.7$$

$$\sigma = 40.3 \pm 0.6$$

WF (4185 entries)

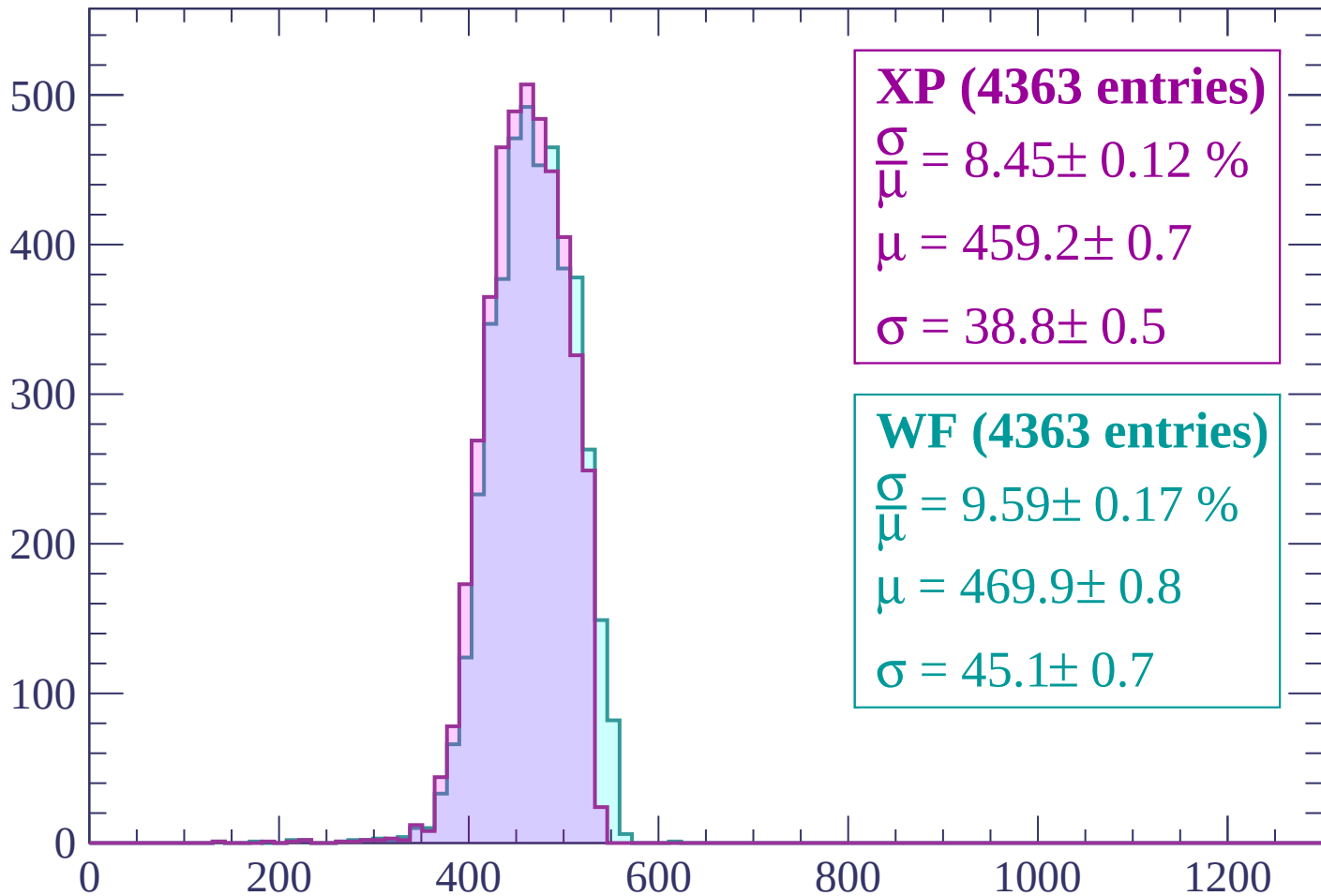
$$\frac{\sigma}{\mu} = 8.87 \pm 0.12 \%$$

$$\mu = 468.2 \pm 0.7$$

$$\sigma = 41.5 \pm 0.5$$

Energy loss | -1560 < p < -1500

Count



XP (4363 entries)

$$\frac{\sigma}{\mu} = 8.45 \pm 0.12 \%$$

$$\mu = 459.2 \pm 0.7$$

$$\sigma = 38.8 \pm 0.5$$

WF (4363 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.17 \%$$

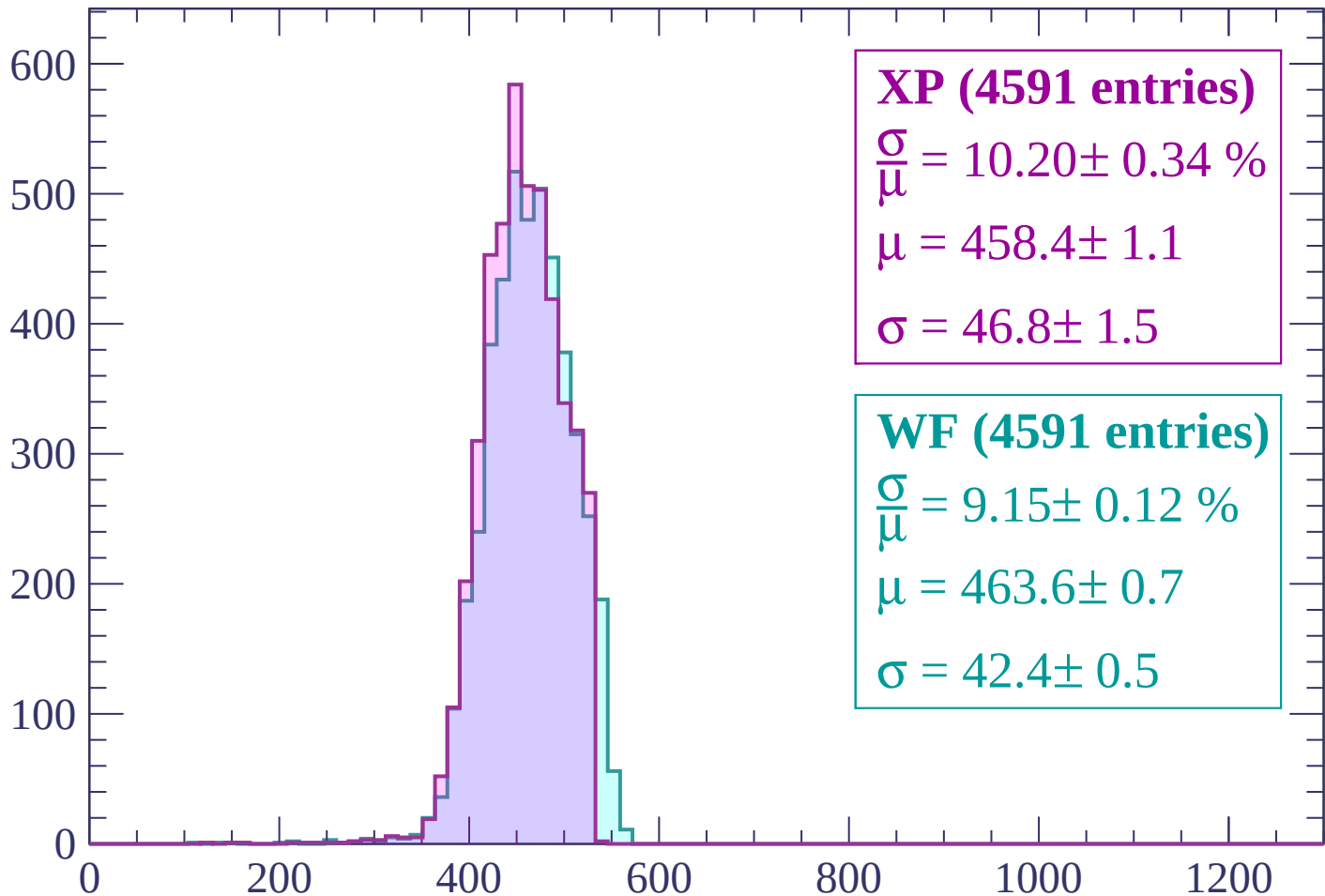
$$\mu = 469.9 \pm 0.8$$

$$\sigma = 45.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (4591 entries)

$\frac{\sigma}{\mu} = 10.20 \pm 0.34 \%$

$\mu = 458.4 \pm 1.1$

$\sigma = 46.8 \pm 1.5$

WF (4591 entries)

$\frac{\sigma}{\mu} = 9.15 \pm 0.12 \%$

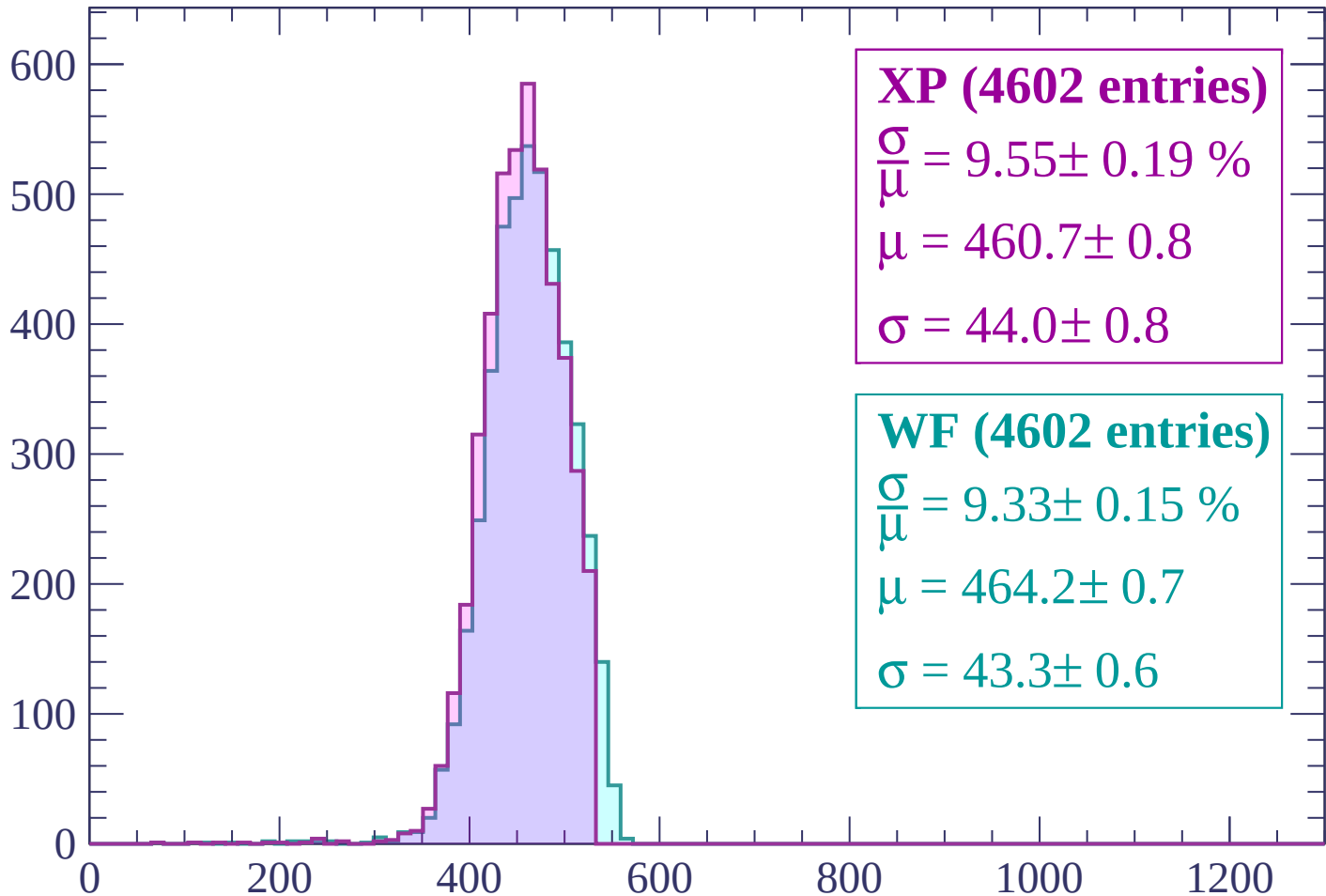
$\mu = 463.6 \pm 0.7$

$\sigma = 42.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (4602 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.19 \%$$

$$\mu = 460.7 \pm 0.8$$

$$\sigma = 44.0 \pm 0.8$$

WF (4602 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.15 \%$$

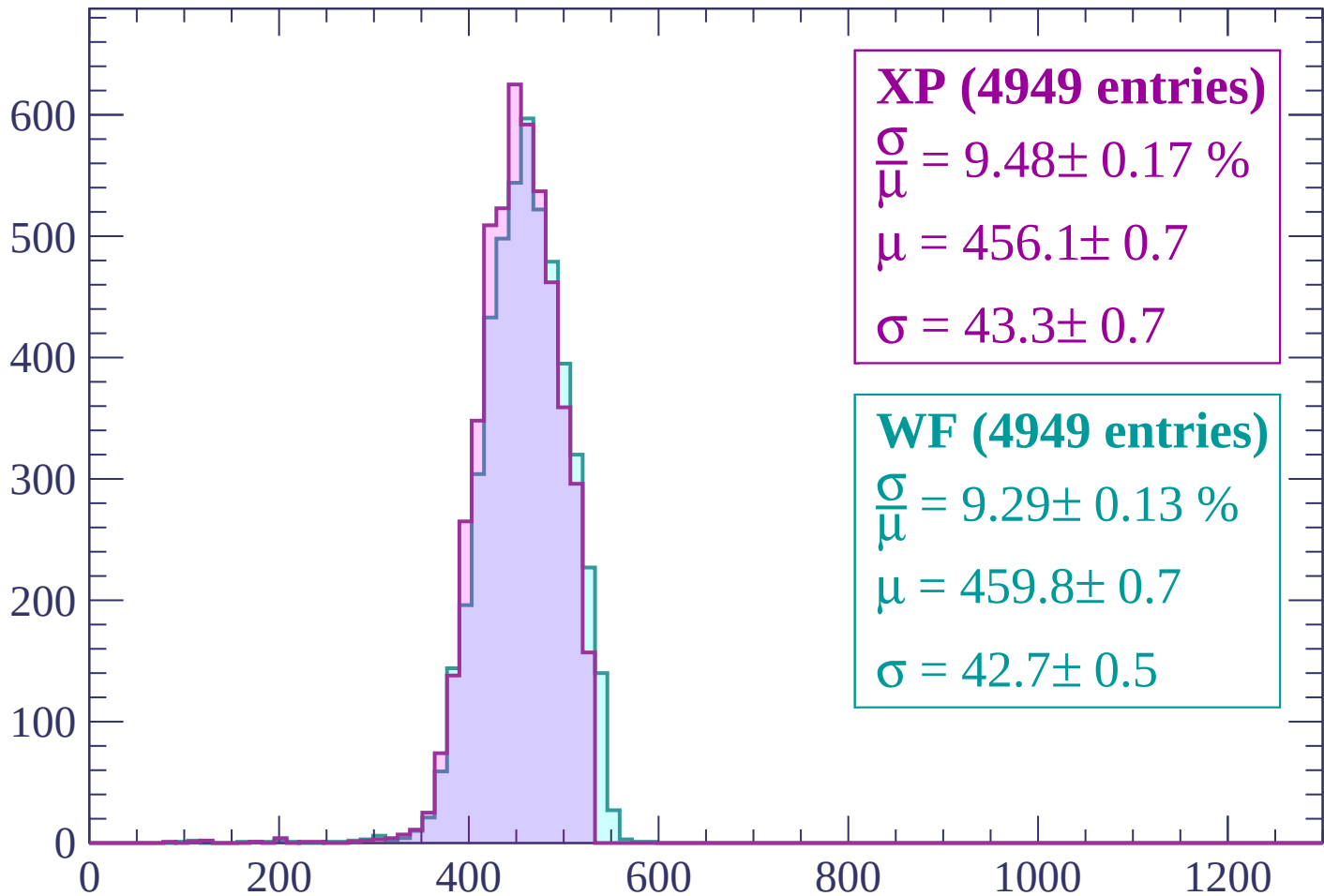
$$\mu = 464.2 \pm 0.7$$

$$\sigma = 43.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (4949 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.17 \%$$

$$\mu = 456.1 \pm 0.7$$

$$\sigma = 43.3 \pm 0.7$$

WF (4949 entries)

$$\frac{\sigma}{\mu} = 9.29 \pm 0.13 \%$$

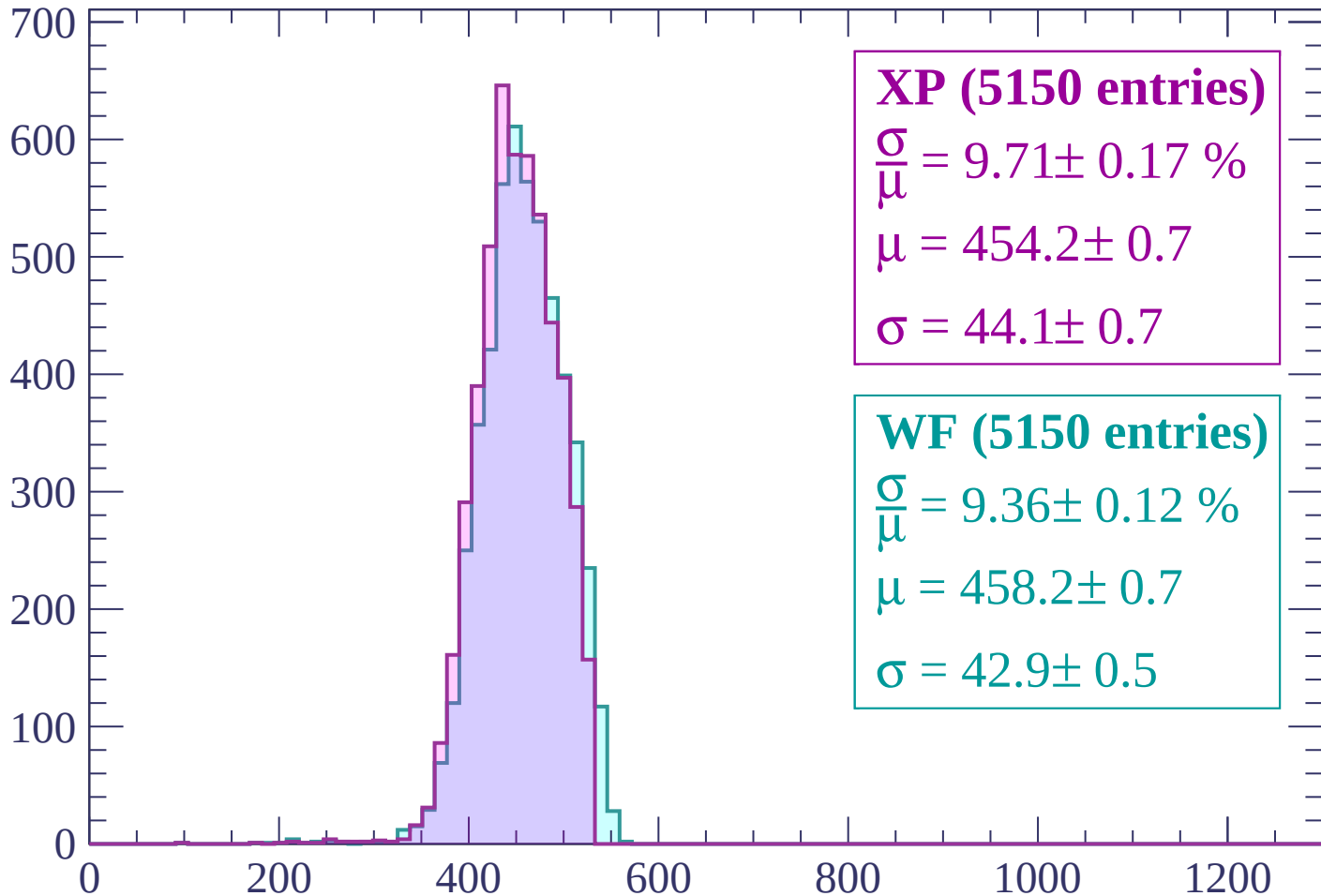
$$\mu = 459.8 \pm 0.7$$

$$\sigma = 42.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count



XP (5150 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.17 \%$$

$$\mu = 454.2 \pm 0.7$$

$$\sigma = 44.1 \pm 0.7$$

WF (5150 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.12 \%$$

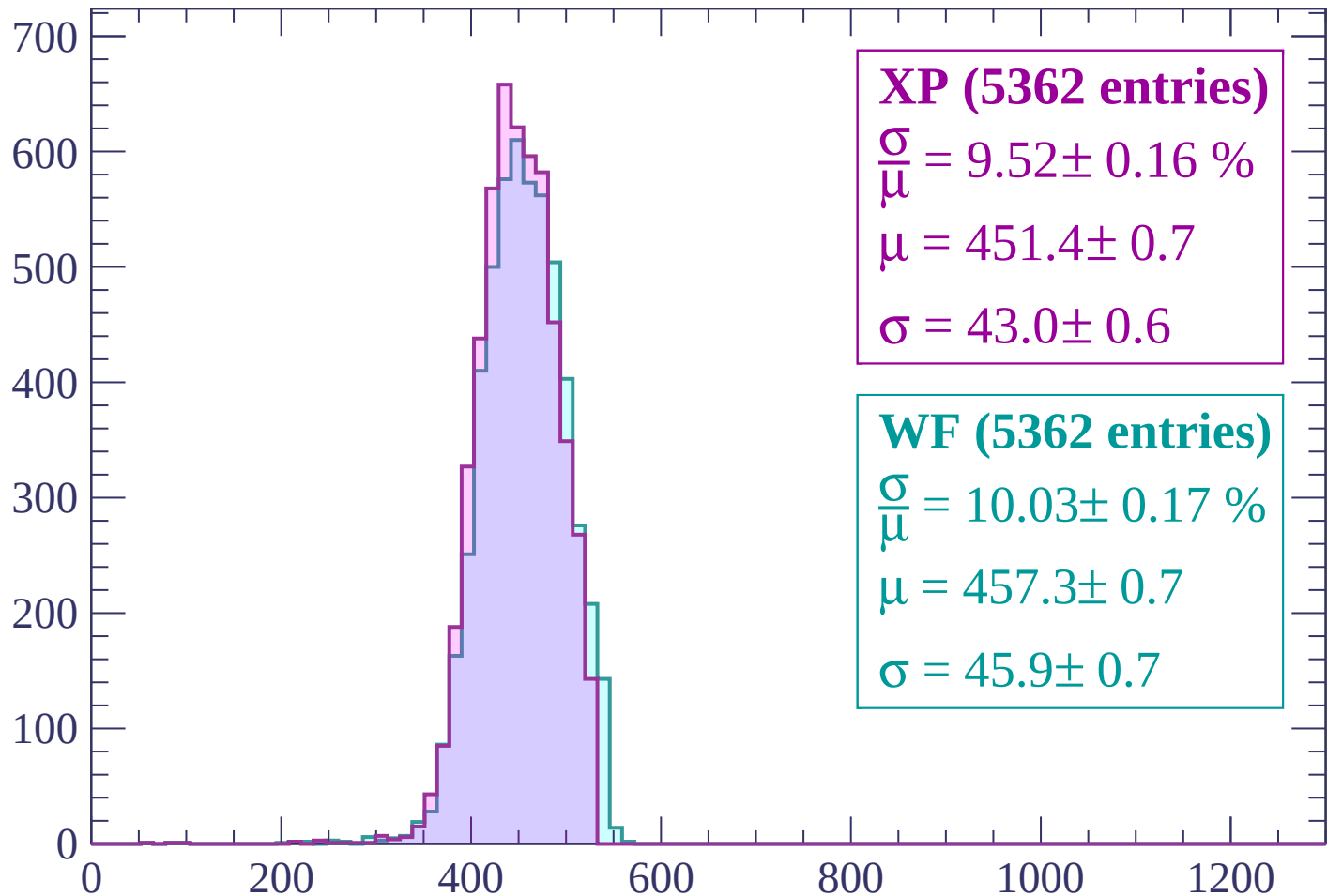
$$\mu = 458.2 \pm 0.7$$

$$\sigma = 42.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1260 < p < -1200$

Count



XP (5362 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.16 \%$$

$$\mu = 451.4 \pm 0.7$$

$$\sigma = 43.0 \pm 0.6$$

WF (5362 entries)

$$\frac{\sigma}{\mu} = 10.03 \pm 0.17 \%$$

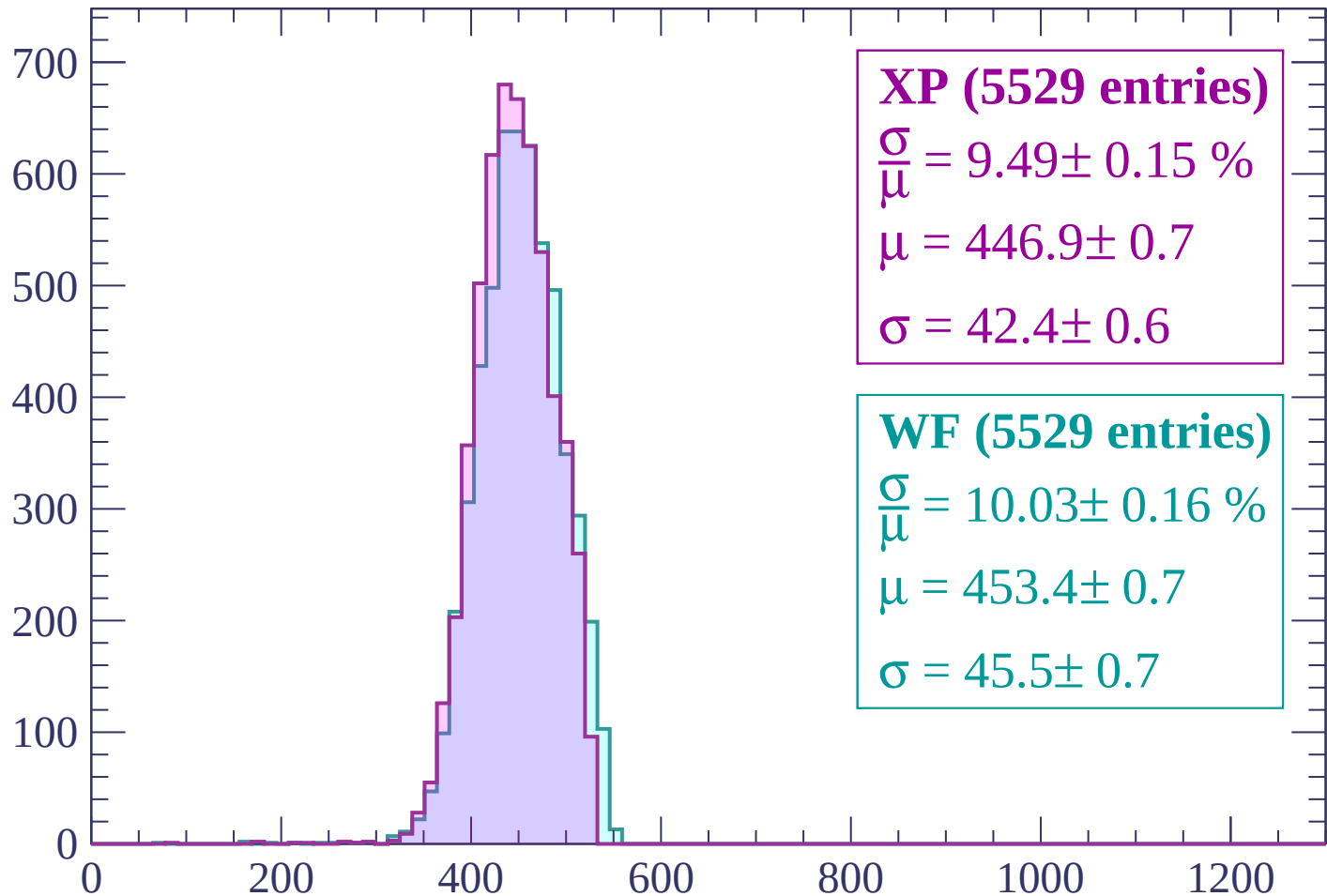
$$\mu = 457.3 \pm 0.7$$

$$\sigma = 45.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count



XP (5529 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.15 \%$$

$$\mu = 446.9 \pm 0.7$$

$$\sigma = 42.4 \pm 0.6$$

WF (5529 entries)

$$\frac{\sigma}{\mu} = 10.03 \pm 0.16 \%$$

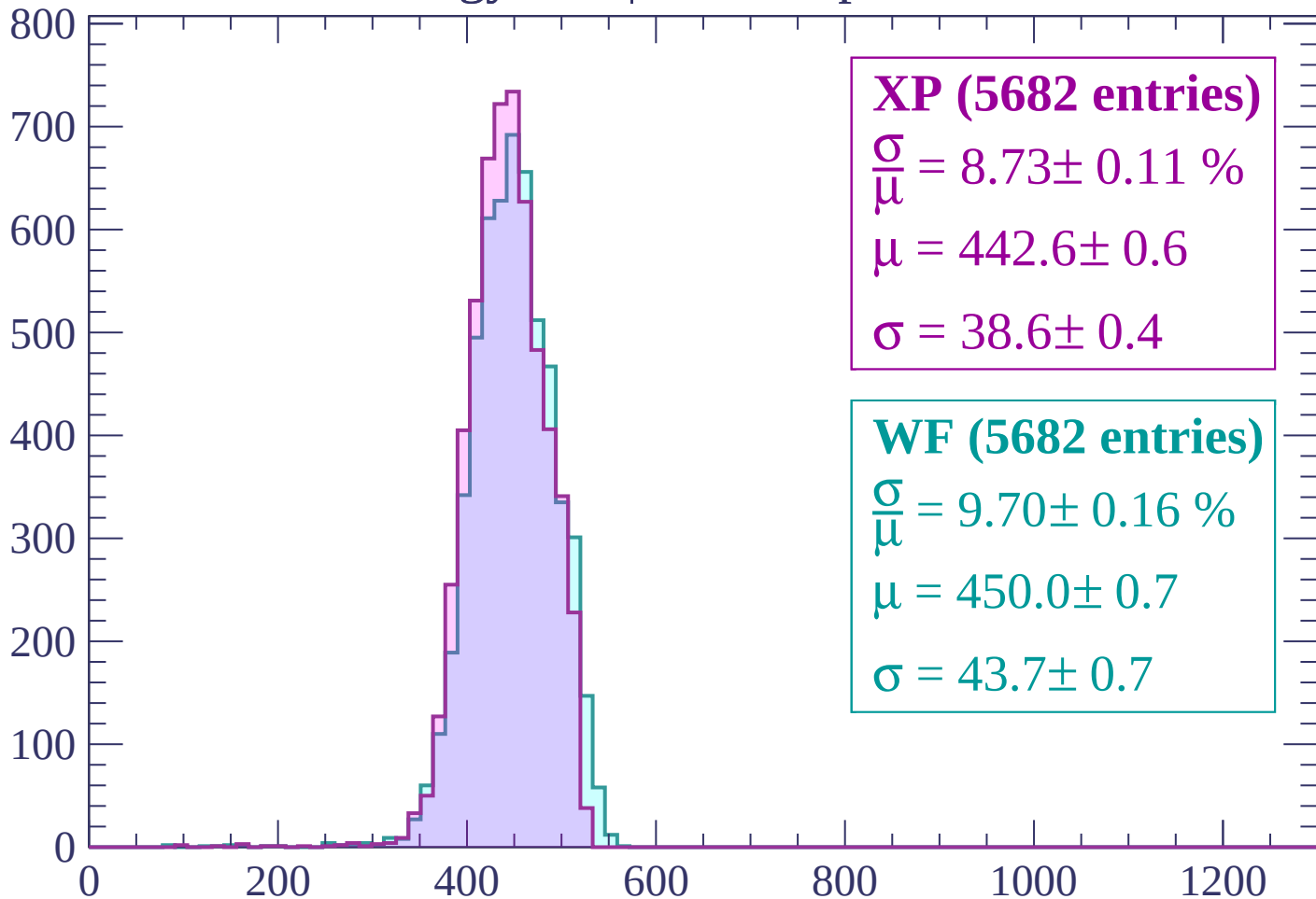
$$\mu = 453.4 \pm 0.7$$

$$\sigma = 45.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count



XP (5682 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.11 \%$$

$$\mu = 442.6 \pm 0.6$$

$$\sigma = 38.6 \pm 0.4$$

WF (5682 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.16 \%$$

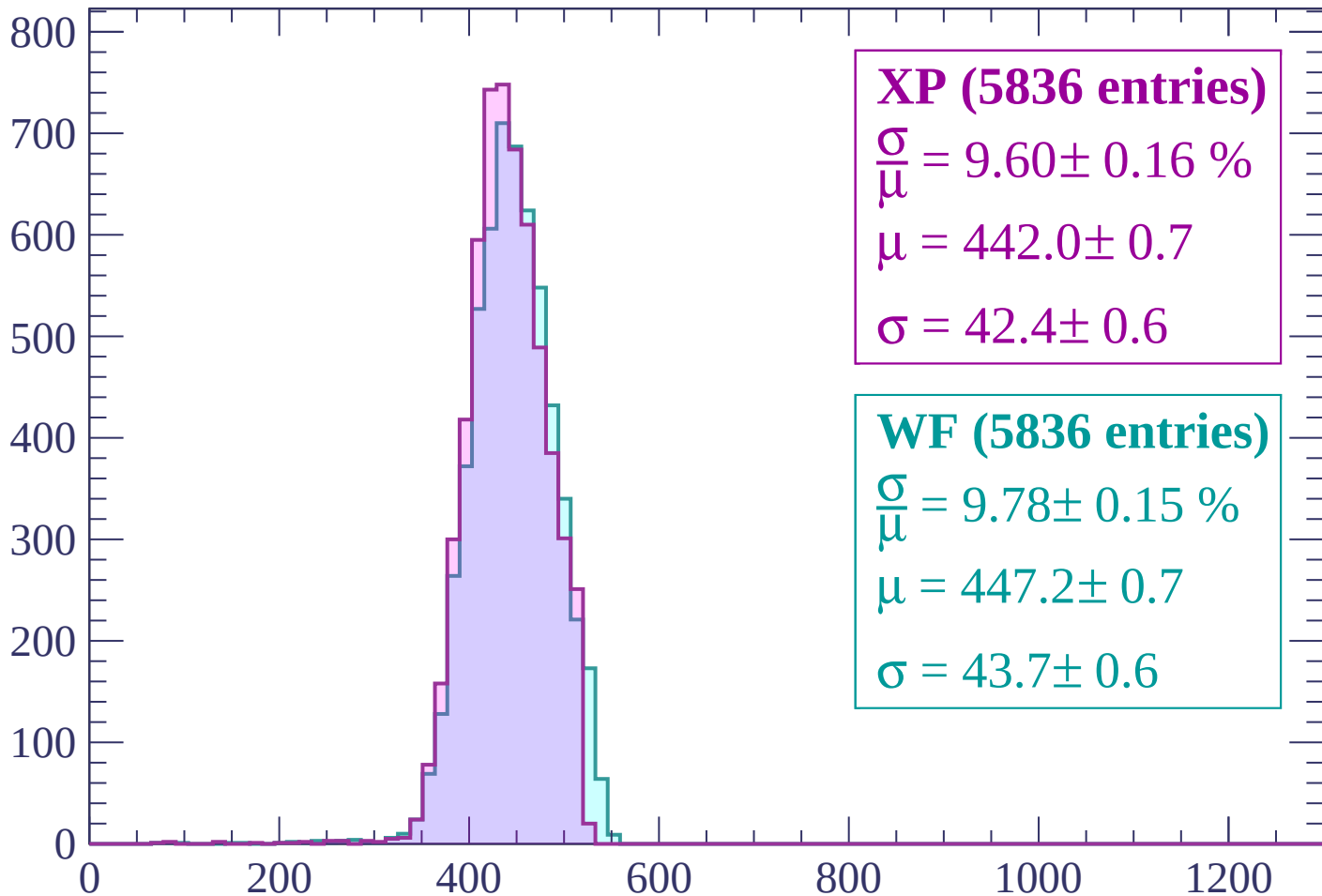
$$\mu = 450.0 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (5836 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$$

$$\mu = 442.0 \pm 0.7$$

$$\sigma = 42.4 \pm 0.6$$

WF (5836 entries)

$$\frac{\sigma}{\mu} = 9.78 \pm 0.15 \%$$

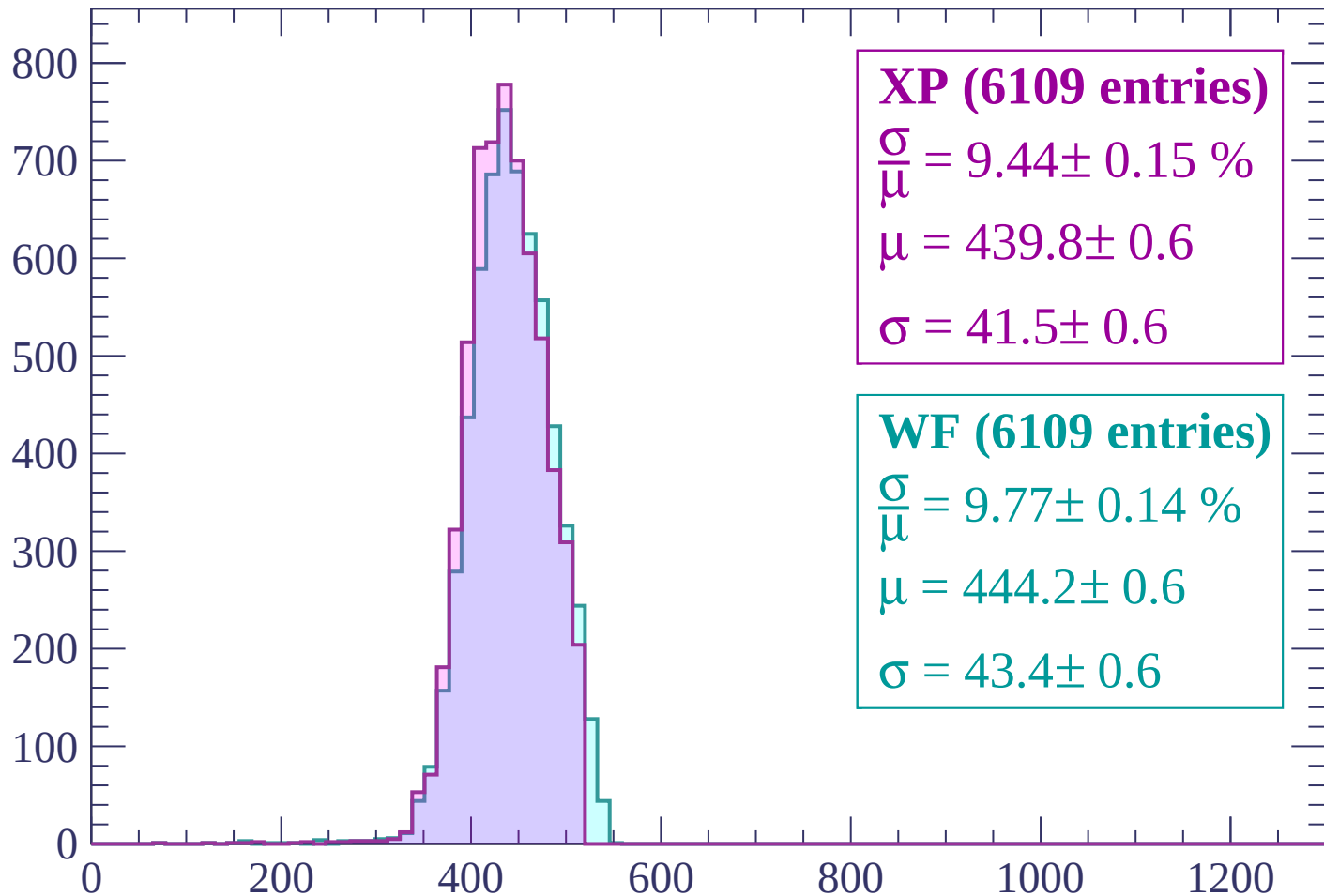
$$\mu = 447.2 \pm 0.7$$

$$\sigma = 43.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (6109 entries)

$$\frac{\sigma}{\mu} = 9.44 \pm 0.15 \%$$

$$\mu = 439.8 \pm 0.6$$

$$\sigma = 41.5 \pm 0.6$$

WF (6109 entries)

$$\frac{\sigma}{\mu} = 9.77 \pm 0.14 \%$$

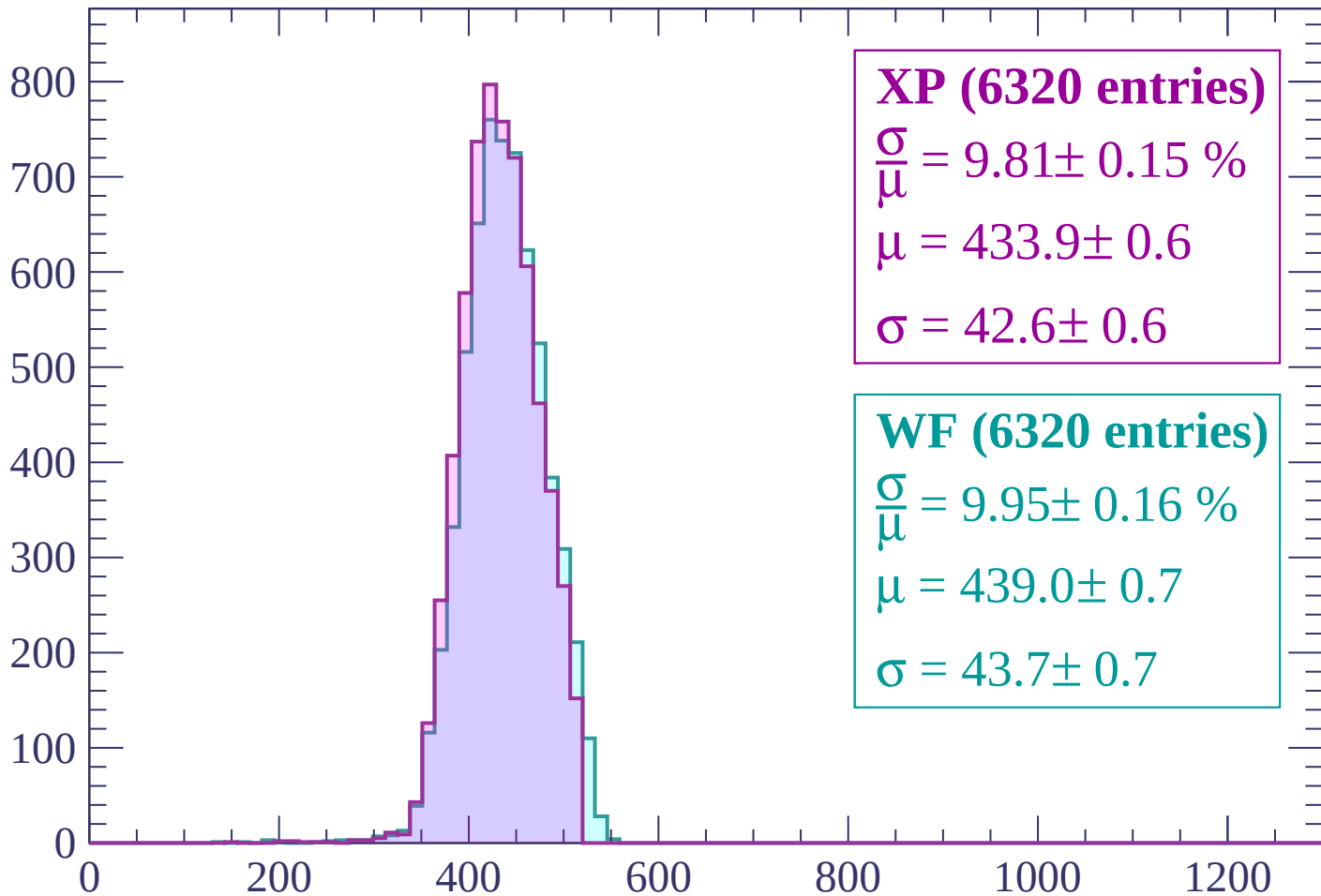
$$\mu = 444.2 \pm 0.6$$

$$\sigma = 43.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (6320 entries)

$$\frac{\sigma}{\mu} = 9.81 \pm 0.15 \%$$

$$\mu = 433.9 \pm 0.6$$

$$\sigma = 42.6 \pm 0.6$$

WF (6320 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.16 \%$$

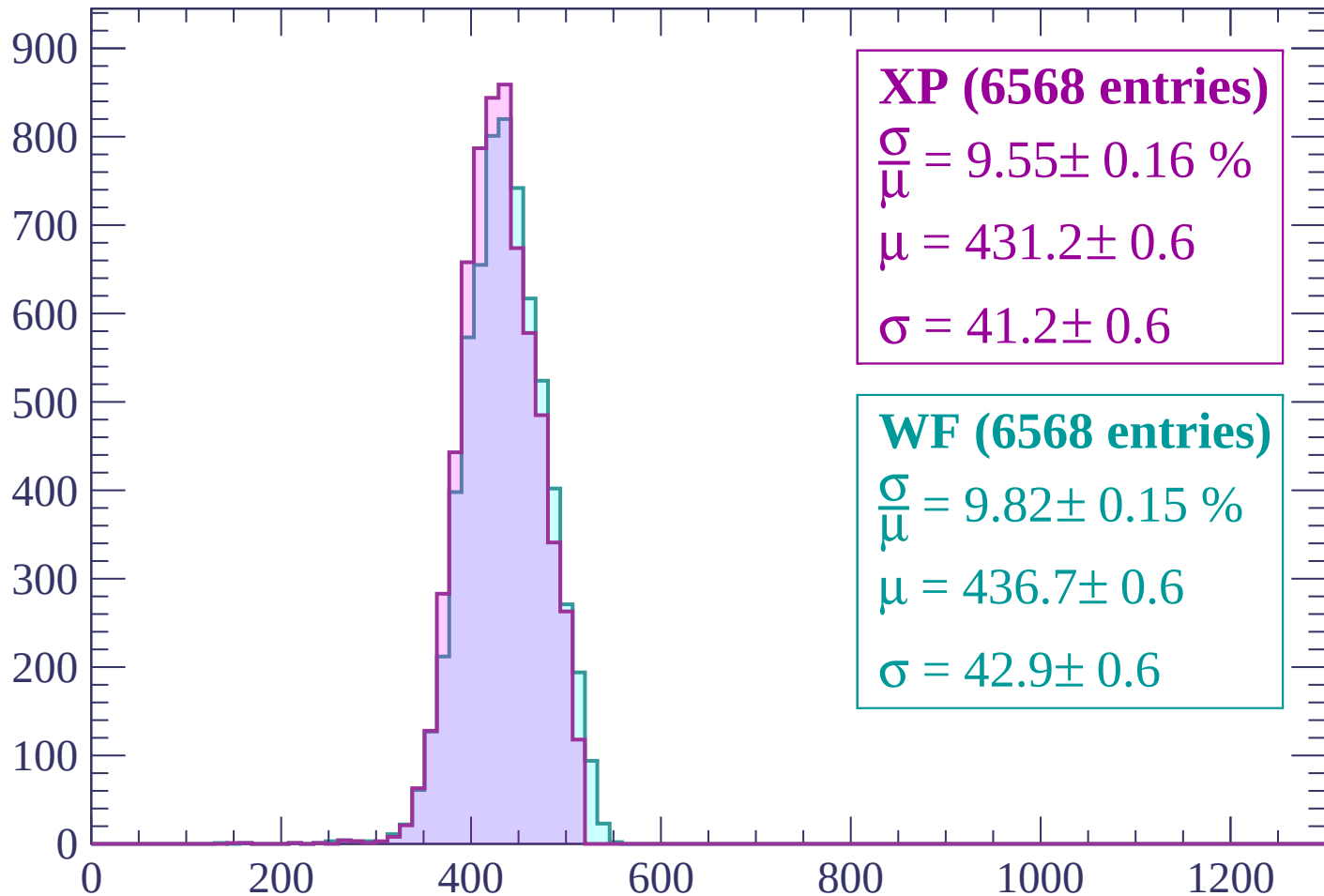
$$\mu = 439.0 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (6568 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.16 \%$$

$$\mu = 431.2 \pm 0.6$$

$$\sigma = 41.2 \pm 0.6$$

WF (6568 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.15 \%$$

$$\mu = 436.7 \pm 0.6$$

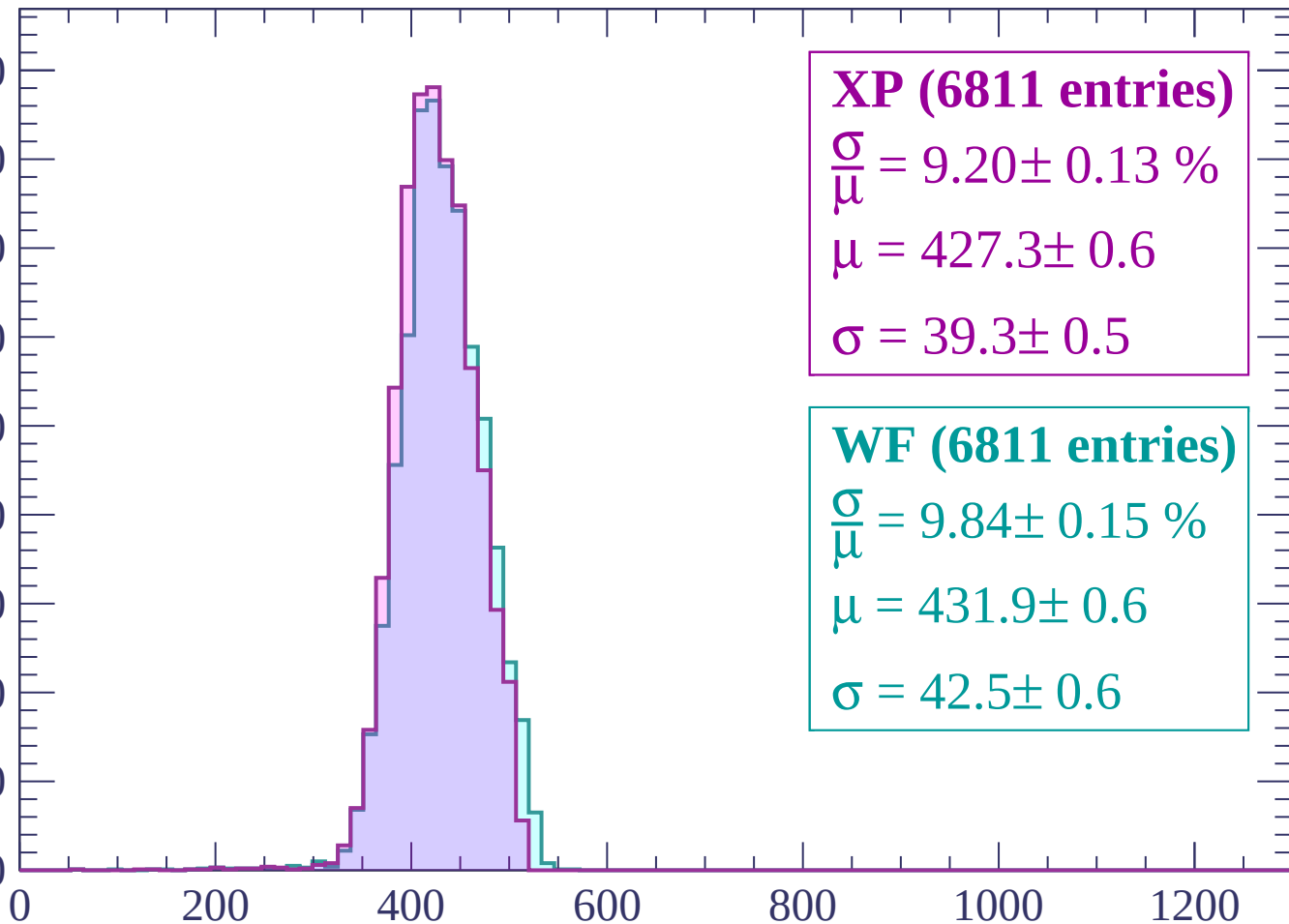
$$\sigma = 42.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

900
800
700
600
500
400
300
200
100
0



XP (6811 entries)

$\frac{\sigma}{\mu} = 9.20 \pm 0.13 \%$

$\mu = 427.3 \pm 0.6$

$\sigma = 39.3 \pm 0.5$

WF (6811 entries)

$\frac{\sigma}{\mu} = 9.84 \pm 0.15 \%$

$\mu = 431.9 \pm 0.6$

$\sigma = 42.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count

1000

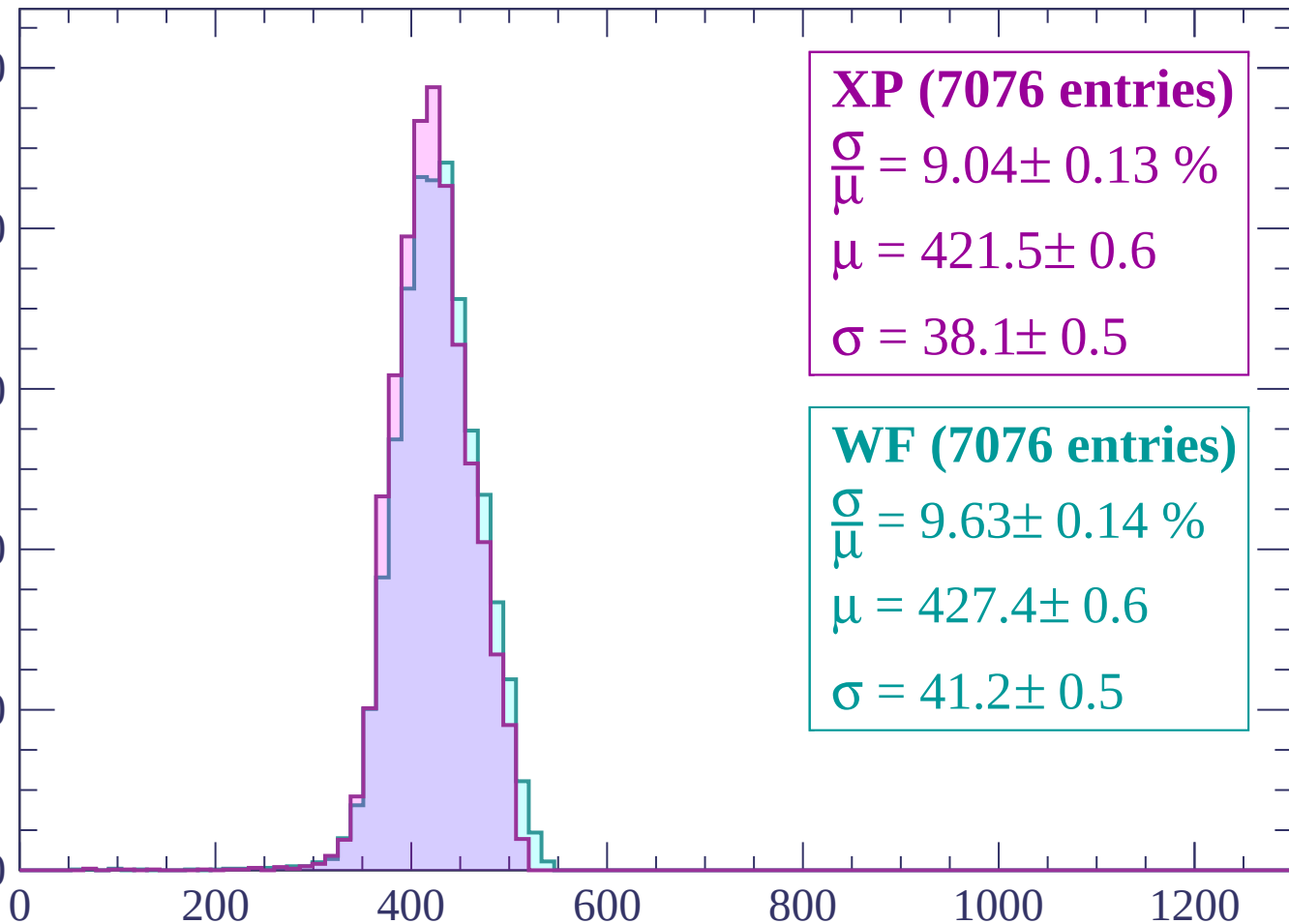
800

600

400

200

0



XP (7076 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.13 \%$

$\mu = 421.5 \pm 0.6$

$\sigma = 38.1 \pm 0.5$

WF (7076 entries)

$\frac{\sigma}{\mu} = 9.63 \pm 0.14 \%$

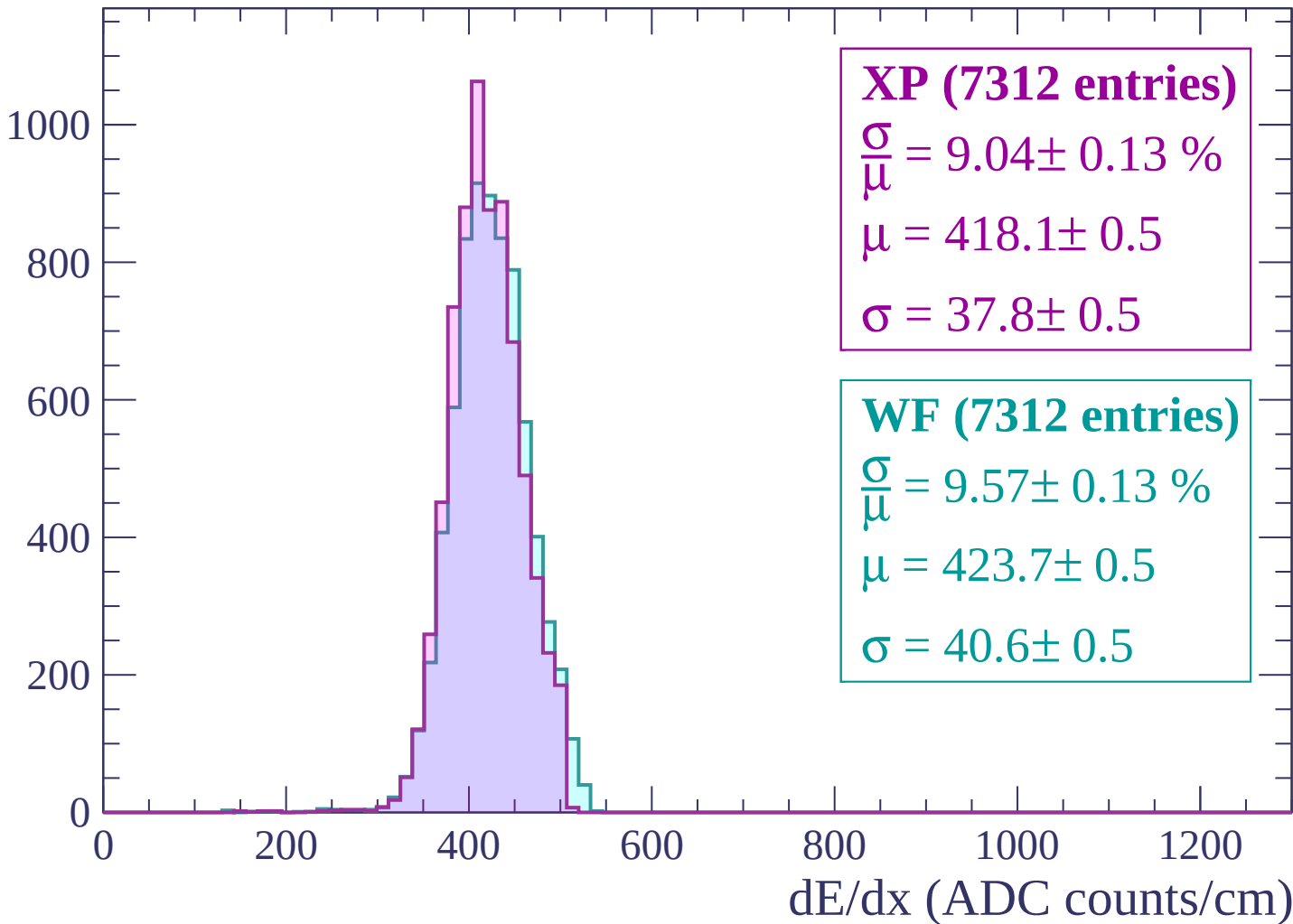
$\mu = 427.4 \pm 0.6$

$\sigma = 41.2 \pm 0.5$

dE/dx (ADC counts/cm)

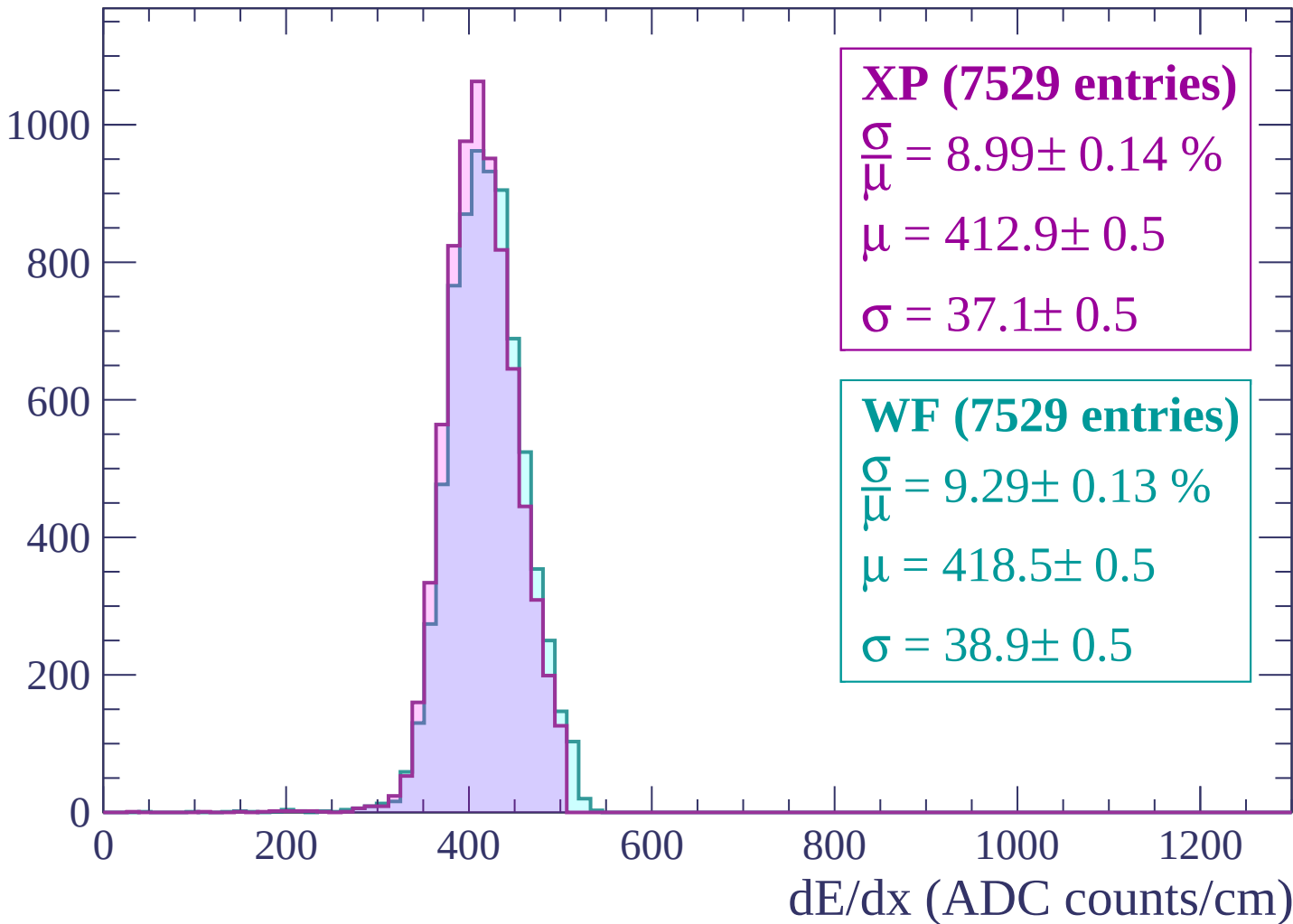
Energy loss | $-720 < p < -660$

Count



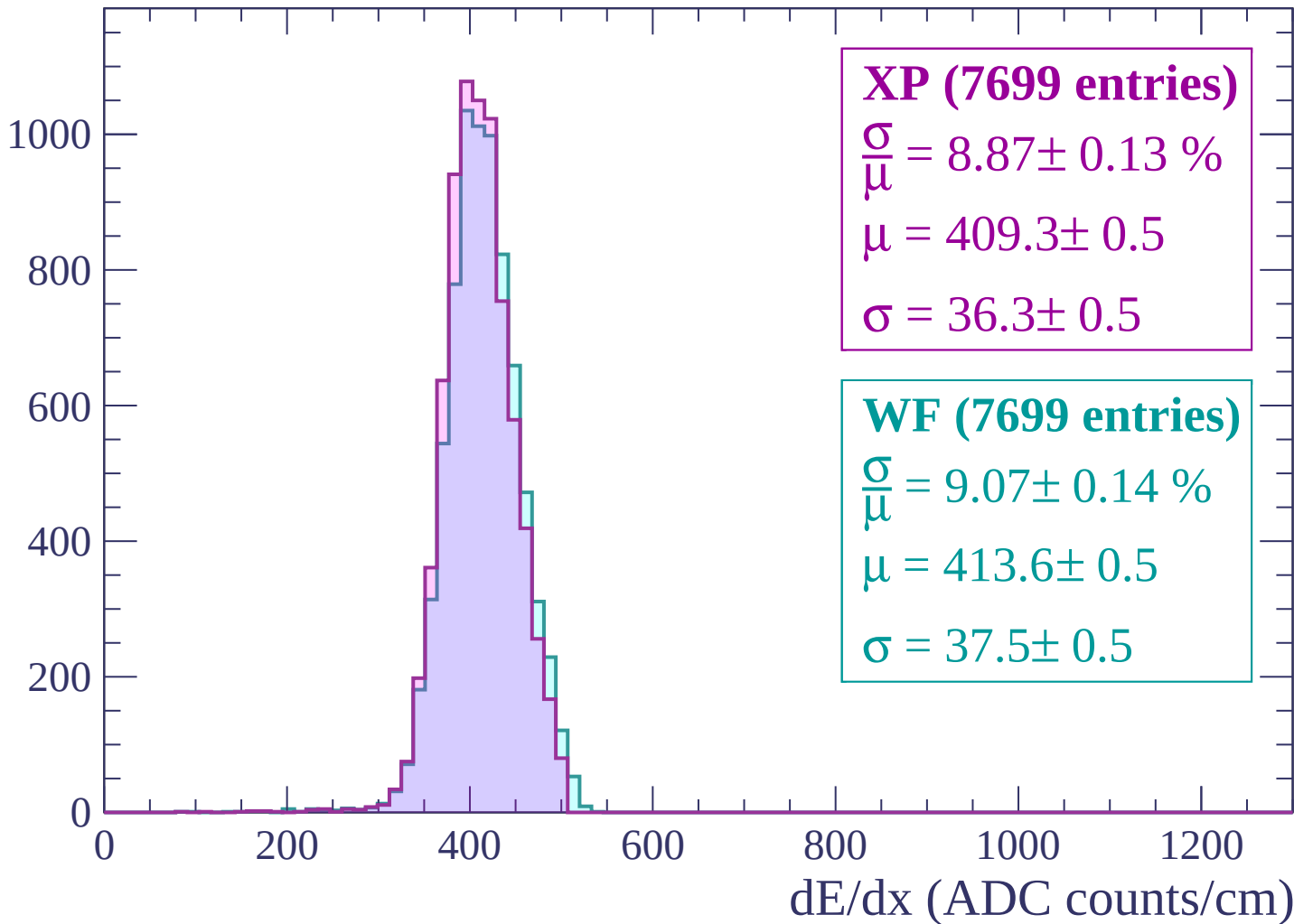
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



Energy loss | $-540 < p < -480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8205 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.14 \%$$

$$\mu = 404.8 \pm 0.5$$

$$\sigma = 35.3 \pm 0.5$$

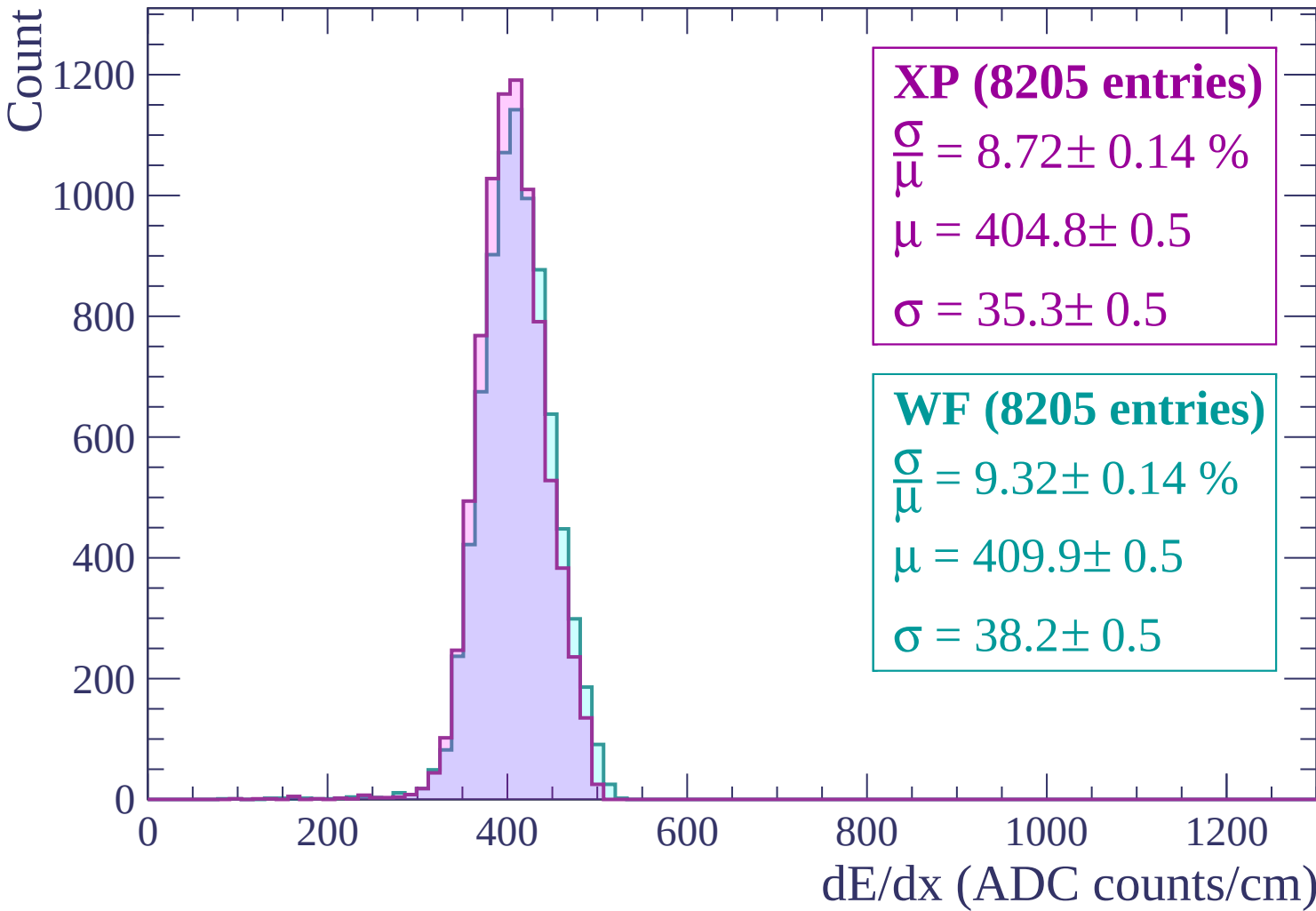
WF (8205 entries)

$$\frac{\sigma}{\mu} = 9.32 \pm 0.14 \%$$

$$\mu = 409.9 \pm 0.5$$

$$\sigma = 38.2 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $-480 < p < -420$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8363 entries)

$$\frac{\sigma}{\mu} = 8.66 \pm 0.13 \%$$

$$\mu = 401.5 \pm 0.5$$

$$\sigma = 34.8 \pm 0.5$$

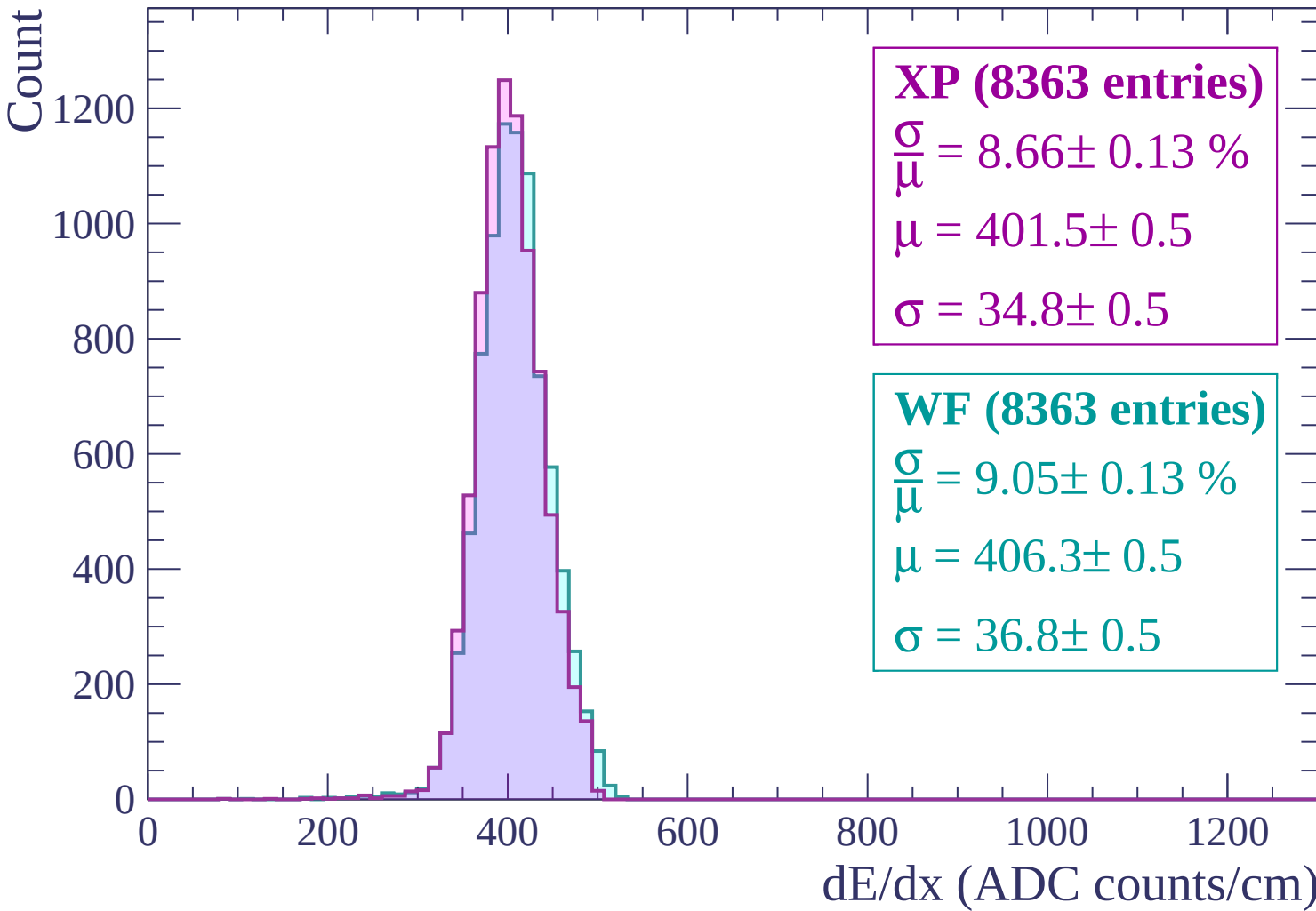
WF (8363 entries)

$$\frac{\sigma}{\mu} = 9.05 \pm 0.13 \%$$

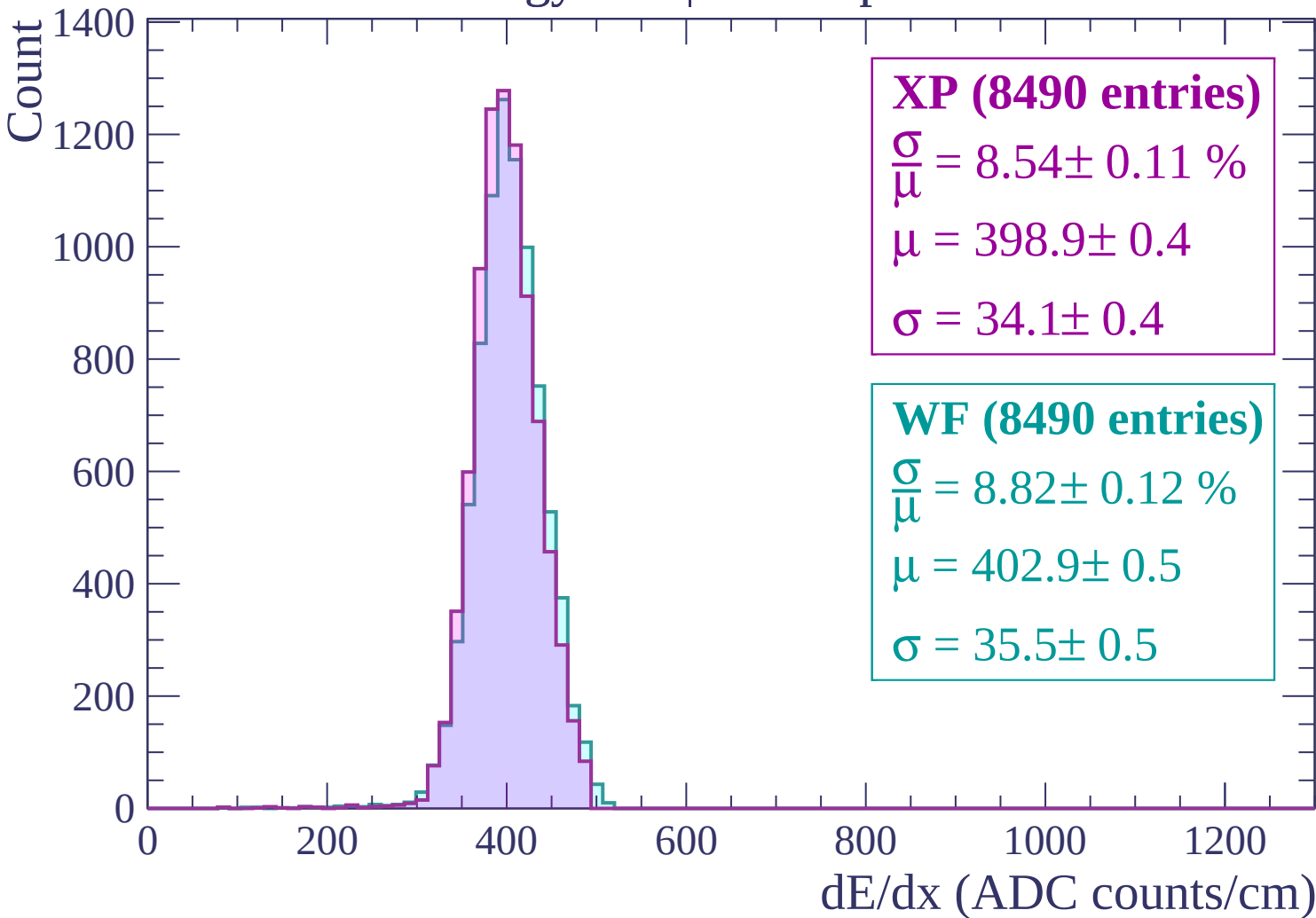
$$\mu = 406.3 \pm 0.5$$

$$\sigma = 36.8 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8460 entries)

$$\frac{\sigma}{\mu} = 8.68 \pm 0.11 \%$$

$$\mu = 397.6 \pm 0.4$$

$$\sigma = 34.5 \pm 0.4$$

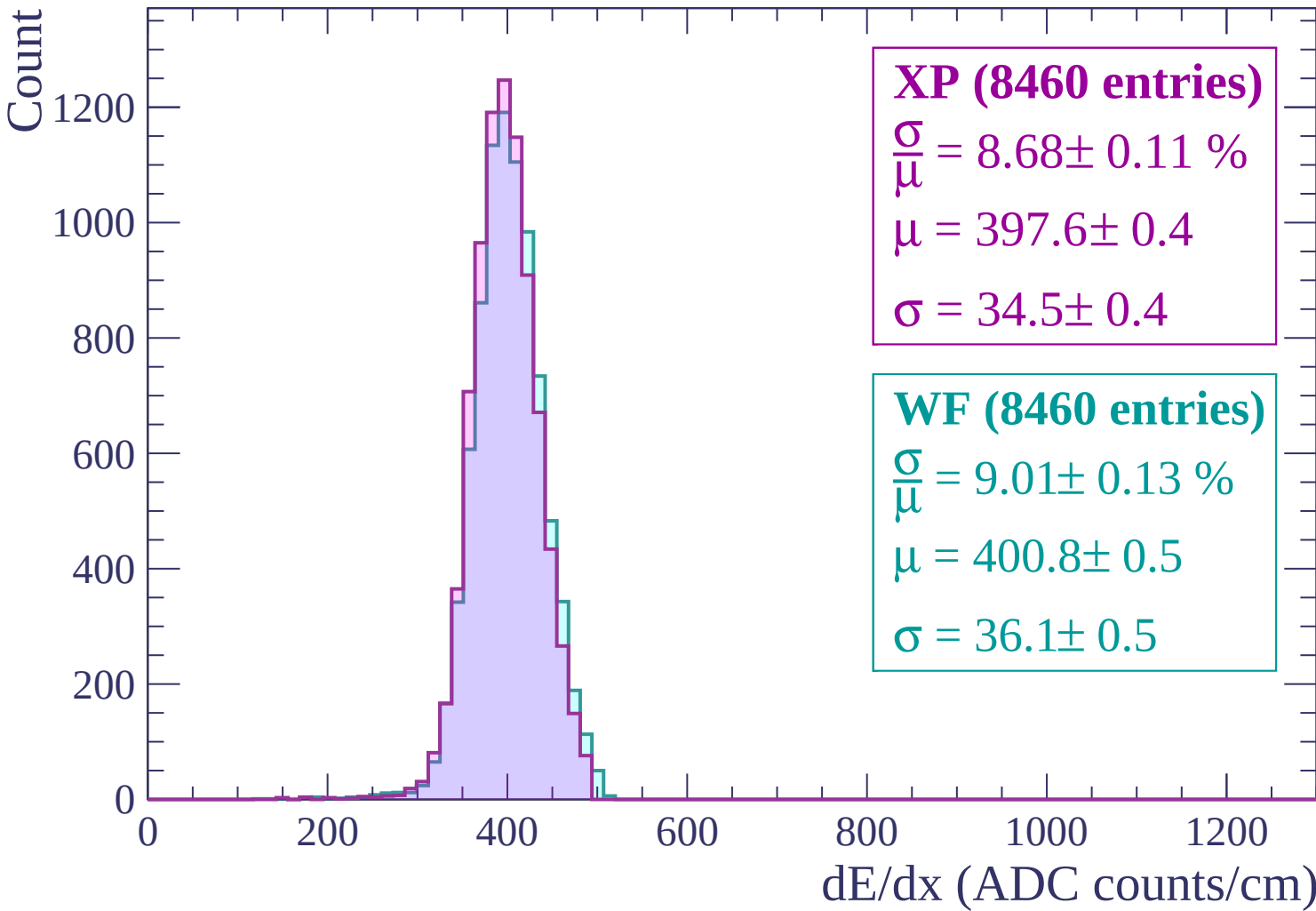
WF (8460 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.13 \%$$

$$\mu = 400.8 \pm 0.5$$

$$\sigma = 36.1 \pm 0.5$$

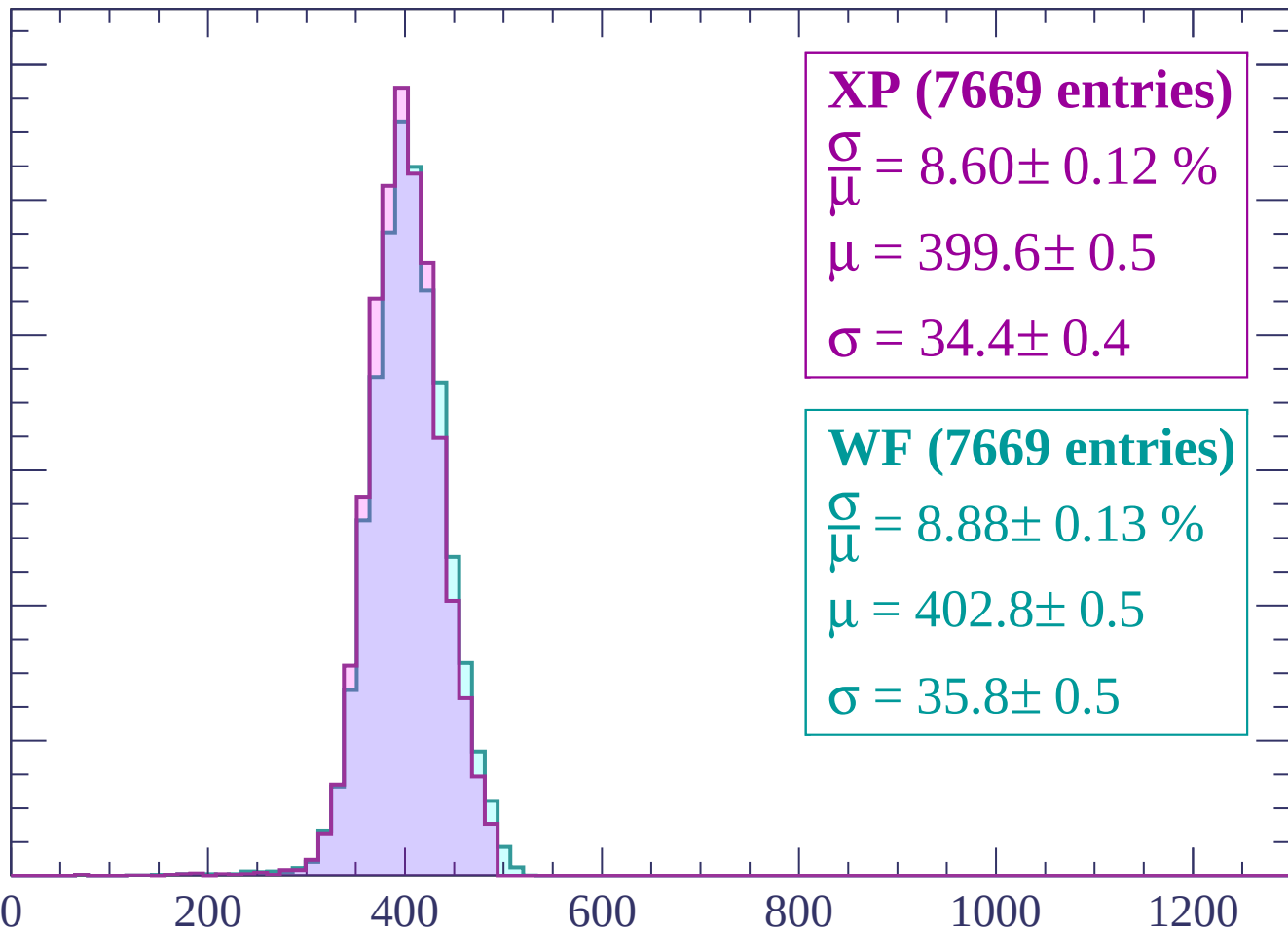
dE/dx (ADC counts/cm)



Energy loss | $-300 < p < -240$

Count

1200
1000
800
600
400
200
0



XP (7669 entries)

$\frac{\sigma}{\mu} = 8.60 \pm 0.12 \%$

$\mu = 399.6 \pm 0.5$

$\sigma = 34.4 \pm 0.4$

WF (7669 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.13 \%$

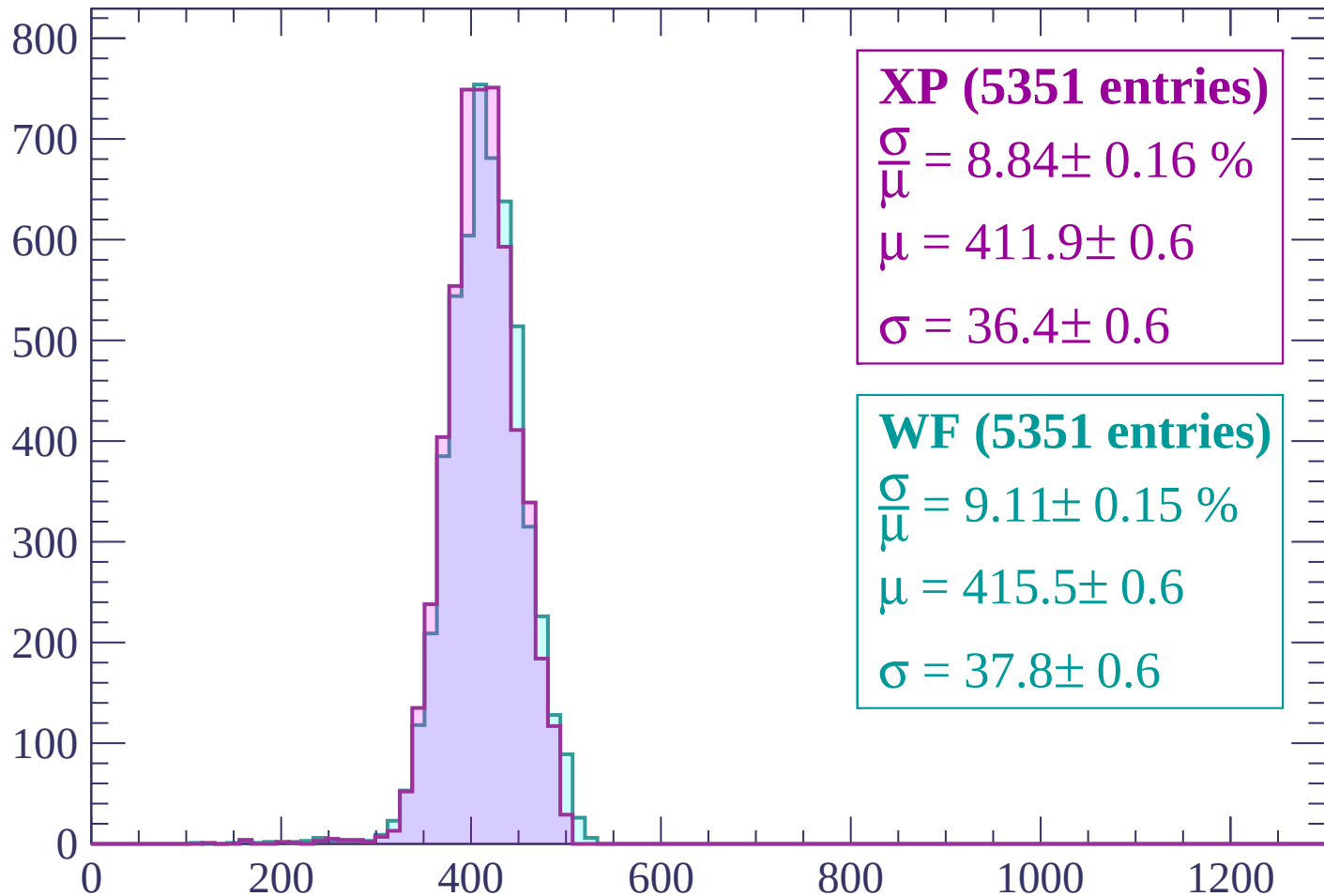
$\mu = 402.8 \pm 0.5$

$\sigma = 35.8 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (5351 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$$

$$\mu = 411.9 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

WF (5351 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.15 \%$$

$$\mu = 415.5 \pm 0.6$$

$$\sigma = 37.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -180 < p < -120

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (3554 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.16 \%$

$\mu = 449.6 \pm 0.7$

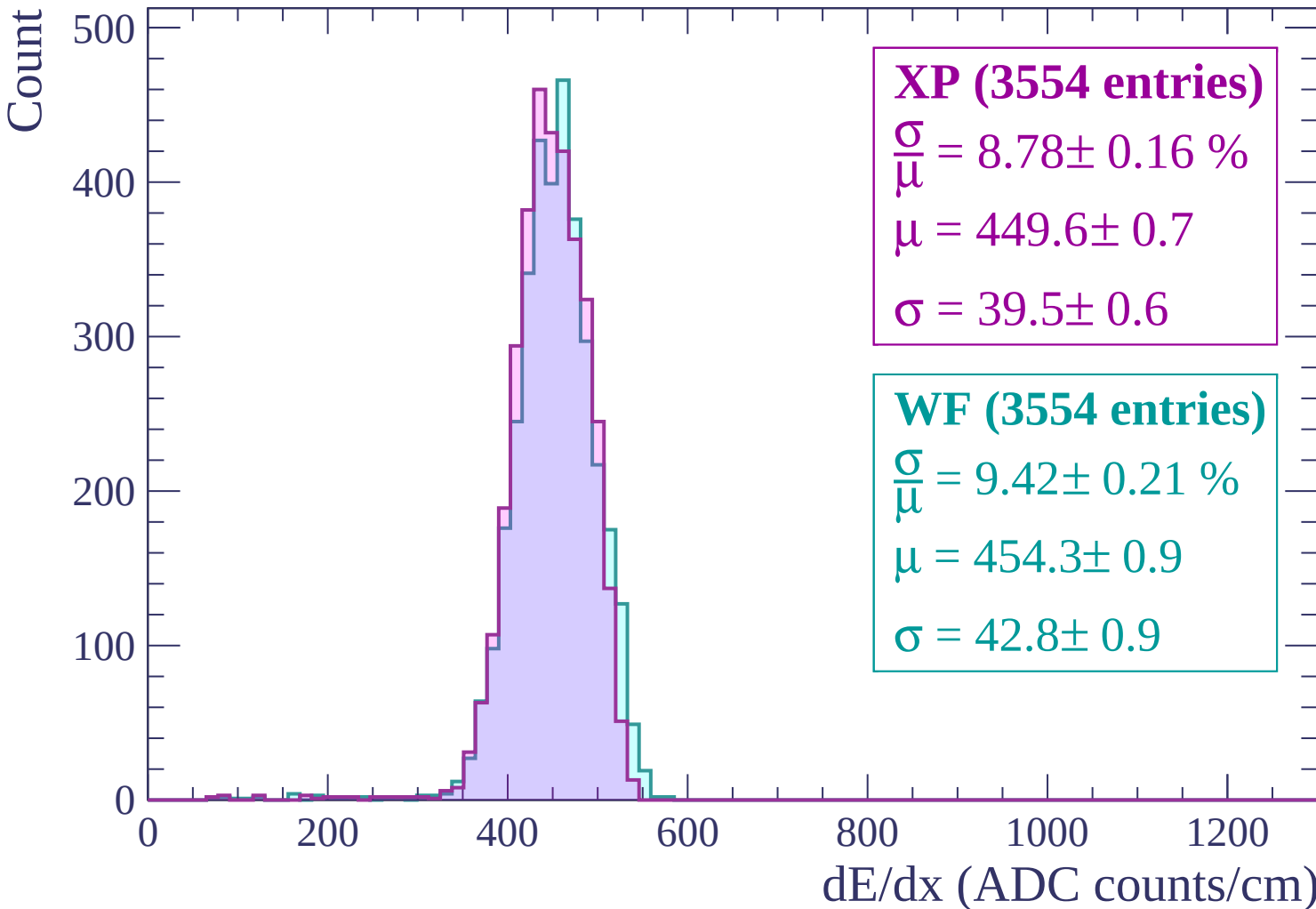
$\sigma = 39.5 \pm 0.6$

WF (3554 entries)

$\frac{\sigma}{\mu} = 9.42 \pm 0.21 \%$

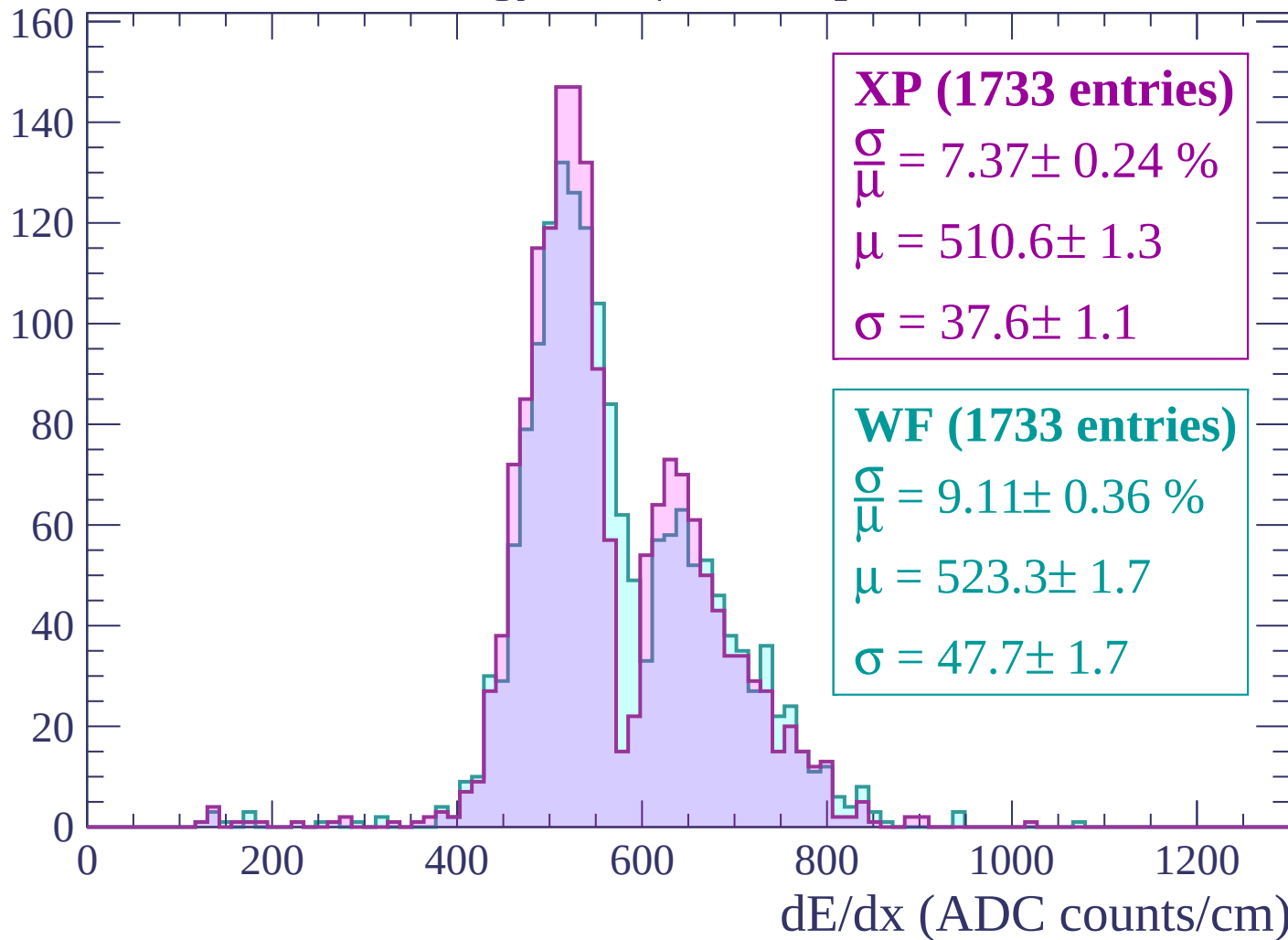
$\mu = 454.3 \pm 0.9$

$\sigma = 42.8 \pm 0.9$



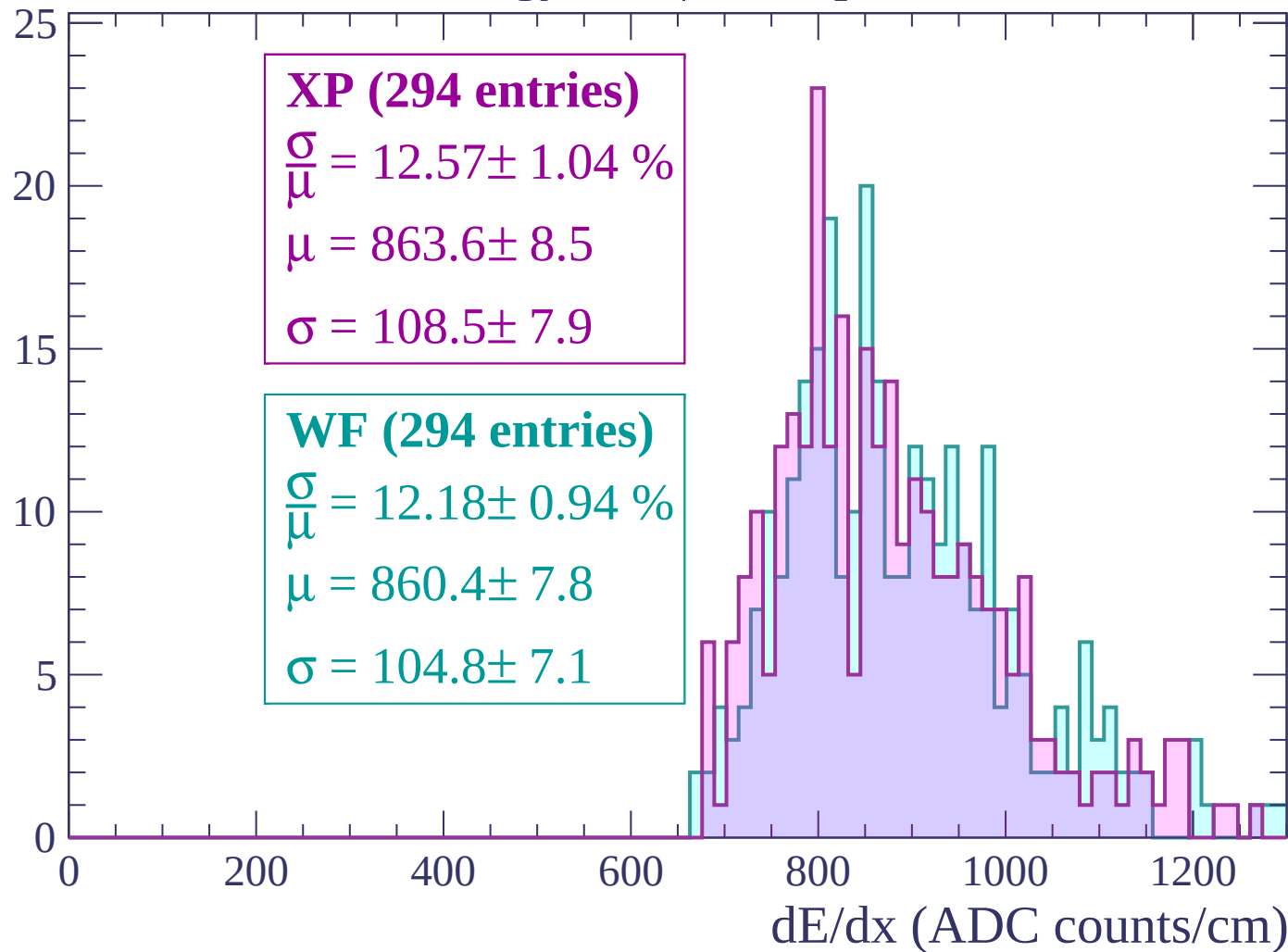
Energy loss | $-120 < p < -60$

Count



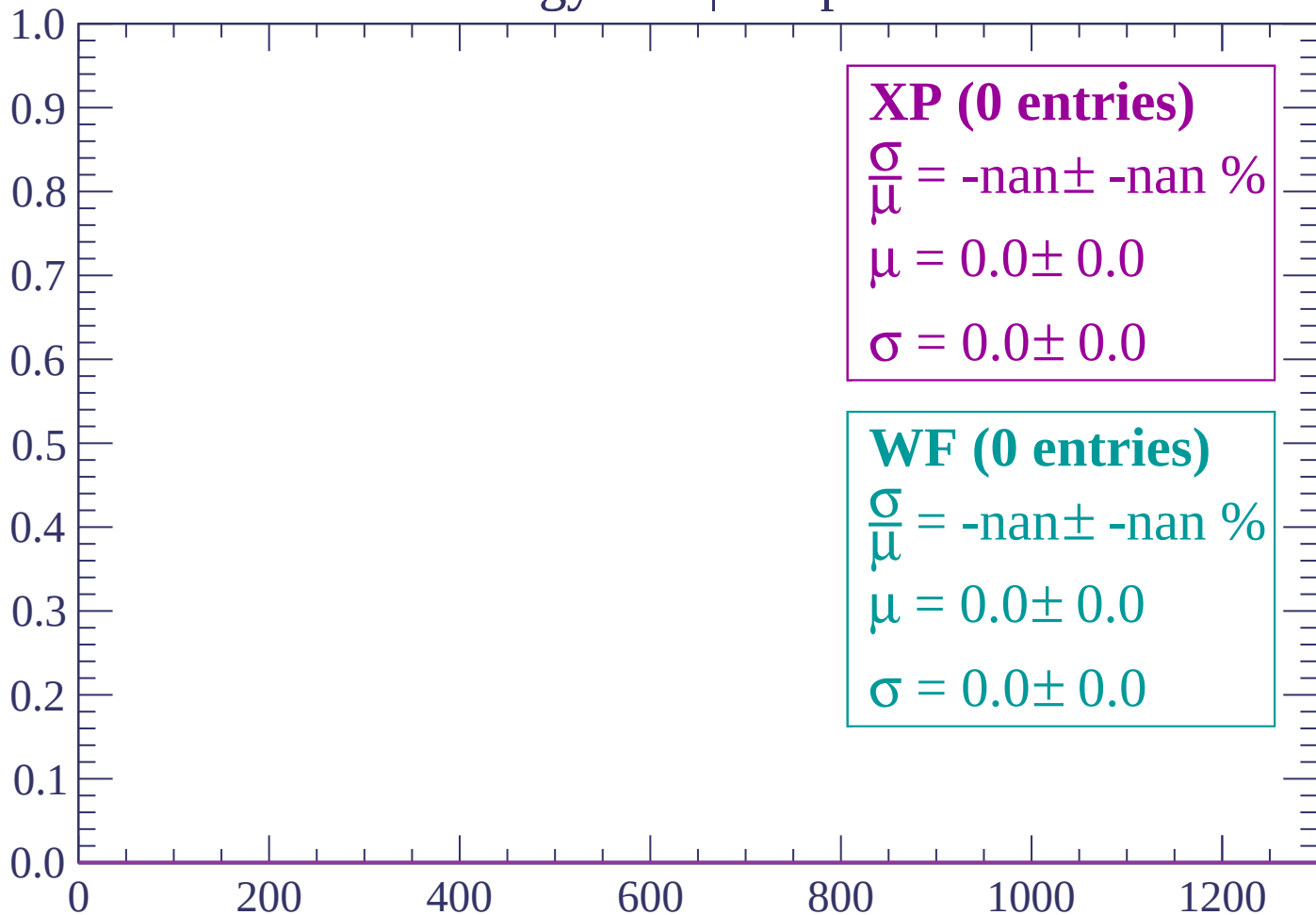
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

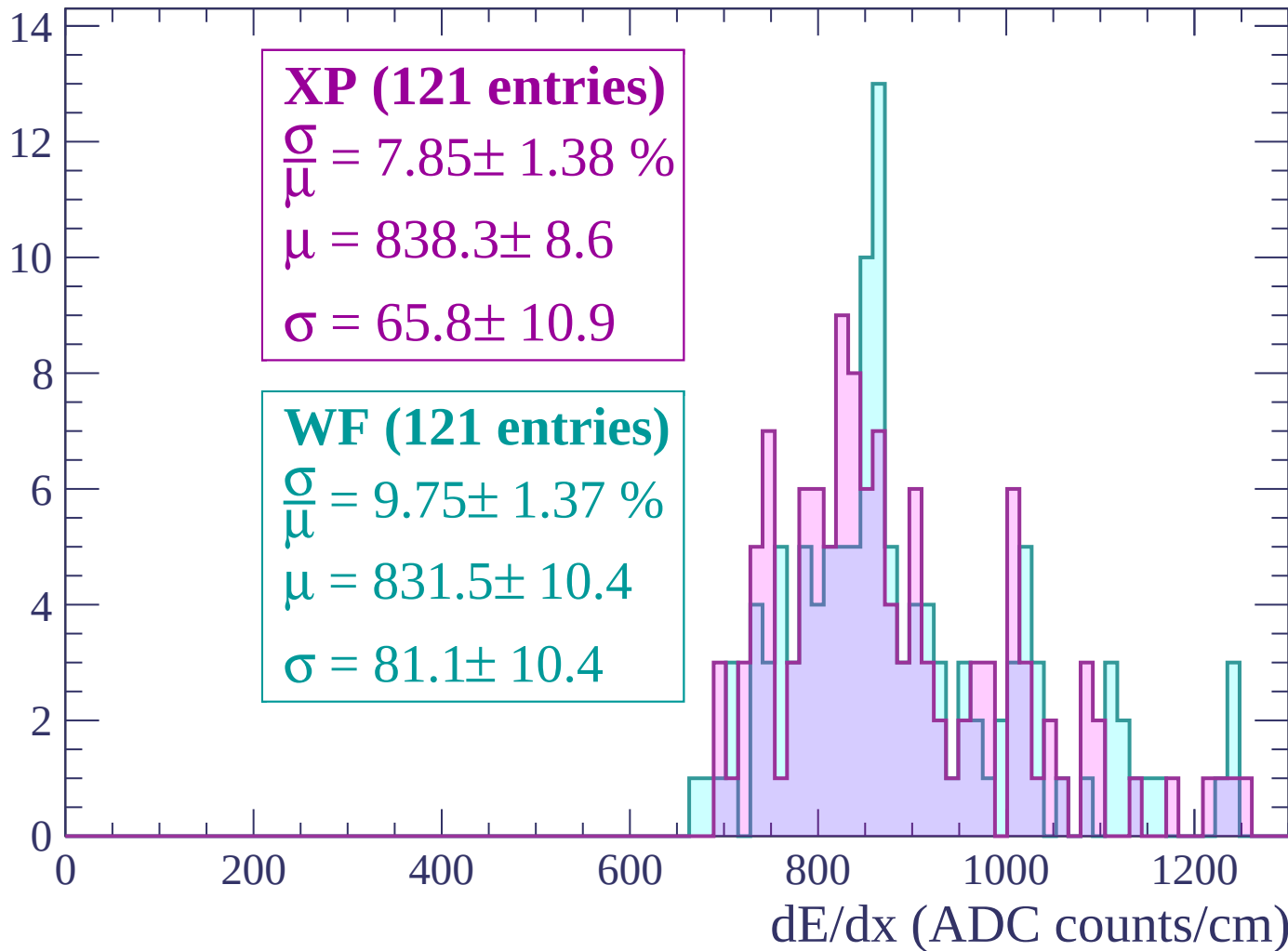
Count



dE/dx (ADC counts/cm)

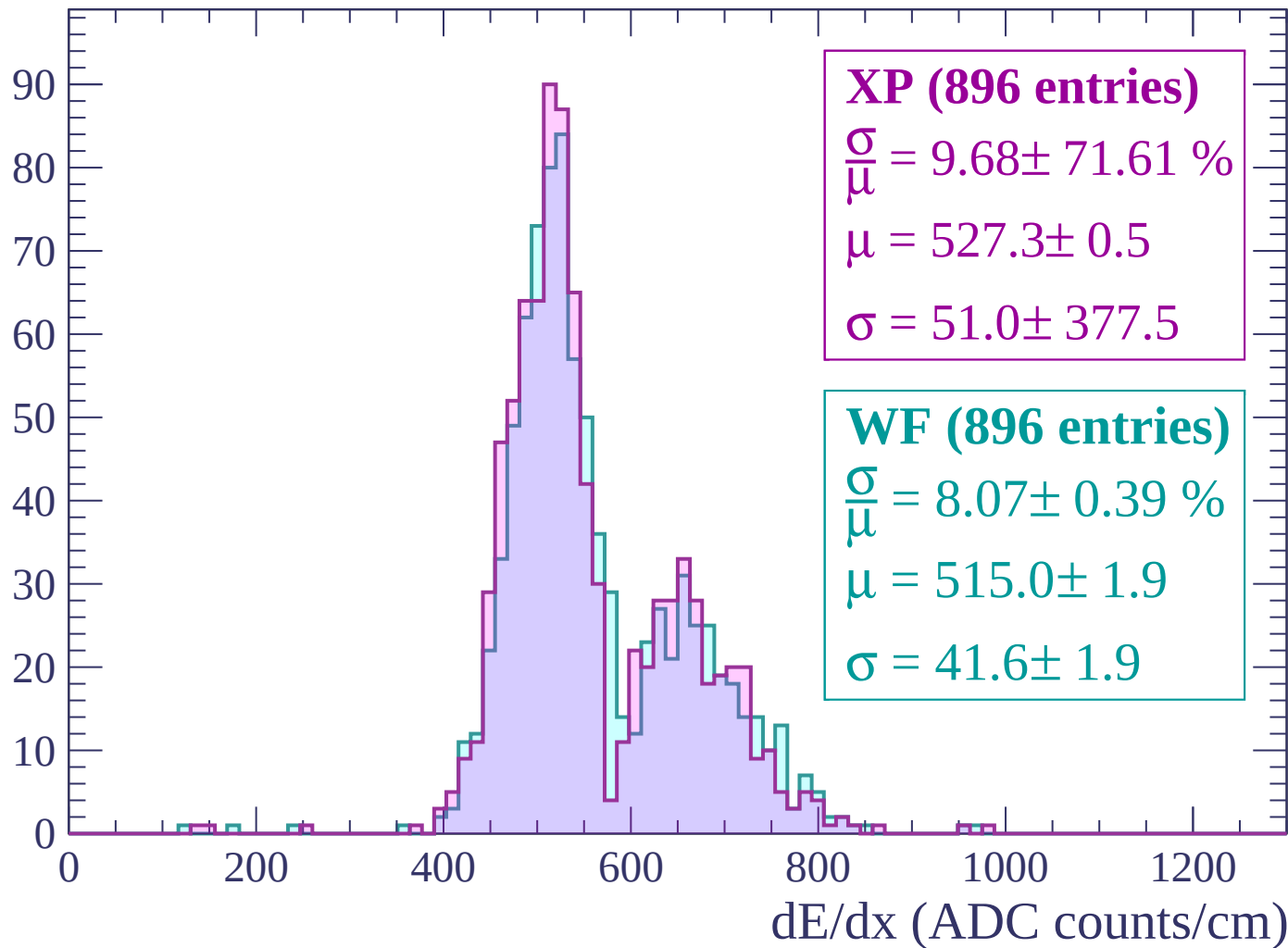
Energy loss | $60 < p < 120$

Count



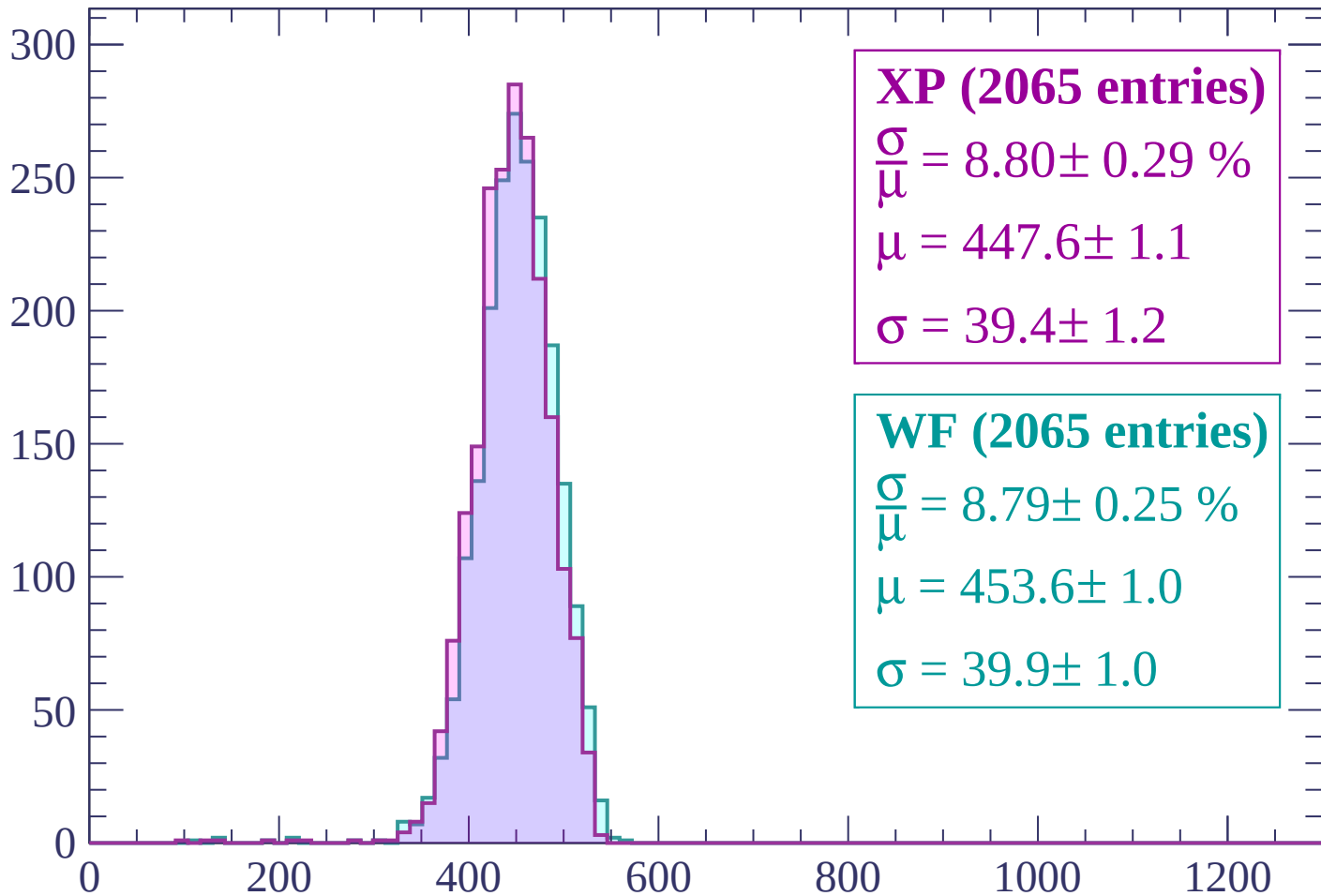
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (2065 entries)

$$\frac{\sigma}{\mu} = 8.80 \pm 0.29 \%$$

$$\mu = 447.6 \pm 1.1$$

$$\sigma = 39.4 \pm 1.2$$

WF (2065 entries)

$$\frac{\sigma}{\mu} = 8.79 \pm 0.25 \%$$

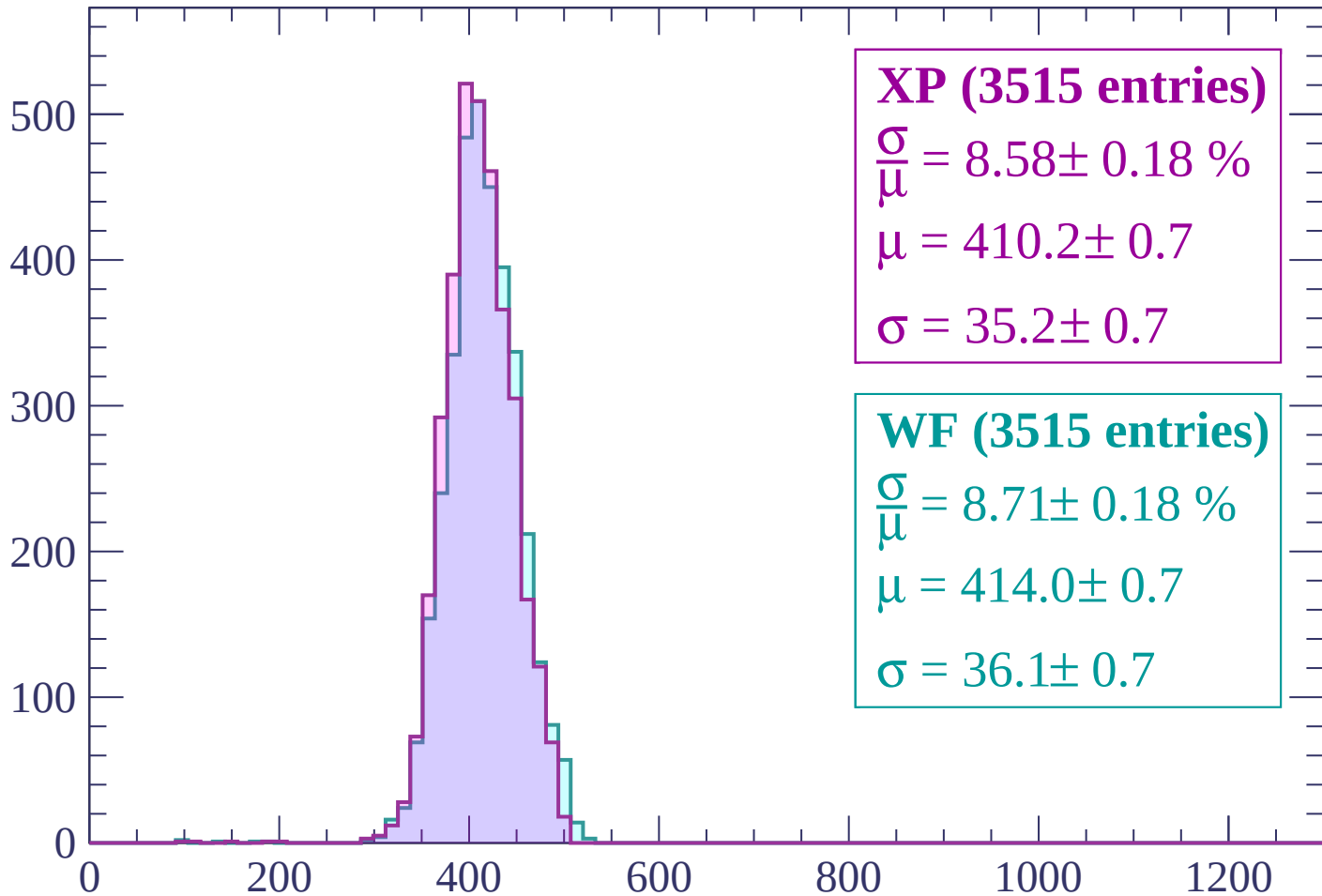
$$\mu = 453.6 \pm 1.0$$

$$\sigma = 39.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (3515 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.18 \%$$

$$\mu = 410.2 \pm 0.7$$

$$\sigma = 35.2 \pm 0.7$$

WF (3515 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.18 \%$$

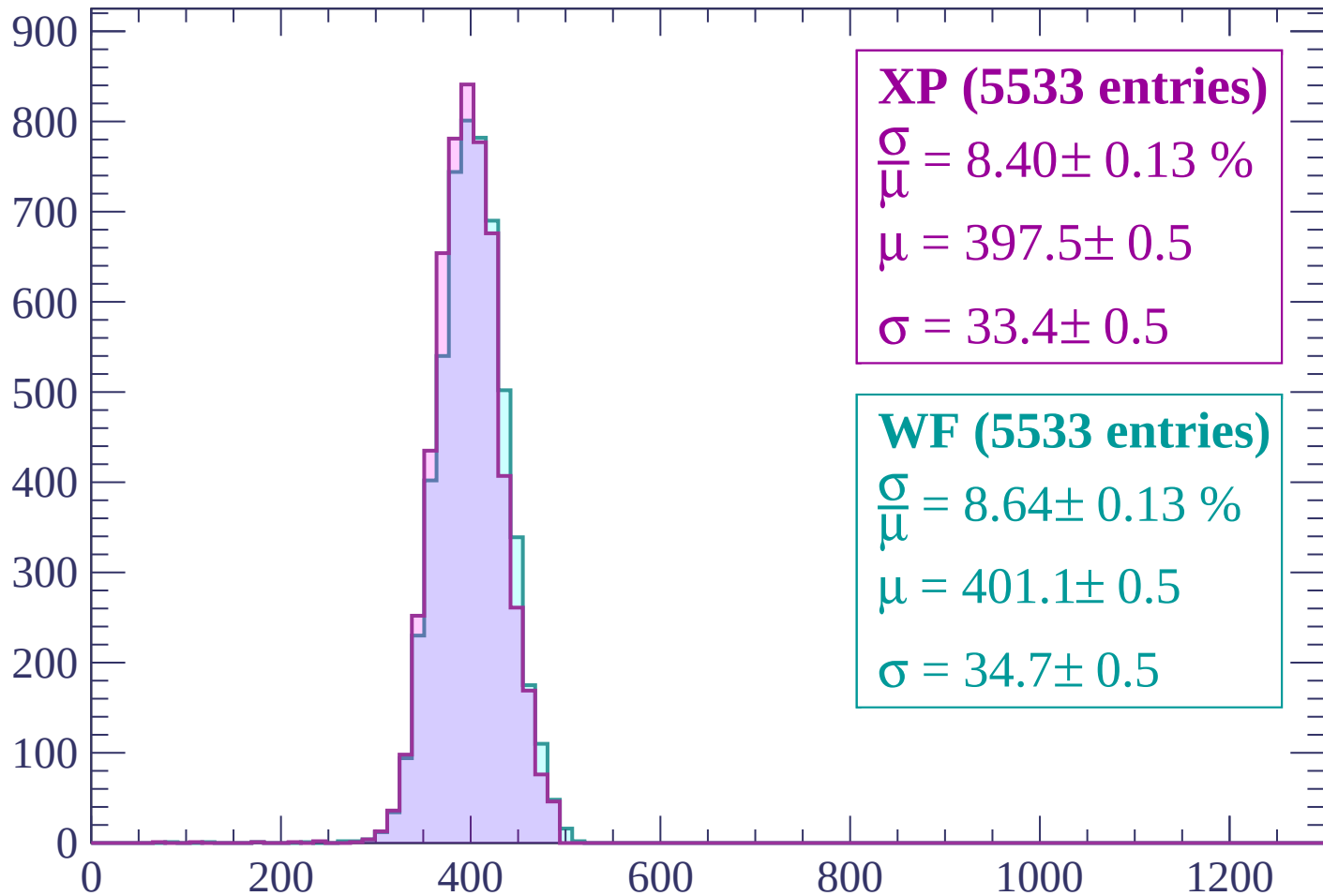
$$\mu = 414.0 \pm 0.7$$

$$\sigma = 36.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (5533 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.13 \%$$

$$\mu = 397.5 \pm 0.5$$

$$\sigma = 33.4 \pm 0.5$$

WF (5533 entries)

$$\frac{\sigma}{\mu} = 8.64 \pm 0.13 \%$$

$$\mu = 401.1 \pm 0.5$$

$$\sigma = 34.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count

1000

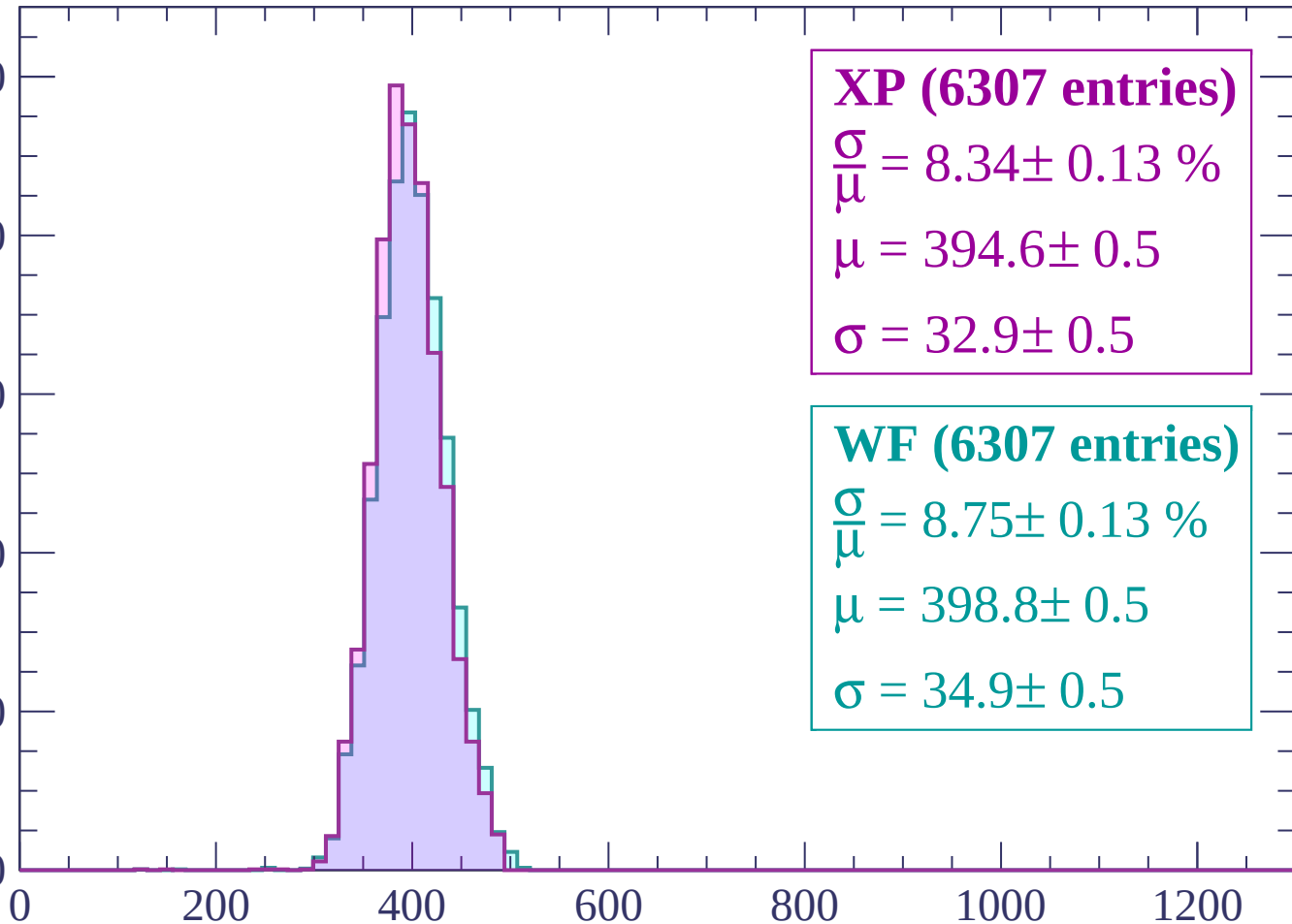
800

600

400

200

0



XP (6307 entries)

$\frac{\sigma}{\mu} = 8.34 \pm 0.13 \%$

$\mu = 394.6 \pm 0.5$

$\sigma = 32.9 \pm 0.5$

WF (6307 entries)

$\frac{\sigma}{\mu} = 8.75 \pm 0.13 \%$

$\mu = 398.8 \pm 0.5$

$\sigma = 34.9 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (6980 entries)

$\frac{\sigma}{\mu} = 8.66 \pm 0.12 \%$

$\mu = 396.8 \pm 0.5$

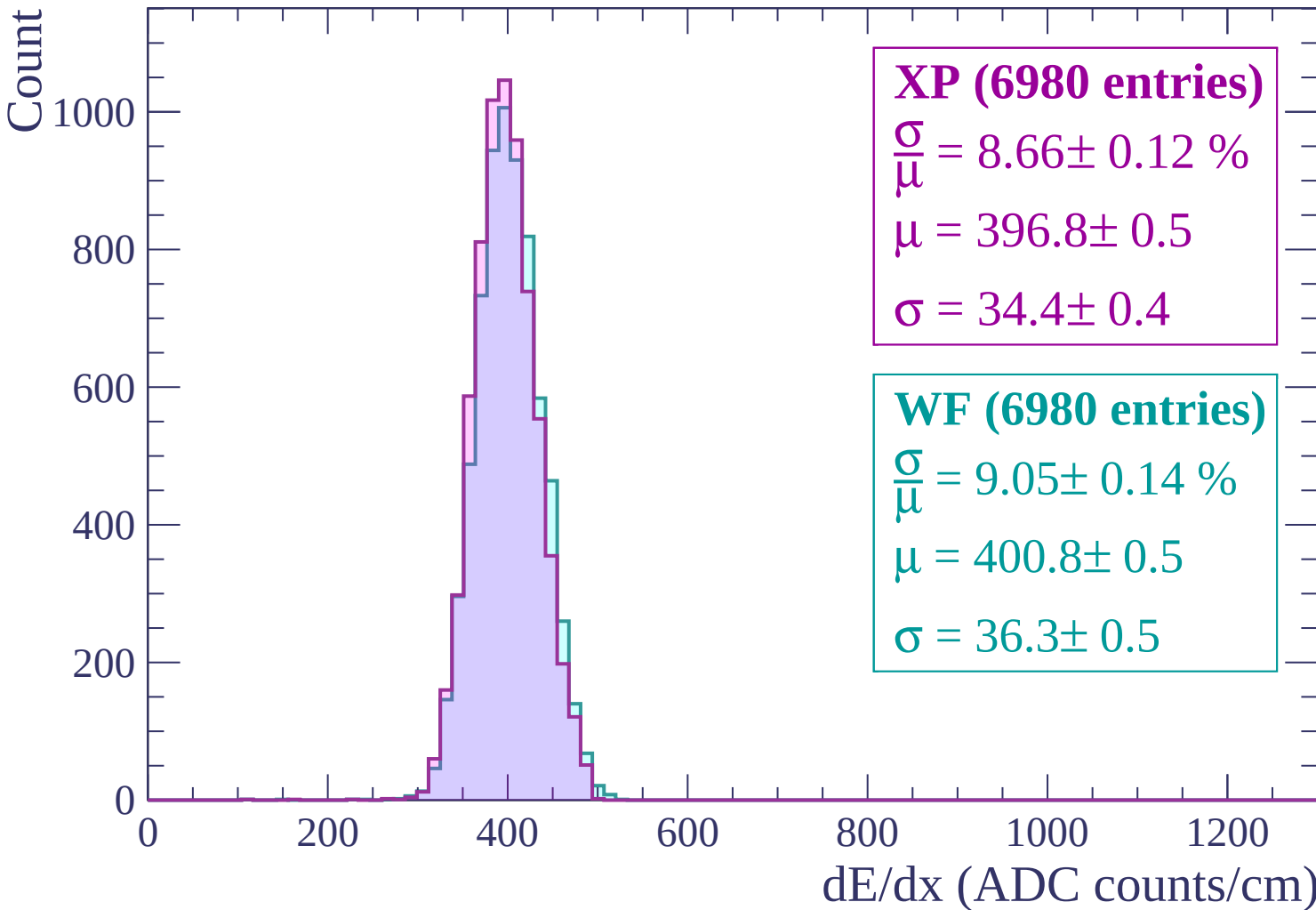
$\sigma = 34.4 \pm 0.4$

WF (6980 entries)

$\frac{\sigma}{\mu} = 9.05 \pm 0.14 \%$

$\mu = 400.8 \pm 0.5$

$\sigma = 36.3 \pm 0.5$



Energy loss | $480 < p < 540$

Count

1000

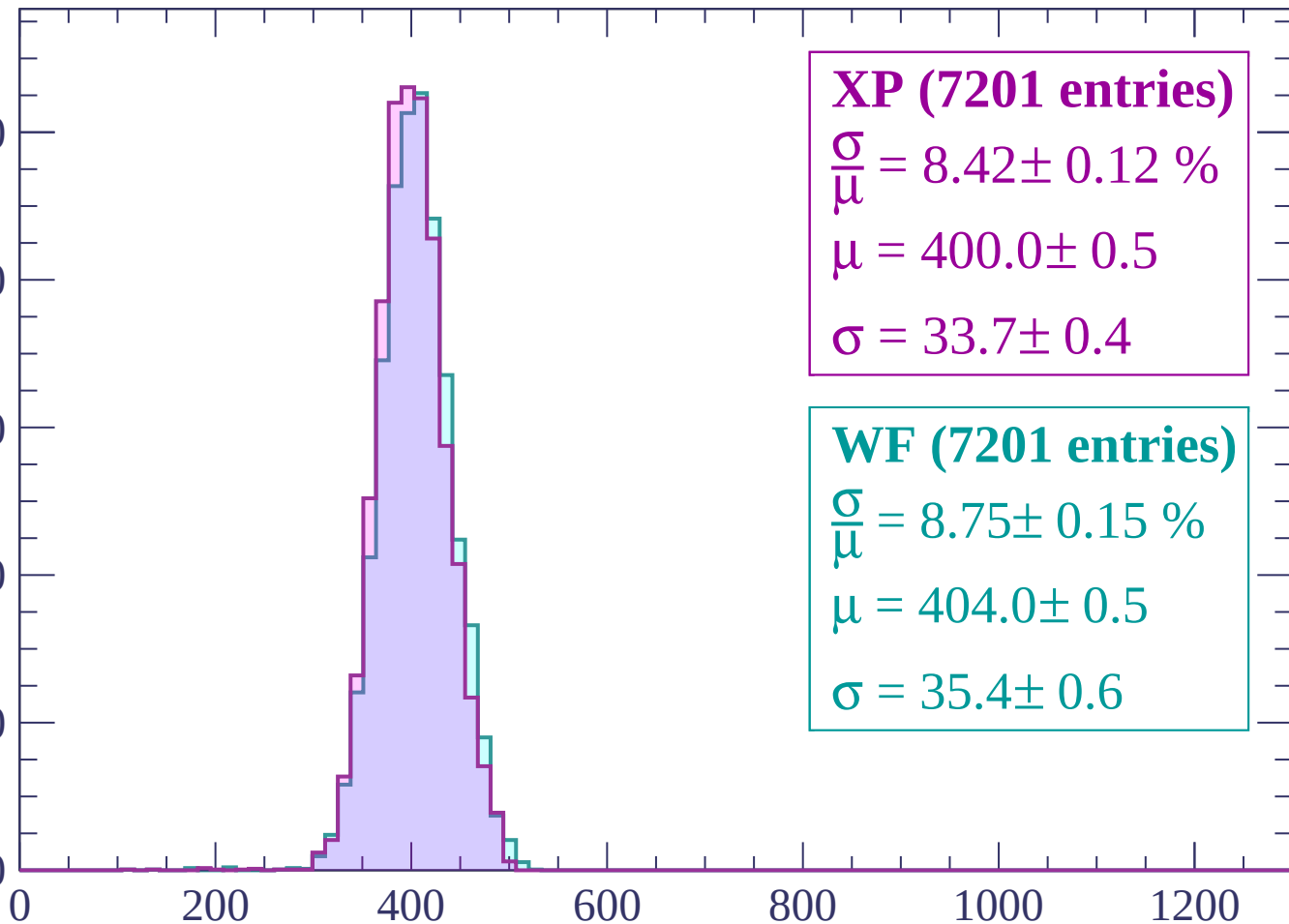
800

600

400

200

0



XP (7201 entries)

$\frac{\sigma}{\mu} = 8.42 \pm 0.12 \%$

$\mu = 400.0 \pm 0.5$

$\sigma = 33.7 \pm 0.4$

WF (7201 entries)

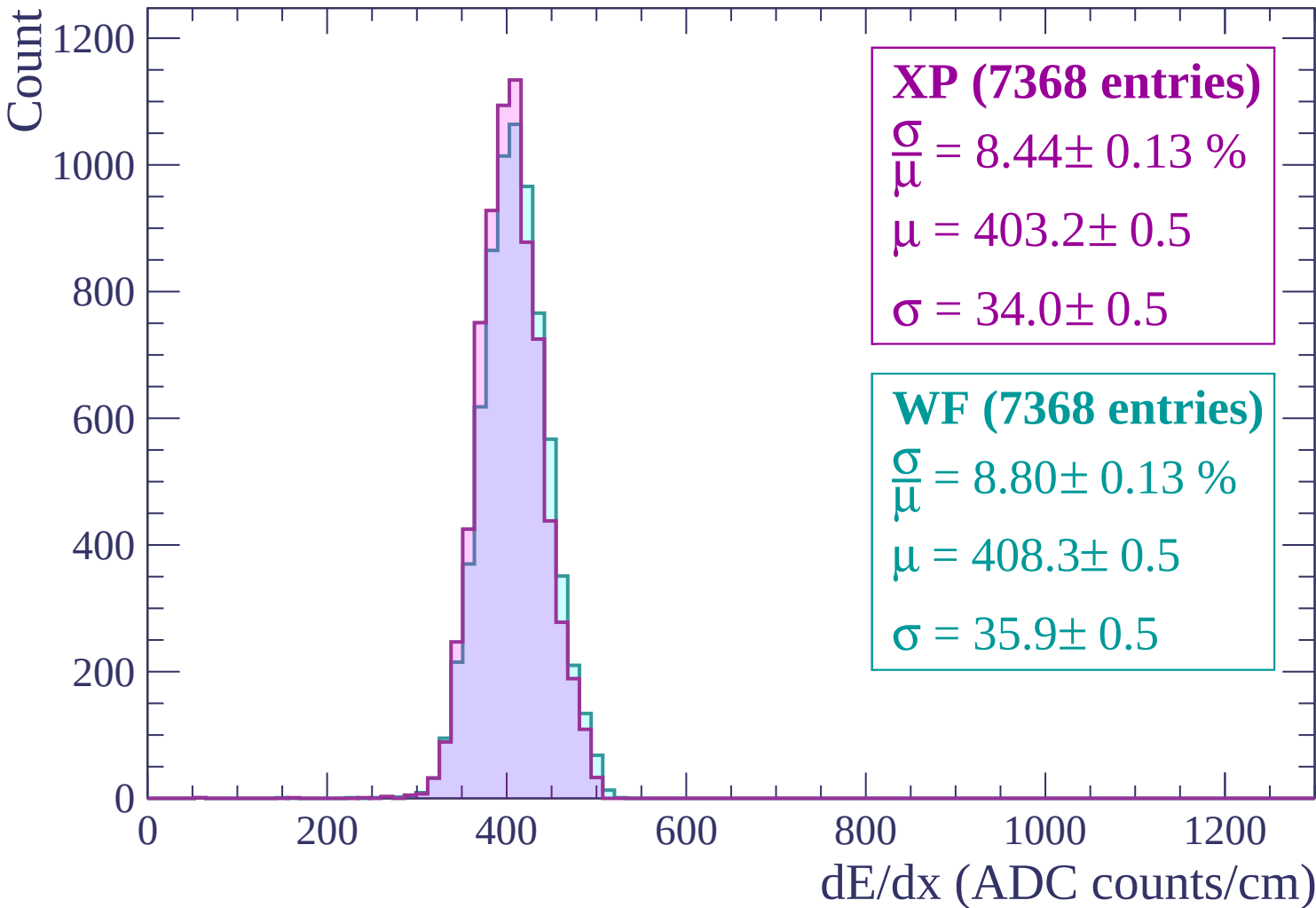
$\frac{\sigma}{\mu} = 8.75 \pm 0.15 \%$

$\mu = 404.0 \pm 0.5$

$\sigma = 35.4 \pm 0.6$

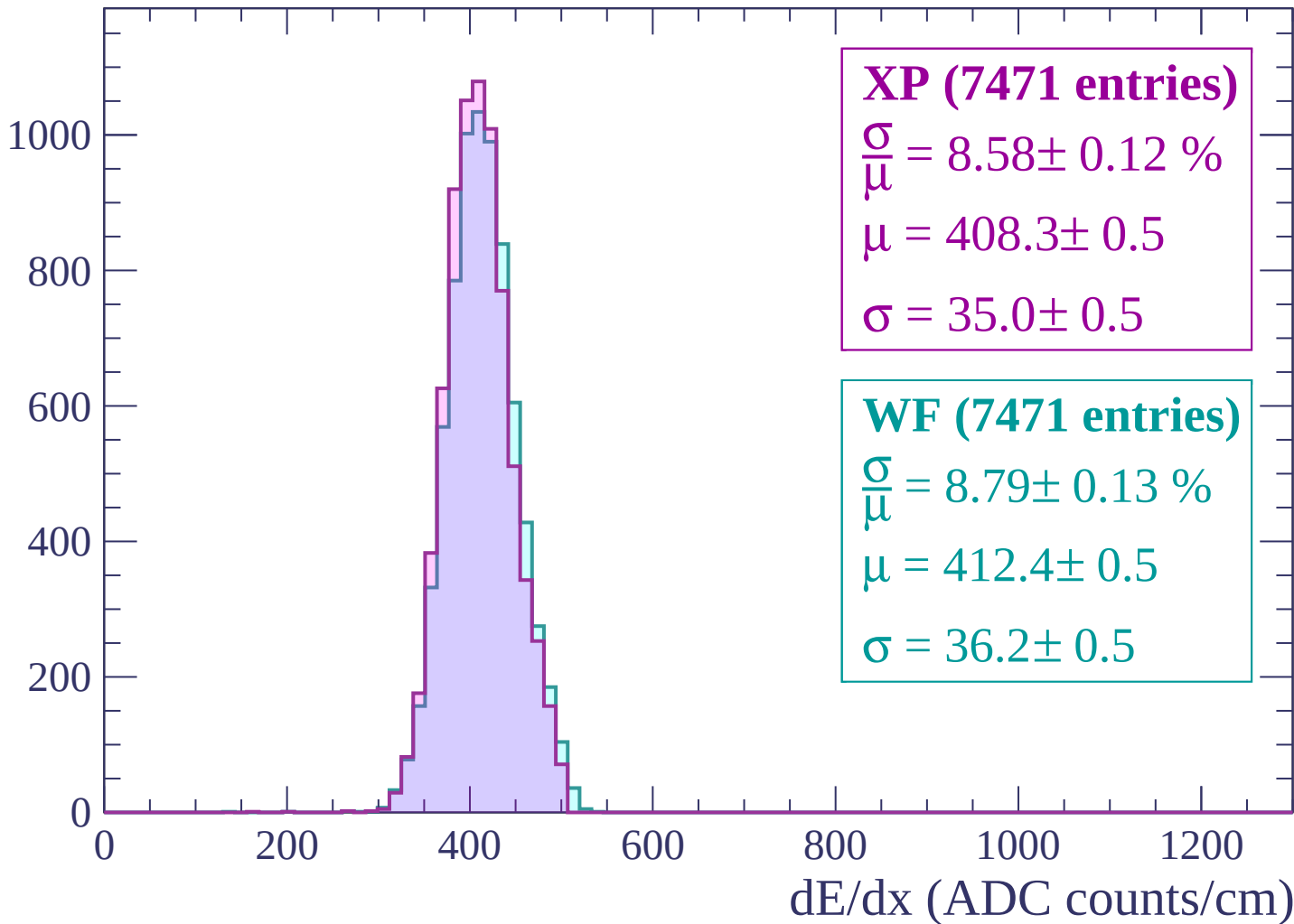
dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$



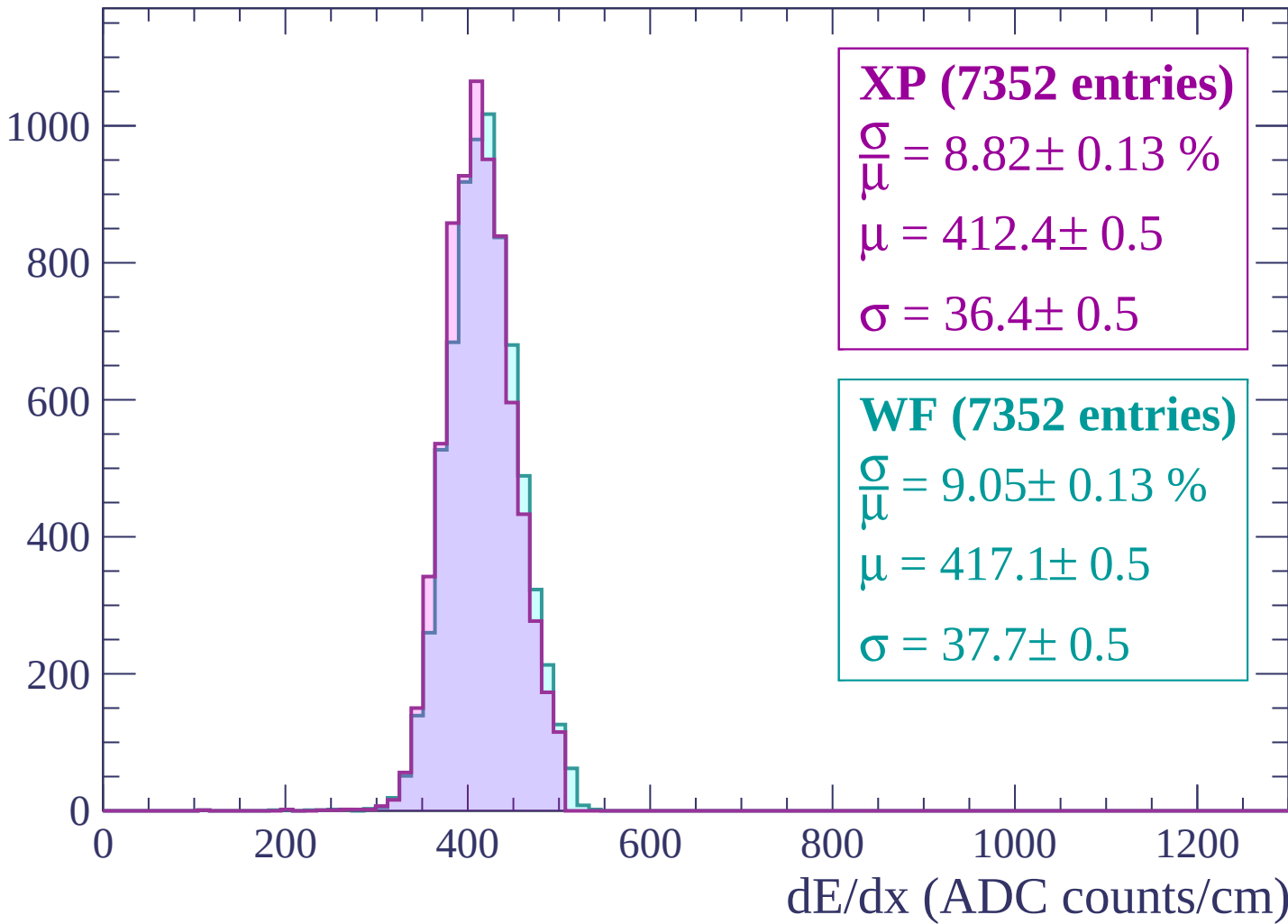
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count

1000

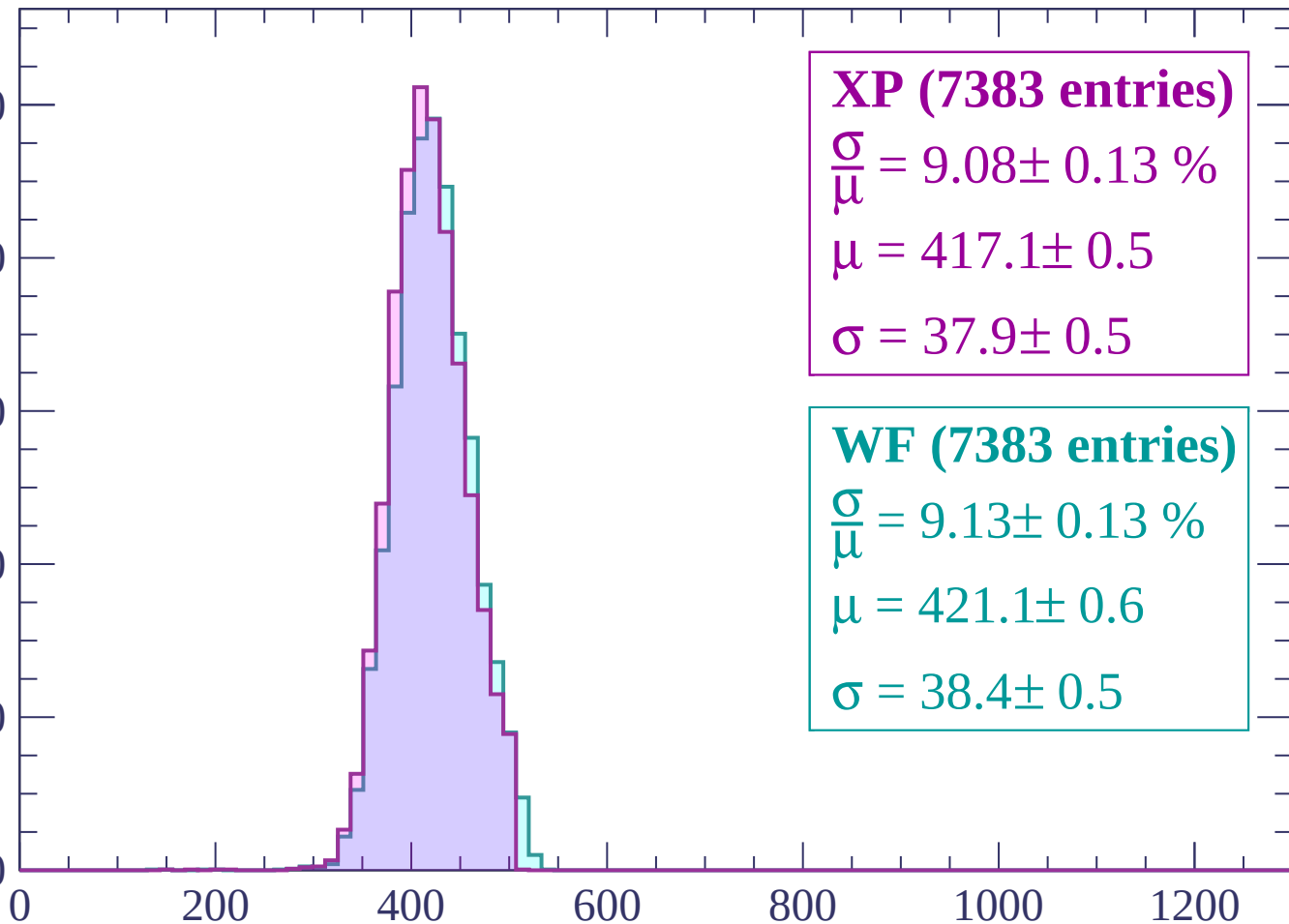
800

600

400

200

0



XP (7383 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.13 \%$

$\mu = 417.1 \pm 0.5$

$\sigma = 37.9 \pm 0.5$

WF (7383 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.13 \%$

$\mu = 421.1 \pm 0.6$

$\sigma = 38.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

1000

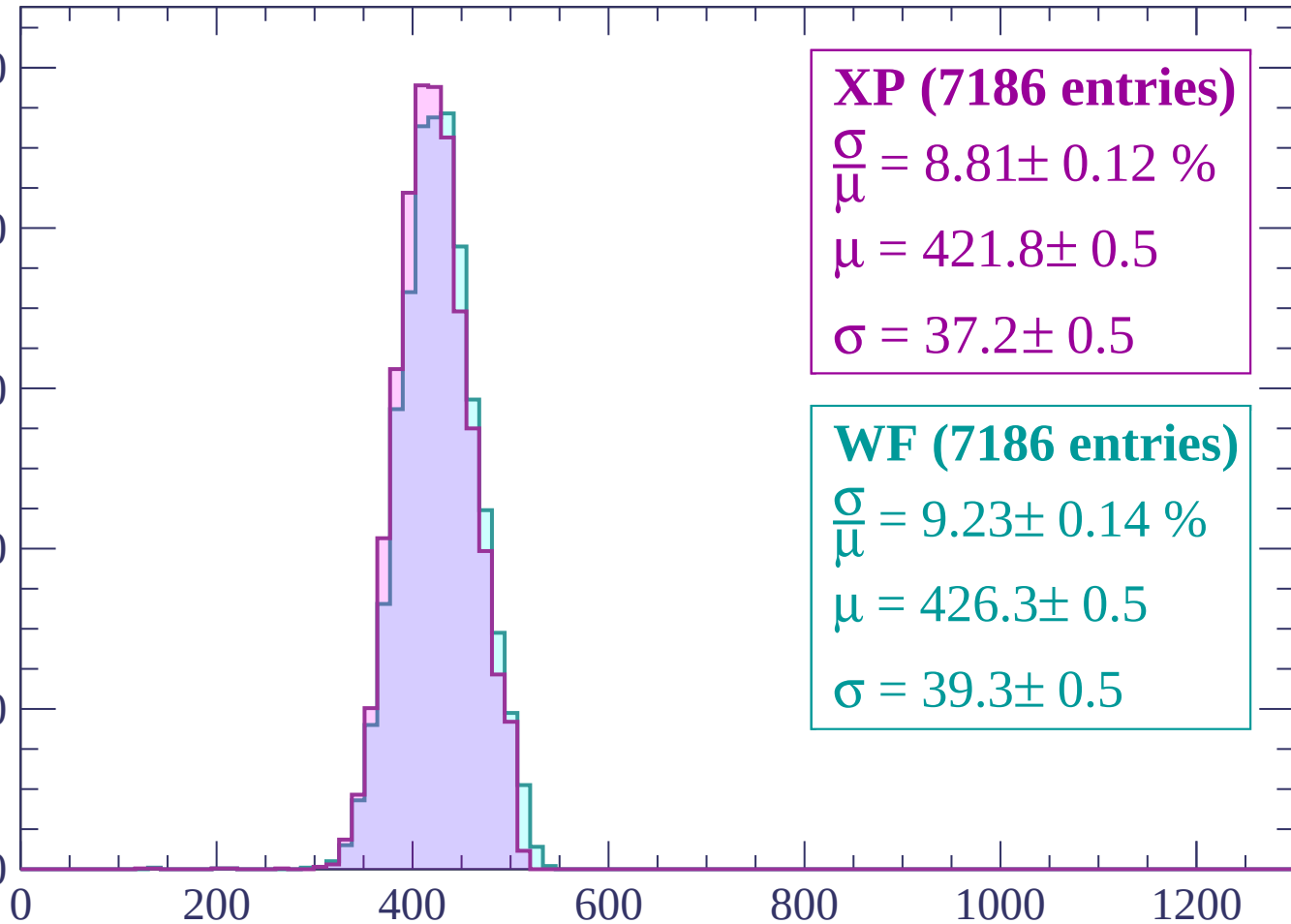
800

600

400

200

0



XP (7186 entries)

$\frac{\sigma}{\mu} = 8.81 \pm 0.12 \%$

$\mu = 421.8 \pm 0.5$

$\sigma = 37.2 \pm 0.5$

WF (7186 entries)

$\frac{\sigma}{\mu} = 9.23 \pm 0.14 \%$

$\mu = 426.3 \pm 0.5$

$\sigma = 39.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count

1000

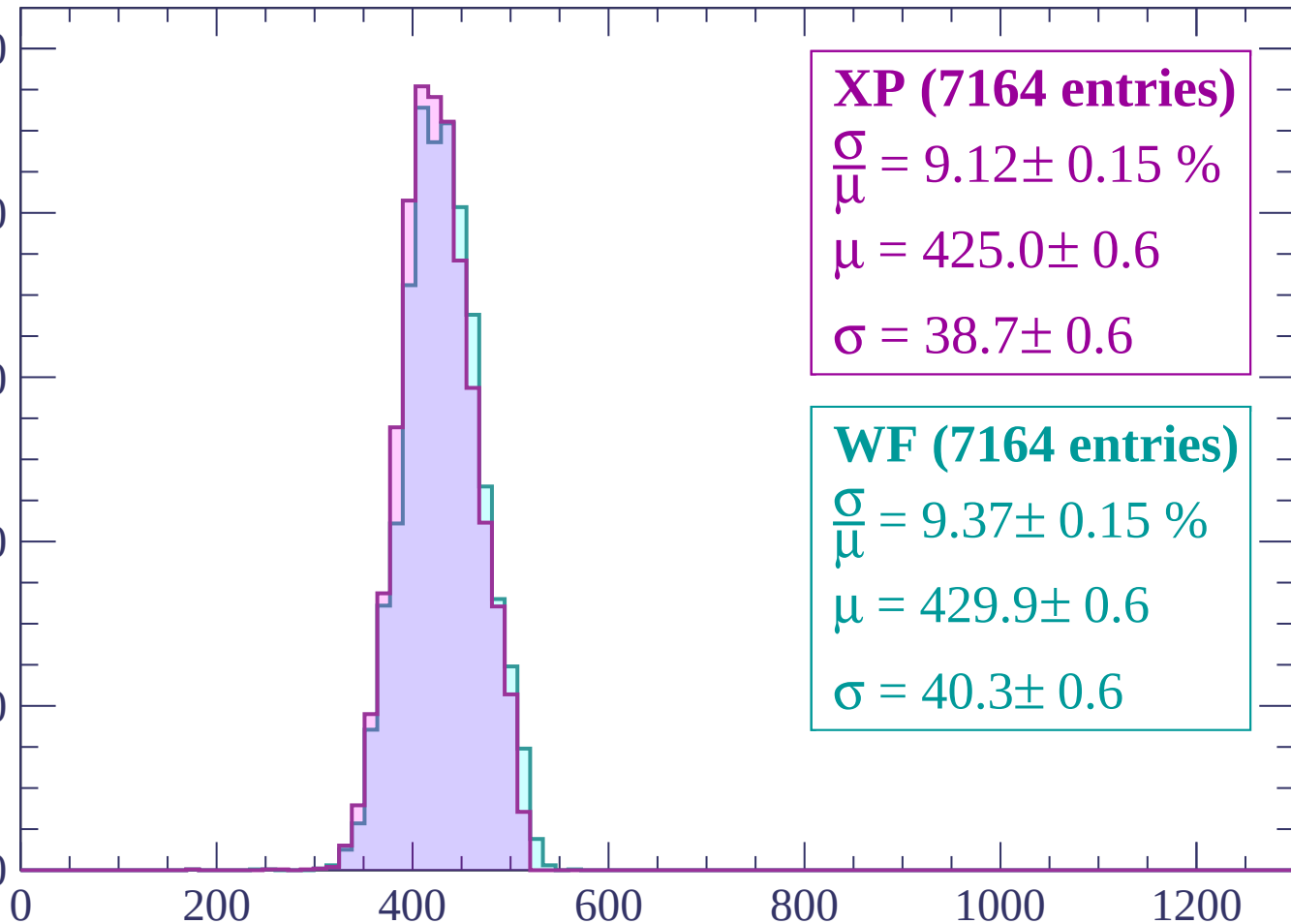
800

600

400

200

0



XP (7164 entries)

$\frac{\sigma}{\mu} = 9.12 \pm 0.15 \%$

$\mu = 425.0 \pm 0.6$

$\sigma = 38.7 \pm 0.6$

WF (7164 entries)

$\frac{\sigma}{\mu} = 9.37 \pm 0.15 \%$

$\mu = 429.9 \pm 0.6$

$\sigma = 40.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count

1000

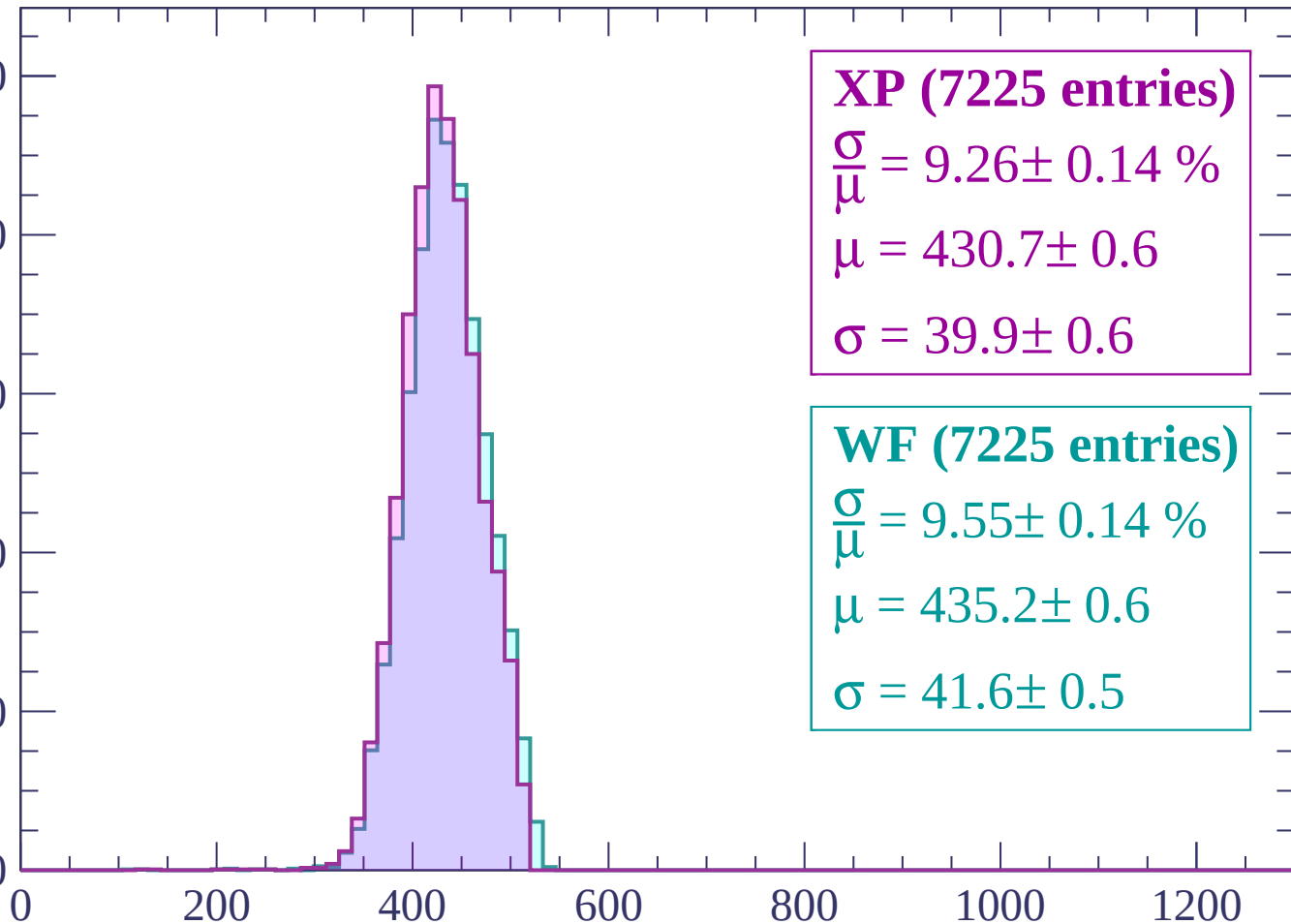
800

600

400

200

0



XP (7225 entries)

$\frac{\sigma}{\mu} = 9.26 \pm 0.14 \%$

$\mu = 430.7 \pm 0.6$

$\sigma = 39.9 \pm 0.6$

WF (7225 entries)

$\frac{\sigma}{\mu} = 9.55 \pm 0.14 \%$

$\mu = 435.2 \pm 0.6$

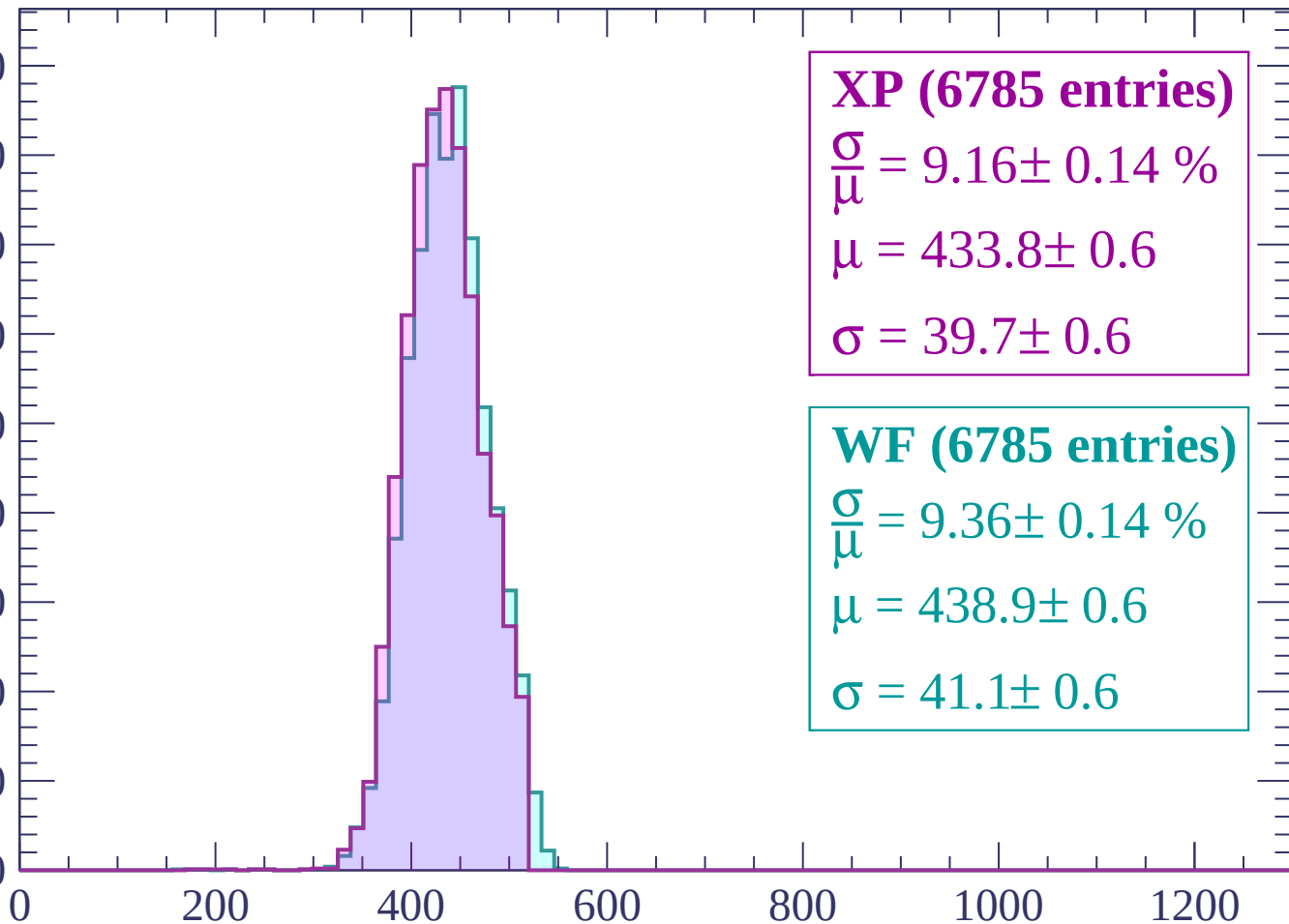
$\sigma = 41.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count

900
800
700
600
500
400
300
200
100
0



XP (6785 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.14 \%$$

$$\mu = 433.8 \pm 0.6$$

$$\sigma = 39.7 \pm 0.6$$

WF (6785 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.14 \%$$

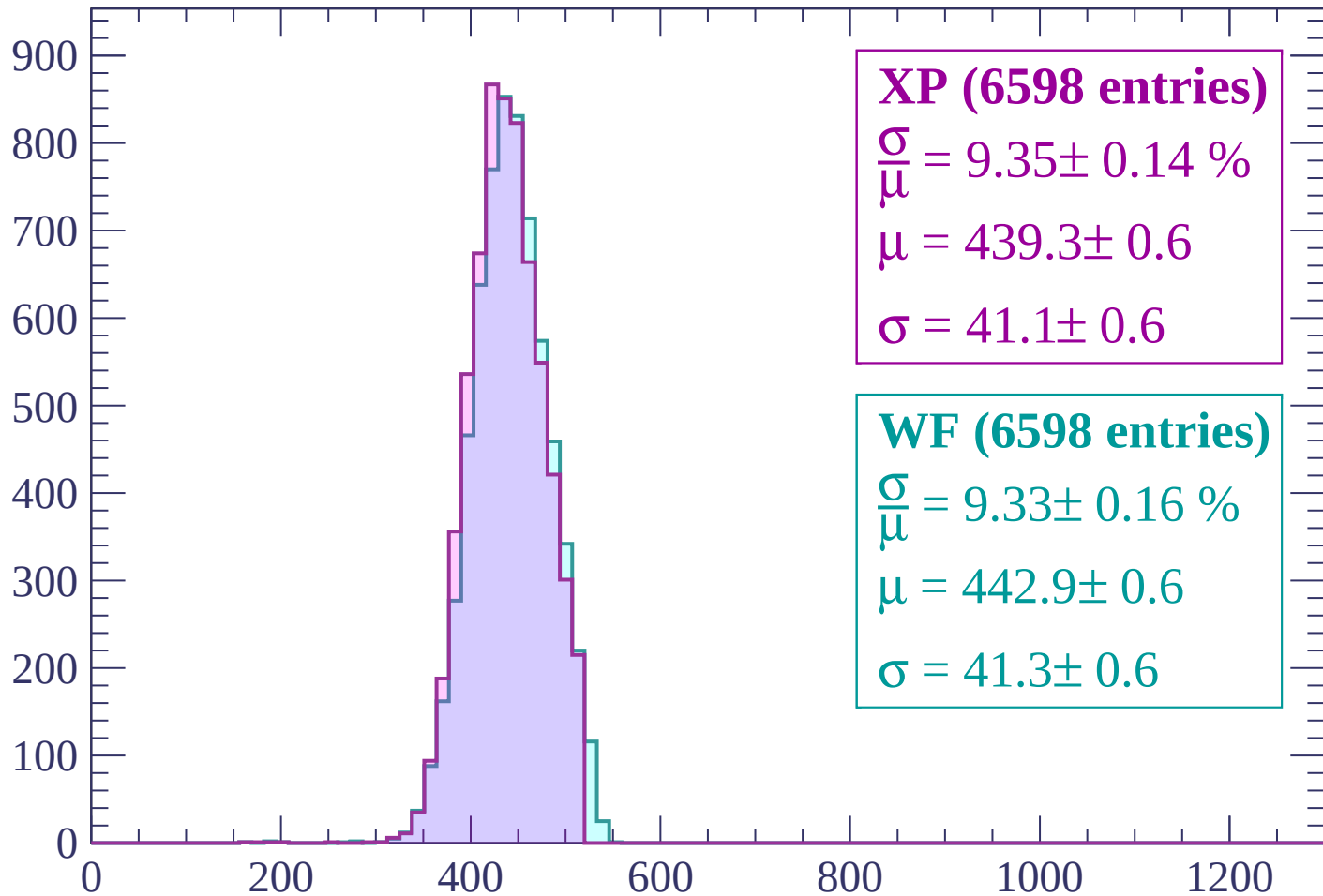
$$\mu = 438.9 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (6598 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.14 \%$$

$$\mu = 439.3 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

WF (6598 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.16 \%$$

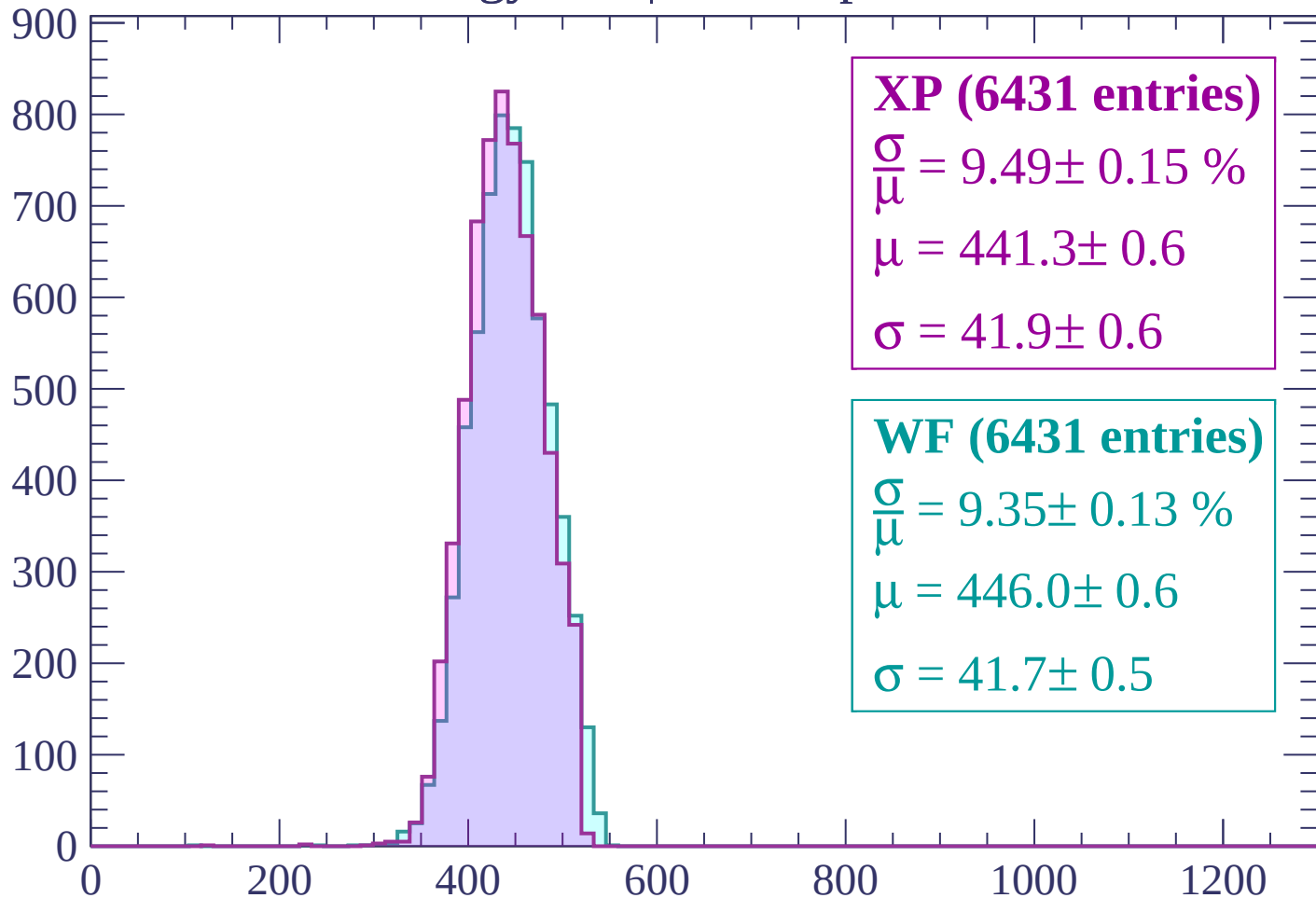
$$\mu = 442.9 \pm 0.6$$

$$\sigma = 41.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (6431 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.15 \%$$

$$\mu = 441.3 \pm 0.6$$

$$\sigma = 41.9 \pm 0.6$$

WF (6431 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.13 \%$$

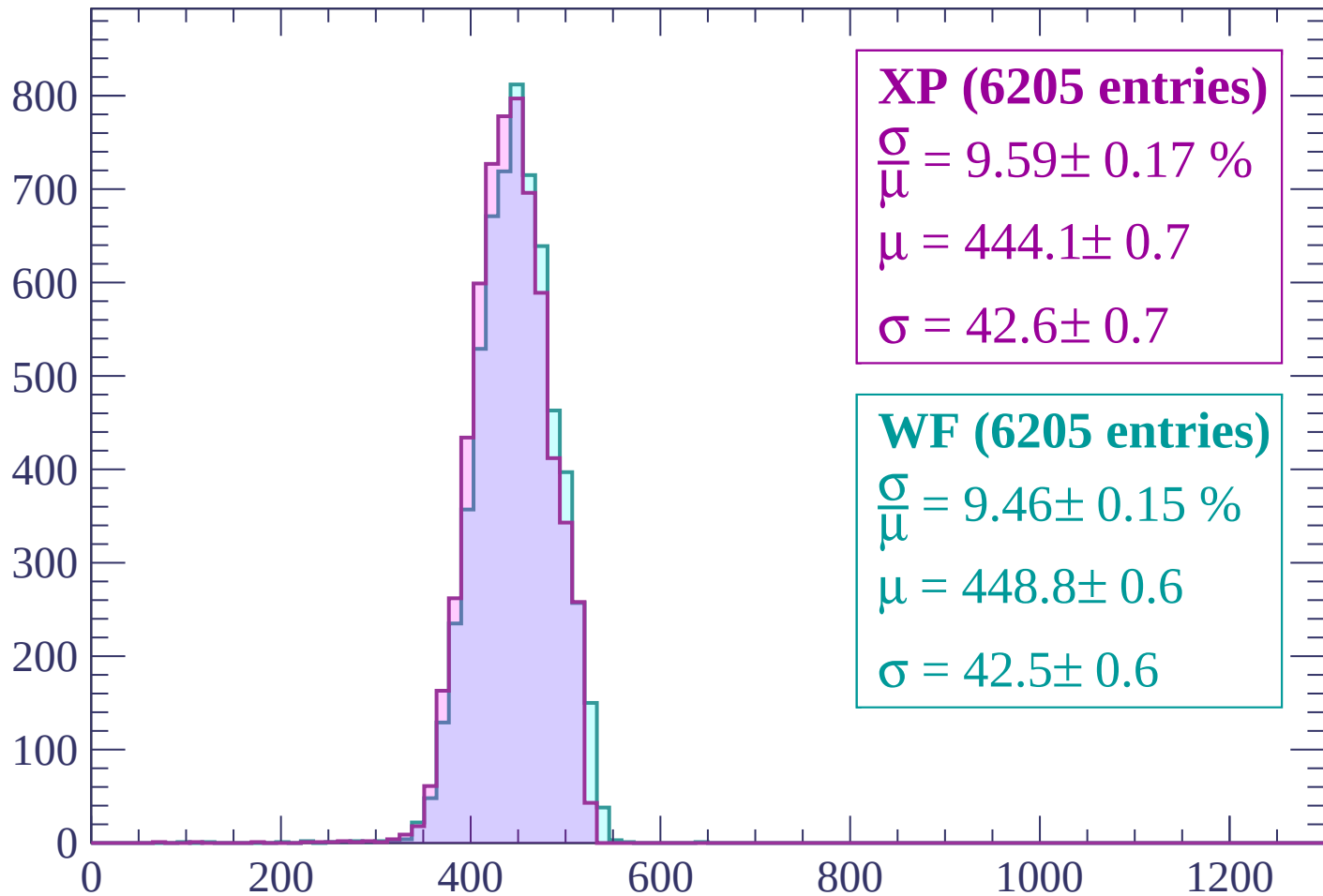
$$\mu = 446.0 \pm 0.6$$

$$\sigma = 41.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (6205 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.17 \%$$

$$\mu = 444.1 \pm 0.7$$

$$\sigma = 42.6 \pm 0.7$$

WF (6205 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.15 \%$$

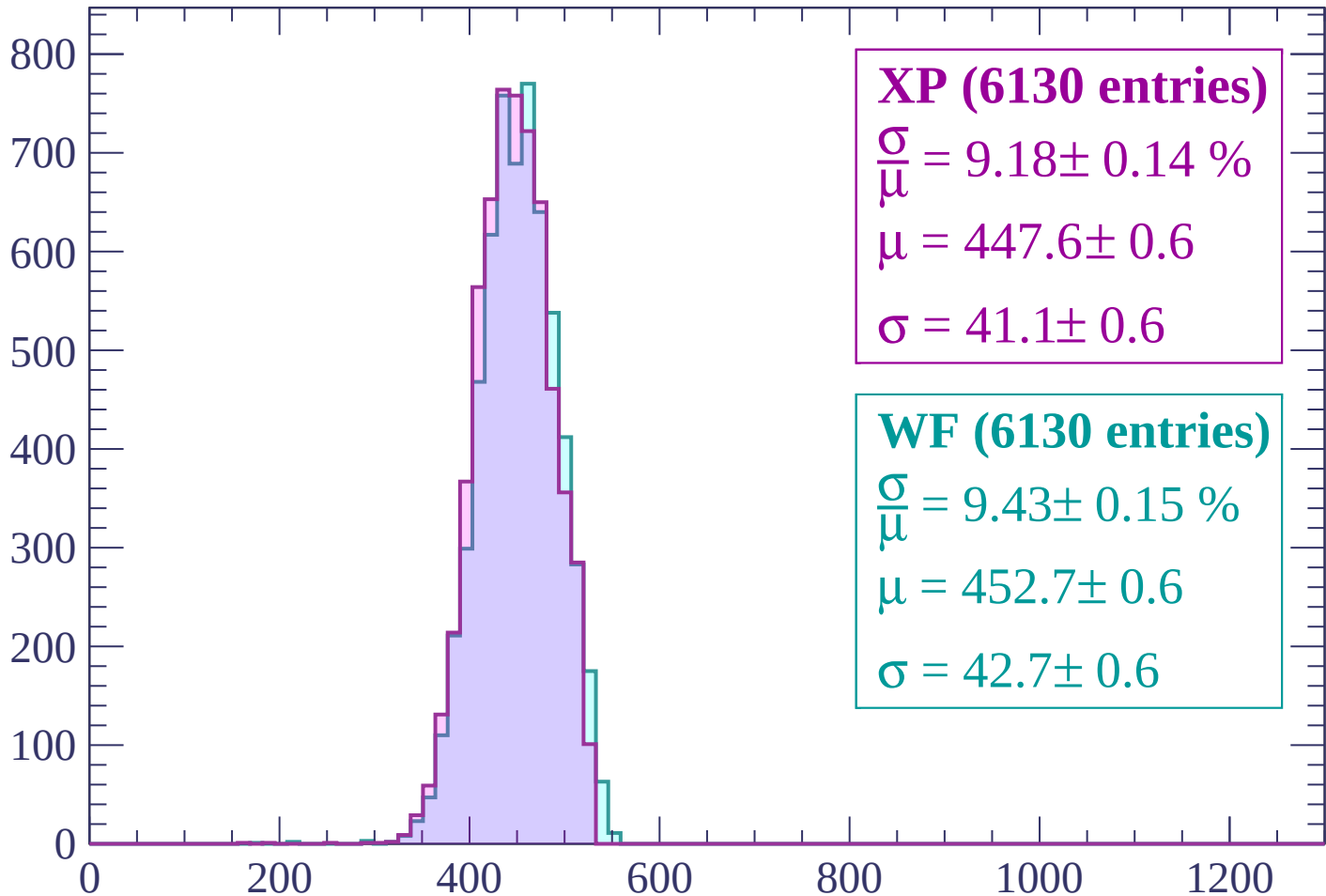
$$\mu = 448.8 \pm 0.6$$

$$\sigma = 42.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (6130 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.14 \%$$

$$\mu = 447.6 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

WF (6130 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.15 \%$$

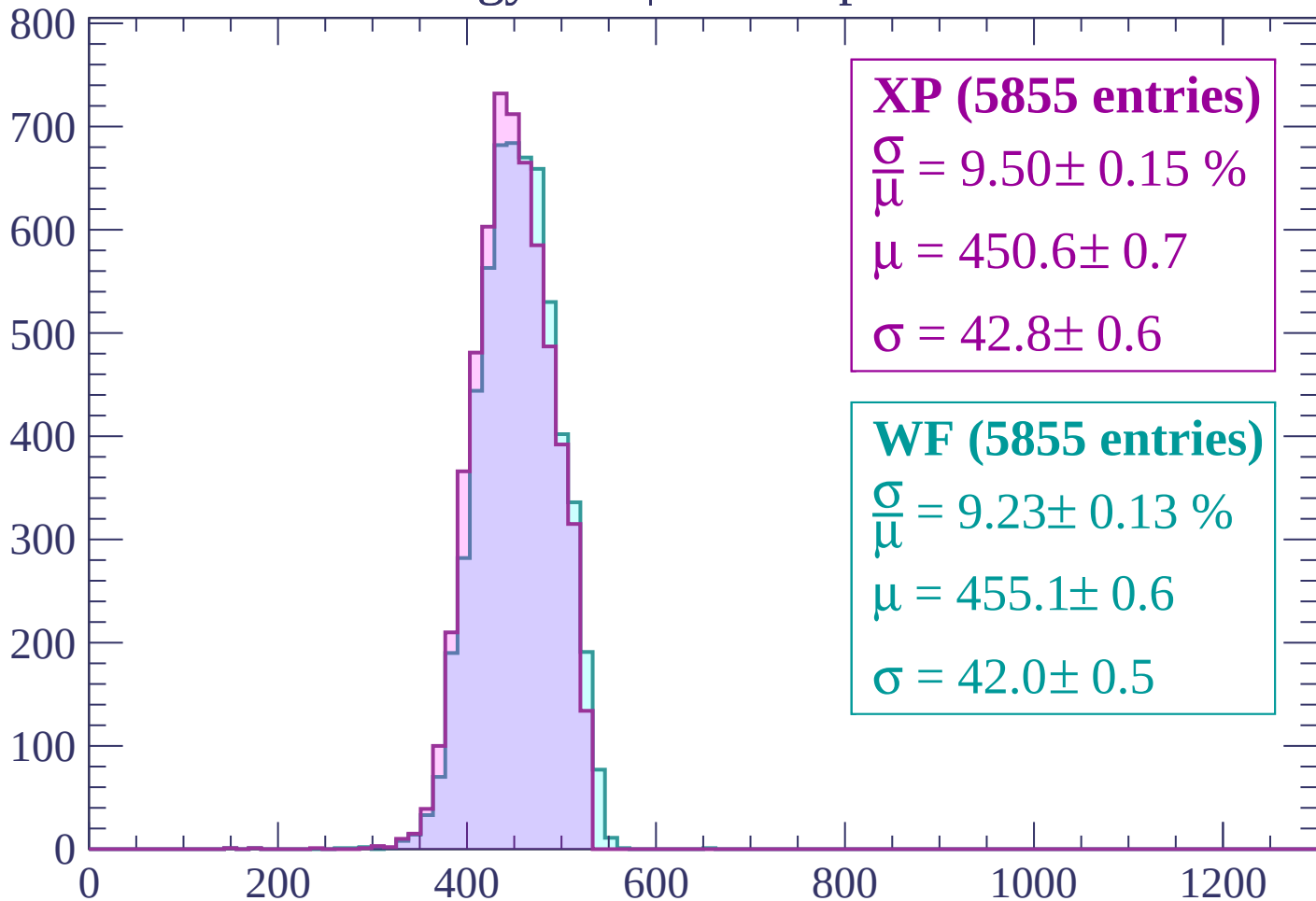
$$\mu = 452.7 \pm 0.6$$

$$\sigma = 42.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (5855 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.15 \%$$

$$\mu = 450.6 \pm 0.7$$

$$\sigma = 42.8 \pm 0.6$$

WF (5855 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.13 \%$$

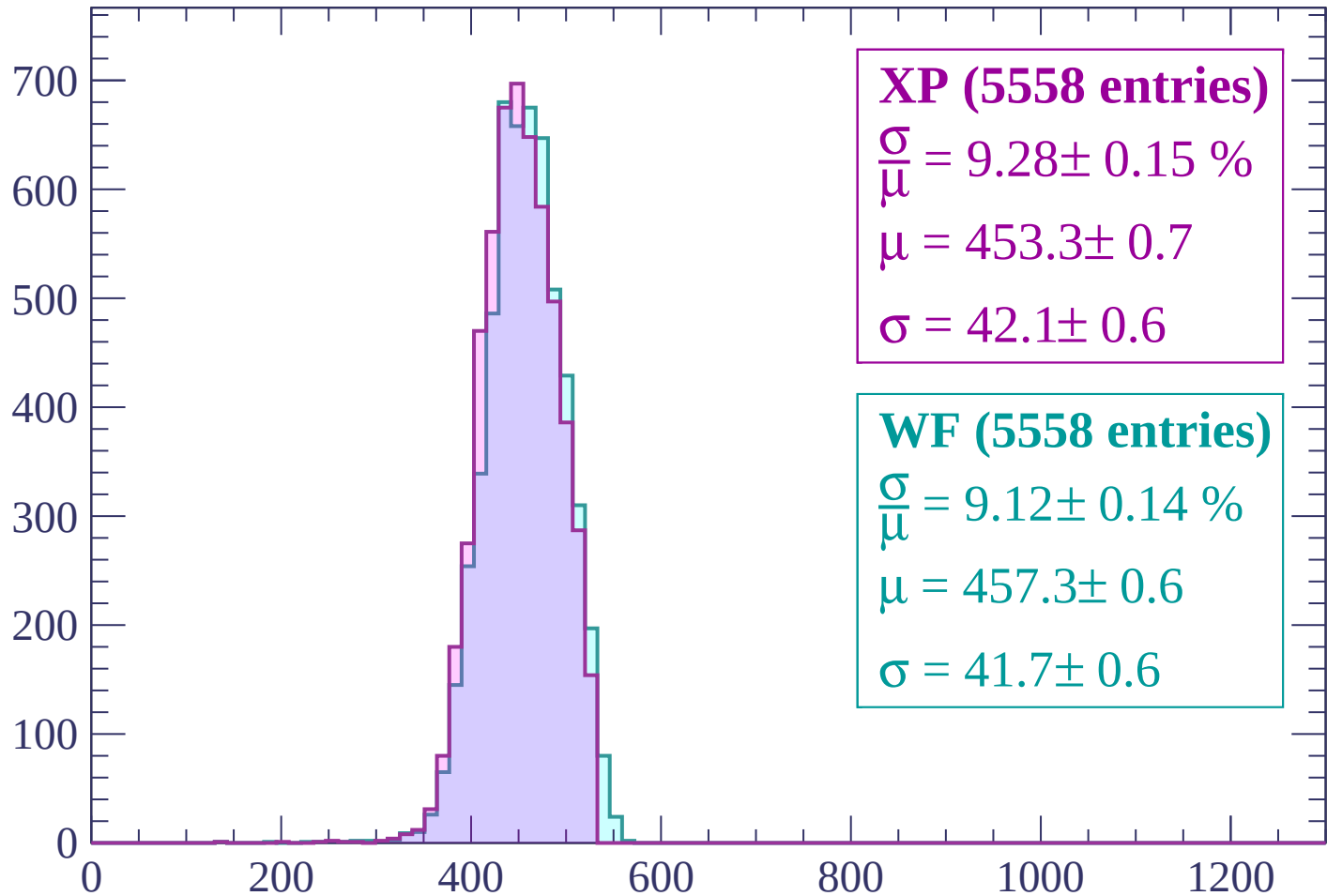
$$\mu = 455.1 \pm 0.6$$

$$\sigma = 42.0 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (5558 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.15 \%$$

$$\mu = 453.3 \pm 0.7$$

$$\sigma = 42.1 \pm 0.6$$

WF (5558 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 0.14 \%$$

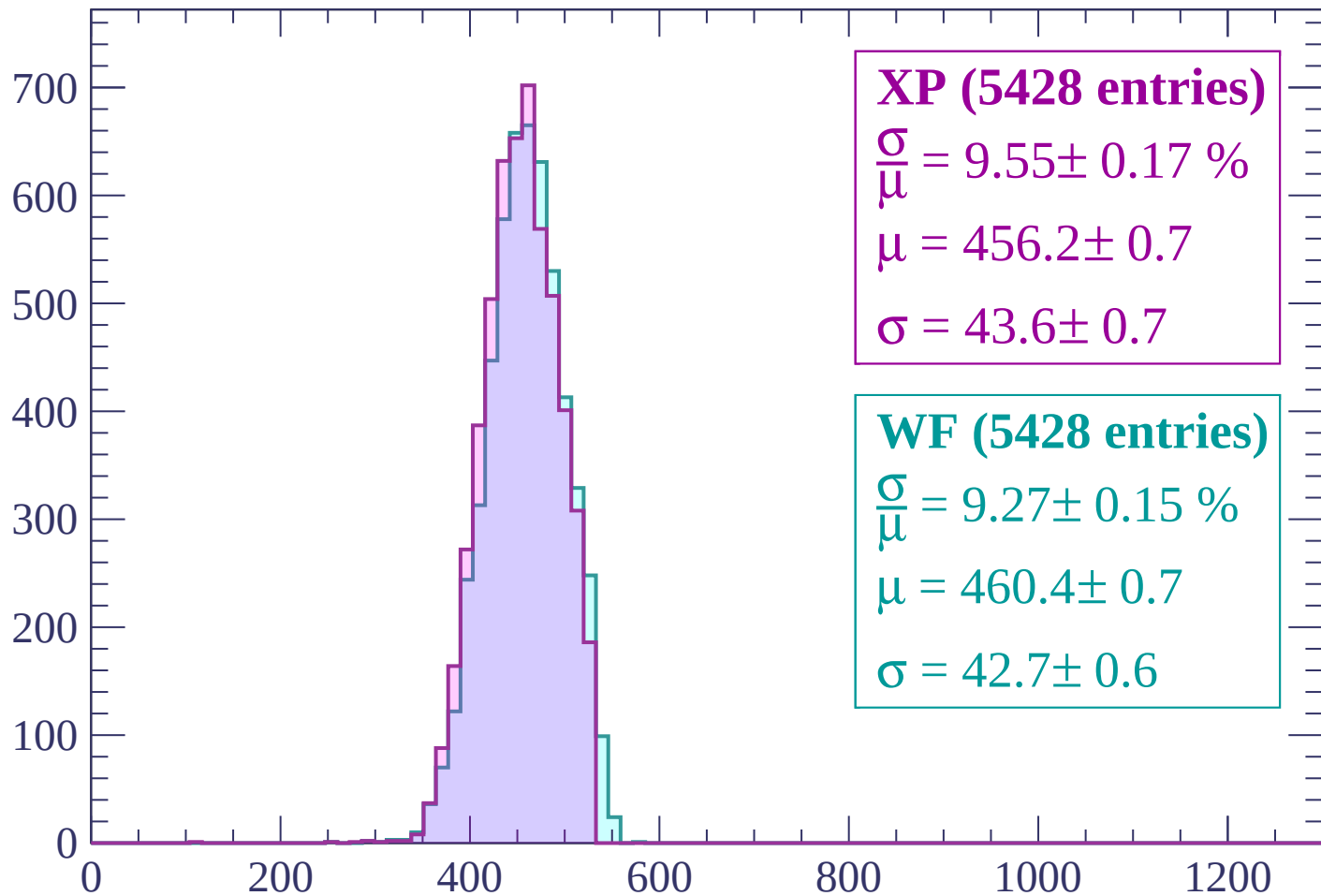
$$\mu = 457.3 \pm 0.6$$

$$\sigma = 41.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (5428 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.17 \%$$

$$\mu = 456.2 \pm 0.7$$

$$\sigma = 43.6 \pm 0.7$$

WF (5428 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.15 \%$$

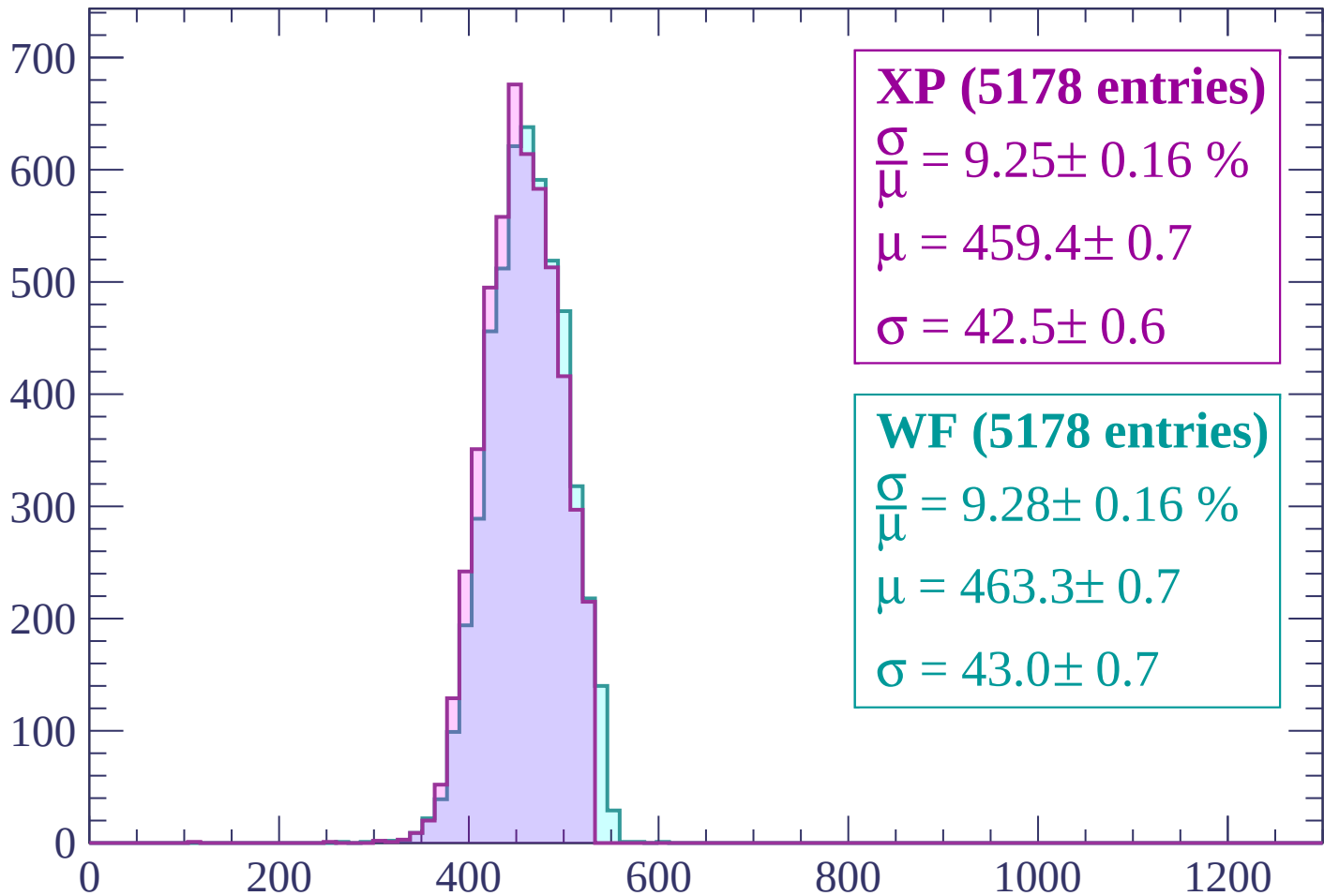
$$\mu = 460.4 \pm 0.7$$

$$\sigma = 42.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (5178 entries)

$$\frac{\sigma}{\mu} = 9.25 \pm 0.16 \%$$

$$\mu = 459.4 \pm 0.7$$

$$\sigma = 42.5 \pm 0.6$$

WF (5178 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.16 \%$$

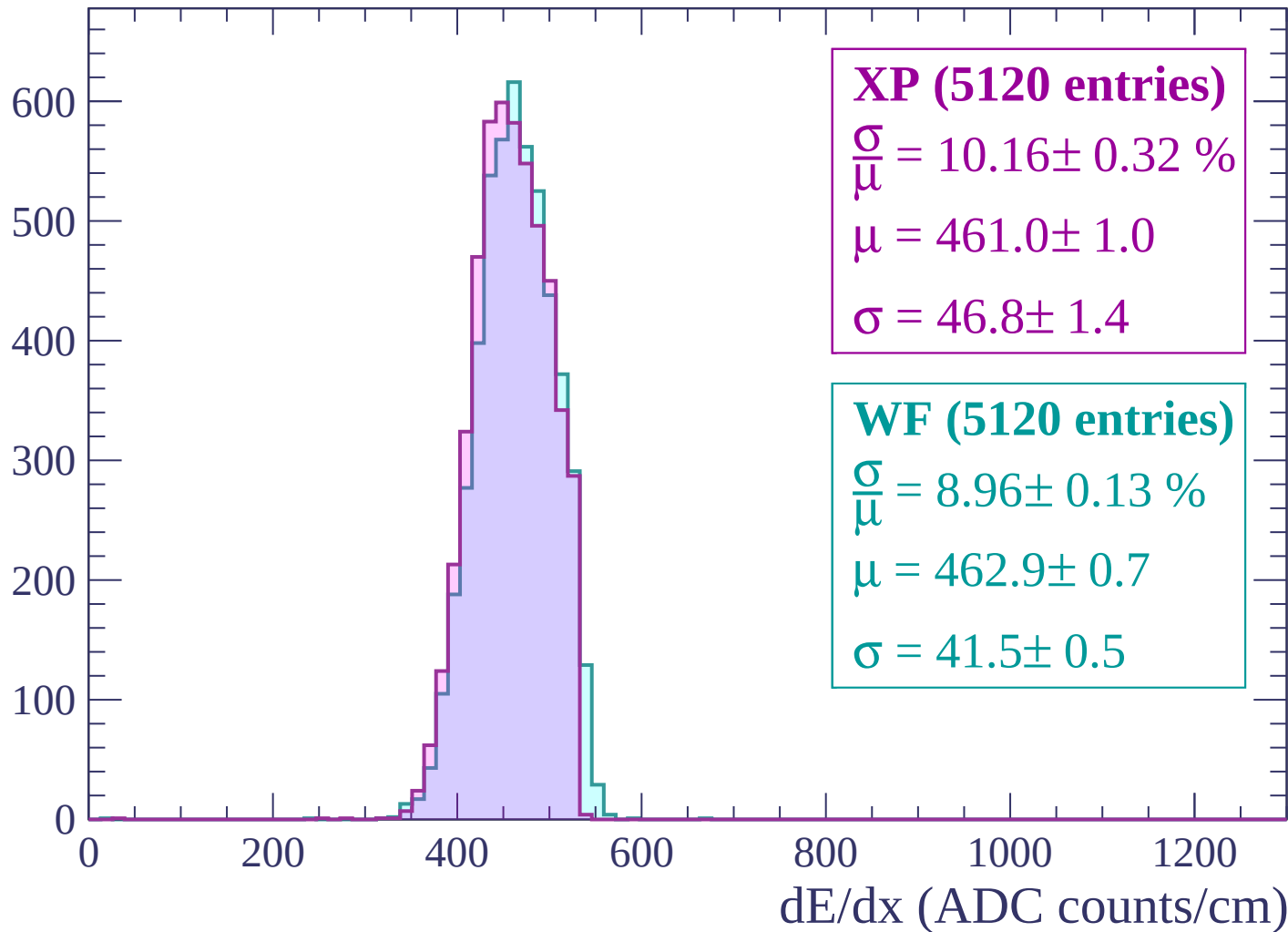
$$\mu = 463.3 \pm 0.7$$

$$\sigma = 43.0 \pm 0.7$$

dE/dx (ADC counts/cm)

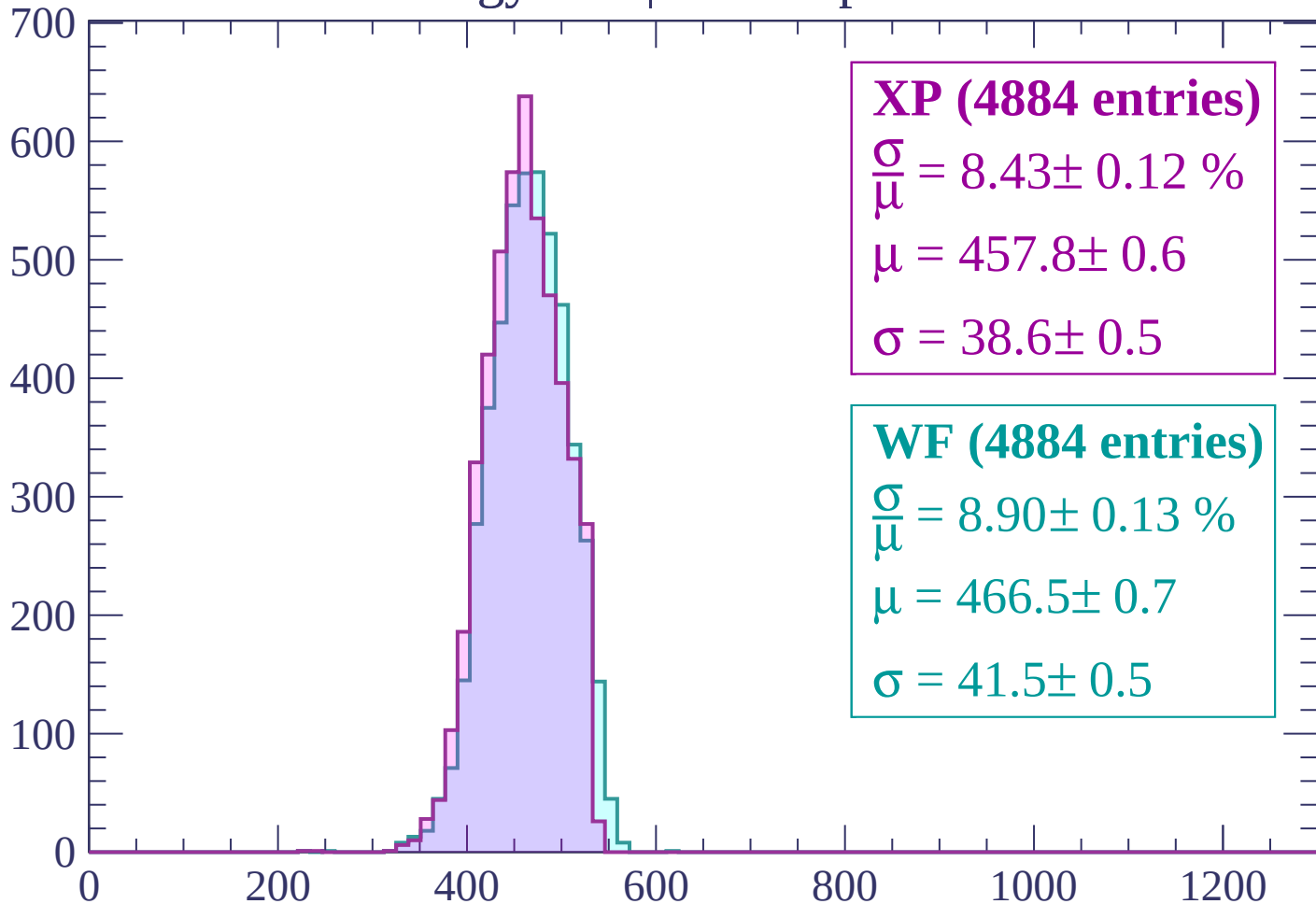
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (4884 entries)

$$\frac{\sigma}{\mu} = 8.43 \pm 0.12 \%$$

$$\mu = 457.8 \pm 0.6$$

$$\sigma = 38.6 \pm 0.5$$

WF (4884 entries)

$$\frac{\sigma}{\mu} = 8.90 \pm 0.13 \%$$

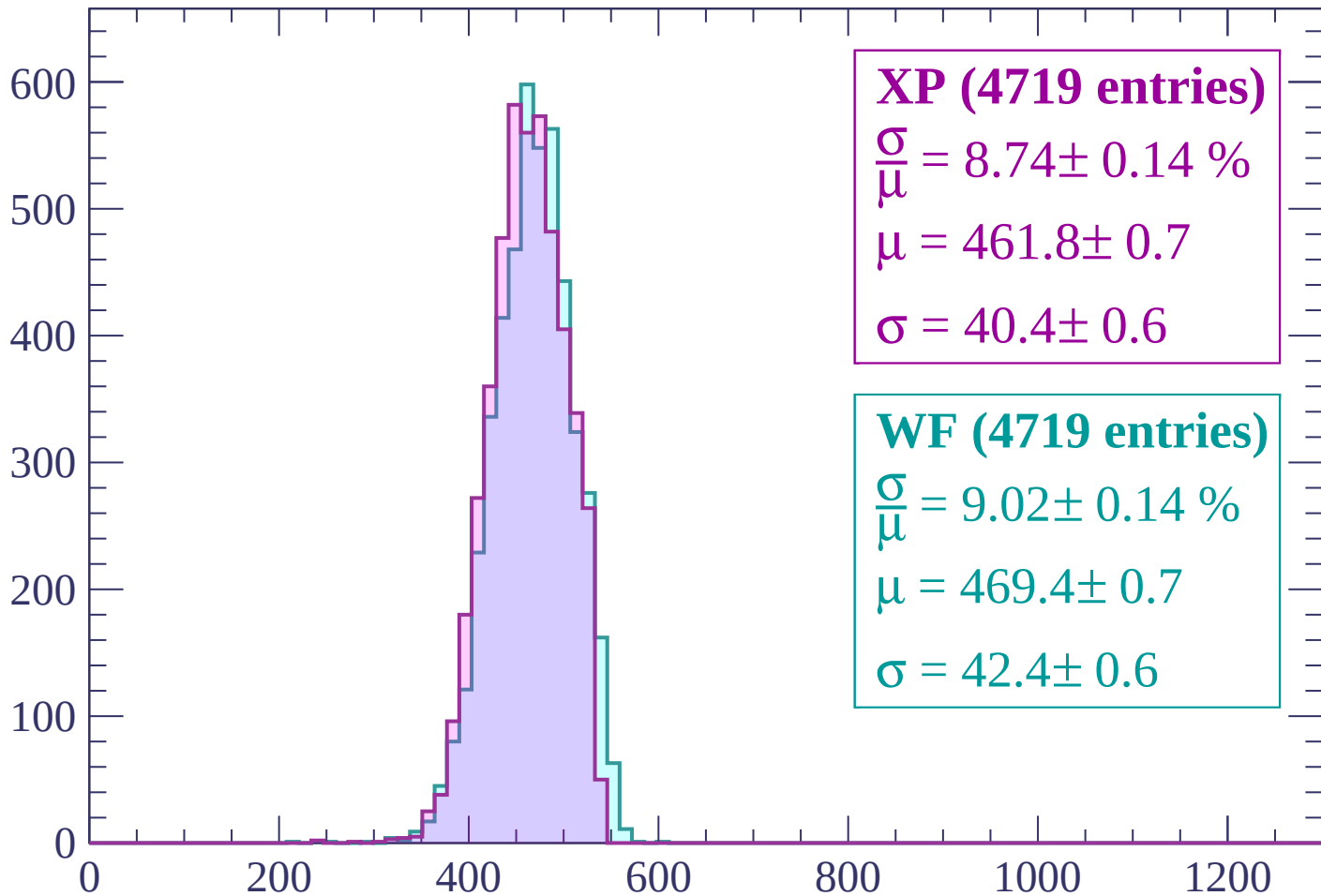
$$\mu = 466.5 \pm 0.7$$

$$\sigma = 41.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (4719 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.14 \%$$

$$\mu = 461.8 \pm 0.7$$

$$\sigma = 40.4 \pm 0.6$$

WF (4719 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.14 \%$$

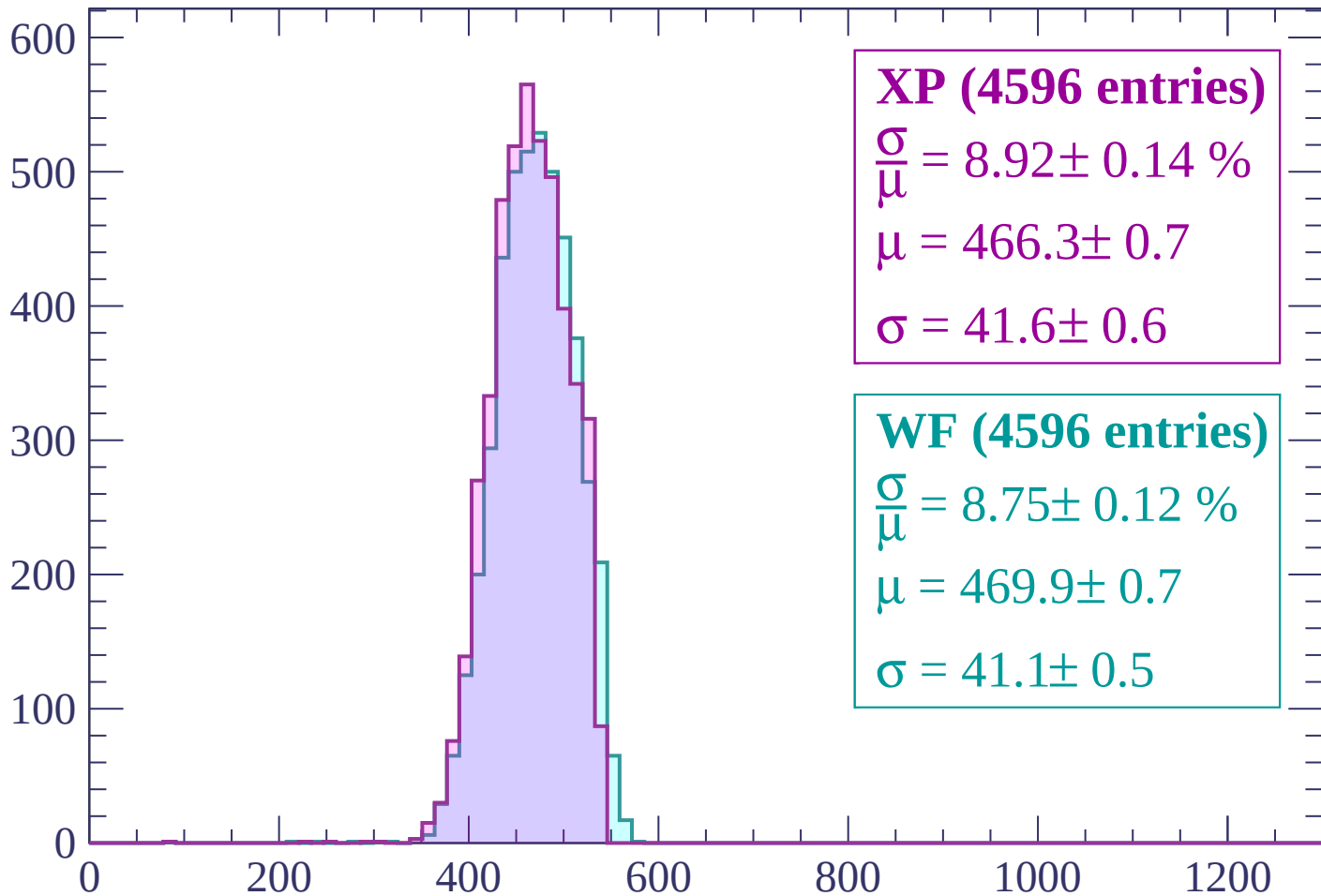
$$\mu = 469.4 \pm 0.7$$

$$\sigma = 42.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (4596 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.14 \%$$

$$\mu = 466.3 \pm 0.7$$

$$\sigma = 41.6 \pm 0.6$$

WF (4596 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.12 \%$$

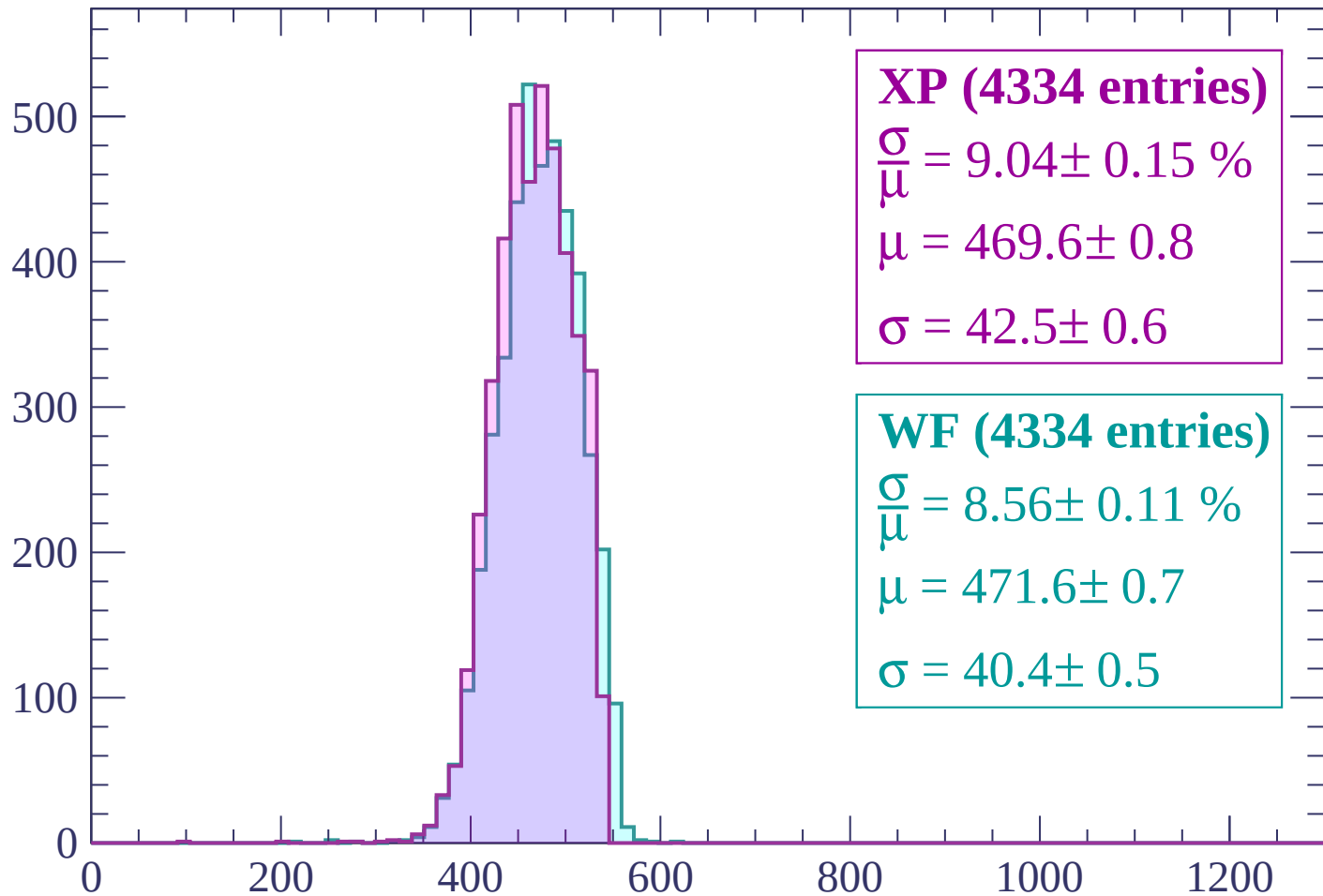
$$\mu = 469.9 \pm 0.7$$

$$\sigma = 41.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (4334 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.15 \%$$

$$\mu = 469.6 \pm 0.8$$

$$\sigma = 42.5 \pm 0.6$$

WF (4334 entries)

$$\frac{\sigma}{\mu} = 8.56 \pm 0.11 \%$$

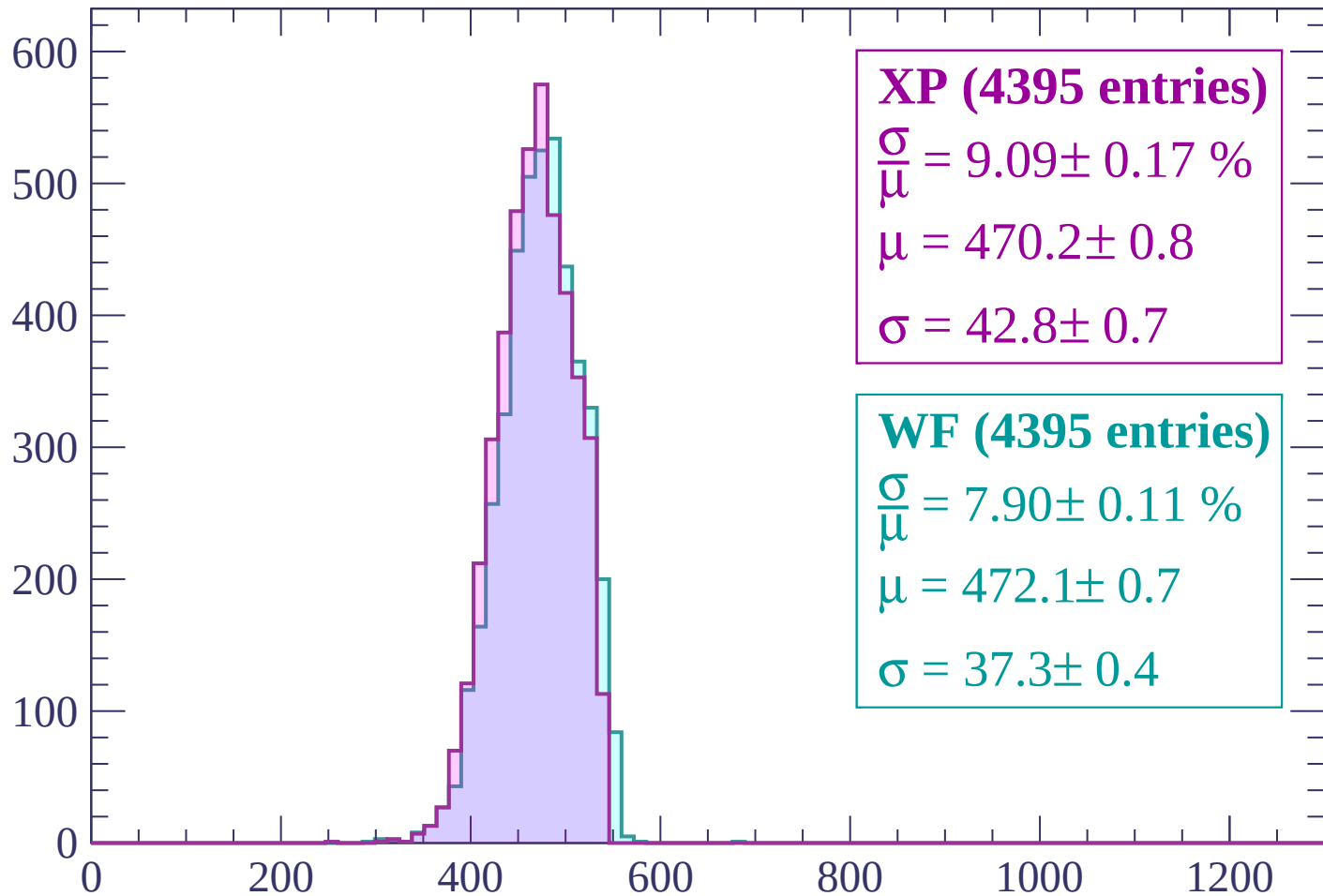
$$\mu = 471.6 \pm 0.7$$

$$\sigma = 40.4 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (4395 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.17 \%$$

$$\mu = 470.2 \pm 0.8$$

$$\sigma = 42.8 \pm 0.7$$

WF (4395 entries)

$$\frac{\sigma}{\mu} = 7.90 \pm 0.11 \%$$

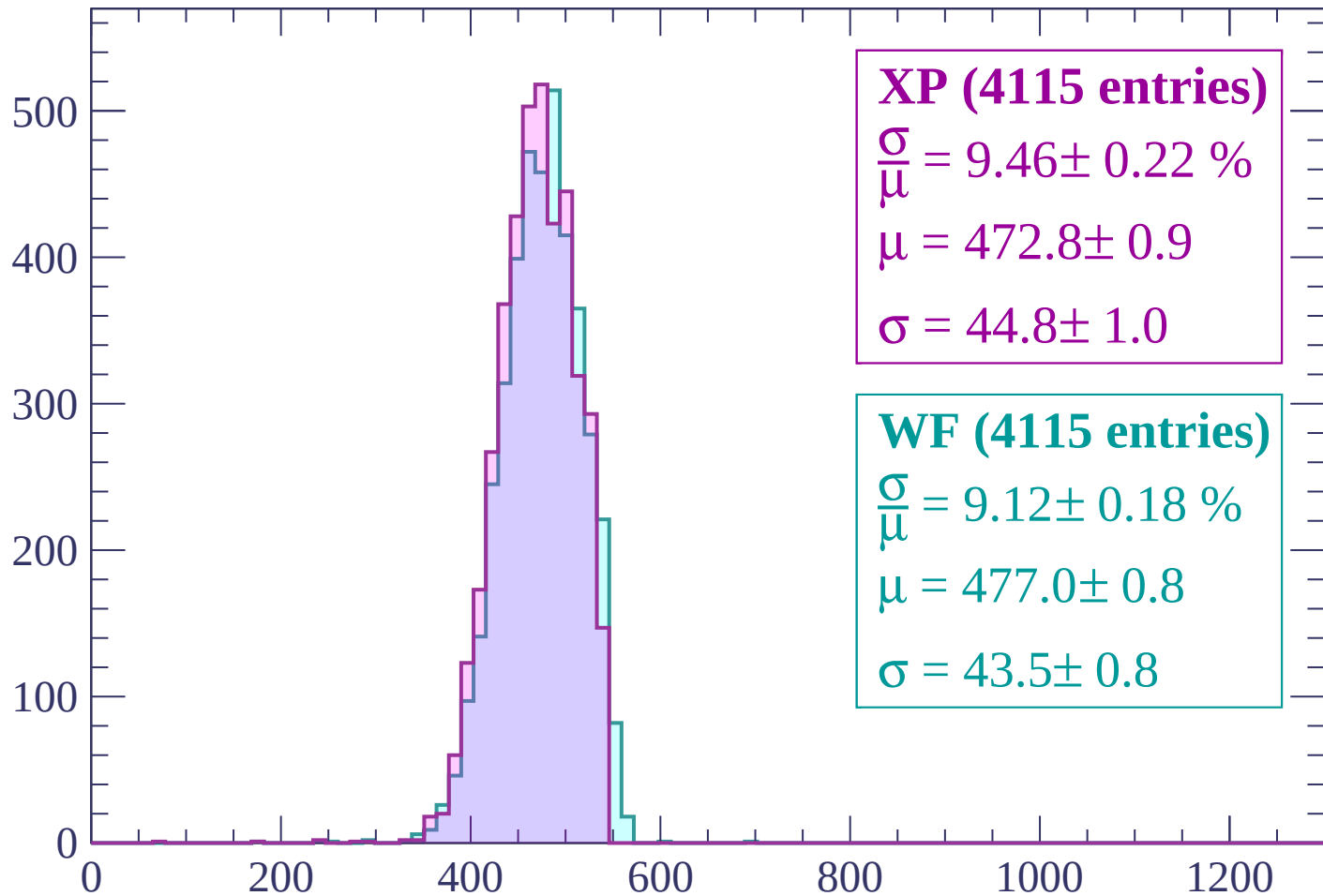
$$\mu = 472.1 \pm 0.7$$

$$\sigma = 37.3 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count



XP (4115 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.22 \%$$

$$\mu = 472.8 \pm 0.9$$

$$\sigma = 44.8 \pm 1.0$$

WF (4115 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 0.18 \%$$

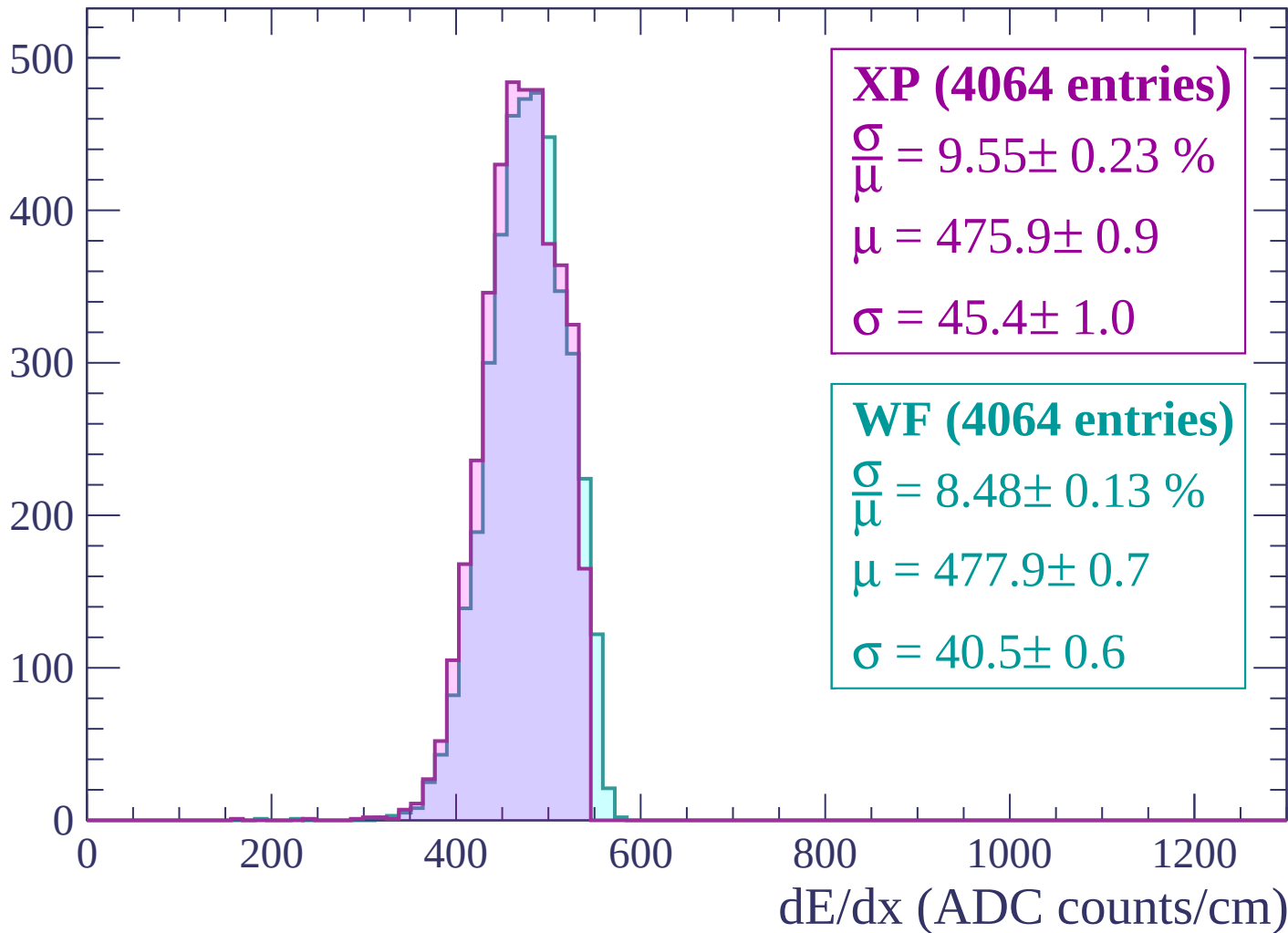
$$\mu = 477.0 \pm 0.8$$

$$\sigma = 43.5 \pm 0.8$$

dE/dx (ADC counts/cm)

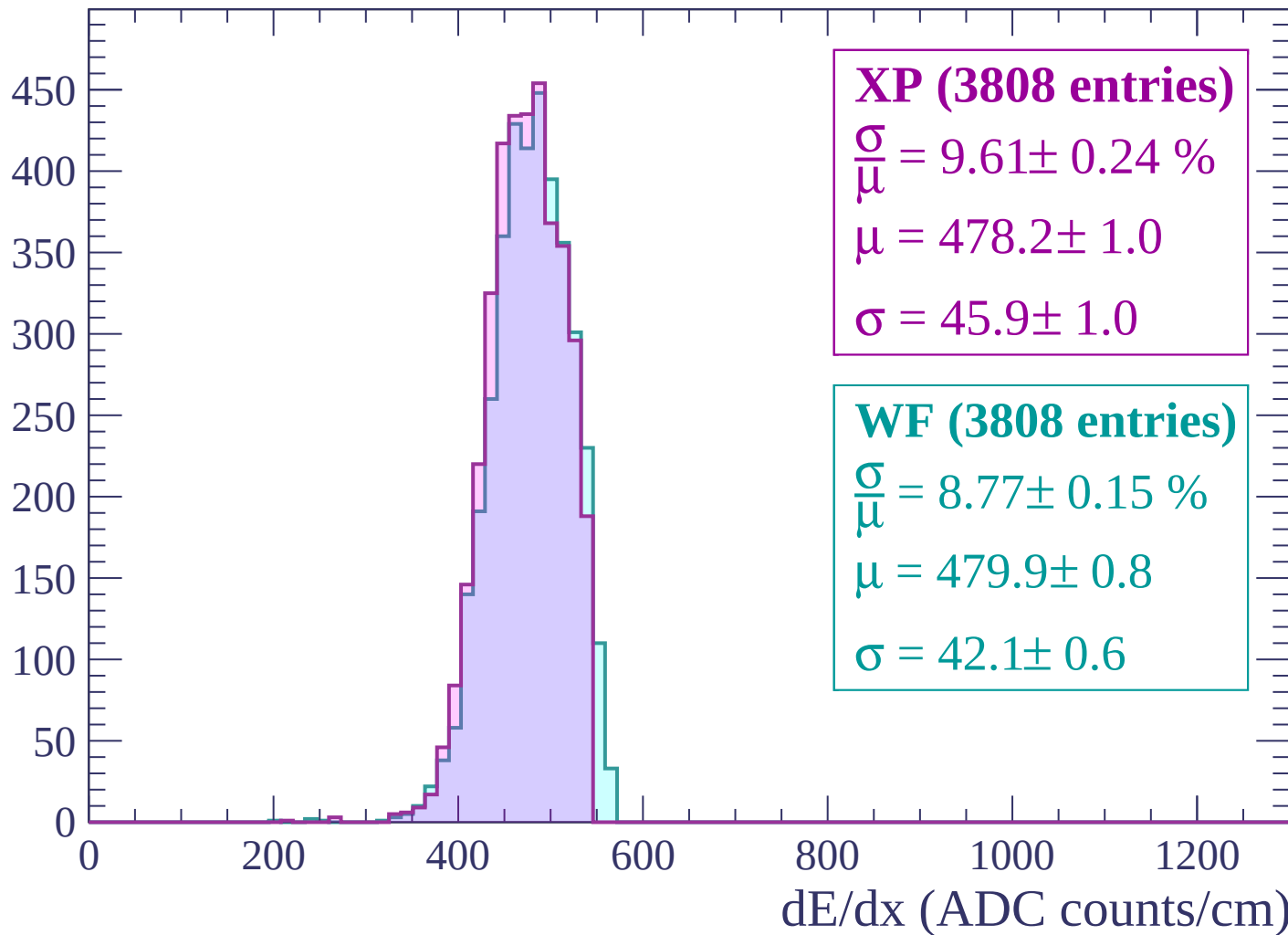
Energy loss | $1920 < p < 1980$

Count



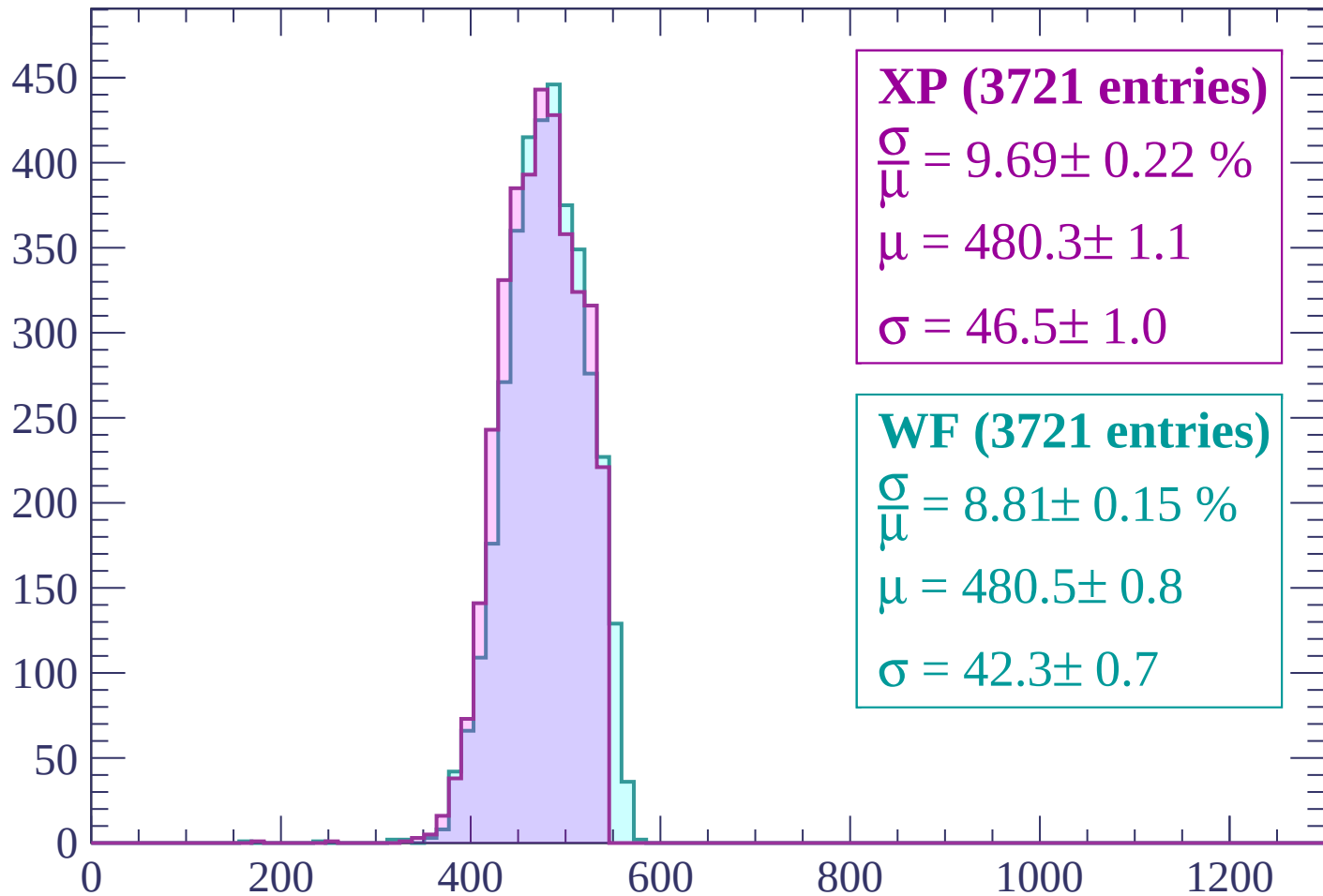
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (3721 entries)

$$\frac{\sigma}{\mu} = 9.69 \pm 0.22 \%$$

$$\mu = 480.3 \pm 1.1$$

$$\sigma = 46.5 \pm 1.0$$

WF (3721 entries)

$$\frac{\sigma}{\mu} = 8.81 \pm 0.15 \%$$

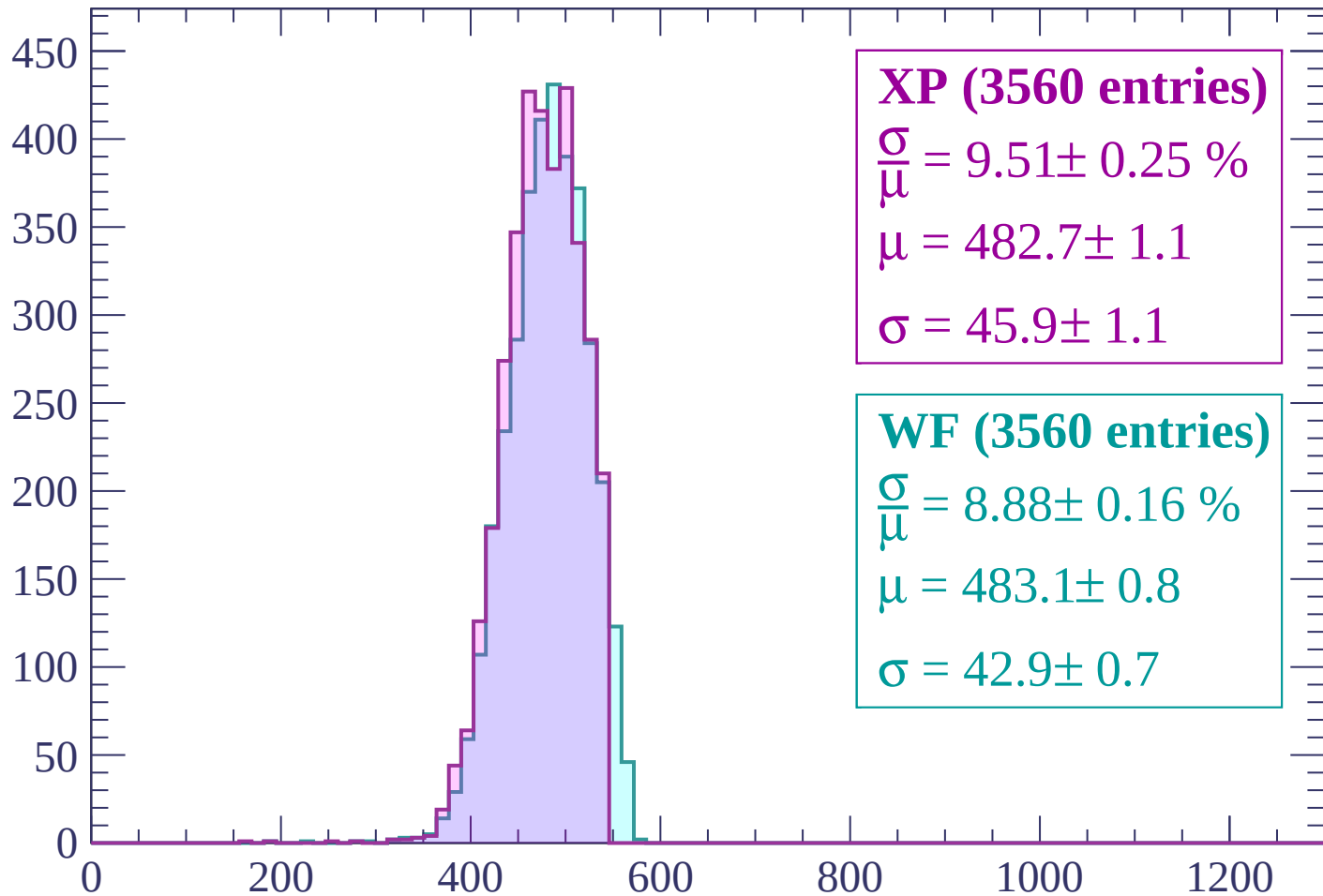
$$\mu = 480.5 \pm 0.8$$

$$\sigma = 42.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (3560 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.25 \%$$

$$\mu = 482.7 \pm 1.1$$

$$\sigma = 45.9 \pm 1.1$$

WF (3560 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.16 \%$$

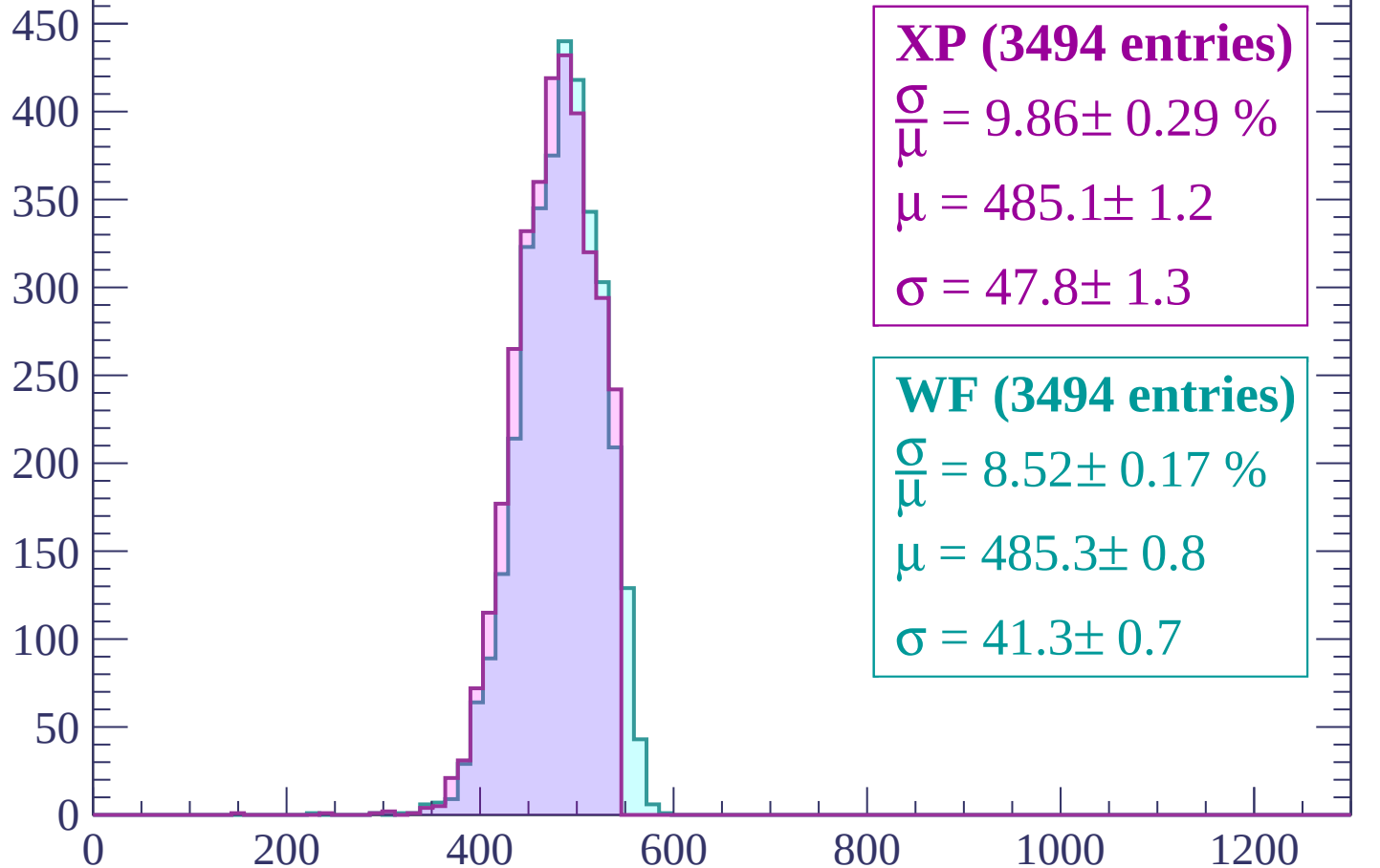
$$\mu = 483.1 \pm 0.8$$

$$\sigma = 42.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (3494 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.29 \%$$

$$\mu = 485.1 \pm 1.2$$

$$\sigma = 47.8 \pm 1.3$$

WF (3494 entries)

$$\frac{\sigma}{\mu} = 8.52 \pm 0.17 \%$$

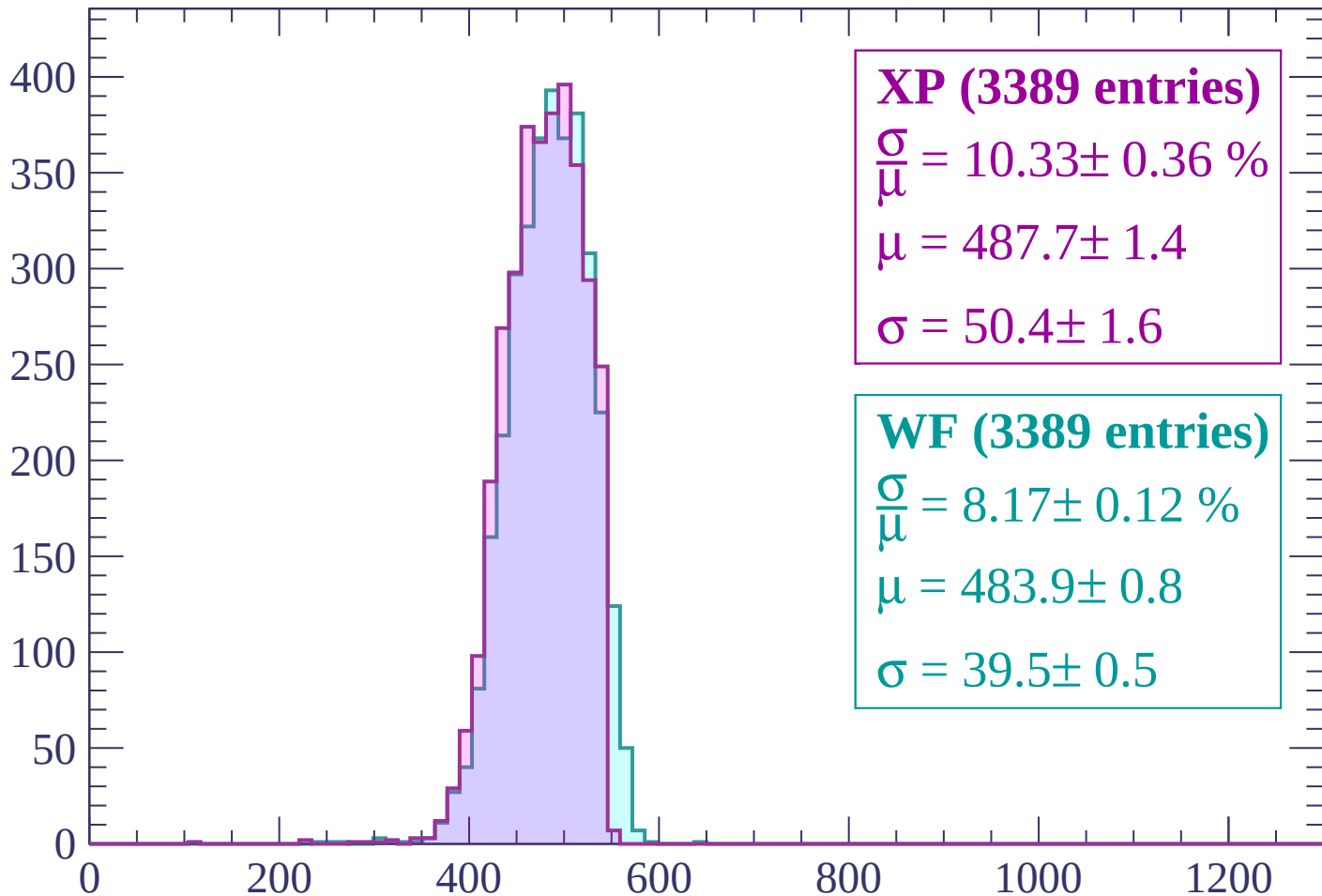
$$\mu = 485.3 \pm 0.8$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (3389 entries)

$\frac{\sigma}{\mu} = 10.33 \pm 0.36 \%$

$\mu = 487.7 \pm 1.4$

$\sigma = 50.4 \pm 1.6$

WF (3389 entries)

$\frac{\sigma}{\mu} = 8.17 \pm 0.12 \%$

$\mu = 483.9 \pm 0.8$

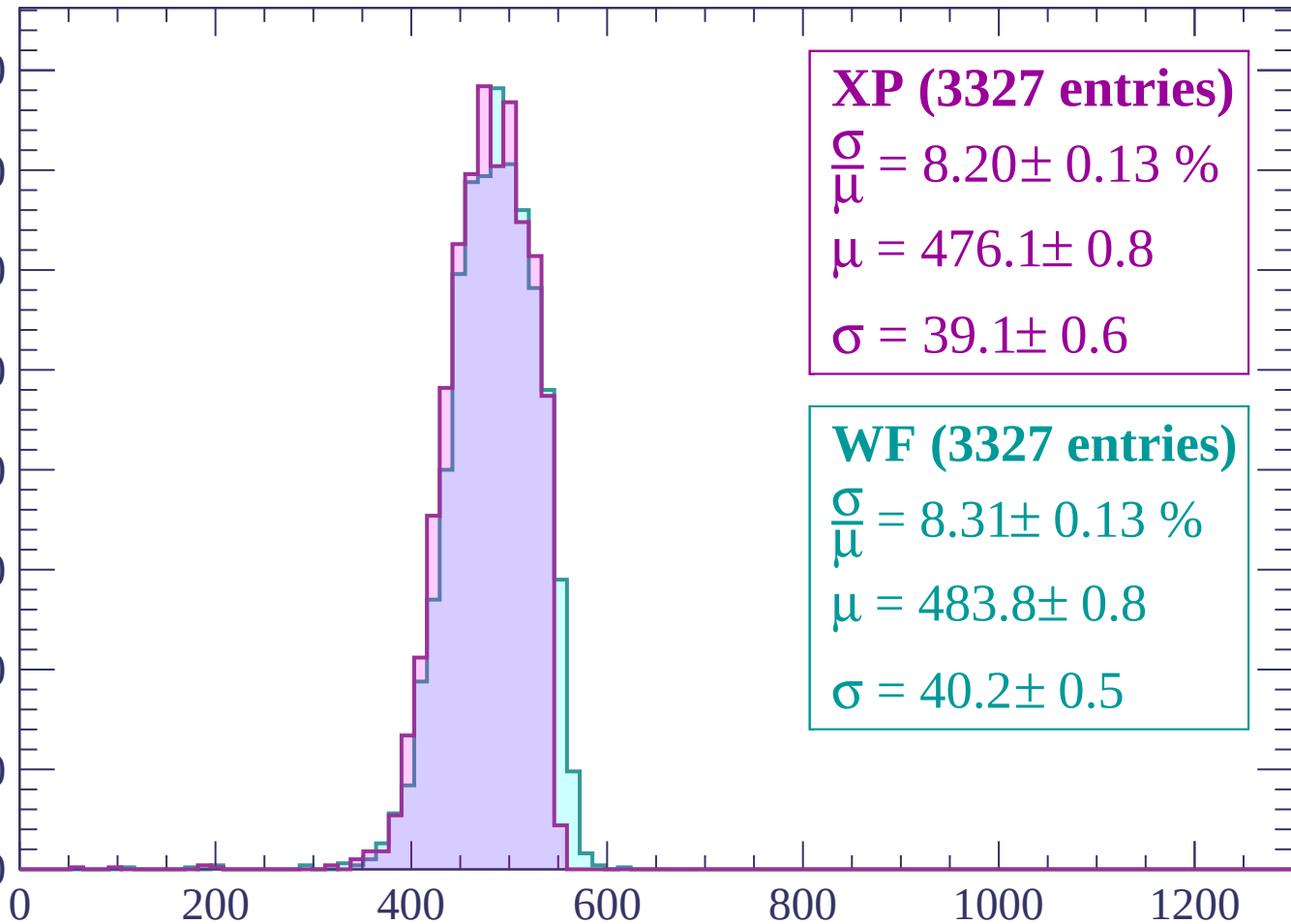
$\sigma = 39.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count

400
350
300
250
200
150
100
50
0



XP (3327 entries)

$\frac{\sigma}{\mu} = 8.20 \pm 0.13 \%$

$\mu = 476.1 \pm 0.8$

$\sigma = 39.1 \pm 0.6$

WF (3327 entries)

$\frac{\sigma}{\mu} = 8.31 \pm 0.13 \%$

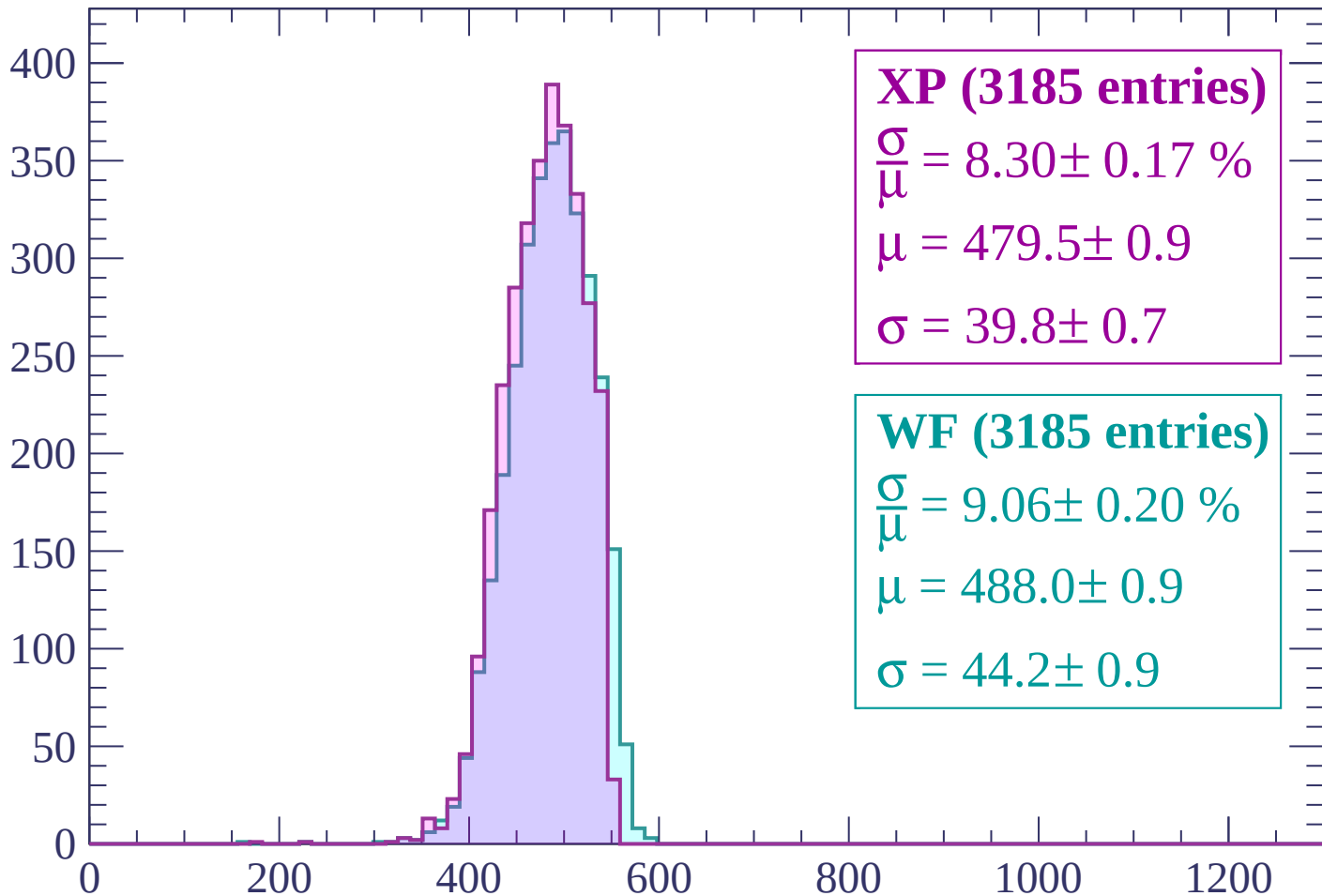
$\mu = 483.8 \pm 0.8$

$\sigma = 40.2 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (3185 entries)

$$\frac{\sigma}{\mu} = 8.30 \pm 0.17 \%$$

$$\mu = 479.5 \pm 0.9$$

$$\sigma = 39.8 \pm 0.7$$

WF (3185 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.20 \%$$

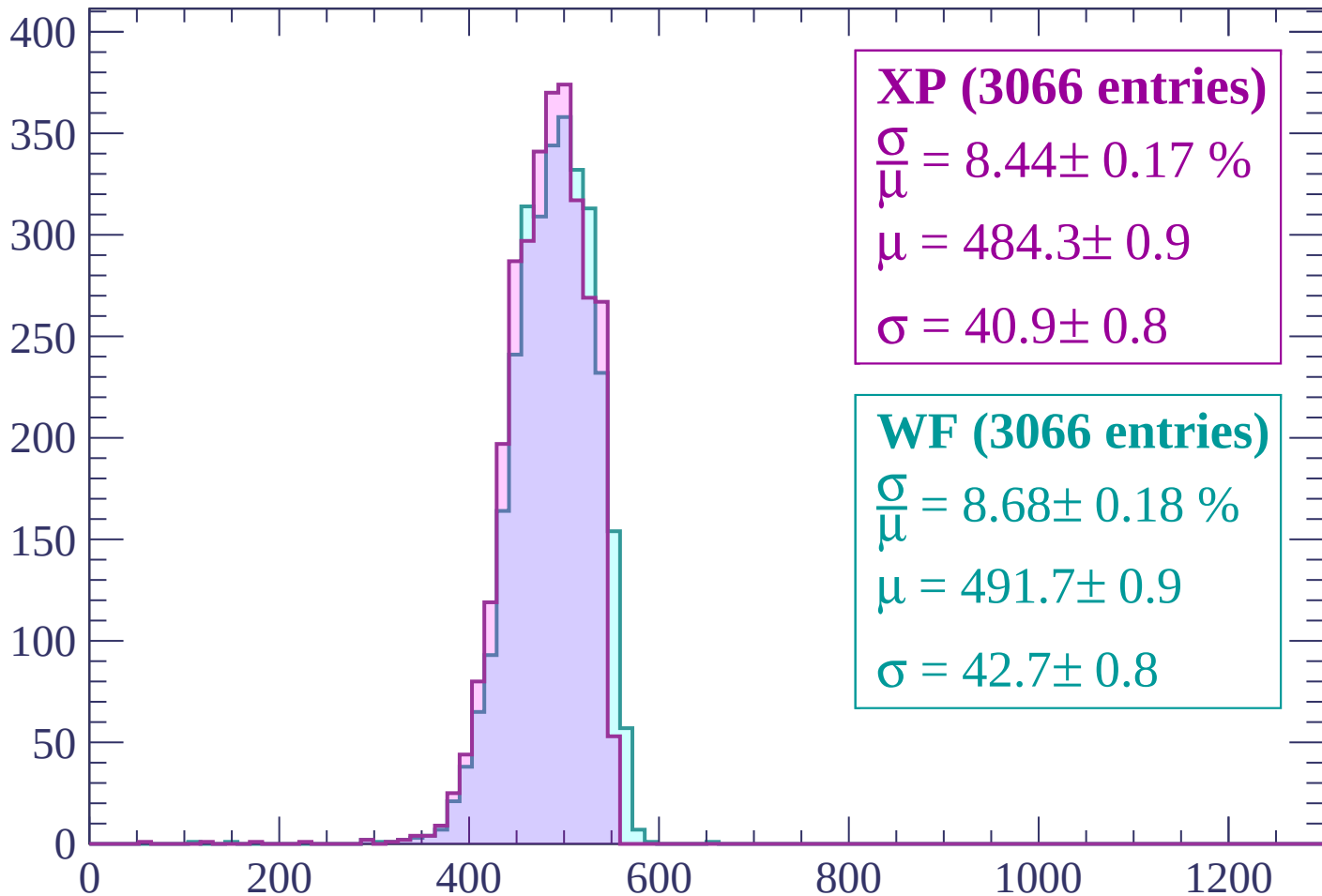
$$\mu = 488.0 \pm 0.9$$

$$\sigma = 44.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (3066 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.17 \%$$

$$\mu = 484.3 \pm 0.9$$

$$\sigma = 40.9 \pm 0.8$$

WF (3066 entries)

$$\frac{\sigma}{\mu} = 8.68 \pm 0.18 \%$$

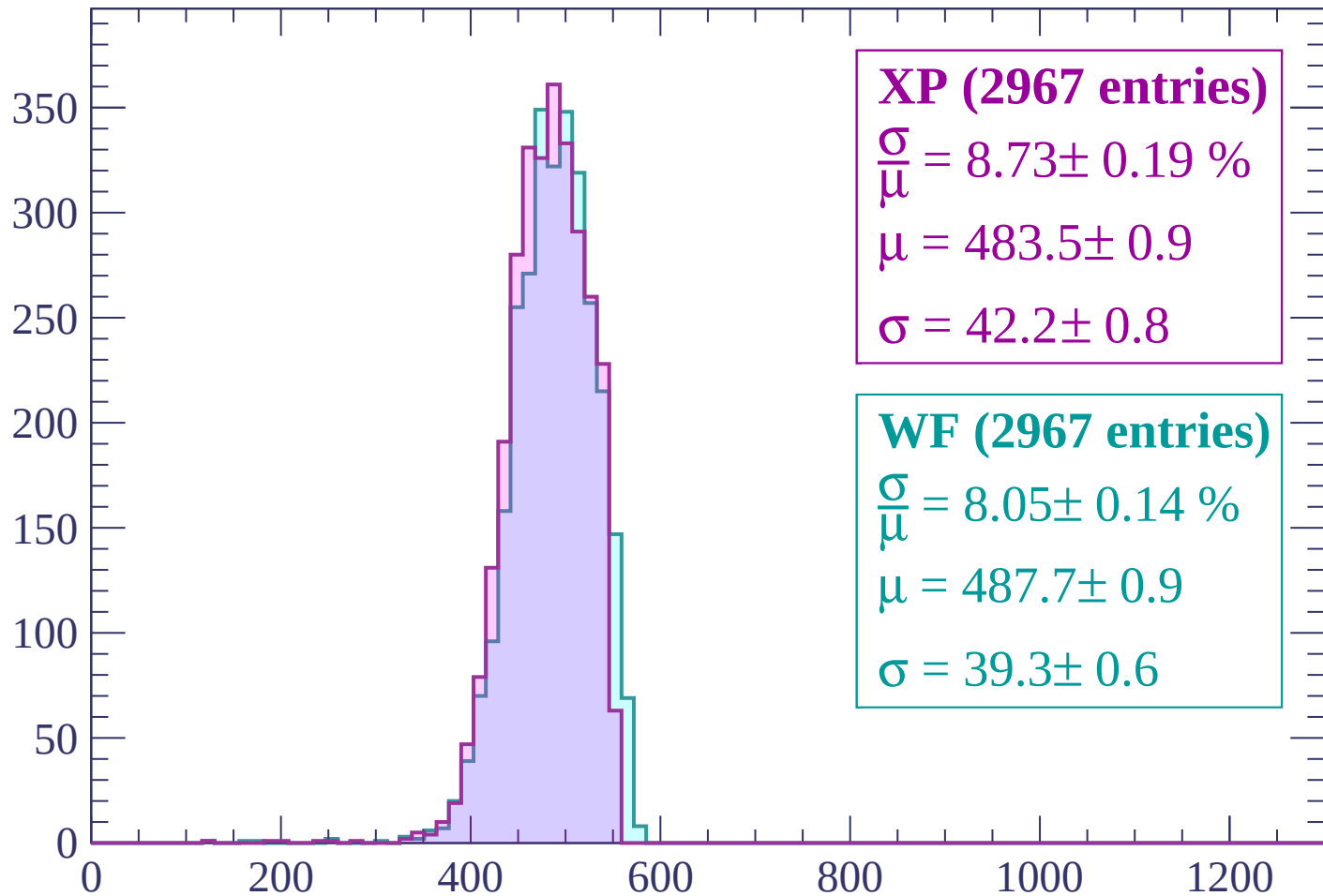
$$\mu = 491.7 \pm 0.9$$

$$\sigma = 42.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (2967 entries)

$\frac{\sigma}{\mu} = 8.73 \pm 0.19 \%$

$\mu = 483.5 \pm 0.9$

$\sigma = 42.2 \pm 0.8$

WF (2967 entries)

$\frac{\sigma}{\mu} = 8.05 \pm 0.14 \%$

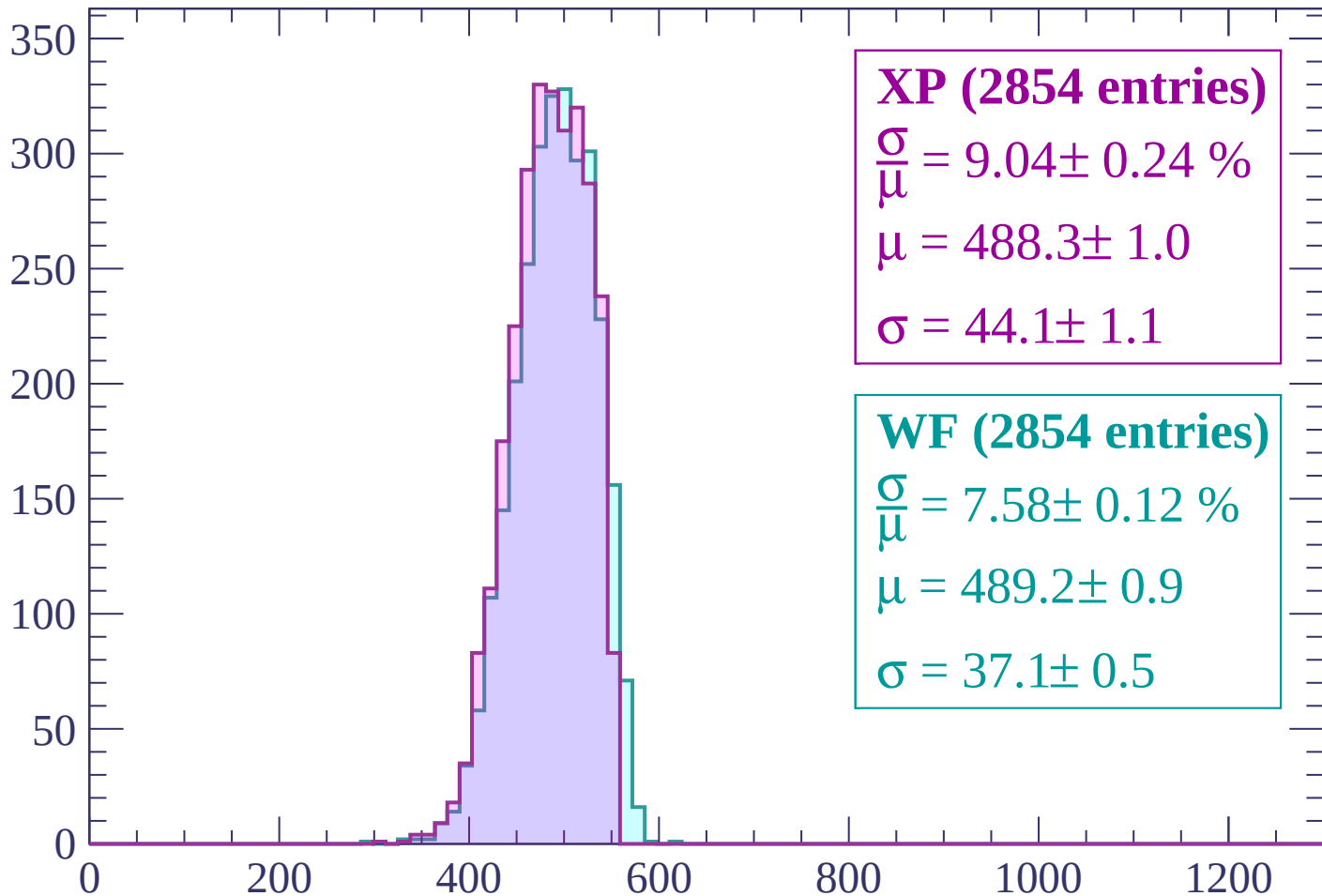
$\mu = 487.7 \pm 0.9$

$\sigma = 39.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP (2854 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.24 \%$$

$$\mu = 488.3 \pm 1.0$$

$$\sigma = 44.1 \pm 1.1$$

WF (2854 entries)

$$\frac{\sigma}{\mu} = 7.58 \pm 0.12 \%$$

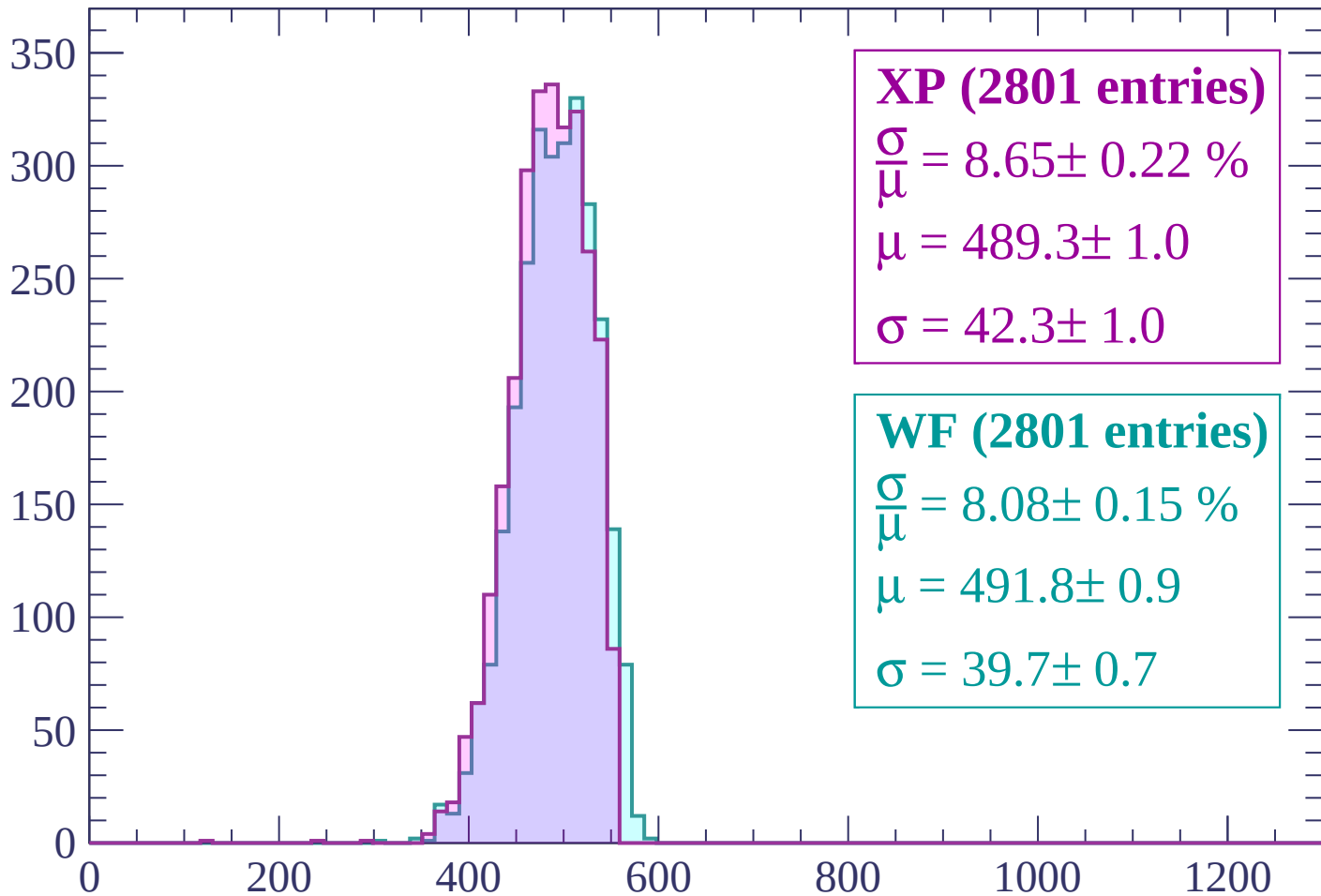
$$\mu = 489.2 \pm 0.9$$

$$\sigma = 37.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (2801 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.22 \%$$

$$\mu = 489.3 \pm 1.0$$

$$\sigma = 42.3 \pm 1.0$$

WF (2801 entries)

$$\frac{\sigma}{\mu} = 8.08 \pm 0.15 \%$$

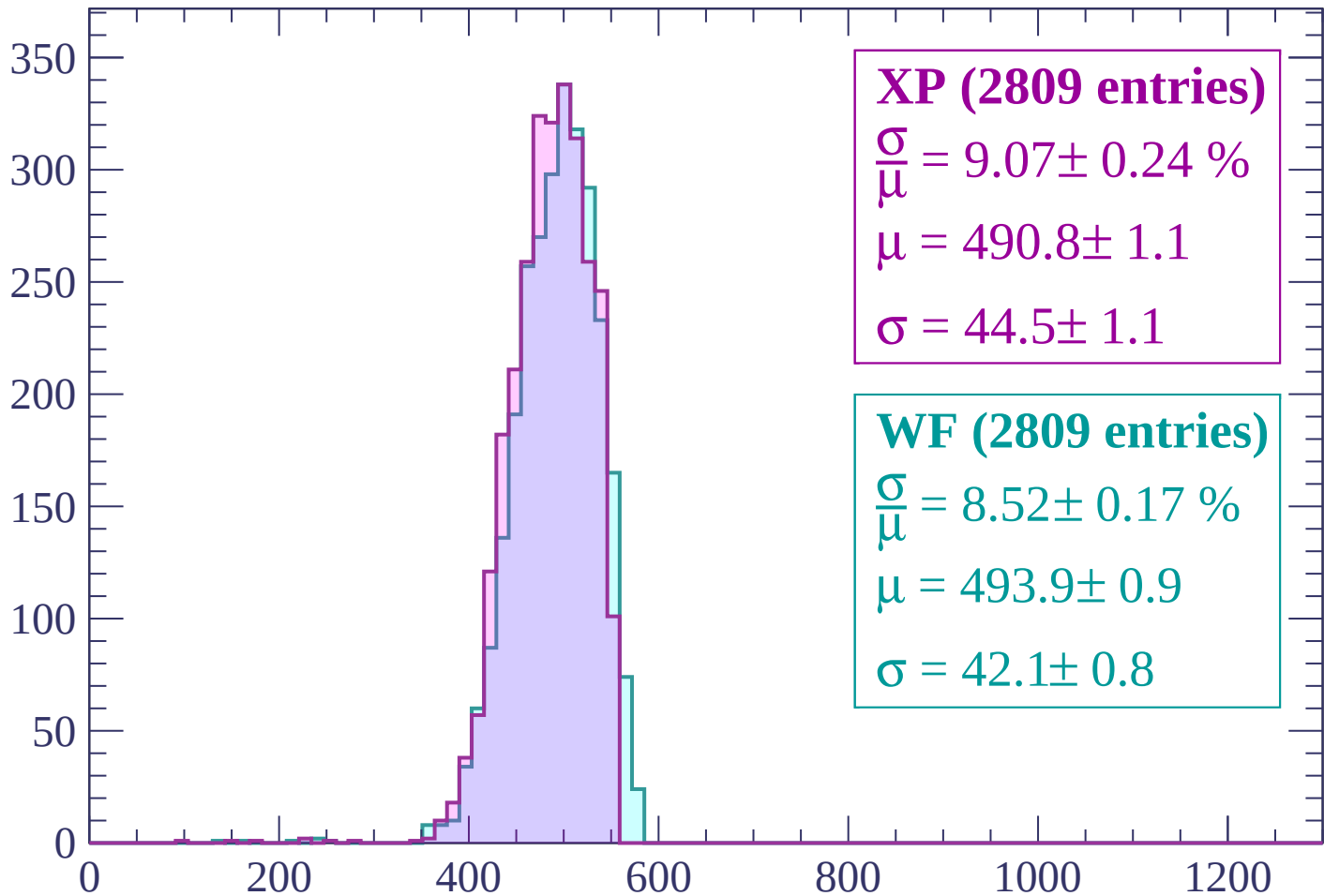
$$\mu = 491.8 \pm 0.9$$

$$\sigma = 39.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (2809 entries)

$$\frac{\sigma}{\mu} = 9.07 \pm 0.24 \%$$

$$\mu = 490.8 \pm 1.1$$

$$\sigma = 44.5 \pm 1.1$$

WF (2809 entries)

$$\frac{\sigma}{\mu} = 8.52 \pm 0.17 \%$$

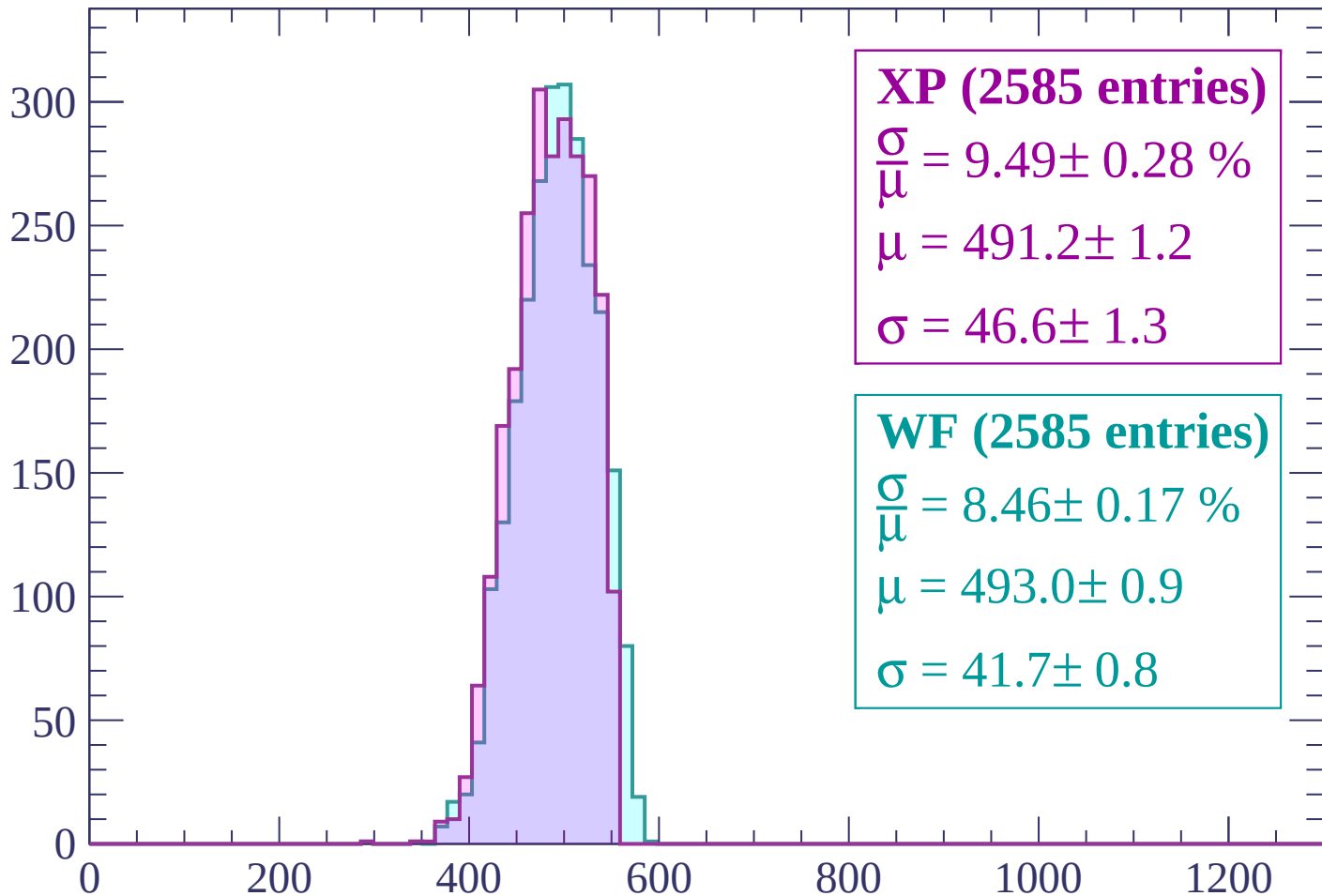
$$\mu = 493.9 \pm 0.9$$

$$\sigma = 42.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (2585 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.28 \%$$

$$\mu = 491.2 \pm 1.2$$

$$\sigma = 46.6 \pm 1.3$$

WF (2585 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

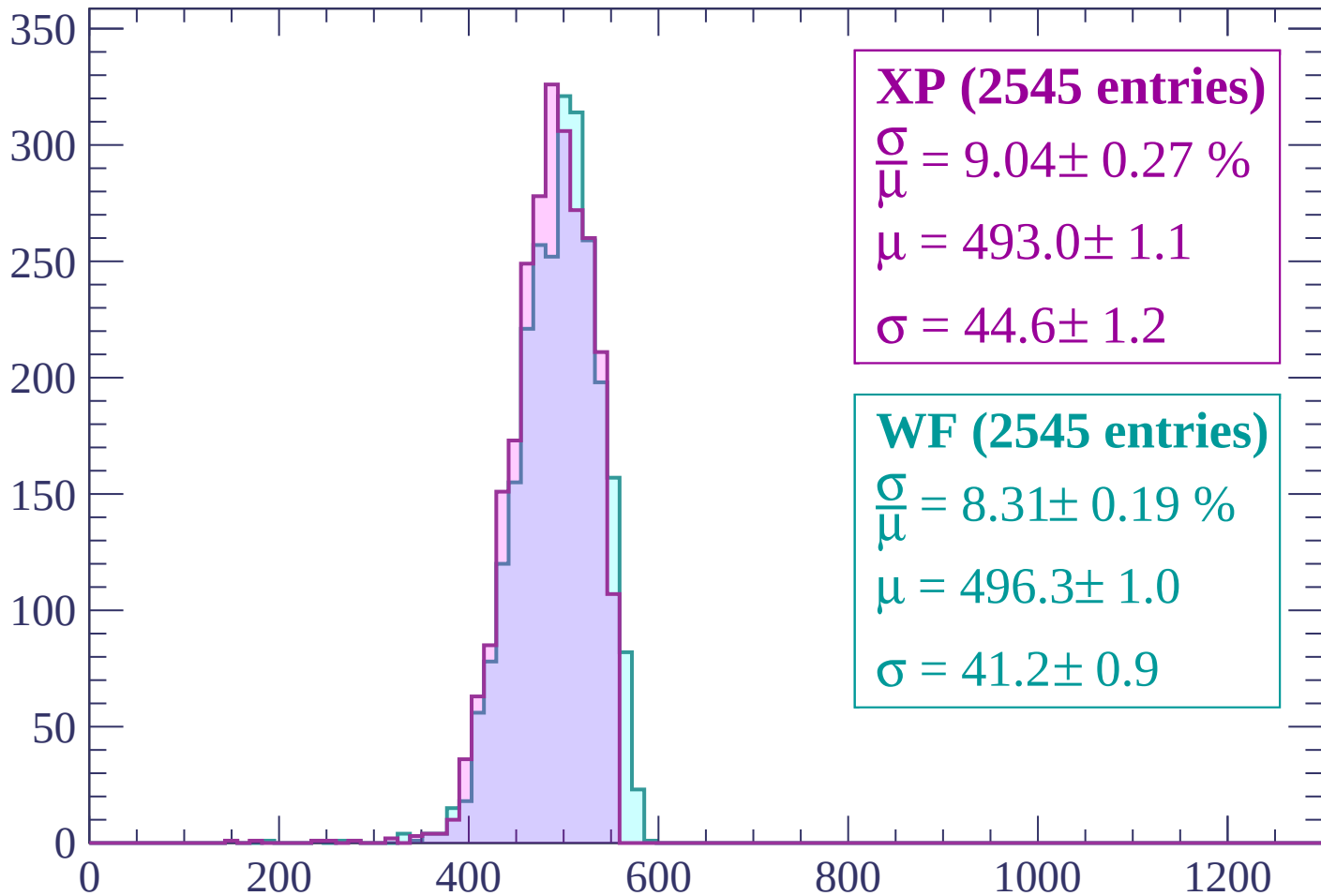
$$\mu = 493.0 \pm 0.9$$

$$\sigma = 41.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2760 < p < 2820$

Count



XP (2545 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.27 \%$$

$$\mu = 493.0 \pm 1.1$$

$$\sigma = 44.6 \pm 1.2$$

WF (2545 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.19 \%$$

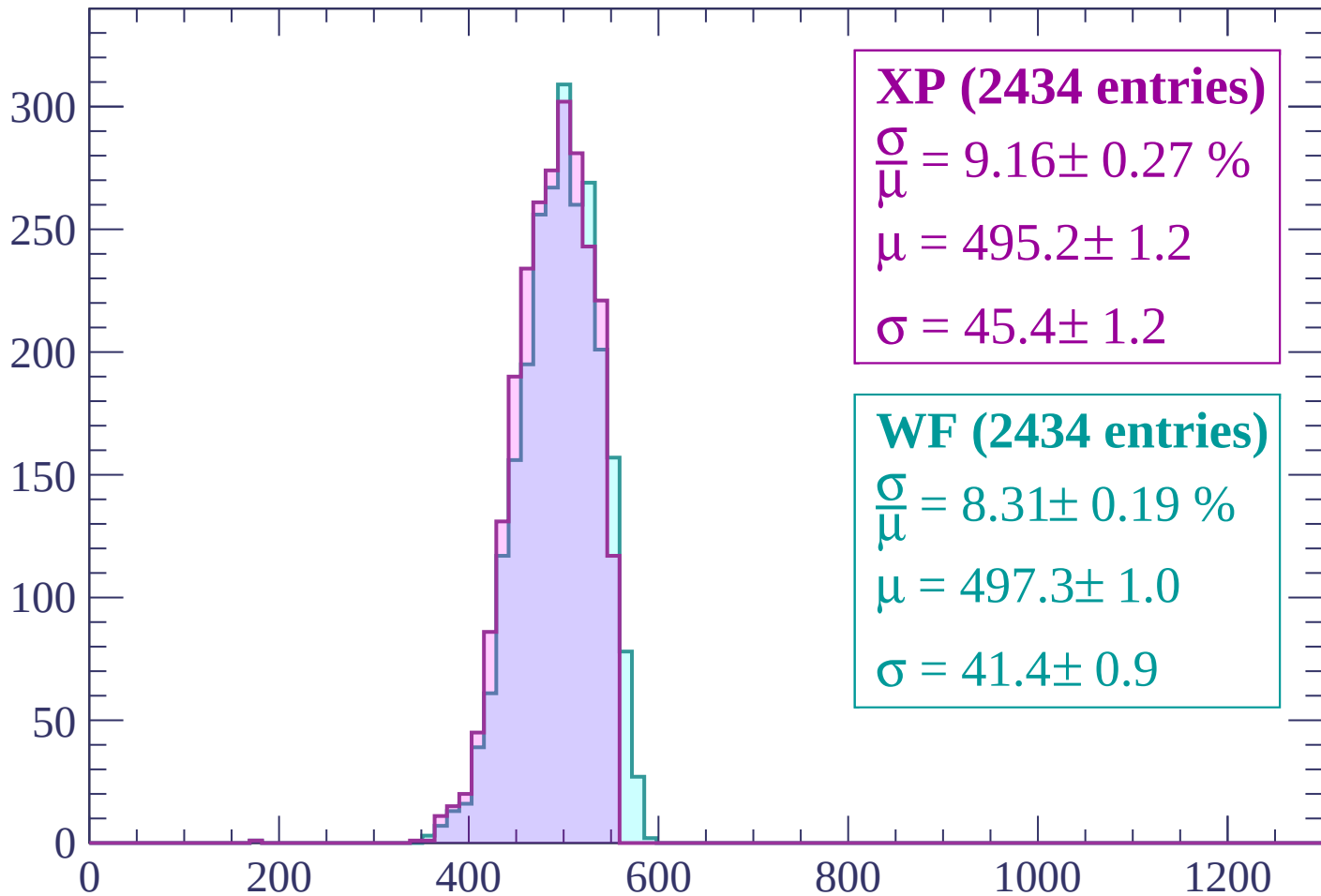
$$\mu = 496.3 \pm 1.0$$

$$\sigma = 41.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP (2434 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.27 \%$$

$$\mu = 495.2 \pm 1.2$$

$$\sigma = 45.4 \pm 1.2$$

WF (2434 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.19 \%$$

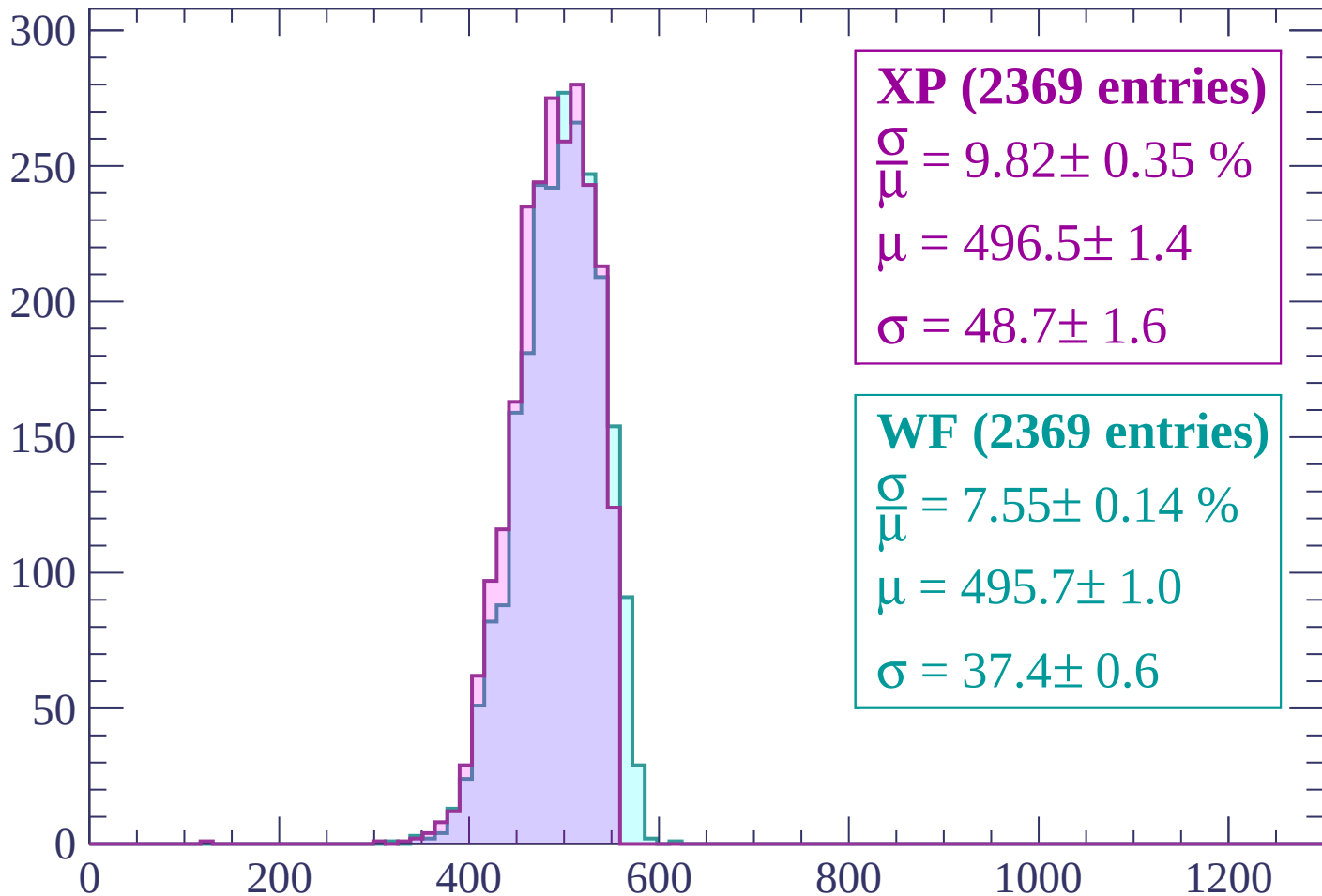
$$\mu = 497.3 \pm 1.0$$

$$\sigma = 41.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (2369 entries)

$\frac{\sigma}{\mu} = 9.82 \pm 0.35 \%$

$\mu = 496.5 \pm 1.4$

$\sigma = 48.7 \pm 1.6$

WF (2369 entries)

$\frac{\sigma}{\mu} = 7.55 \pm 0.14 \%$

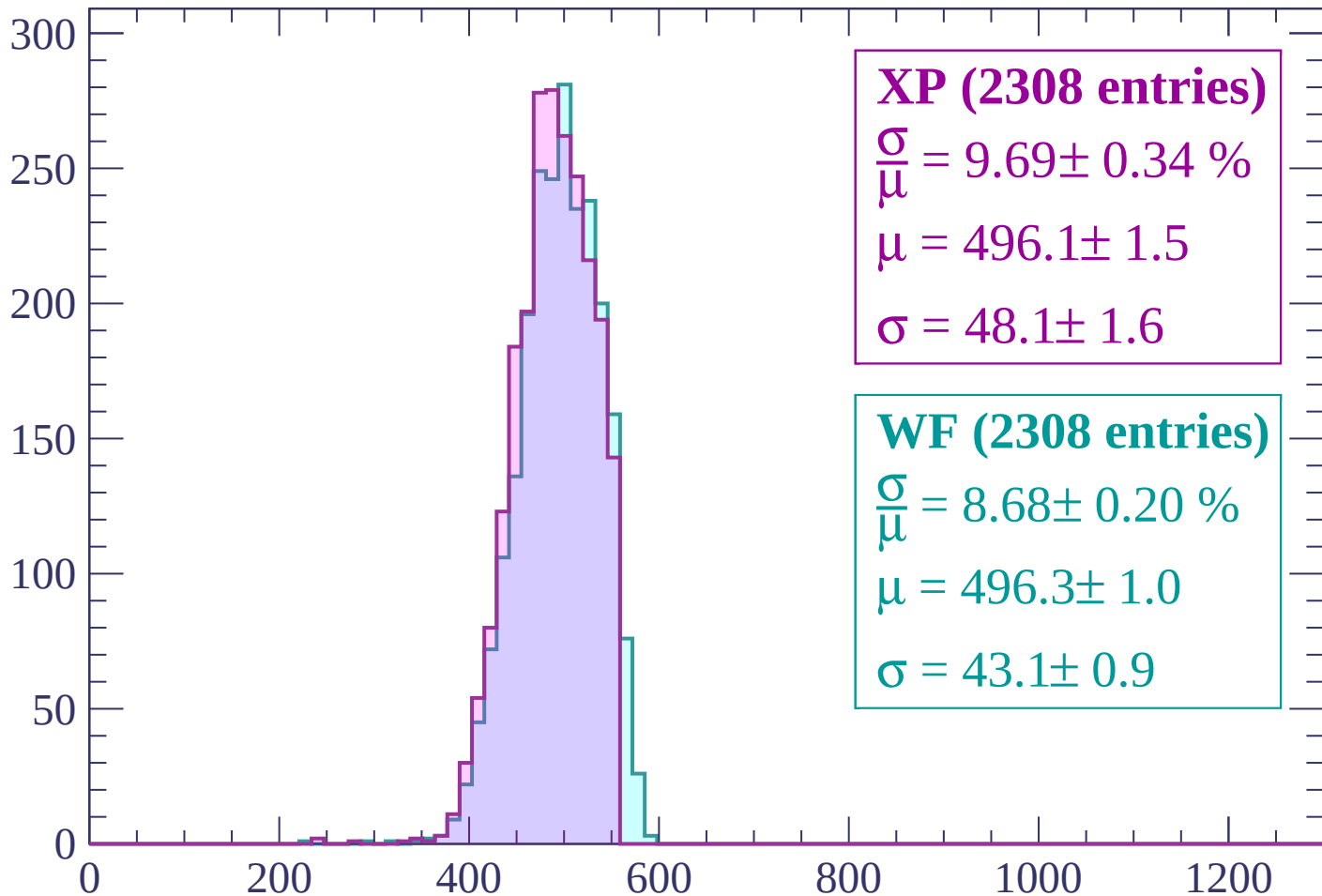
$\mu = 495.7 \pm 1.0$

$\sigma = 37.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (2308 entries)

$\frac{\sigma}{\mu} = 9.69 \pm 0.34 \%$

$\mu = 496.1 \pm 1.5$

$\sigma = 48.1 \pm 1.6$

WF (2308 entries)

$\frac{\sigma}{\mu} = 8.68 \pm 0.20 \%$

$\mu = 496.3 \pm 1.0$

$\sigma = 43.1 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

